

8		7		6		5		4		3		2		1	
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
												10	0008927012	ENGINEERING RELEASED	2017-06-06
D21 MLB - DVT (OK2FAB)															
LAST_MODIFICATION= Tue Jun 6 18:57:04 2017															
PAGE CSA CONTENTS				SYNC		DATE		PAGE CSA CONTENTS				SYNC		DATE	
1	1	TABLE OF CONTENTS		sync	06/19/2016		46	62	I/O: USB PD		sync	04/14/2017			
2	2	SYSTEM: BOM TABLES (1/2)		sync	02/24/2017		47	63	I/O: Hydra		sync	04/14/2017			
3	3	SYSTEM: BOM TABLES (2/2)		test_mlb	03/17/2017		48	64	I/O: B2B DOCK		sync	10/10/2016			
4	4	SYSTEM:MECH COMPONENTS		sync	04/14/2017		49	65	SYSTEM: I2C MAP		sync	04/14/2017			
5	5	SYSTEM: Testpoints		sync	04/14/2017		50	80	RADIOS		D21_MLB_DOE	07/18/2016			
6	6	BOOTSTRAPPING		sync	09/19/2016		51	1	page1						
7	10	SOC: JTAG, USB, XTAL		sync	04/14/2017		52	75	NFC						
8	11	SOC: PCIE		sync	04/14/2017		53	1	FRONT PAGE						
9	12	SOC: MIPI + ISP		sync	04/14/2017		54	2	HIERARCHY						
10	13	SOC: LPDP		sync	04/14/2017		55	3	PMIC: BUCKS & LDOS						
11	14	SOC: SERIAL + SMC		sync	04/14/2017		56	4	PMIC: CLOCKS & CONTROL						
12	15	SOC: GPIO + UART		sync	04/14/2017		57	5	BB: POWER						
13	16	SOC: AOP		sync	04/14/2017		58	6	BB: CONTROL & HS PERIPHERALS [6]						
14	17	SOC: Power (1/3)		sync	04/14/2017		59	7	BB: GPIOs & QLINK						
15	18	SOC: Power (2/3)		sync	04/14/2017		60	8	XCVR						
16	19	SOC: Power (3/3)		sync	04/14/2017		61	9	XCVR						
17	26	NAND		sync	04/14/2017		62	10	QPOET MODULE						
18	27	SYSTEM POWER: PMU Bucks (1/4)		sync	04/14/2017		63	11	LB PAD						
19	28	SYSTEM POWER: PMU Bucks (2/4)		sync	04/14/2017		64	12	HB PAD						
20	29	SYSTEM POWER: PMU LDos (3/4)		sync	04/14/2017		65	13	UHB PAD						
21	30	SYSTEM POWER: PMU (4/4)		sync	04/14/2017		66	14	2G PA						
22	31	SYSTEM POWER: Boost		sync	04/14/2017		67	15	LB DSM						
23	32	SYSTEM POWER: B2B Battery		sync	04/14/2017		68	16	HB DSM						
24	33	SYSTEM POWER: Charger		sync	04/14/2017		69	17	COUPLER2						
25	34	SYSTEM POWER: Iktara		sync	04/14/2017		70	18	LOWER ANTENNA						
26	35	SYSTEM POWER: B2B Cyclone		sync	04/14/2017		71	19	UPPER ANTENNA						
27	36	SENSORS		sync	04/14/2017		72	20	GNSS						
28	37	CAMERA: PMU (1/2)		sync	04/14/2017		73	21	SIM, DEBUG CONN						
29	38	CAMERA: PMU (2/2)		sync	04/14/2017		74	22	TEST POINTS		PROBE POINTS	07/18/2016			
30	39	CAMERA: B2B Wide (KY)		sync	04/14/2017		75	1	FRONT PAGE						
31	40	CAMERA: B2B Tele (TN)		sync	04/14/2017		76	2	METROCIRC						
32	41	CAMERA: Strobe Drivers		sync	04/14/2017		77	3	UAT MATCH AND TUNER						
33	42	CAMERA: FCAM + FOREHEAD		sync	04/14/2017		78	4	RFFE TO GPO TRANSLATOR						
34	43	CAMERA: B2B Strobe + Buttons		sync	04/14/2017		79	1	SymbolPorts						
35	47	AUDIO: CODEC (1/2)		sync	04/14/2017		80	4	Guinness		WIFI	11/28/2016			
36	48	AUDIO: CODEC (2/2)		sync	04/14/2017		81	5	WiFIANTFeeds			11/28/2016			
37	49	AUDIO: Speaker Amp Bottom		sync	04/14/2017										
38	50	AUDIO: Speaker Amp Top		sync	04/14/2017										
39	51	ARC: Driver		sync	04/14/2017										
40	55	CG: Niki		sync	04/14/2017										
41	56	CG: PMUs		sync	04/14/2017										
42	57	CG: B2B DISPLAY + TOUCH		sync	04/14/2017										
43	58	CG: B2B ORB + MESA		sync	04/14/2017										
44	59	I/O: Overvoltage Cut-Off Circuit		sync	04/14/2017										
45	61	I/O: Accessory Buck		sync	04/14/2017										
Sub Designs															
PART#		QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION		SOURCE PROJECT		SUB-DESIGN NAME		VERSION	HARD/ SOFT	SYNC_DATE/TIME		
051-02159		1	SCH, MLB, D21	SCH	?		D21		RADIO_MLB		0.122.0	S	2017_06_06_17:14:31		
020-00846		1	PCBF, MLB, D21	PCB	?		D20		NFC_MLB		0.50.0	S	2017_05_26_07:54:42		
							D21		WIFI_MLB		0.20.0	S	2017_05_16_02:56:32		
							D21		RADIO_MLB_FF		0.35.0	S	2017_06_02_14:56:02		
1 OF 81															

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB, 639-94441, JP, ULTRA)	EEEE_J08R	CRITICAL	EEEE_ULTIMATE
825-7691	1	EEEE FOR (MLB, 639-93377, JP, EXTREME)	EEEE_HNV8	CRITICAL	EEEE_EXTREME
825-7691	1	EEEE FOR (MLB, 639-94442, ROW, ULTRA)	EEEE_J08X	CRITICAL	EEEE_ULTIMATE_ROW
825-7691	1	EEEE FOR (MLB, 639-93389, ROW, EXTREME)	EEEE_HNVF	CRITICAL	EEEE_EXTREME_ROW

D21x Specific

D21x Specific

D21x Specific

D21x Specific

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00416	1	SoC, H, 3GB 23nm DDR, B1	U1000	CRITICAL	COMMON

D21x Specific

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00415	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00417	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00418	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 1xnm DDR, B1
339S00429	339S00416	BOM_TABLE_ALTS	U1000	SoC, H, 3GB 20nm DDR, B1
339S00430	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00431	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00432	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 18nm DDR, B1

SIPs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339M00011	339M00010	BOM_TABLE_ALTS	M5500	NIKI, STATS SIP
339M00013	339M00012	BOM_TABLE_ALTS	M4100	HUNT, STATS SIP

D21x Specific

SOFT-TERM CAP SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00152	685-00151	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM, MLB, CAP, SOFT, D21
138S00049	138S0031	TYPICAL_CAP	ALL	CAP, R0201, 2.2UF, 6.3V, KYOCERA

To enable alt in SUBBOM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00151	1	SUBBOM, MLB, SOFT-TERM, D21	SUBBOM_CAP		
138S00159	9	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2000, C2001, C2003, C2005, C2006, C2007, C2008, C2013		SOFT_CAP
138S0031	9	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2000, C2001, C2003, C2005, C2006, C2007, C2008, C2013		TYPICAL_CAP
138S00159	15	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2051, C2052, C2054, C2055, C2062, C2066, C2068		SOFT_CAP
138S0031	15	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2051, C2052, C2054, C2065, C2062, C2066, C2068		TYPICAL_CAP
138S00159	10	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3070, C3073, C3074, C3076, C3078, C3071, C3080, C3081		SOFT_CAP
138S0031	10	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3070, C3073, C3074, C3076, C3078, C3071, C3080, C3081		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3351, C3352		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3351, C3352		TYPICAL_CAP
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3602, C3611, C3622		SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3602, C3611, C3622		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3791, C3795, C3798		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3791, C3795, C3798		TYPICAL_CAP
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3909, C3925, C3926		SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3909, C3925, C3926		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4250, C4261		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4250, C4261		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4303		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4303		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4805, C4809, C4914, C4926, C4929		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4805, C4809, C4914, C4926, C4929		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5016, C5018, C5134, C5029, C5136		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5016, C5018, C5134, C5029, C5136		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5611		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5611		TYPICAL_CAP
138S00159	4	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5700, C5705, C5716, C5721		SOFT_CAP
138S0031	4	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5700, C5705, C5716, C5721		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5802, C5804, C5811, C5808		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5802, C5804, C5811, C5808		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5890		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5890		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C6320, C6321, C6322, C6323, C6324		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C6320, C6321, C6322, C6323, C6324		TYPICAL_CAP
138S00160	4	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C4907, C4903, C4904, C4931		SOFT_CAP
138S0979	4	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KYO	C4907, C4903, C4904, C4931		TYPICAL_CAP
138S00160	5	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C5600, C5602, C5603, C5702, C5703		SOFT_CAP
138S0979	5	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KYO	C5600, C5602, C5603, C5702, C5703		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4825, C4826		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4825, C4826		TYPICAL_CAP

Default is typical

D21x Specific

D21x Specific

IKTARA DIODES SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00172	685-00171	BOM_TABLE_ALTS	SUBBOM_DS	SUBBOM, MLB, SCHOTTKY DIODES, D21

#31526972: Update Iktara Diodes APNs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00171	1	SUBBOM, MLB, SCHOTTKY DIODES, D100ES, D21	SUBBOM_DS		
371S00133	4	DIODES, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403		DIODES_DS
371S00132	4	ONSEMI, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403		ONSEMI_DS

Footprint is ONSEMI

Primary is D100ES

NAND BOM Options

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00287	1	NAND_H, ULTRA, TLC	U2000	CRITICAL	ULTIMATE
335S00240	1	NAND_H, EXTREME, TLC	U2000	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00284	335S00287	BOM_TABLE_ALTS	U2000	T, 1z, ULTIMATE
335S00285	335S00287	BOM_TABLE_ALTS	U2000	T, B1CS3, ULTIMATE
335S00288	335S00287	BOM_TABLE_ALTS	U2000	S, B1CS3, ULTIMATE
335S00286	335S00287	BOM_TABLE_ALTS	U2000	SS, 30v4, ULTIMATE
335S00228	335S00240	BOM_TABLE_ALTS	U2000	T, B1CS3, EXTREME
335S00247	335S00240	BOM_TABLE_ALTS	U2000	S, B1CS3, EXTREME
335S00276	335S00240	BOM_TABLE_ALTS	U2000	SS, 30v4, EXTREME

Isolator Switch

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S00126	311S00114	BOM_TABLE_ALTS	ALL	SPDT, TI

Mamba LDO

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00932	353S00976	BOM_TABLE_ALTS	U5090	STMICRO LDO

Load Switch

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00999	353S01039	BOM_TABLE_ALTS	U6100	TI, LDO SW, TPS22915C
353S01007	353S01039	BOM_TABLE_ALTS	U6100	ONSEMI, LDO SW, MCP333

Crystal Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0012	197S00118	BOM_TABLE_ALTS	Y1000	NDK, XTAL, 24MHz, 1.6x1.2
197S00120	197S00118	BOM_TABLE_ALTS	Y1000	EPSON, XTAL, 24MHz, 1.6x1.2

CRITICAL PART#	COMMENT
197S00118	XTAL, 24M, 1612, TDK
197S0012	XTAL, 24M, 1612, NDK
197S00120	XTAL, 24M, 1612, EPSON

PMID1 Zener Diode

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S00121	371S00030	BOM_TABLE_ALTS	DZ3300	DIODE, ZENER, 20V

ZRB Caps

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00083	1	1uF, 25V, 0402, X5R, 10%	C3001	CRITICAL	COMMON

CRITICAL PART#	COMMENT
138S00185	MUR, CAP, ZRB, 2.2UF, 25V, 0402
138S00106	MUR, CAP, ZRB, 4.7UF, 25V, 0603

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S00764	118S0717	BOM_TABLE_ALTS	ALL	RES, 3.92K, 0.1%, 0201
138S00648	138S00652	BOM_TABLE_ALTS	ALL	CAP, X5R, 4.7UF, 6.3V, 0.60MM, 0402, TAIYO
138S00024	138S00986	BOM_TABLE_ALTS	ALL	CAP, CER, 3.3UF, 1.5V, 20%, 0402, TAIYO/TK
138S00706	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 10UF, 20%, 16V, X5R, 0502, MURATA
138S00945	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 10UF, 20%, 16V, X5R, 0502, KYOCERA
138S00739	138S0706	BOM_TABLE_ALTS	ALL	CAP, CER, X5R, 0.220F, 20%, 6.3V, 20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP, X5R, X5R, 0.220F, 20%, 6.3V, 0505
138S00133	138S00128	BOM_TABLE_ALTS	ALL	CAP, CER, X5R, 0.47UF, 20%, 6.3V, 01005
138S00148	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 15UF, 4V, KYOCERA
138S00149	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 14UF, 4V, MURATA
138S00151	138S00150	BOM_TABLE_ALTS	ALL	CAP, 0402, 15UF, 4V, TAIYO
138S00144	138S00143	BOM_TABLE_ALTS	ALL	CAP, 0402, 20UF, 4V, MURATA
138S00143	138S00144	BOM_TABLE_ALTS	ALL	CAP, 0402, 22UF, 4V, KYOCERA
138S00139	138S00138	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 4V, MURATA
138S00138	138S00139	BOM_TABLE_ALTS	ALL	CAP, 0201, 4.7UF, 4V, KYOCERA
138S00145	138S00146	BOM_TABLE_ALTS	ALL	CAP, 0402, 20UF, 6.3V, KYOCERA
138S00141	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 6.3V, MURATA
138S00142	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, SAMSUNG
138S00142	138S00141	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, SAMSUNG
138S00049	138S00031	BOM_TABLE_ALTS	ALL	CAP, 0201, 2.2UF, 6.3V, KYOCERA
138S00163	138S00143	BOM_TABLE_ALTS	ALL	CAP, 0402, 22UF, 4V, TY
138S00164	138S00138	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 4V, TY
138S00165	138S00146	BOM_TABLE_ALTS	ALL	CAP, 0402, 20UF, 6.3V, TY
138S00166	138S00140	BOM_TABLE_ALTS	ALL	CAP, 0201, 4UF, 6.3V, TY
138S00140	138S00141	BOM_TABLE_ALTS	ALL	CAP, 0201, 3UF, 6.3V, KYO
138S00048	138S00003	BOM_TABLE_ALTS	ALL	CAP, X5R, 150F, 6.3V, 0.60MM, 0402, KYO
132S0639	132S00088	BOM_TABLE_ALTS	ALL	CAP, X5R, 0.47UF, 25V, 0.20MM, 0502, MUR

CRITICAL PART#	COMMENT
118S00717	RES, 3.92K, 0.1%, 0201
138S00652	CAP, X5R, 4.7UF, 6.3V, 0.65MM, 0402
138S00706	CAP, CER, X5R, 1UF, 20%, 6.3V, 0201
132S0400	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 01005
138S00031	CAP, CER, X5R, 2.2UF, 20%, 6.3V, 0201
138S00128	CAP, CER, X5R, 0.47UF, 20%, 6.3V, 01005

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

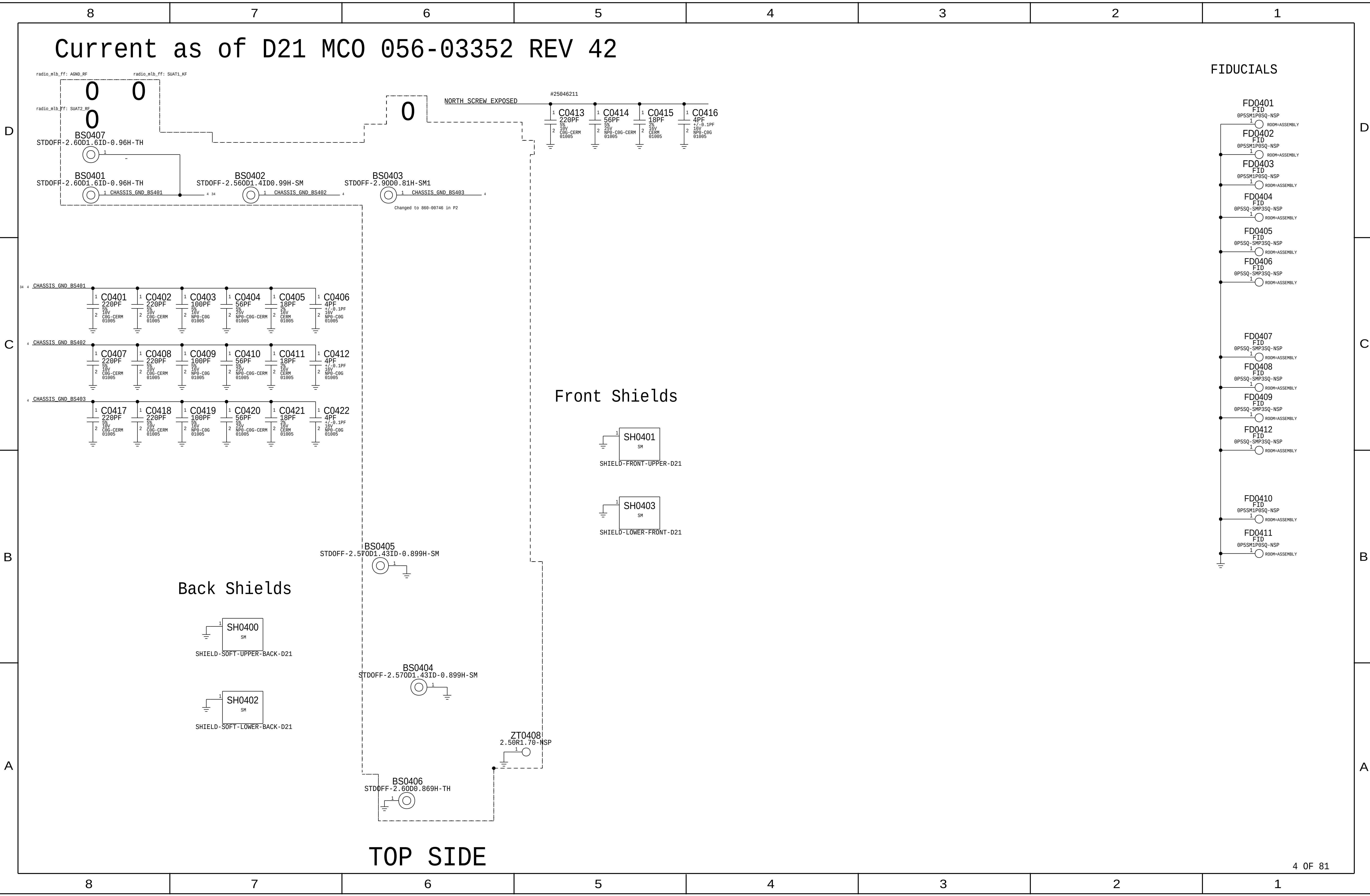
CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

Multi-Vendor Criticals

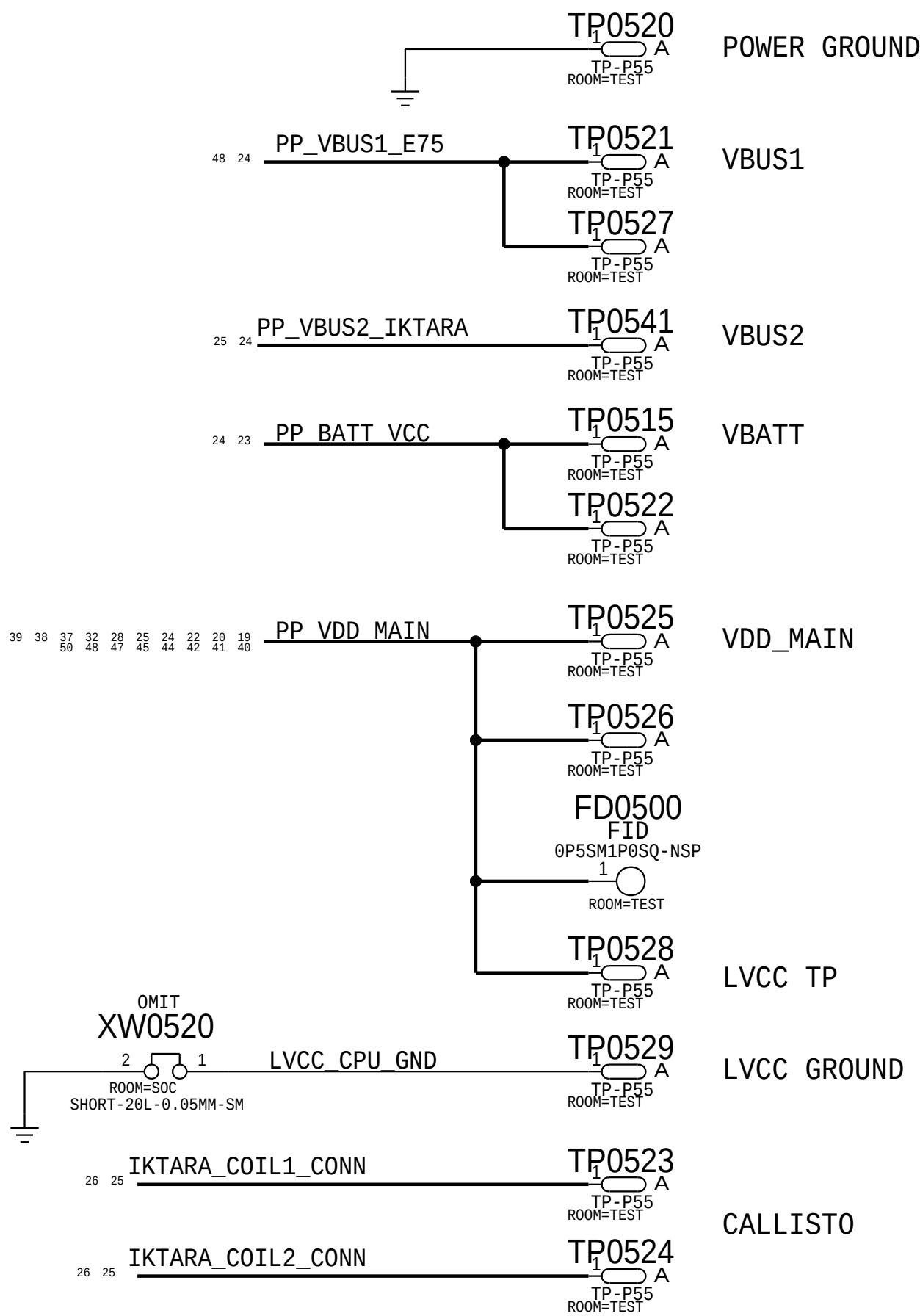
CRITICAL PART#	COMMENT	CRITICAL PART#	COMMENT
377S0106	SUPPR, TRANS, VARISTOR, 12V, 33PF, 01005	132S0208	CAP, CER, X5R, 0.1UF, 10%, 16V, 0201
197S0446	XTAL, 24MHZ, 30PPM, 9.5PF, 00 OHM MAX, 1612	132S0275	CAP, CER, X5R, 470PF, 10%, 10V, 01005
155S00576	FERR BD, 10 OHM, 50%, 750MA, 0.07 DCR, 01005	132S0249	CAP, CER, X7R, 220PF, 10%, 10V, 01005
155S00168	FLTR, NOISE, 65 OHMZ, 3.40MM, 0.7-20HZ, 0605	132S0245	CAP, CER, X5R, 0.01UF, 10%, 6.3V, 01005
130S00979	CAP, CER, X5R, 10UF, 20%, 10V, 0402, H=0.65MM	132S00093	CAP, X5R, 0.022UF, 20%, 6.3V, 01005
138S00692	CAP, CER, X5R, 1UF, 20%, 6.3V, 0201	132S00025	CAP, CER, X5R, 0.047UF, 20%, 6.3V, 01005
138S00683	CAP, CER, X5R, 1UF, 10%, 25V, 0402	132S00008	CAP, CER, 0.1UF, 10%, 50V, X7R, 0402
138S0652	CAP, CER, X5R, 4.7UF, 20%, 6.3V, H=0.65MM, 0402	131S00083	CAP, CER, NP0/C0G, 220PF, 2%, 50V, 0201
130S00070	CAP, X5R, 4.7UF, 20%, 25V, 0402	131S00004	CAP, CER, 27PF, 5%, C0G, 25V, 0201
138S00014	CAP, CER, 1UF, 20%, 16V, X5R, 0201, H=0.39MM	131S00307	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005
132S00664	CAP, CER, 0.047UF, 10%, 25V, X5R, 0201	131S02225	CAP, CER, NP0/C0G, 15PF, 5%, 16V, 01005
132S00663	CAP, CER, X5R, 1UF, 10%, 25V, 0402	131S02223	CAP, CER, NP0/C0G, 12PF, 5%, 16V, 01005
132S00534	CAP, CER, X5R, 0.1UF, 10%, 25V, 0201	131S02220	CAP, CER, NP0/C0G, 12PF, 5%, 16V, 01005
132S0436	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 01005	131S02216	CAP, CER, NP0/C0G, 47PF, 5%, 16V, 01005
132S0396	CAP, CER, X5R, 1000PF, 10%, 10V, 01005	131S00053	CAP, CER, C0G, 220PF, 5%, 10V, 01005
132S0316	CAP, CER, X5R, 0.1UF, 20%, 6.3V, 01005	118S00068	RES, MF, 1.3 MOHM, 1%, 200PPM, 1/20W, 0201
132S0304	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 0201	117S00055	RES, MF, 1/20W, 2M OHM, 5, 0201, SMD
132S0296	CAP, CER, X5R, 1000PF, 10%, 6.3V, 01005	107S0257	THERMISTOR, NTC, 10K OHM, 1%, B=3435, 01005

Current as of D21 MC0 056-03352 REV 42

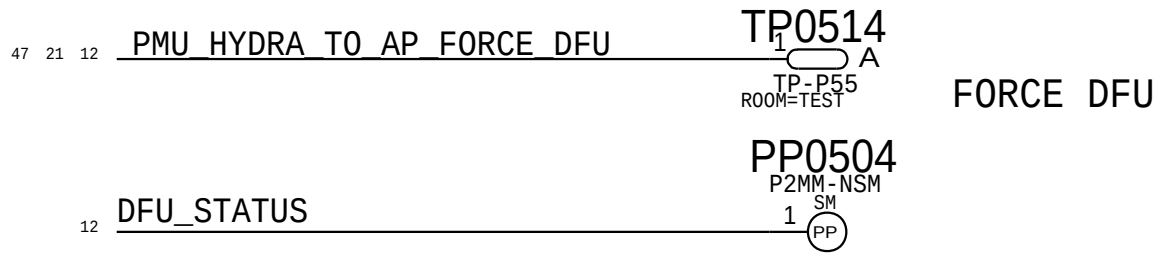


Test Points

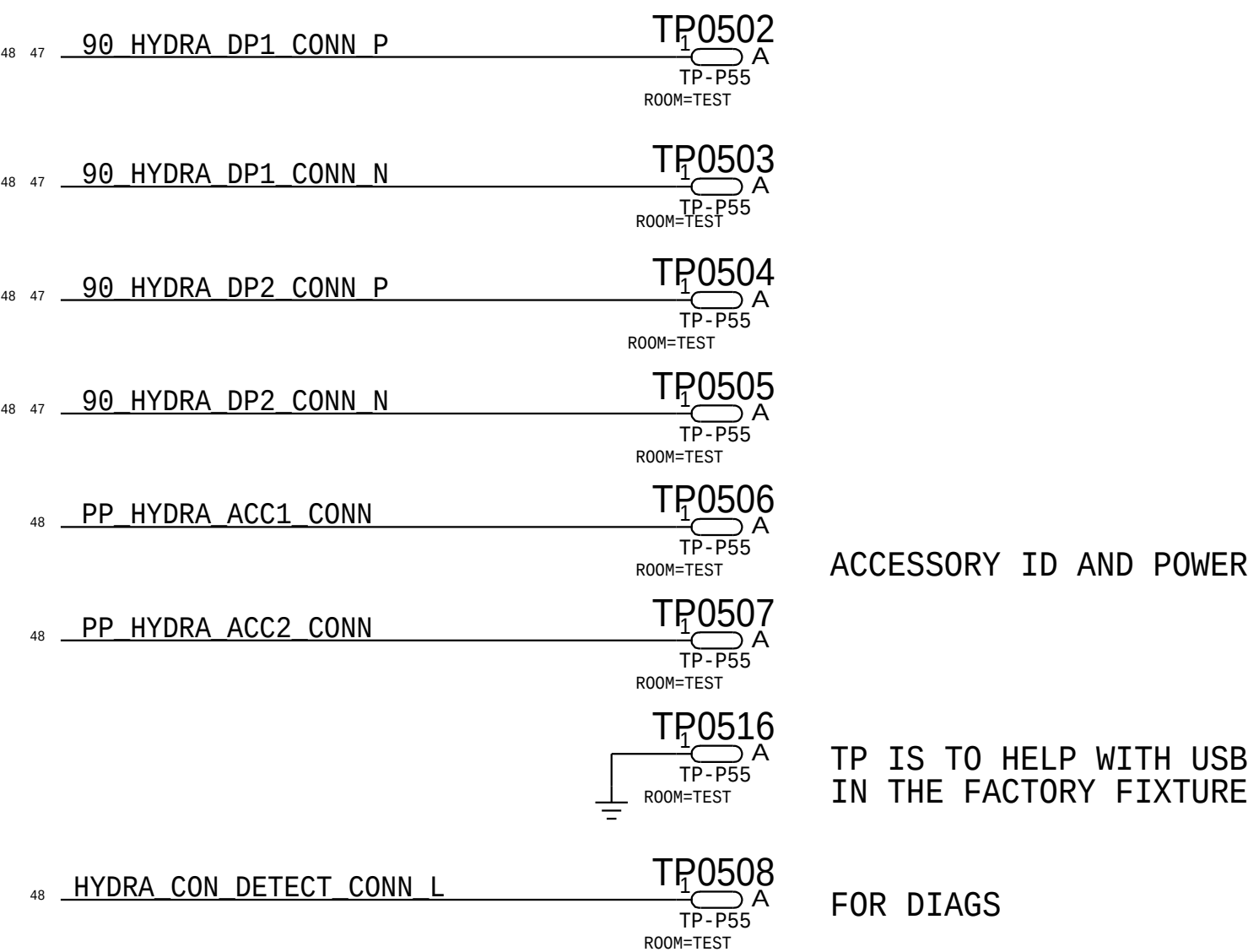
POWER



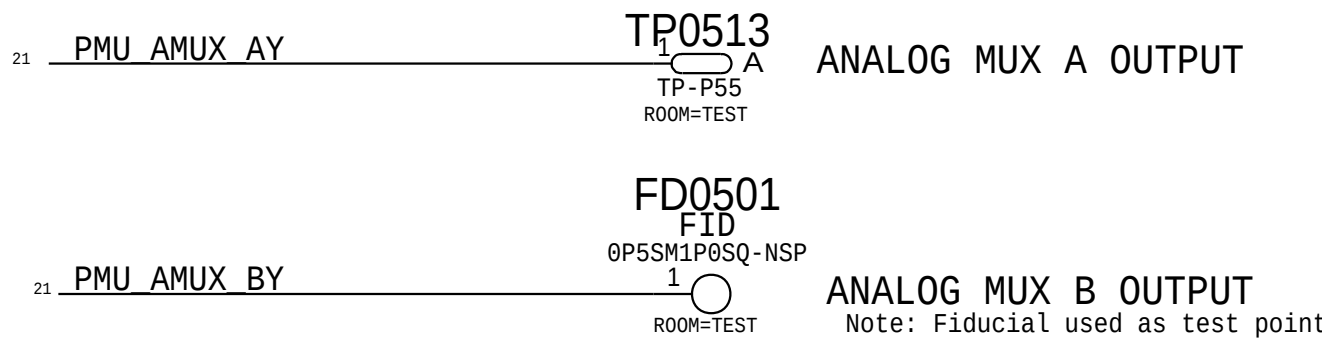
DFU



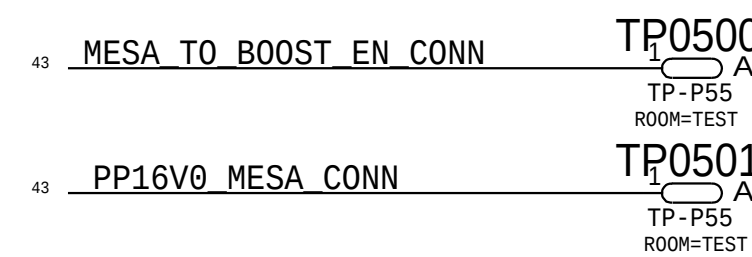
E75



AMUX



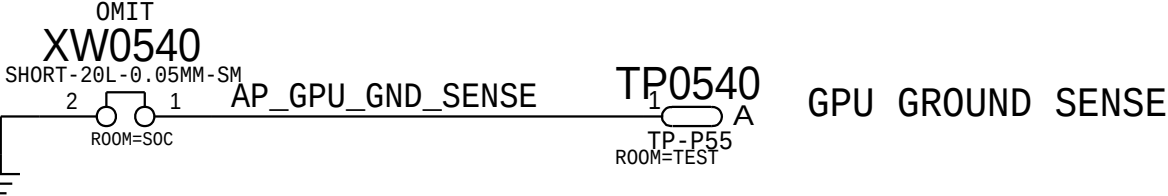
MOJAVE



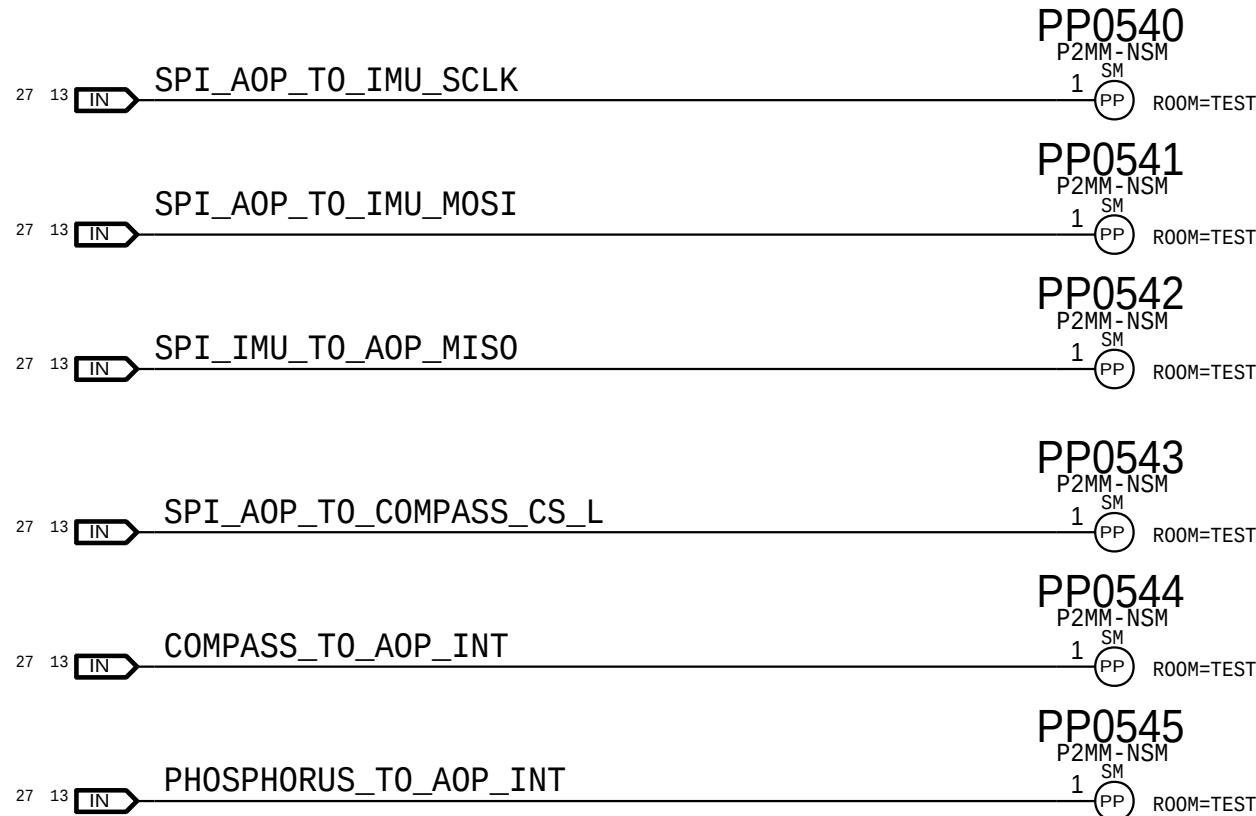
LCM



SENSE TP

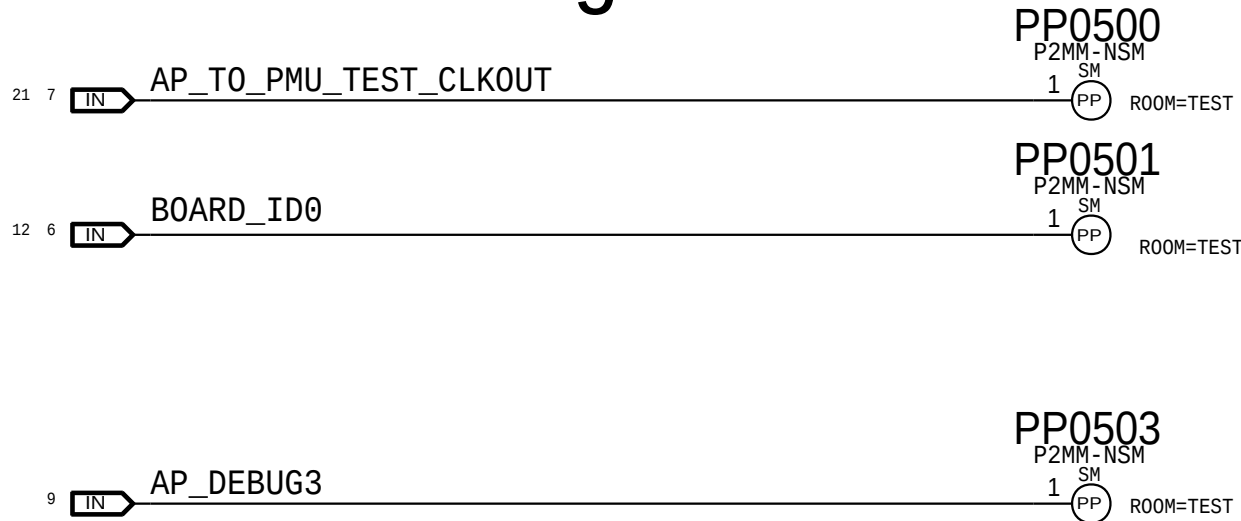


Sensor SPI

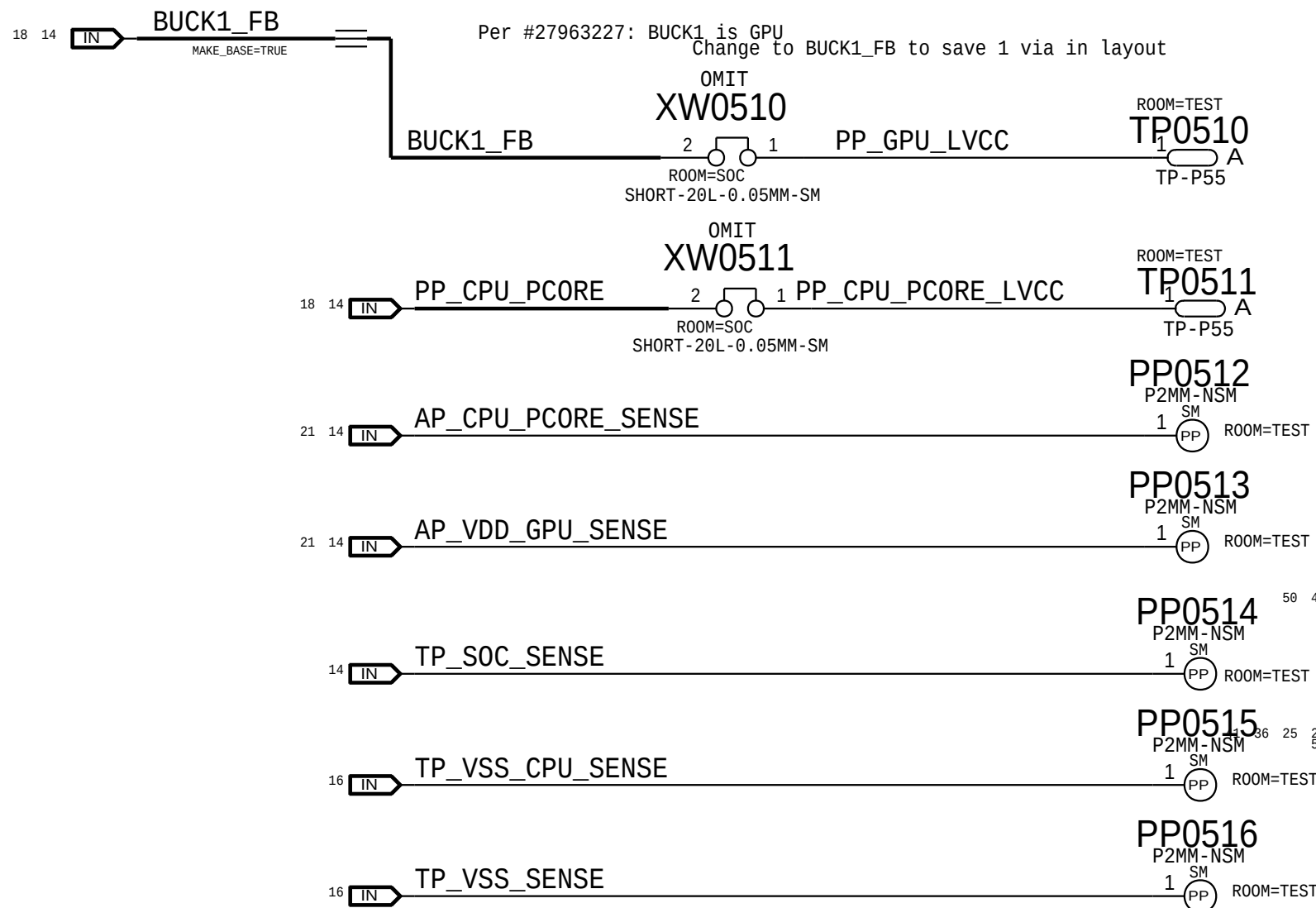


Probe Points

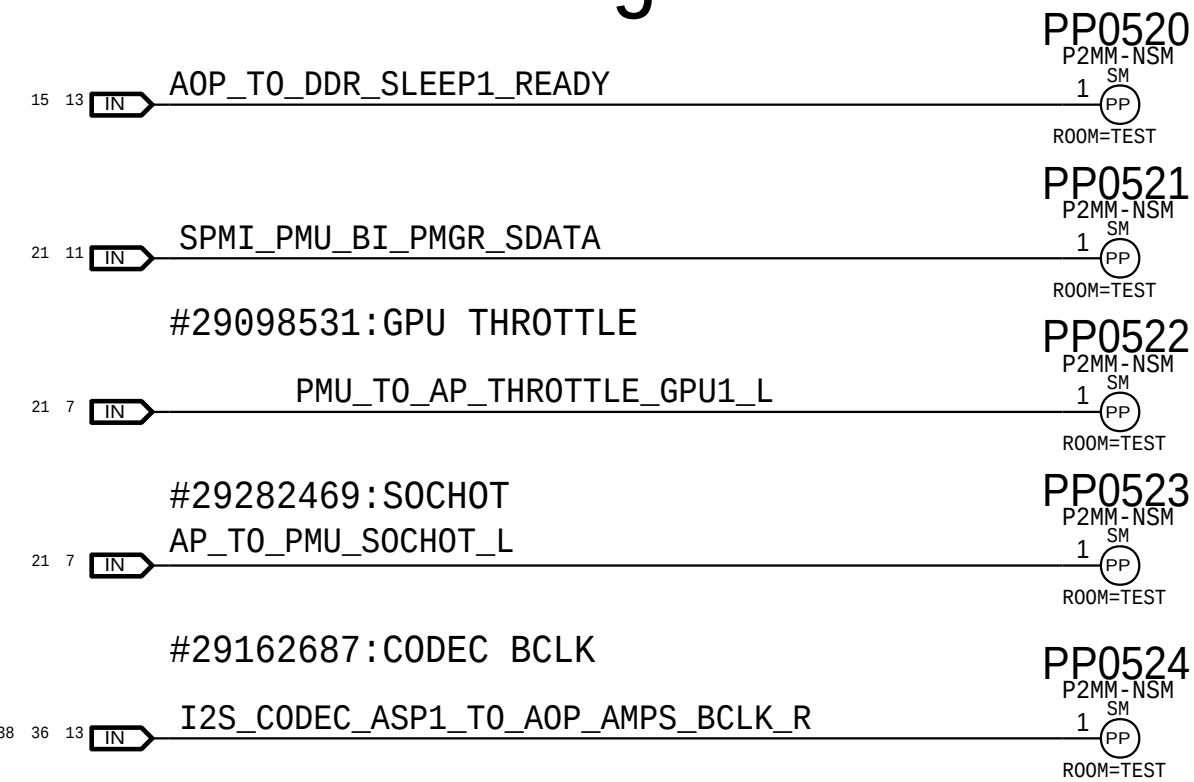
SOC Debug



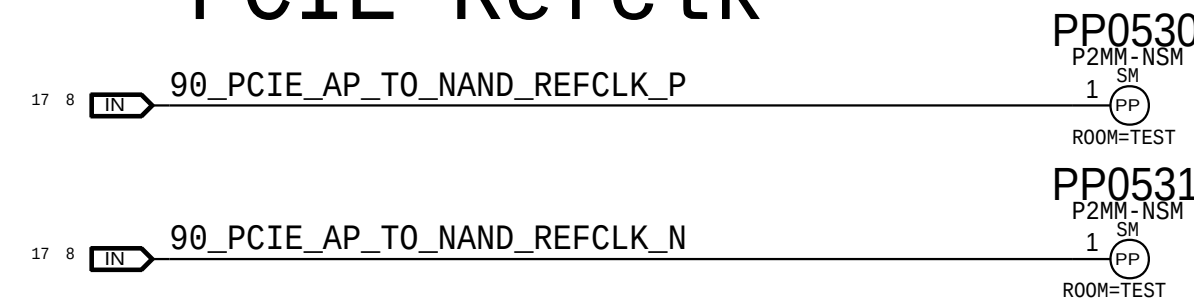
SOC CPU/GPU



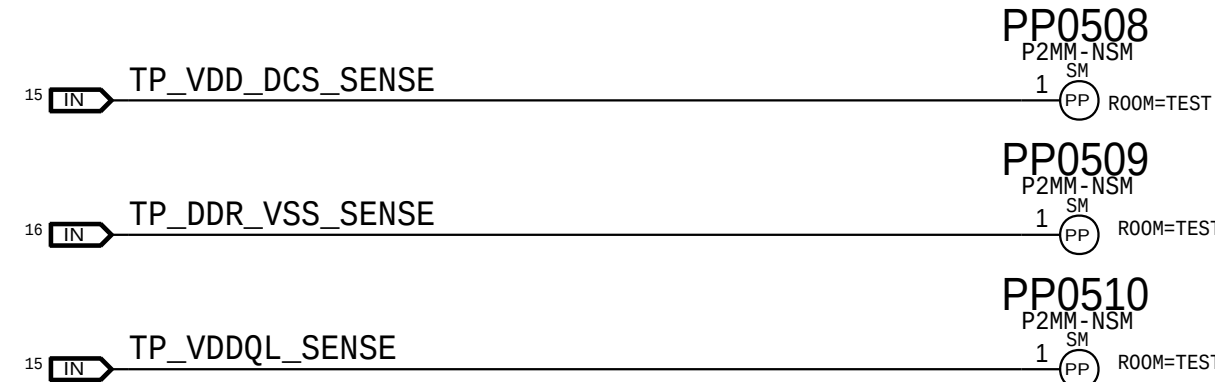
PMU Debug



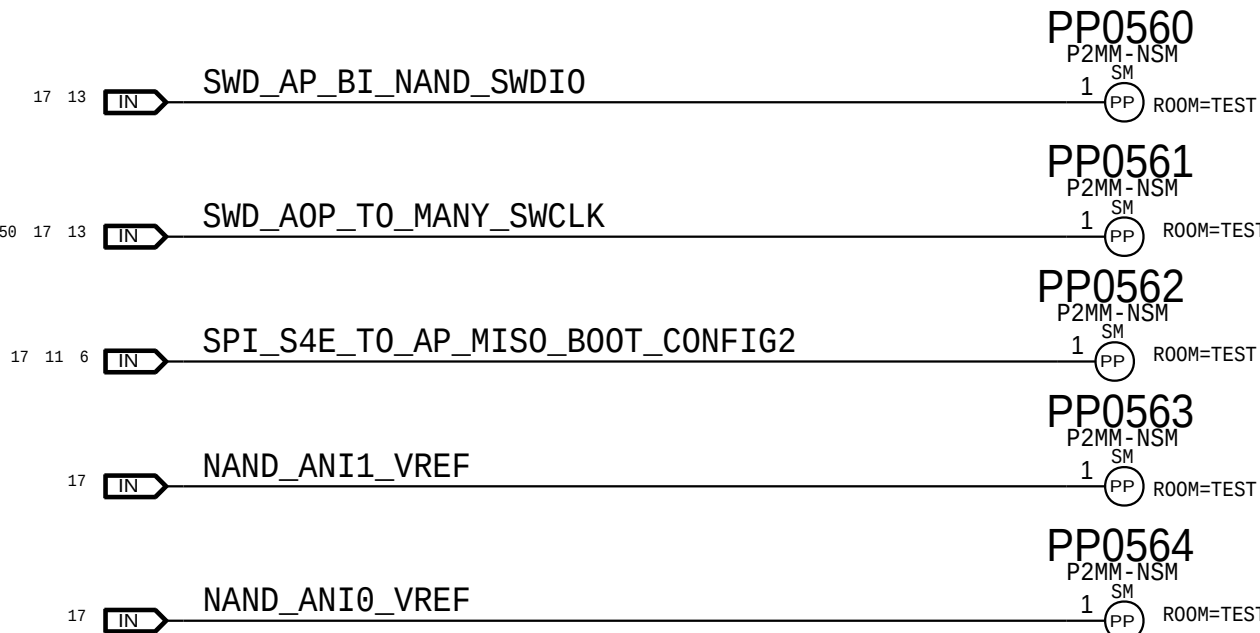
PCIE Refclk



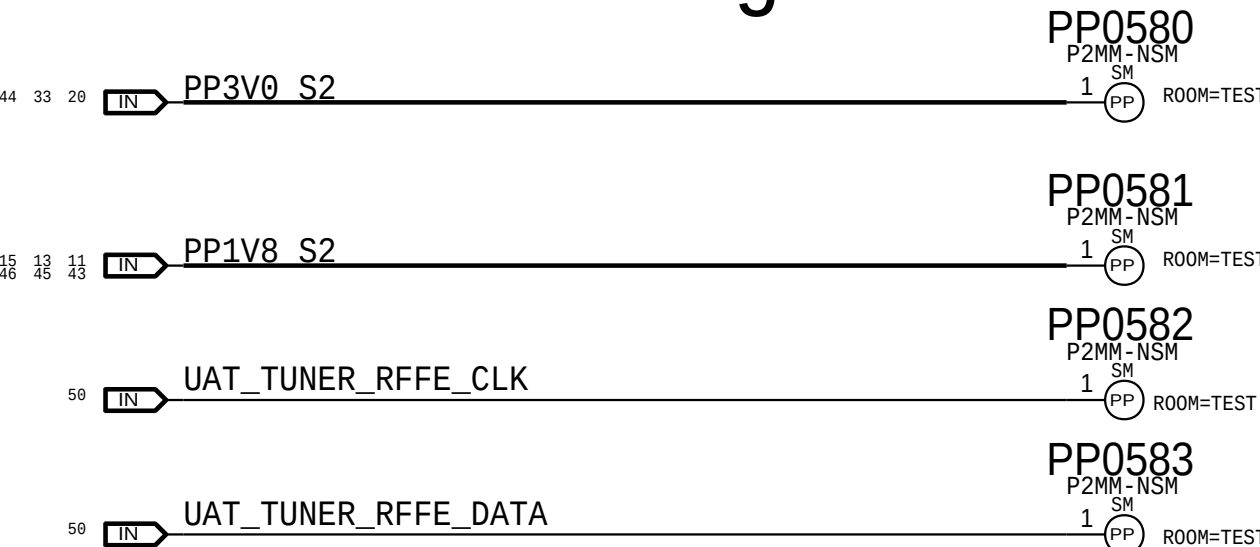
SOC DDR



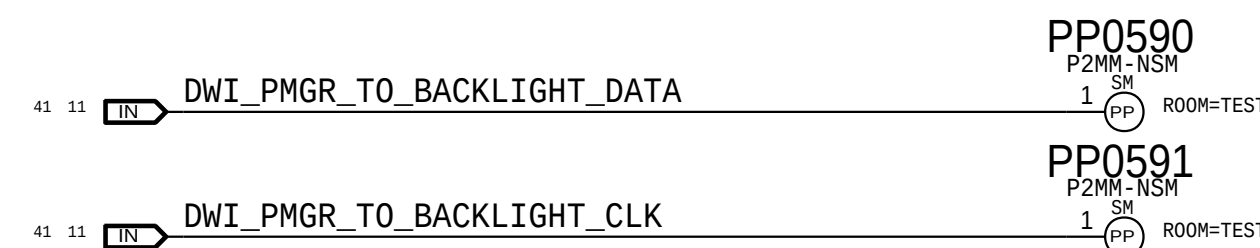
NAND Debug



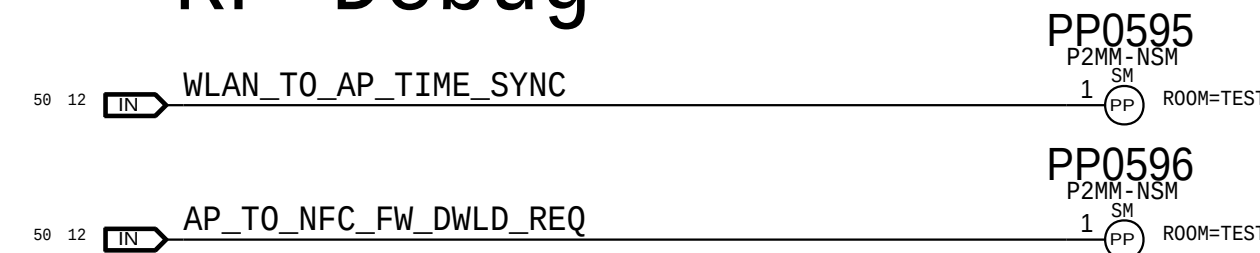
BB UAT Debug

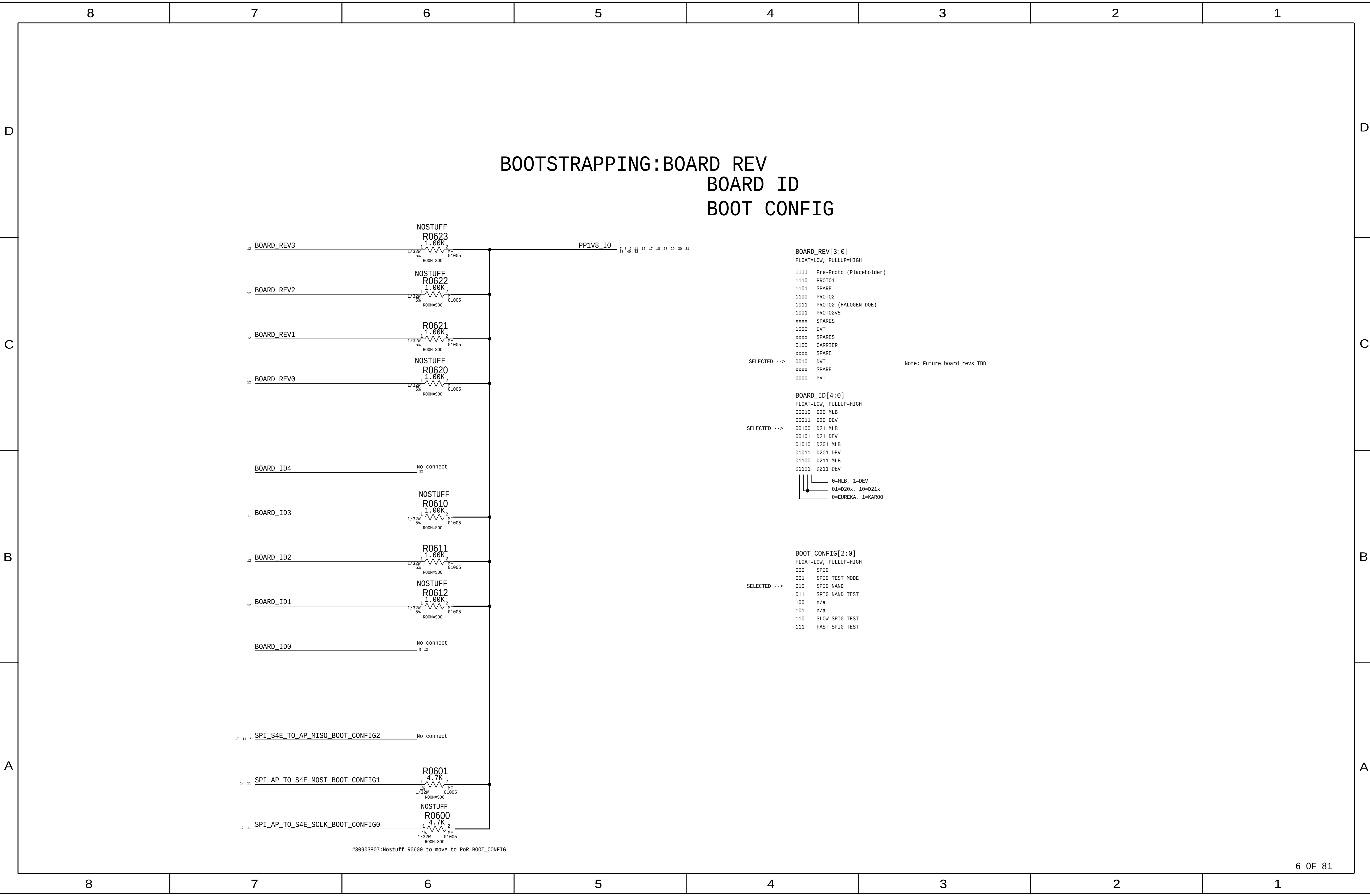


BACKLIGHT Debug

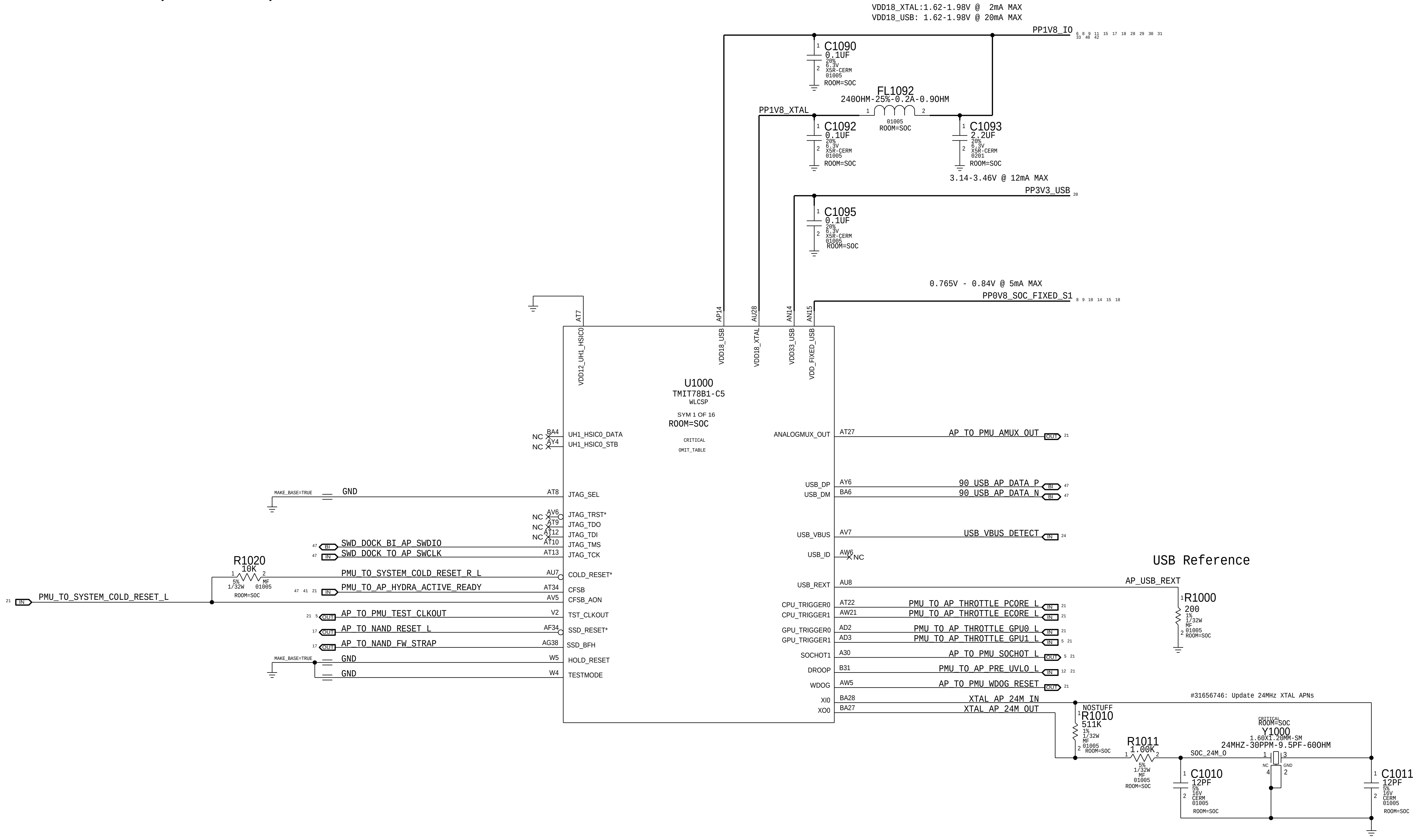


RF Debug

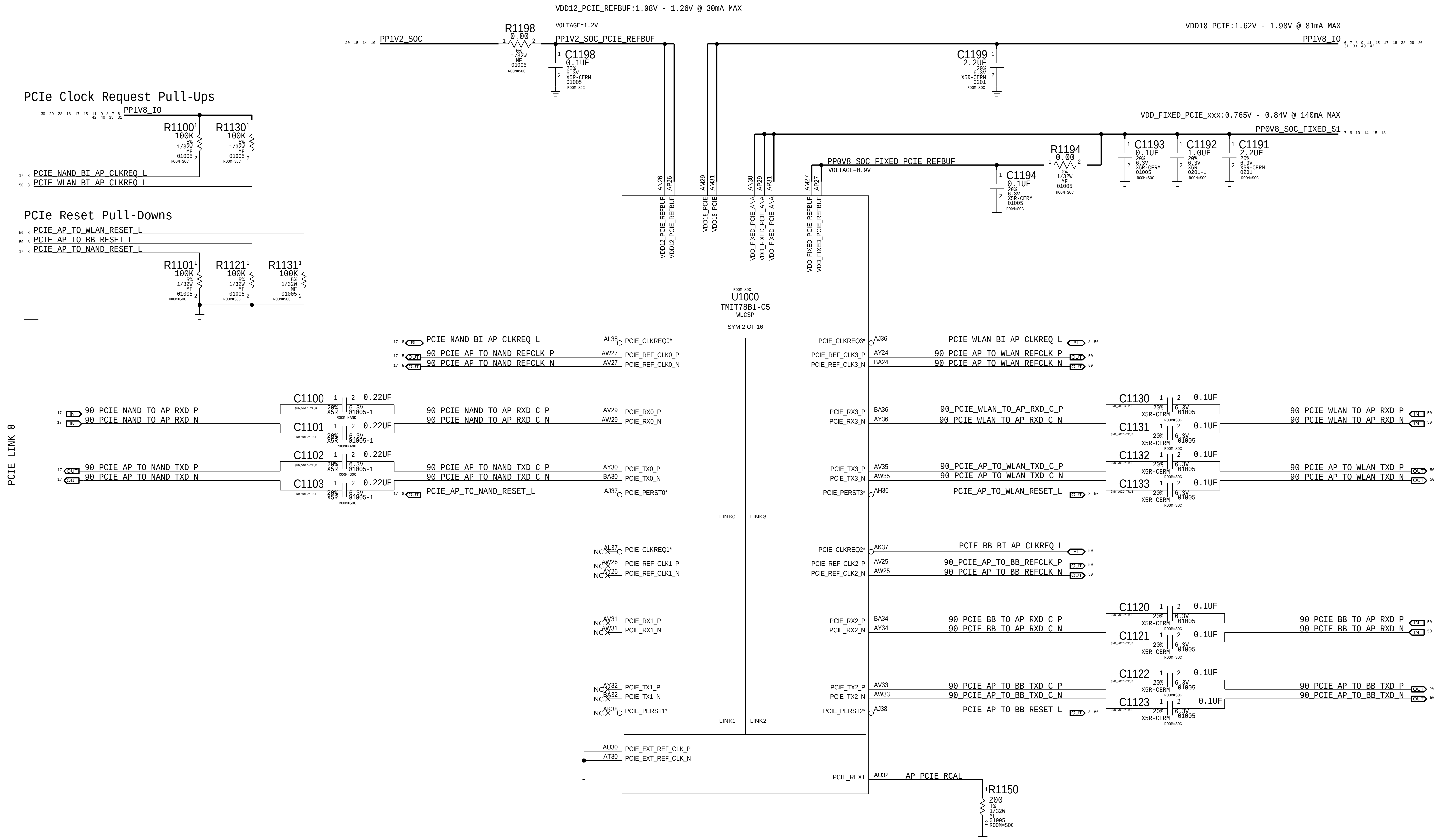




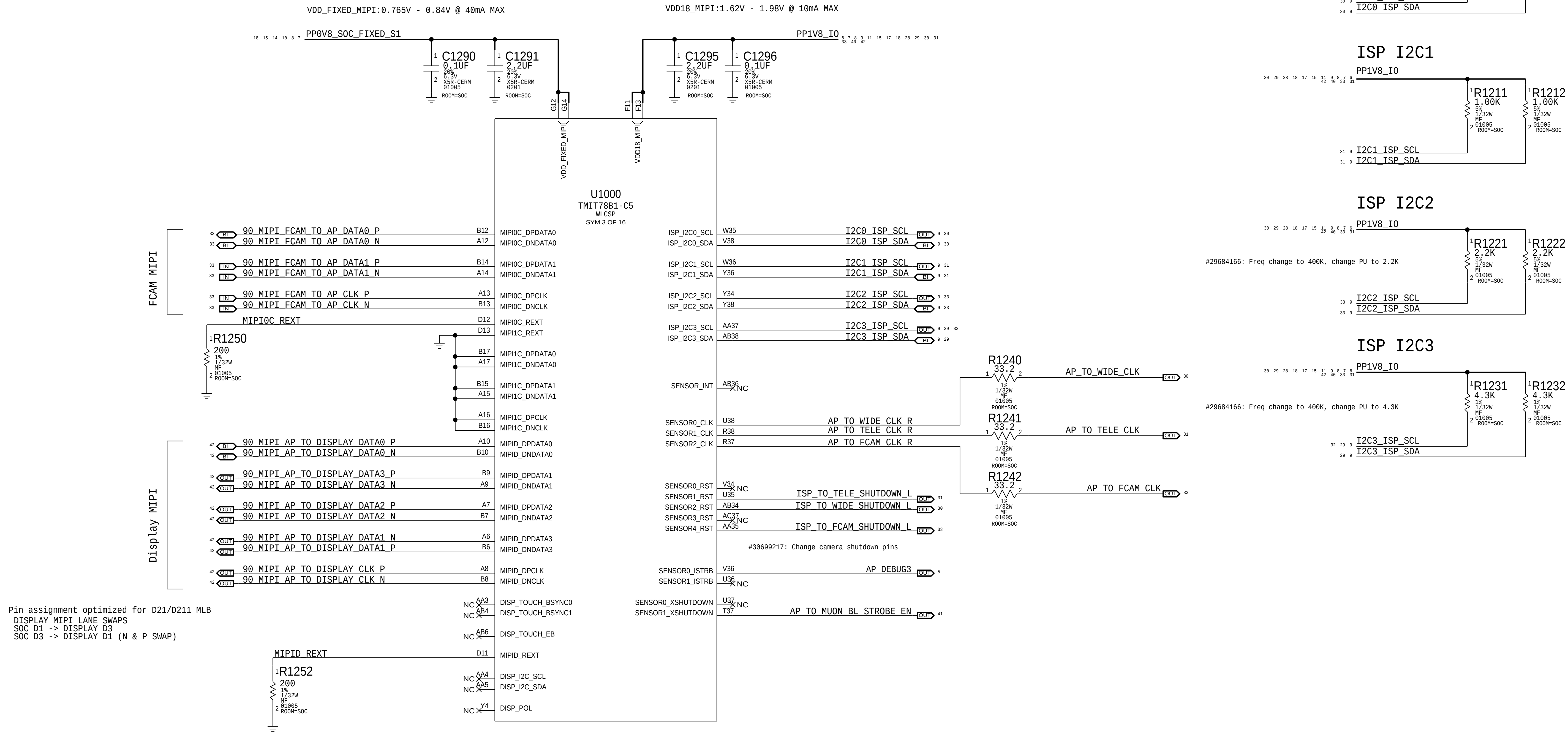
SOC - USB, JTAG, XTAL



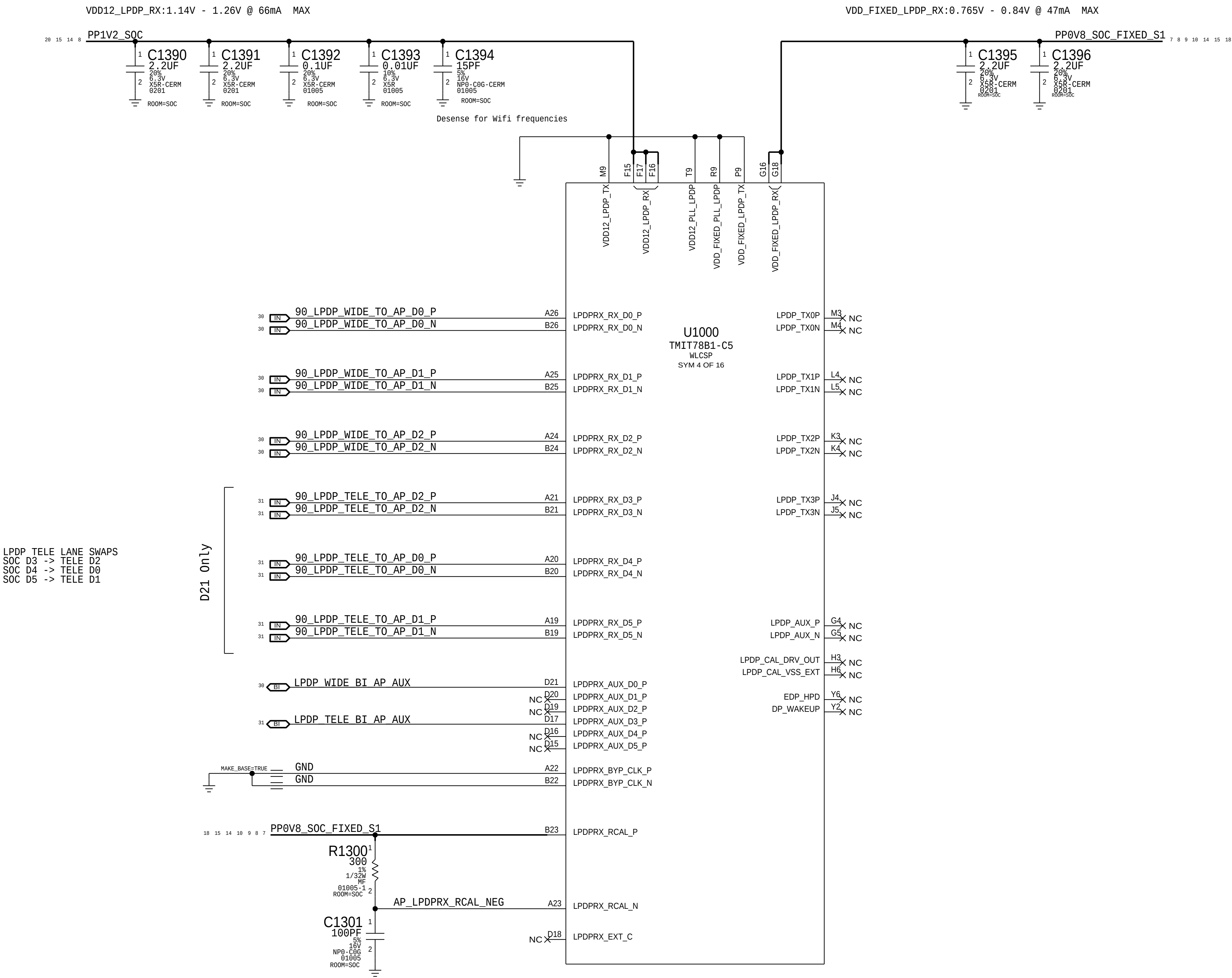
SOC - PCIE INTERFACES



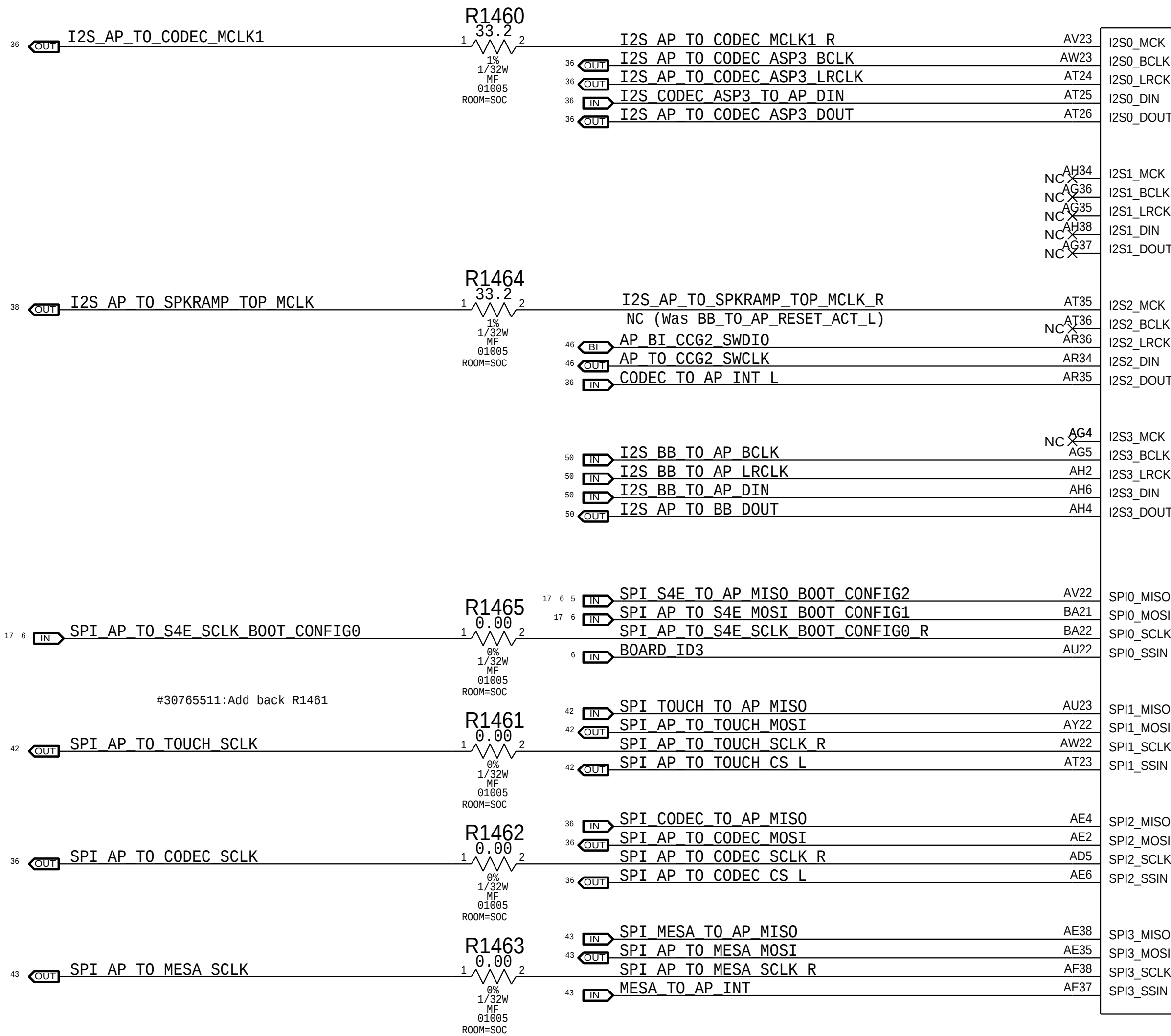
SOC - MIPI & ISP INTERFACES



SOC - LPDP INTERFACES

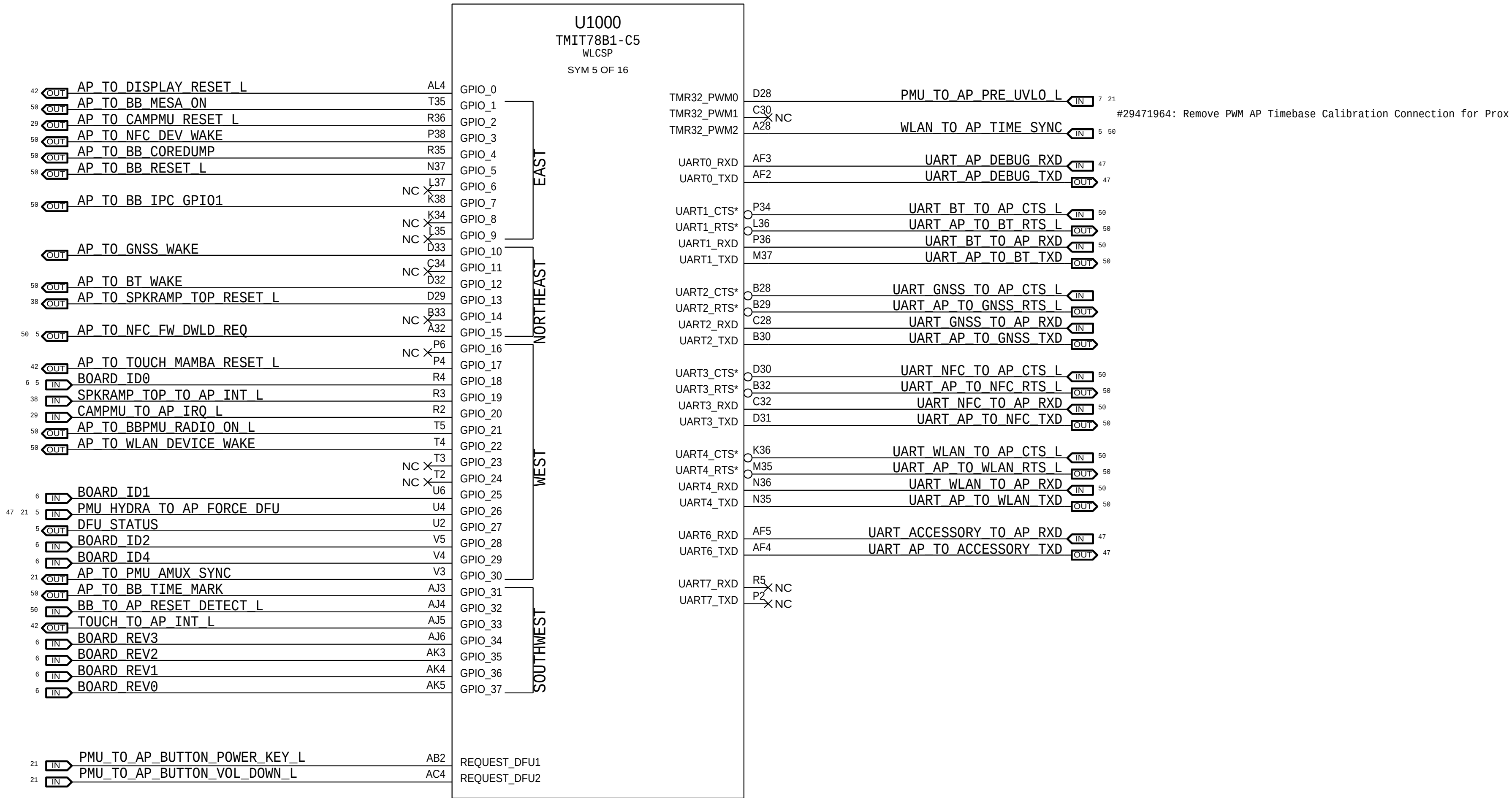


SOC - SERIAL INTERFACES



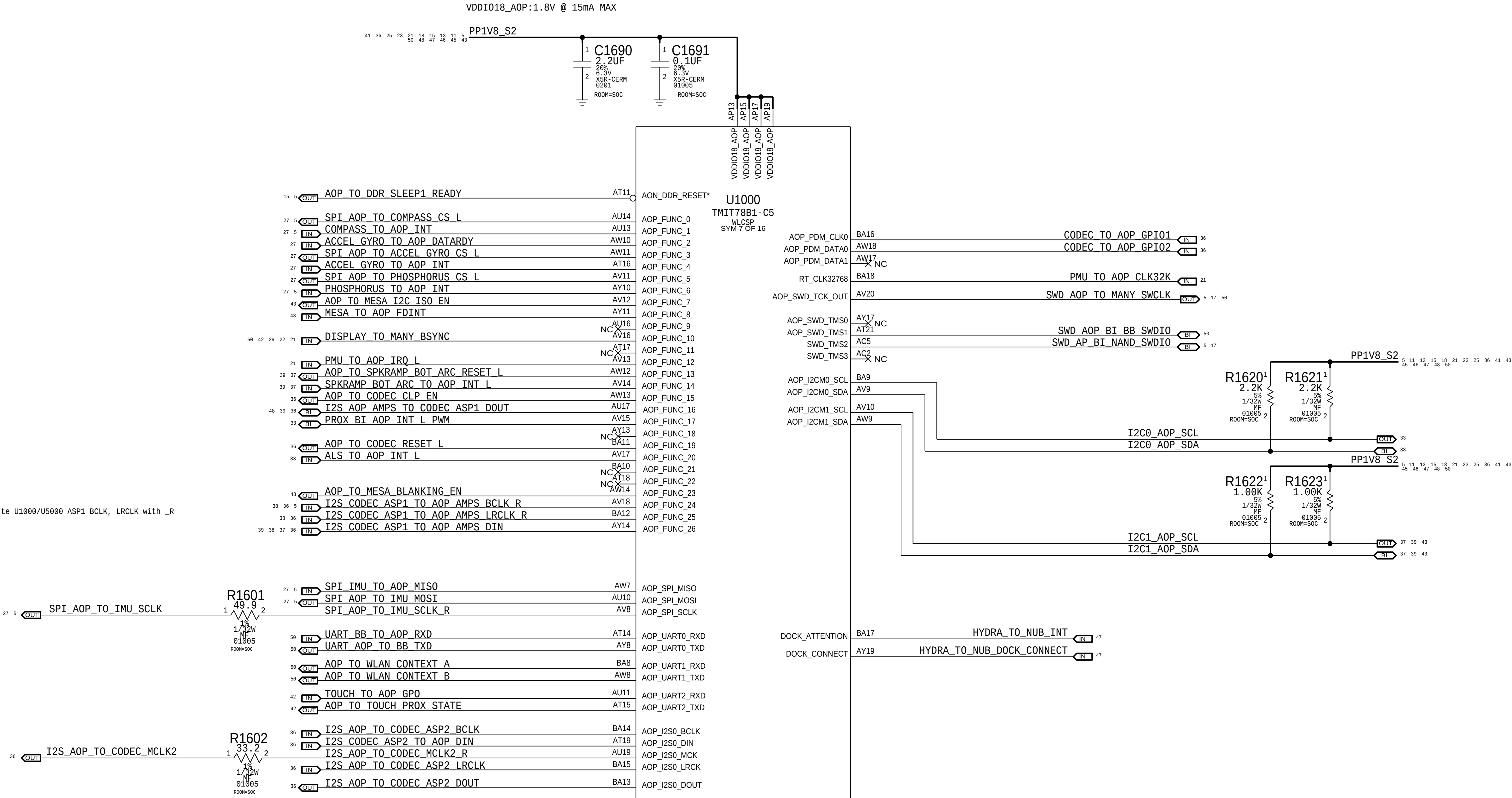
SPI: Route as Daisy-Chain (No T's Allowed)

SOC - GPIO INTERFACES



SOC - AOP

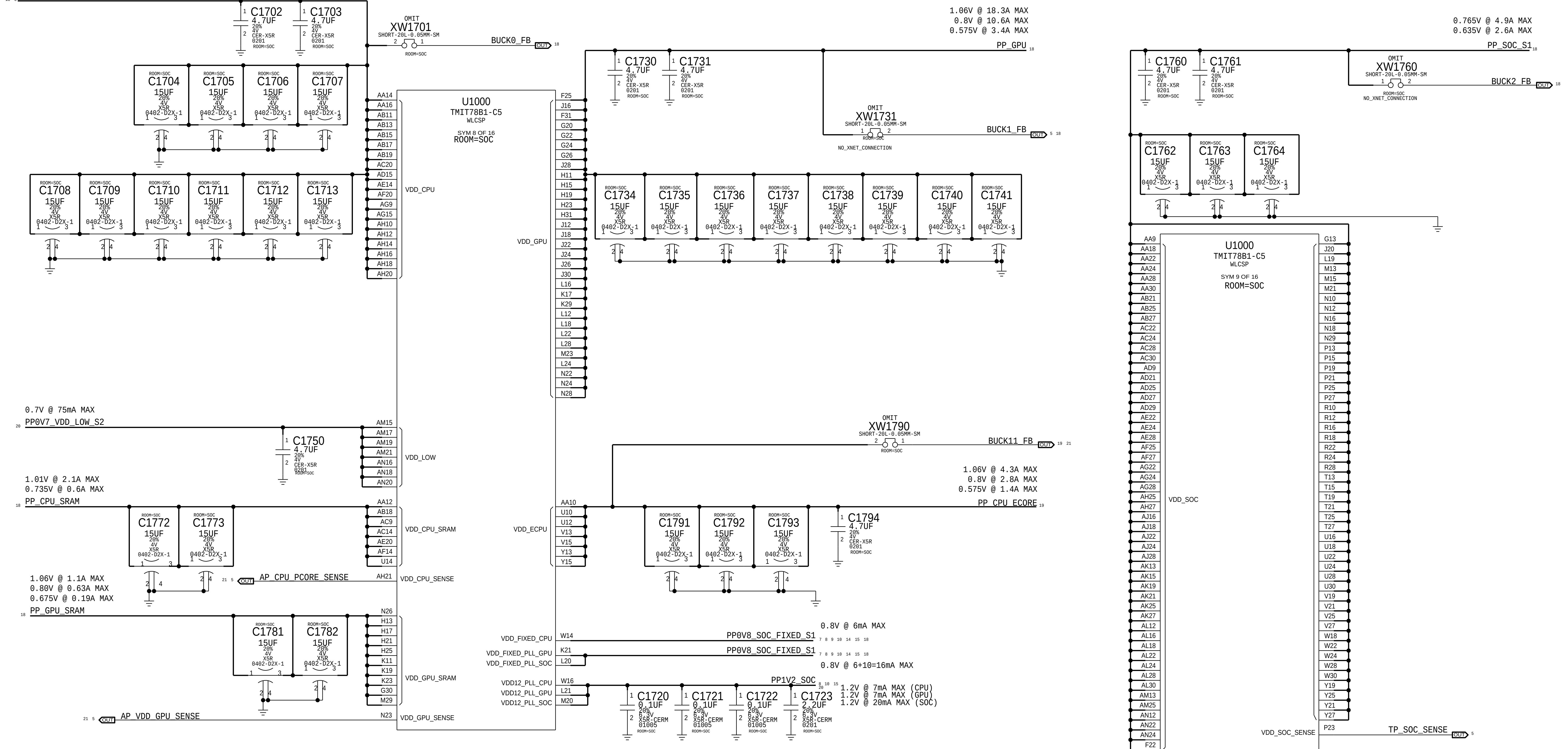
#38638490: Route U1000/U5000 ASP1 BCLK, LRCLK with _R



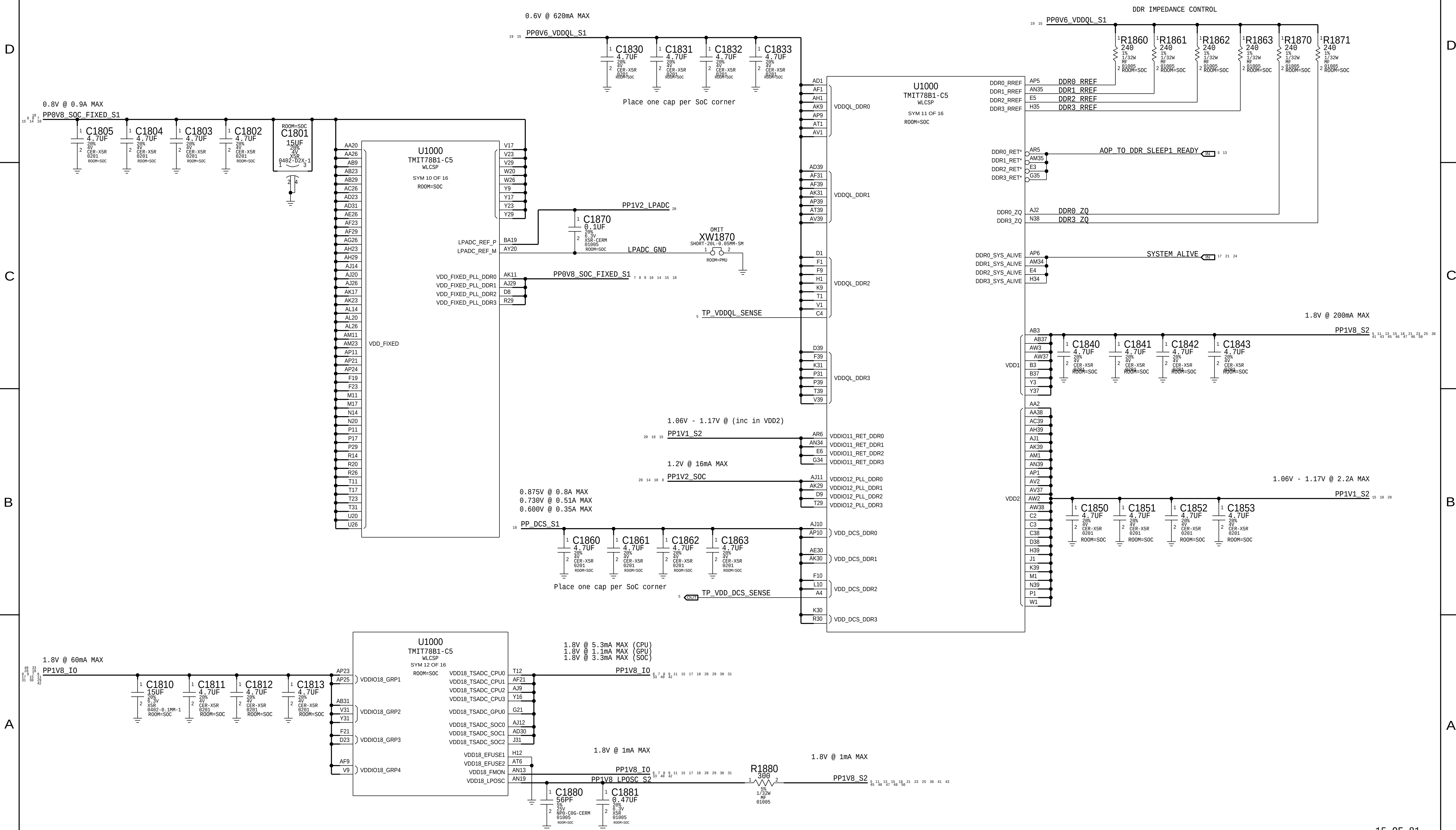
SOC - CPU, GPU & SOC RAILS

1.06V @ 11.0A MAX
0.8V @ 6A MAX
0.575V @ 2.7A MAX

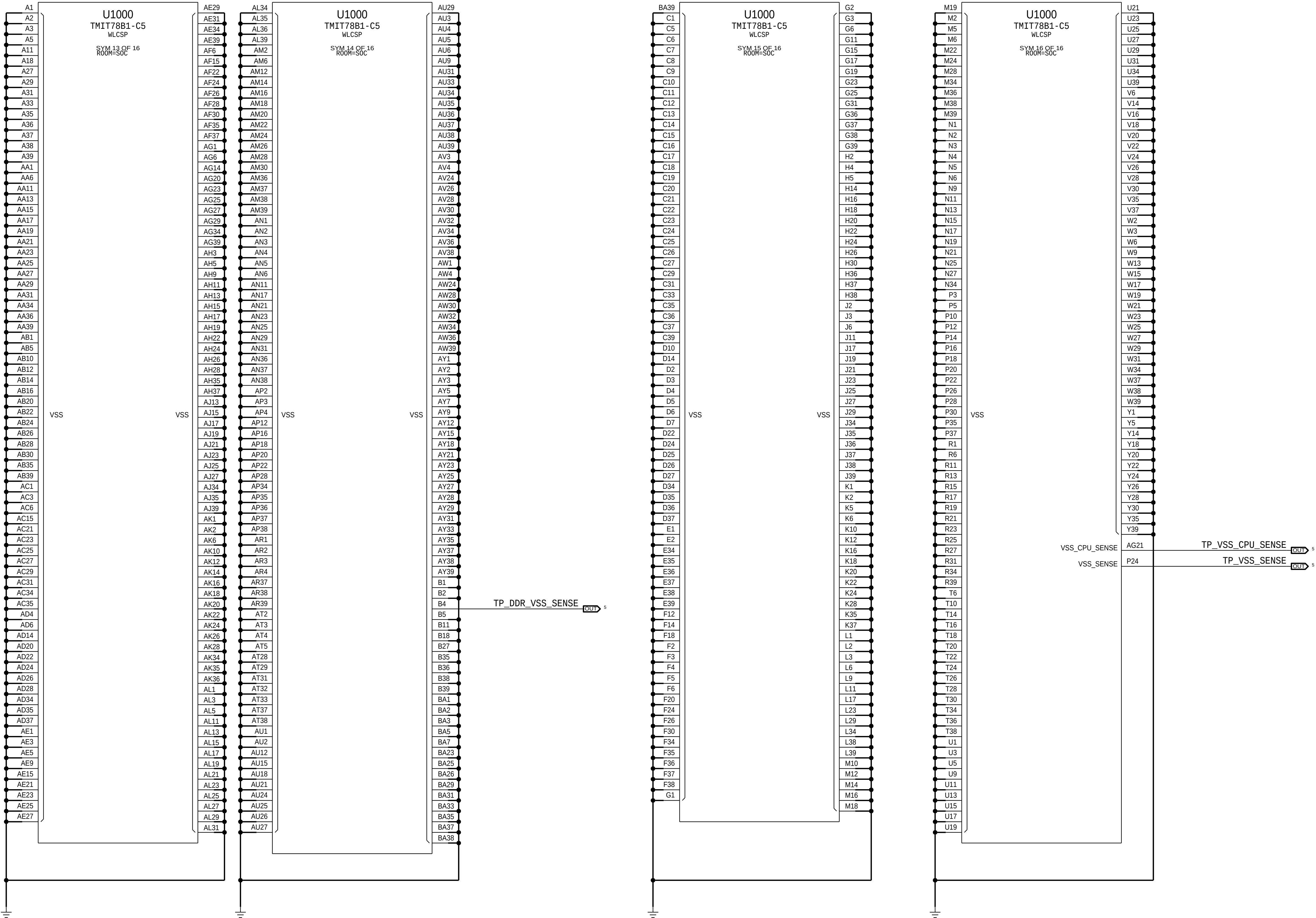
5 PP_CPU_PCORE

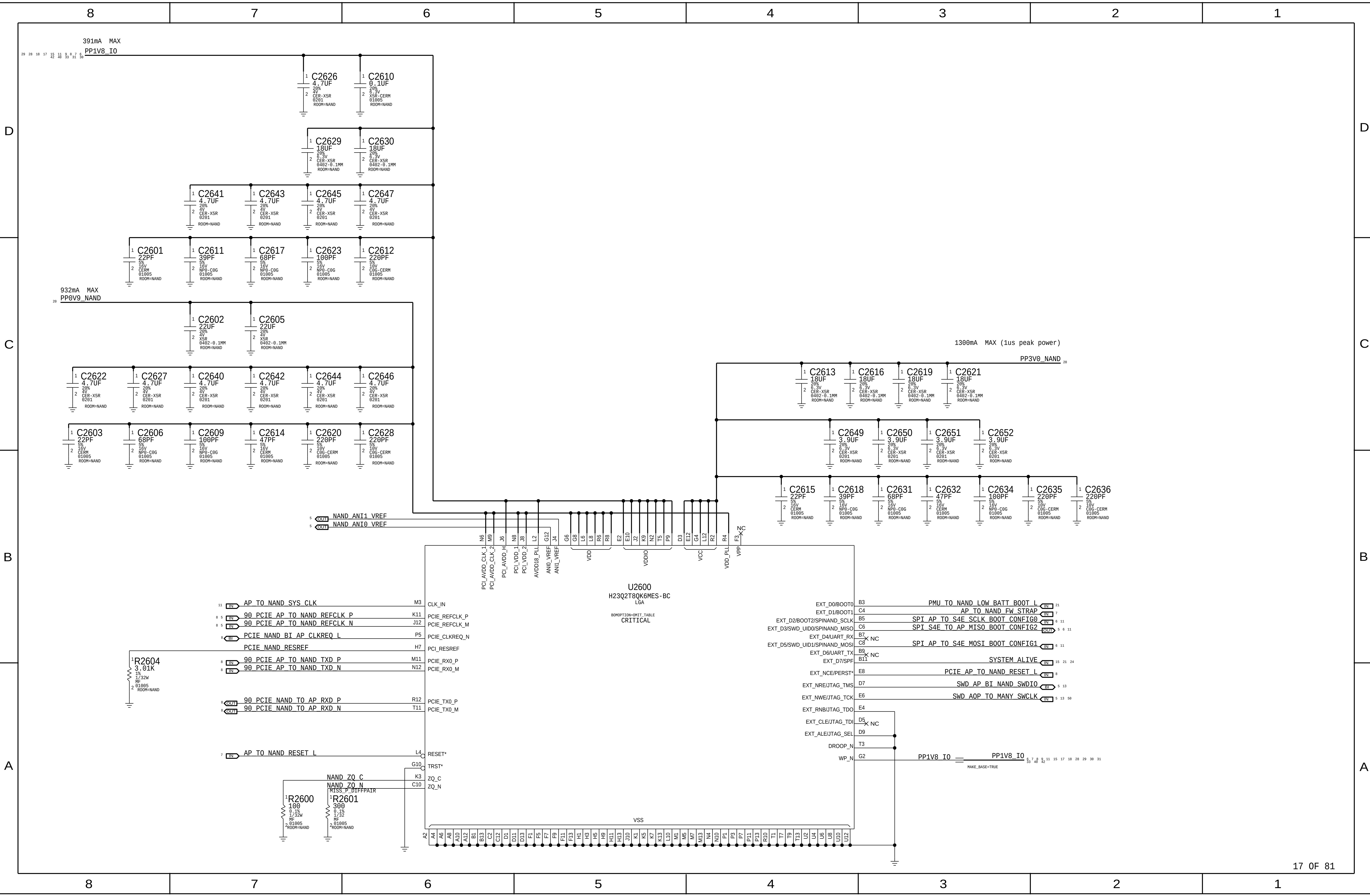


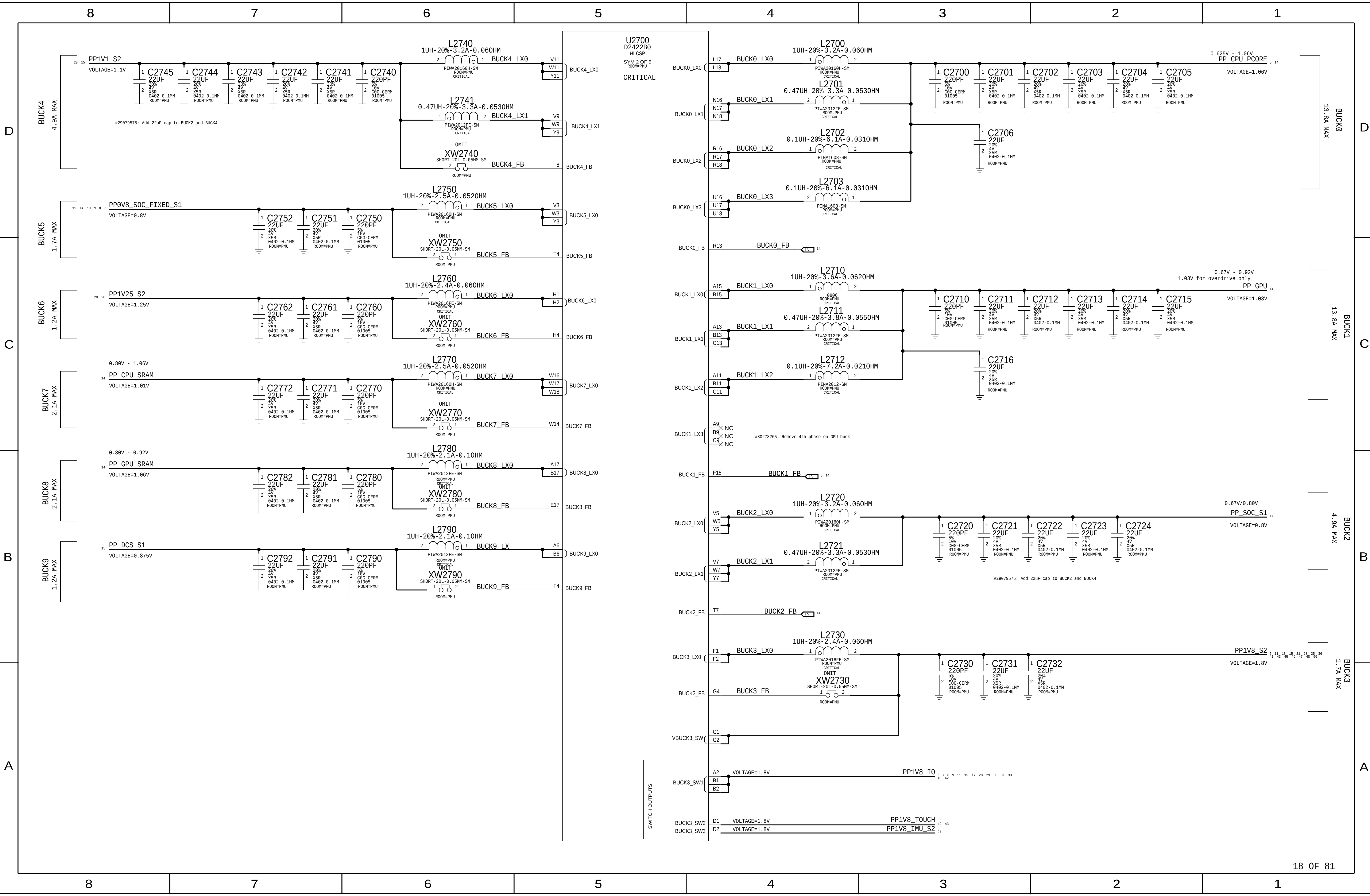
SOC - POWER SUPPLIES



SOC - POWER SUPPLIES







D

C

B

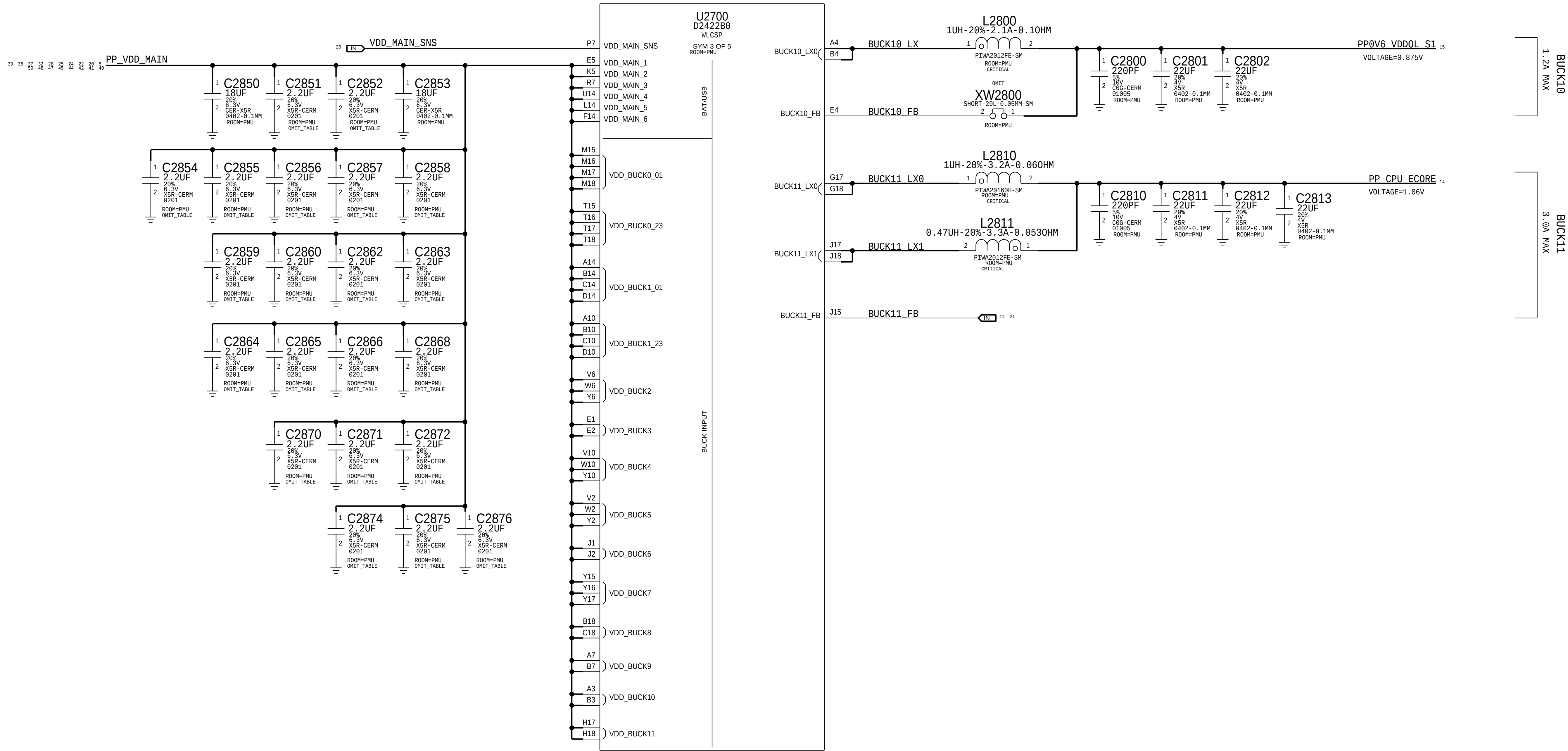
A

D

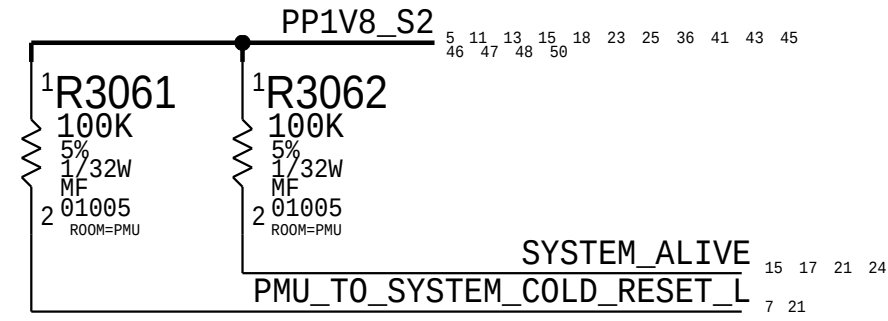
C

B

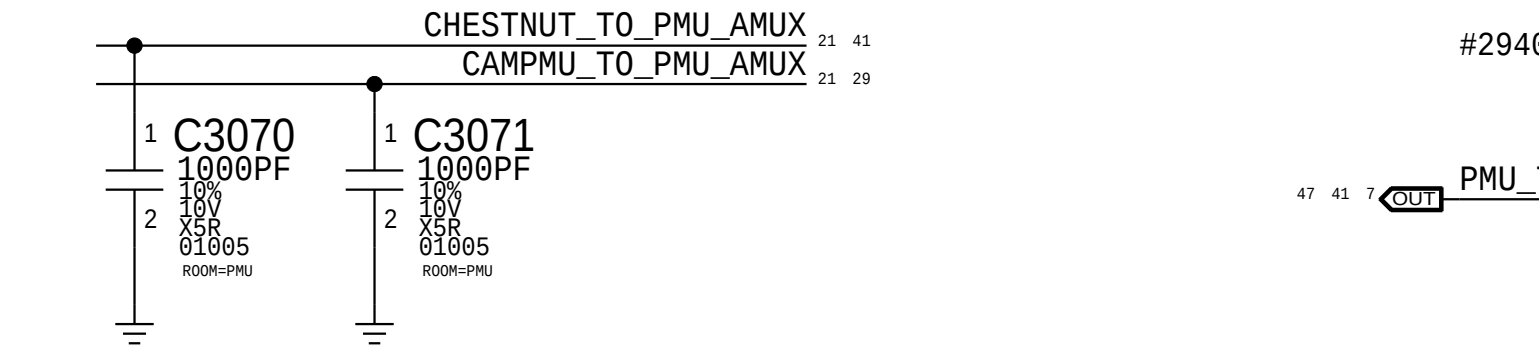
A



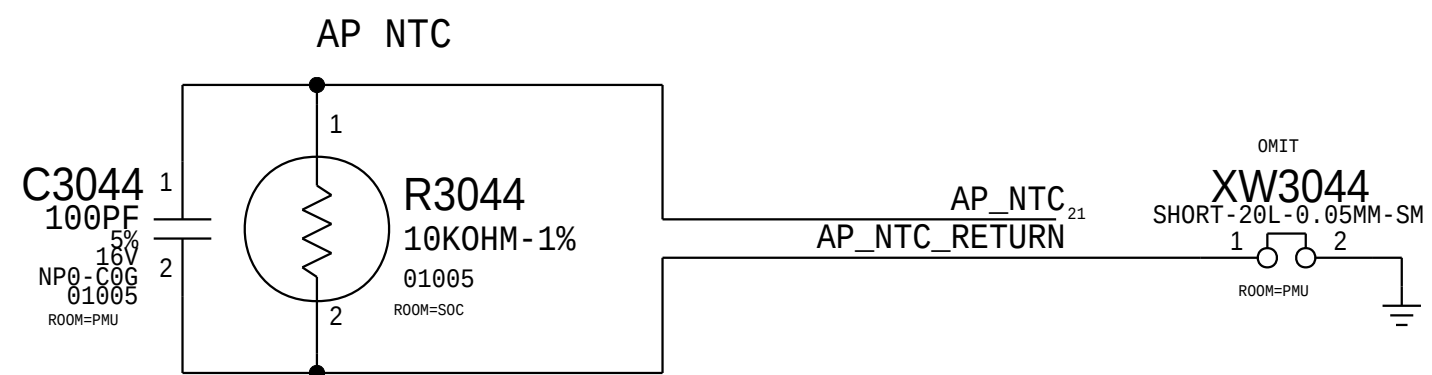
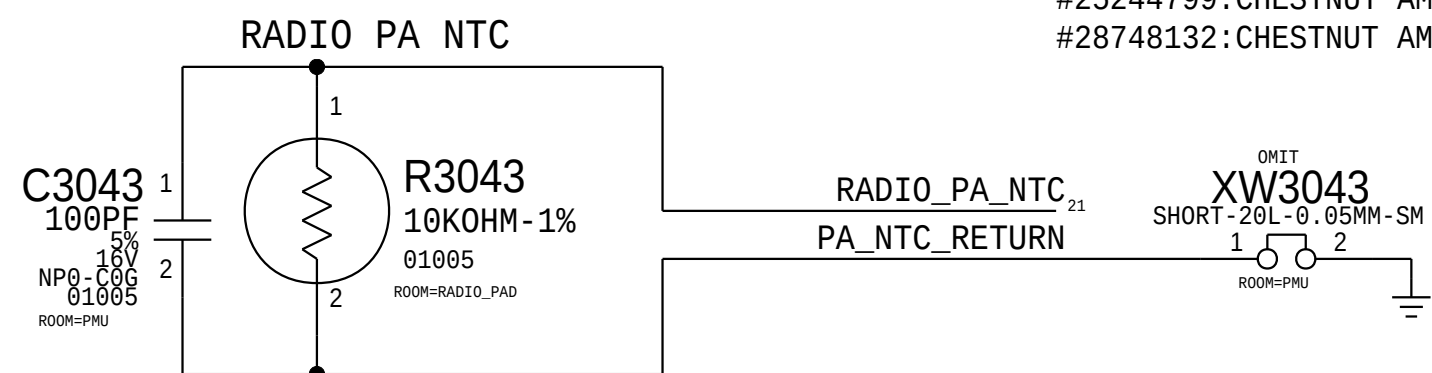
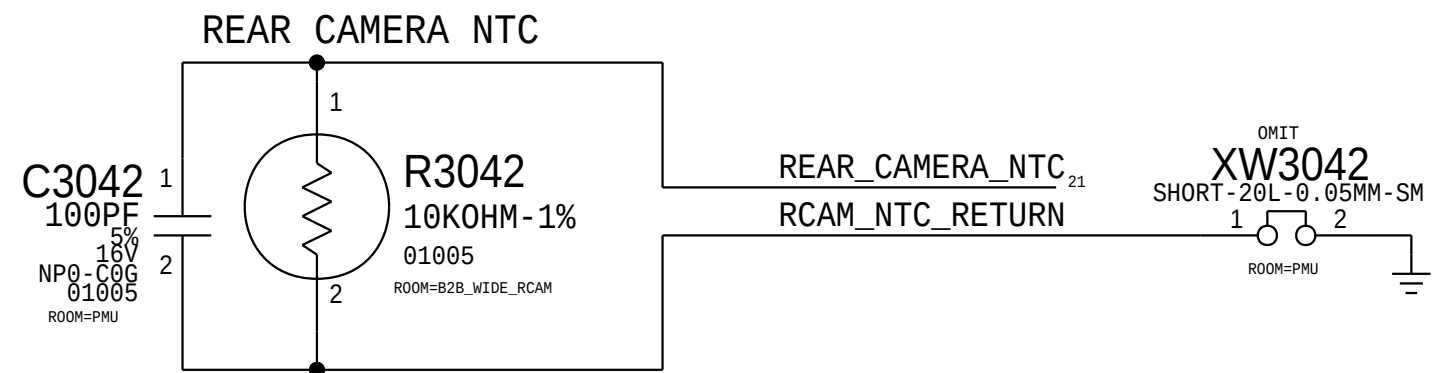
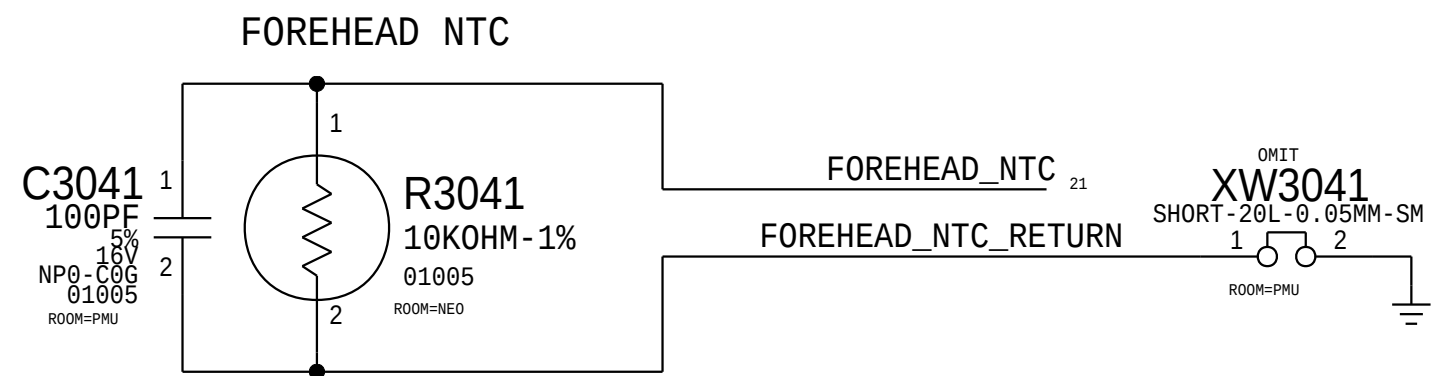
COLD_RESET & SYSTEM_ALIVE



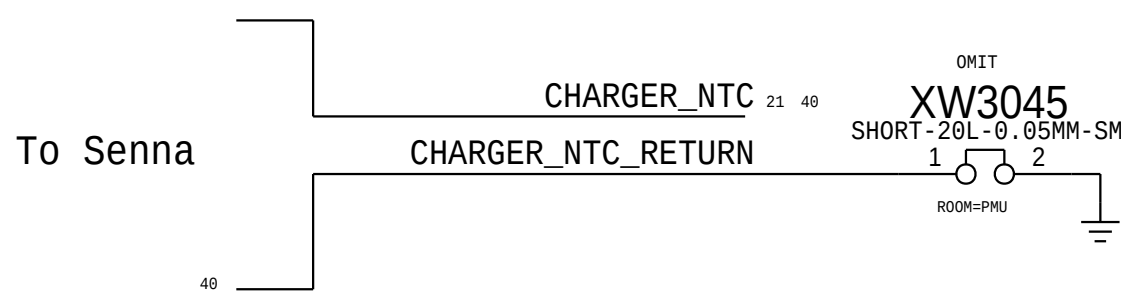
Chestnut / Ansel AMUX



NTCs



CHARGER NTC (IN SIP)



NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC

#29404057:Damping R for ACTIVE_RDY

#29300440:Connect PMU crash to SoC

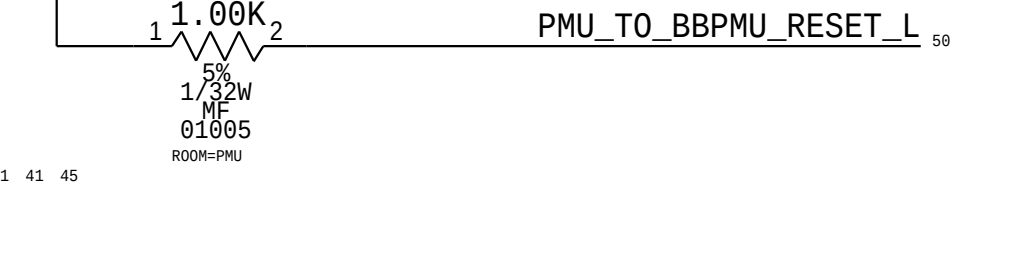
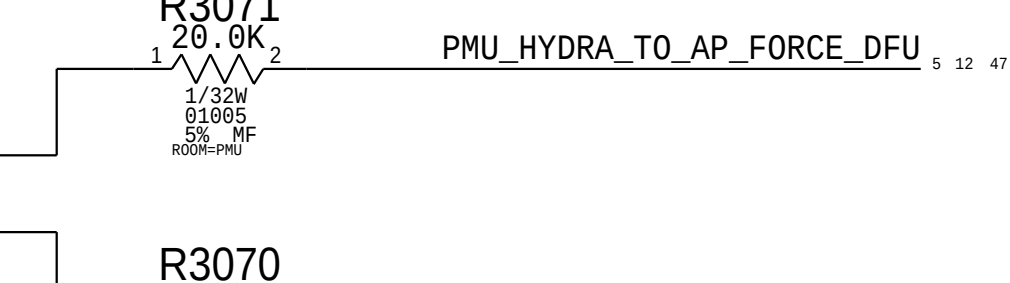
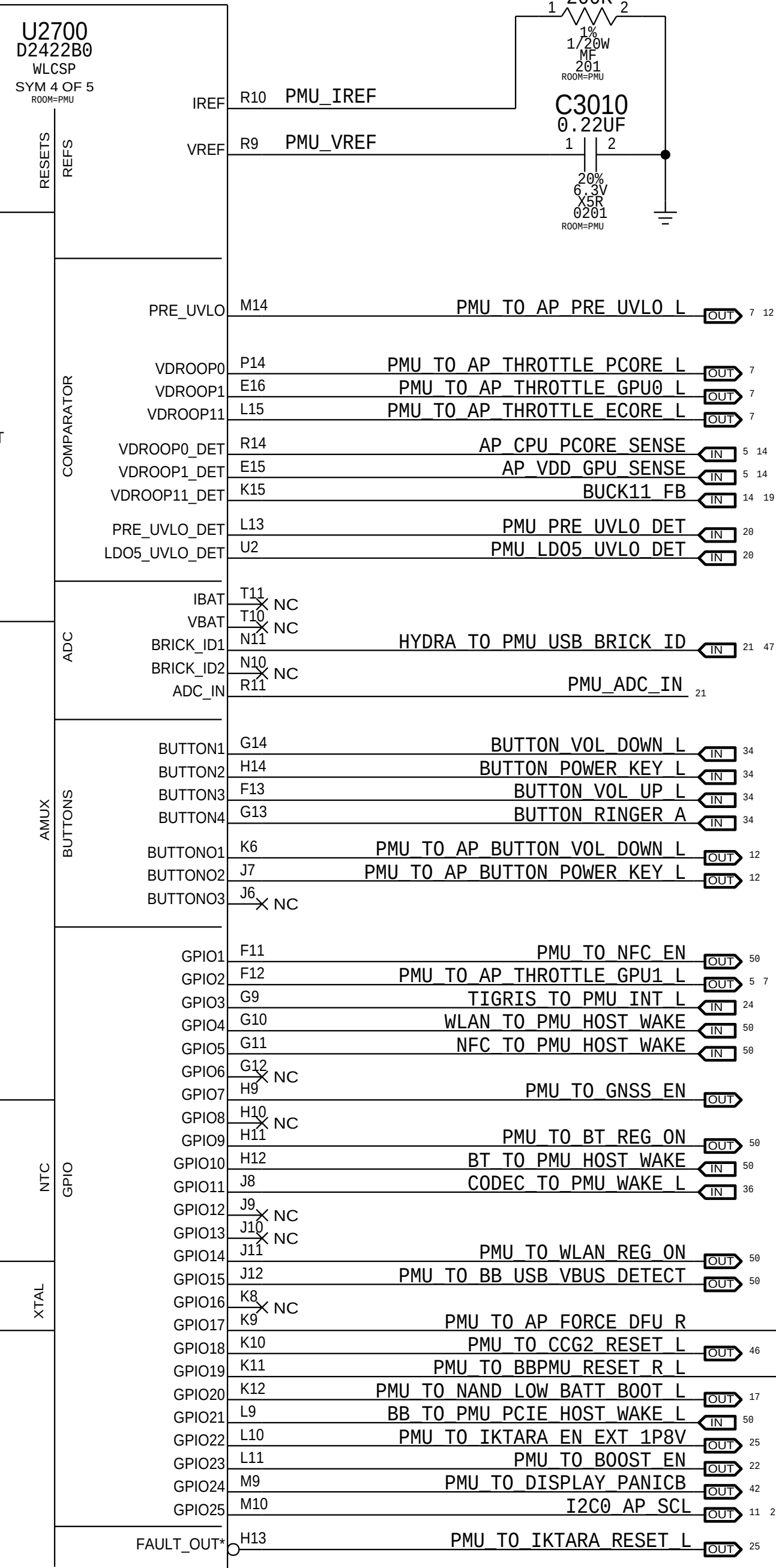
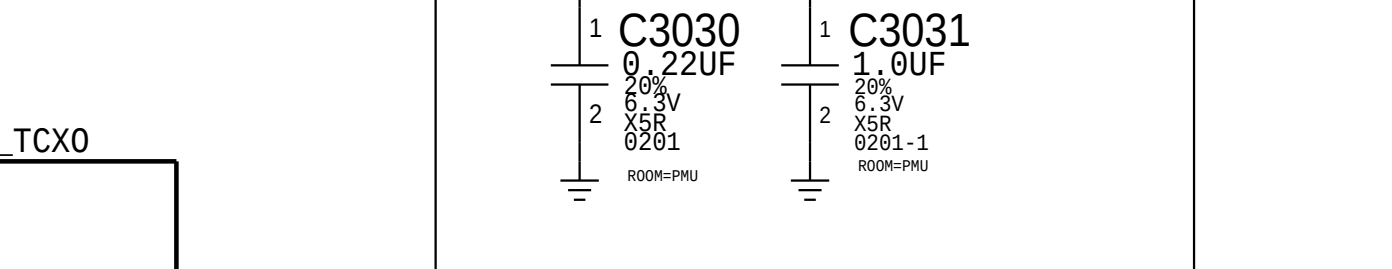
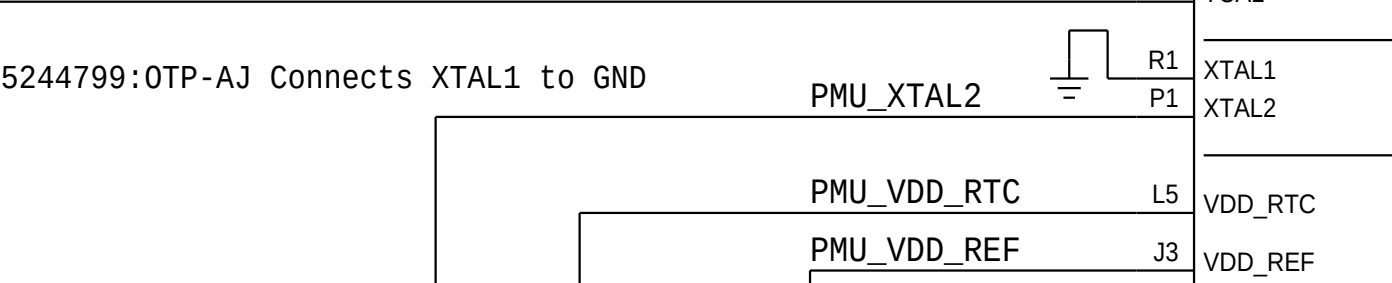
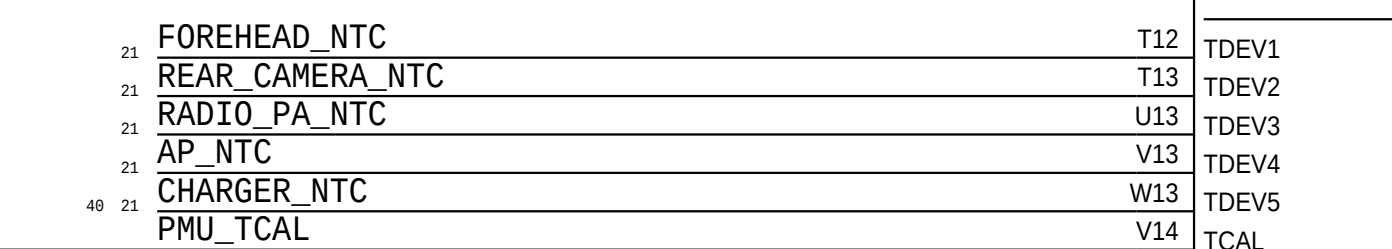
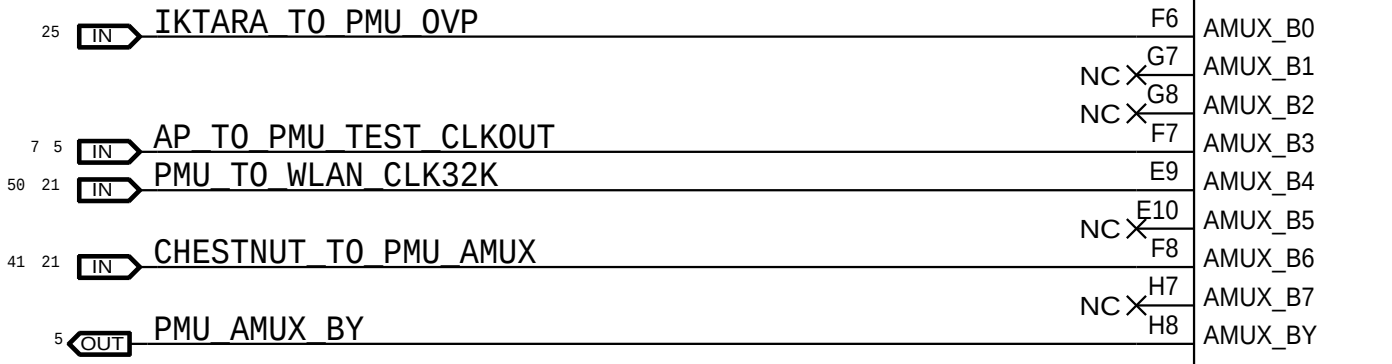
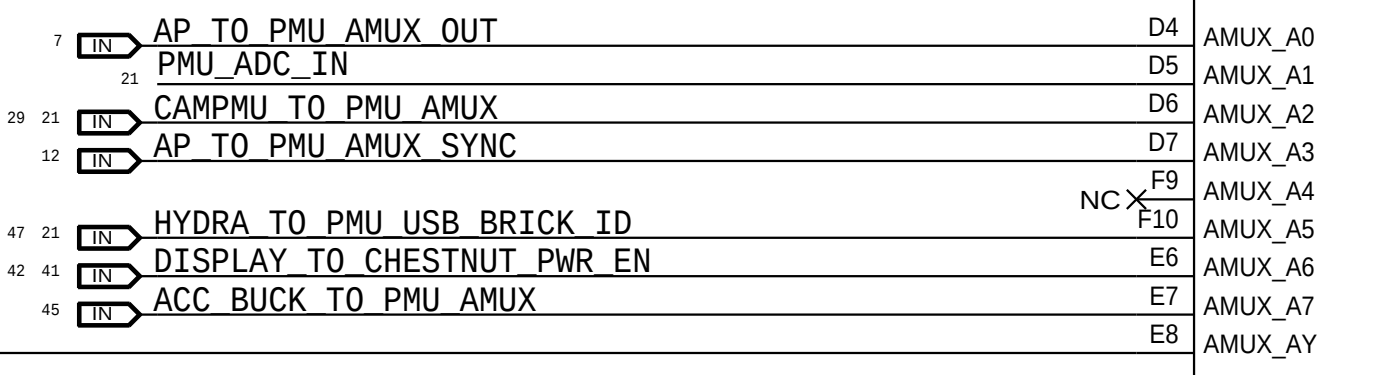
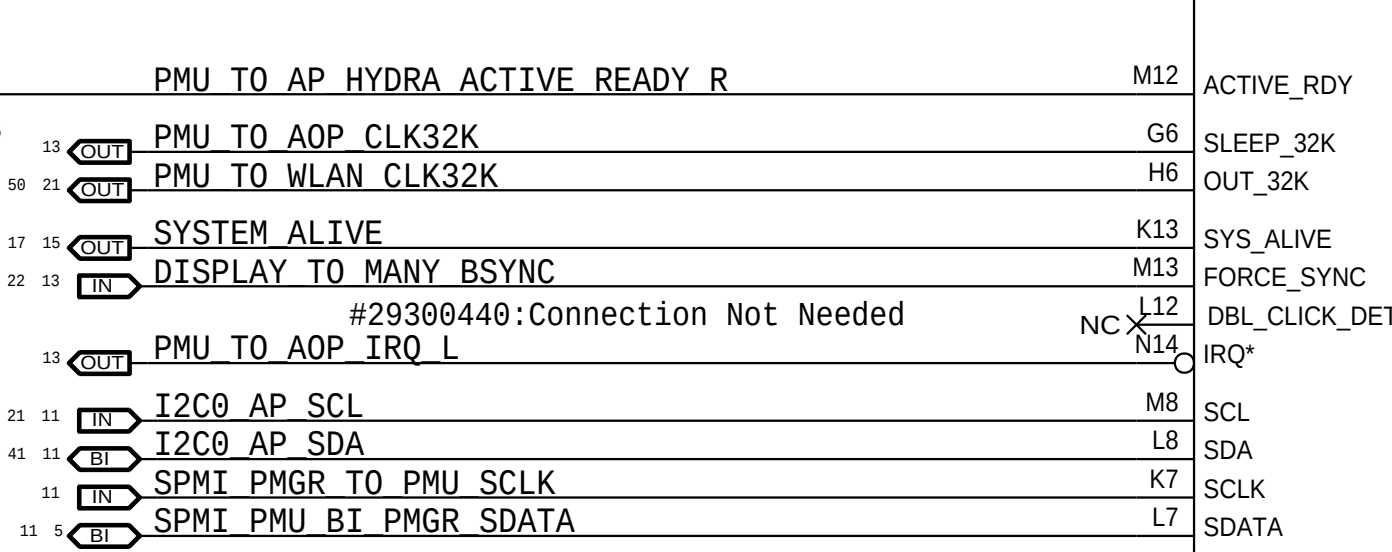
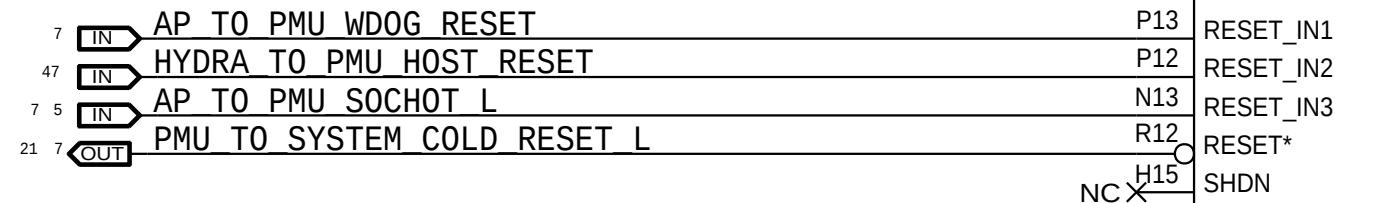
#27852679:No SDA R, PoR is SPMI

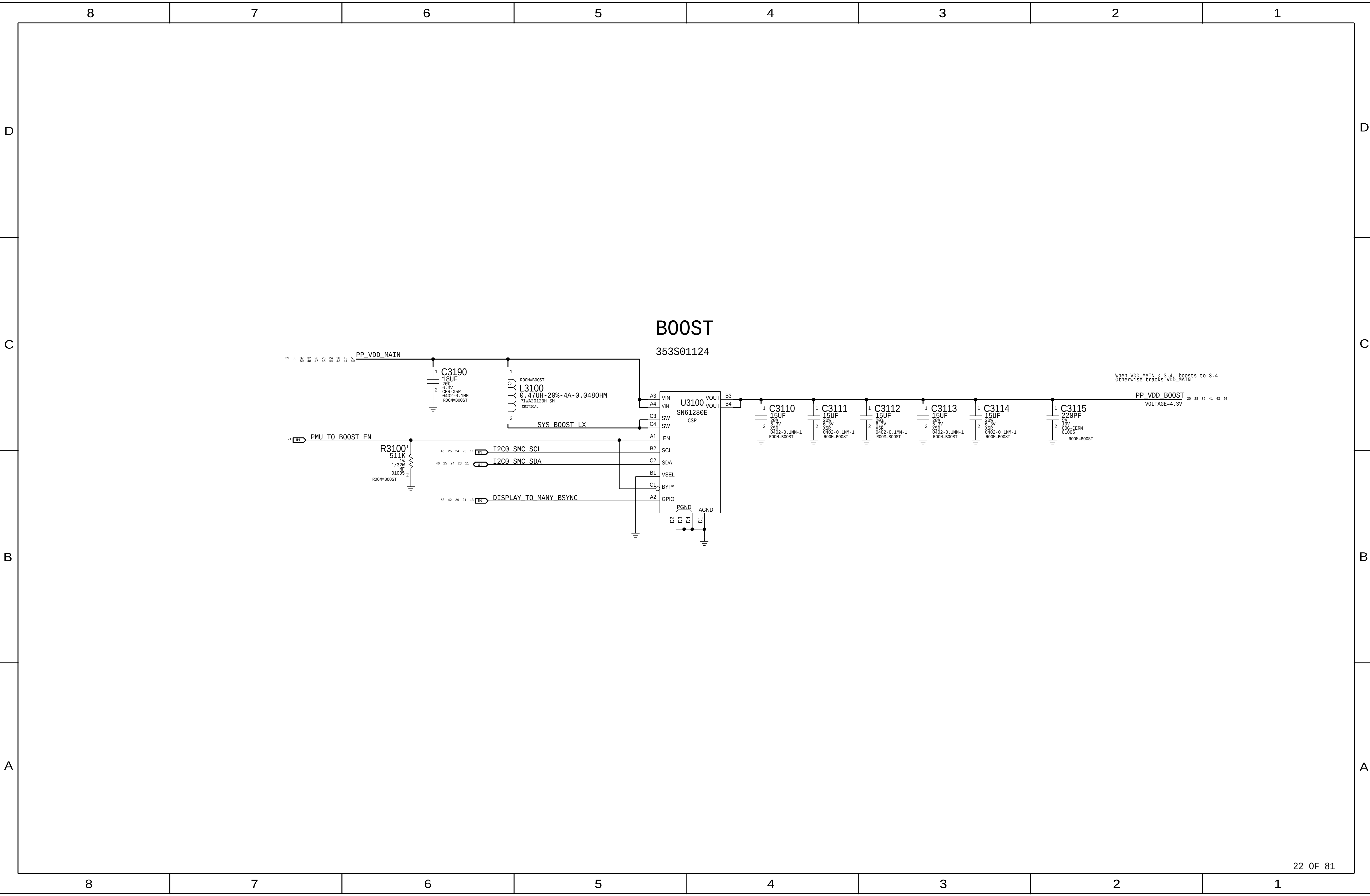
#25244799:200K pull-down

#25244799:CHESTNUT AMUX AND PWR_EN ON SAME CHANNEL
#28748132:CHESTNUT AMUX AND PWR_EN SWAP BETWEEN A6/B6

#25244799:0TP-AJ Connects XTAL1 to GND

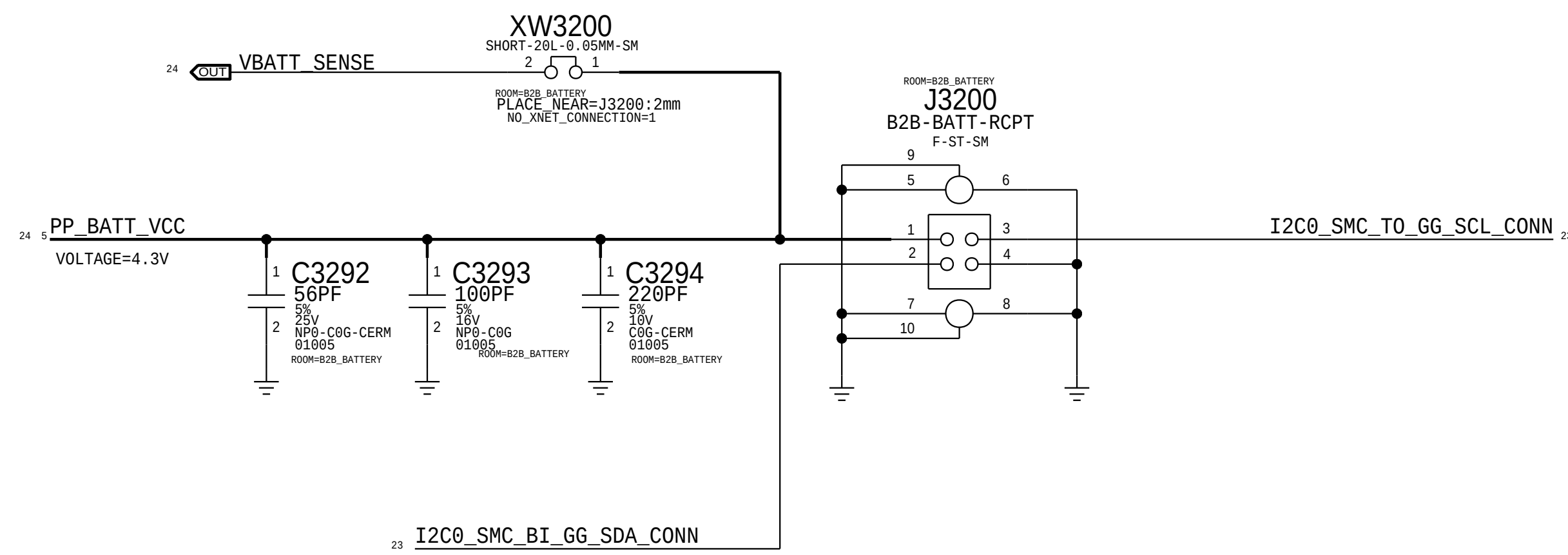
#29312542:REPLACE PMU XTAL WITH XO



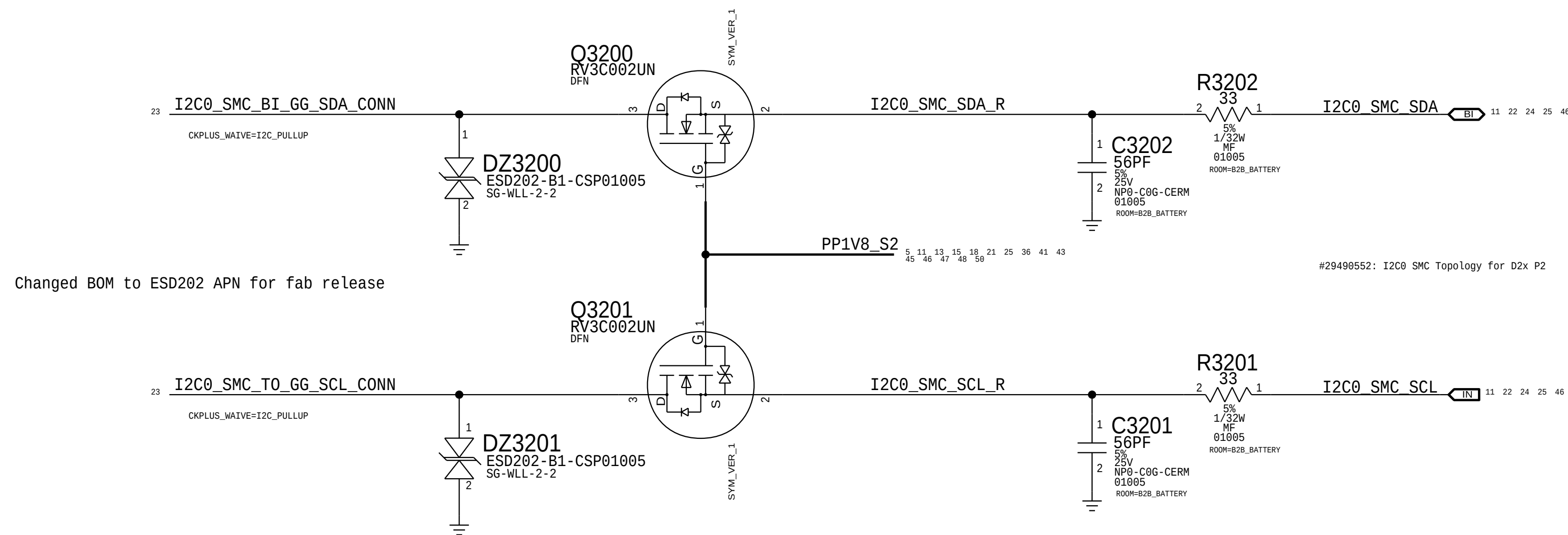


Battery Connector

THIS ONE ON MLB ---> 516S00232

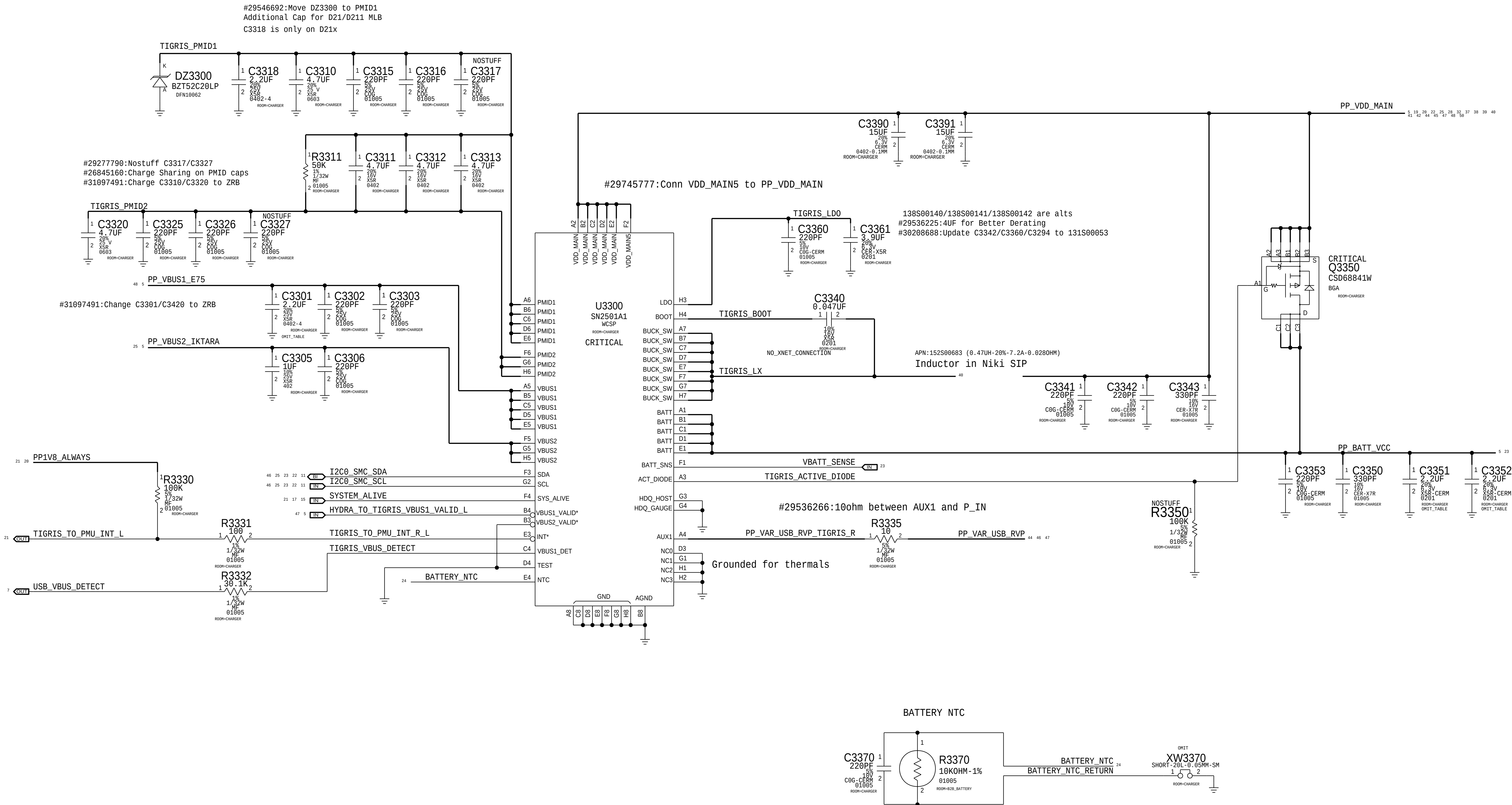


```
#32515053:Add DZ3200/DZ3201 on I2C CONN with 377S00001
#31734657:Replace battery I2C FETs with 376S00134
  Gas gauge I2C level translator
```



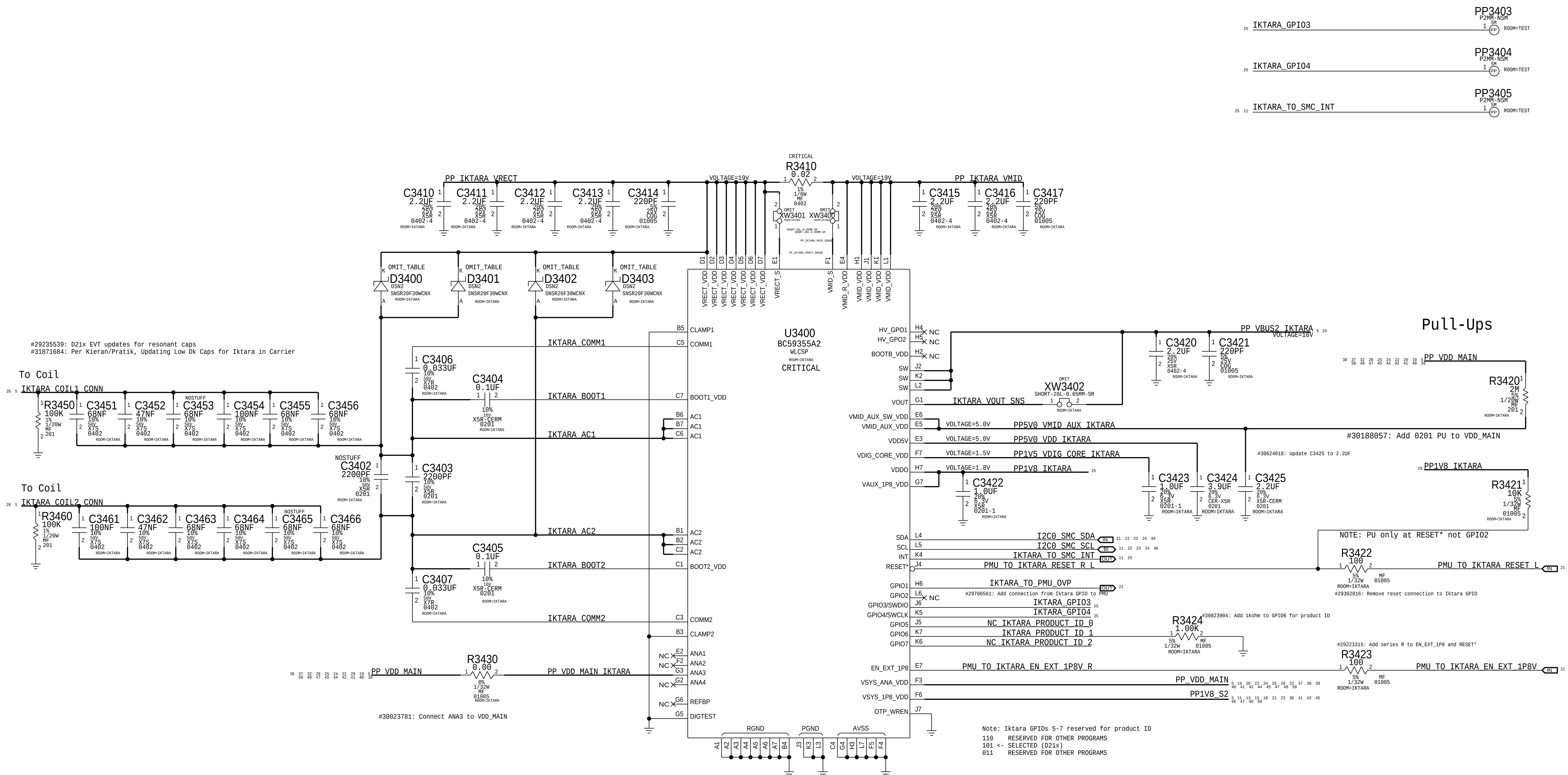
Tigris 2

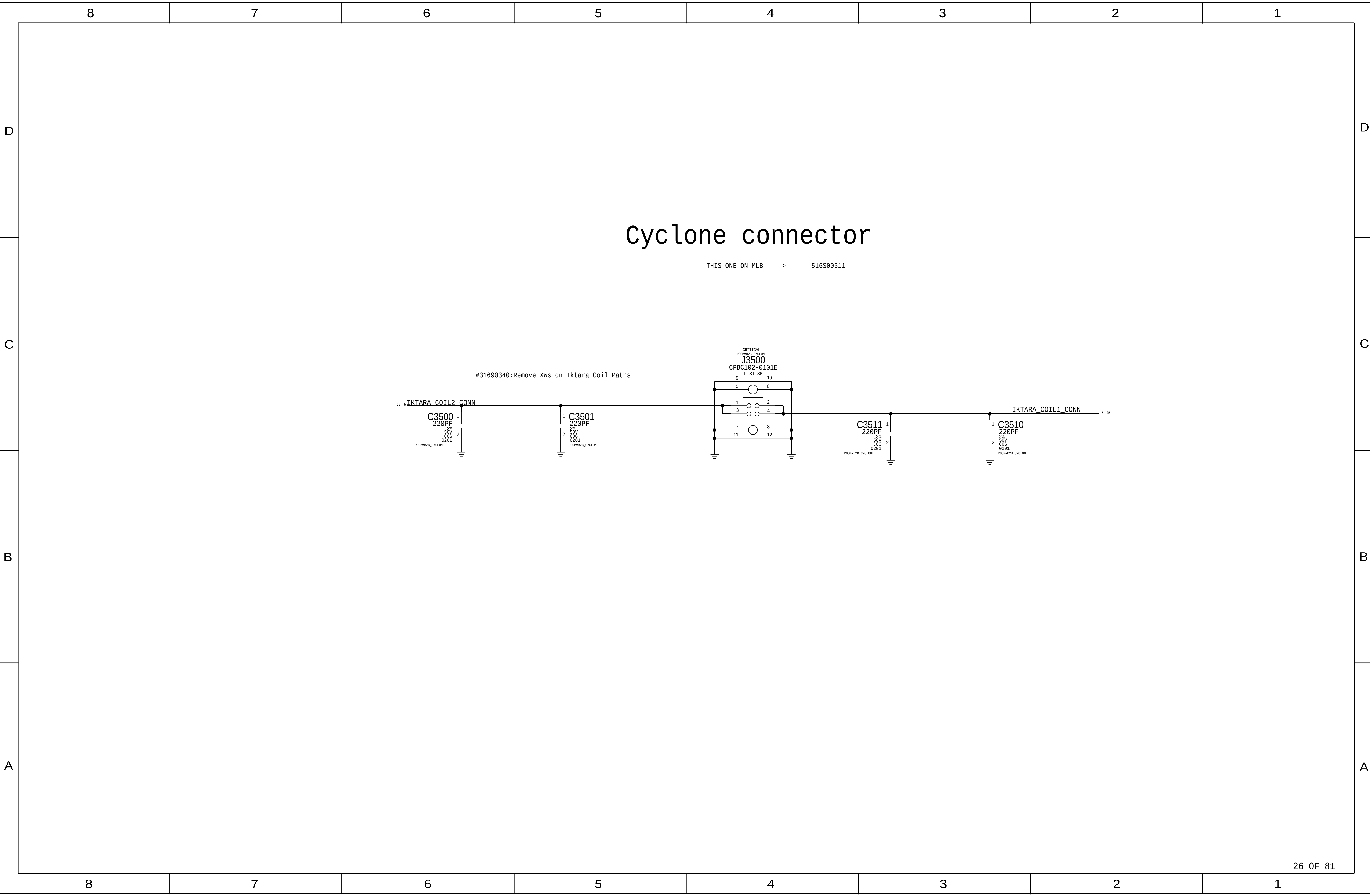
APN:343S00170



Iktara

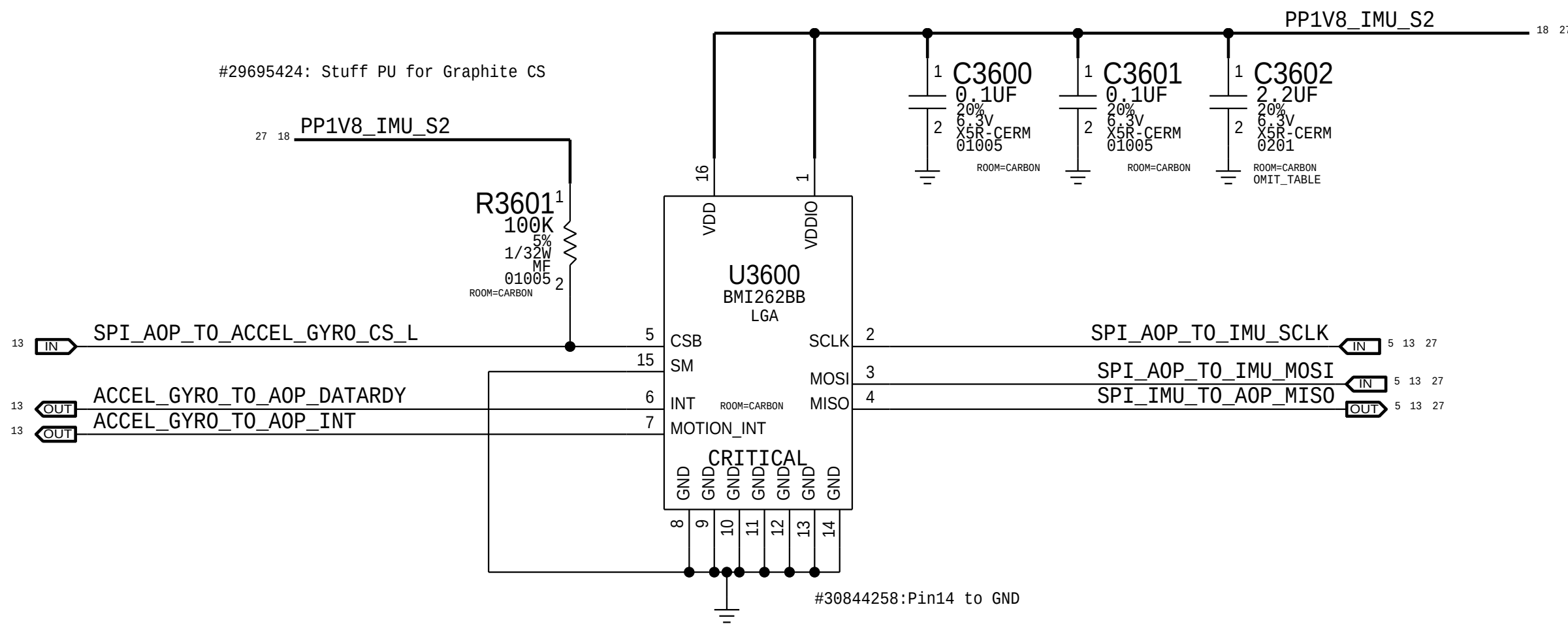
Iktara Debug





Graphite - Accel & Gyro

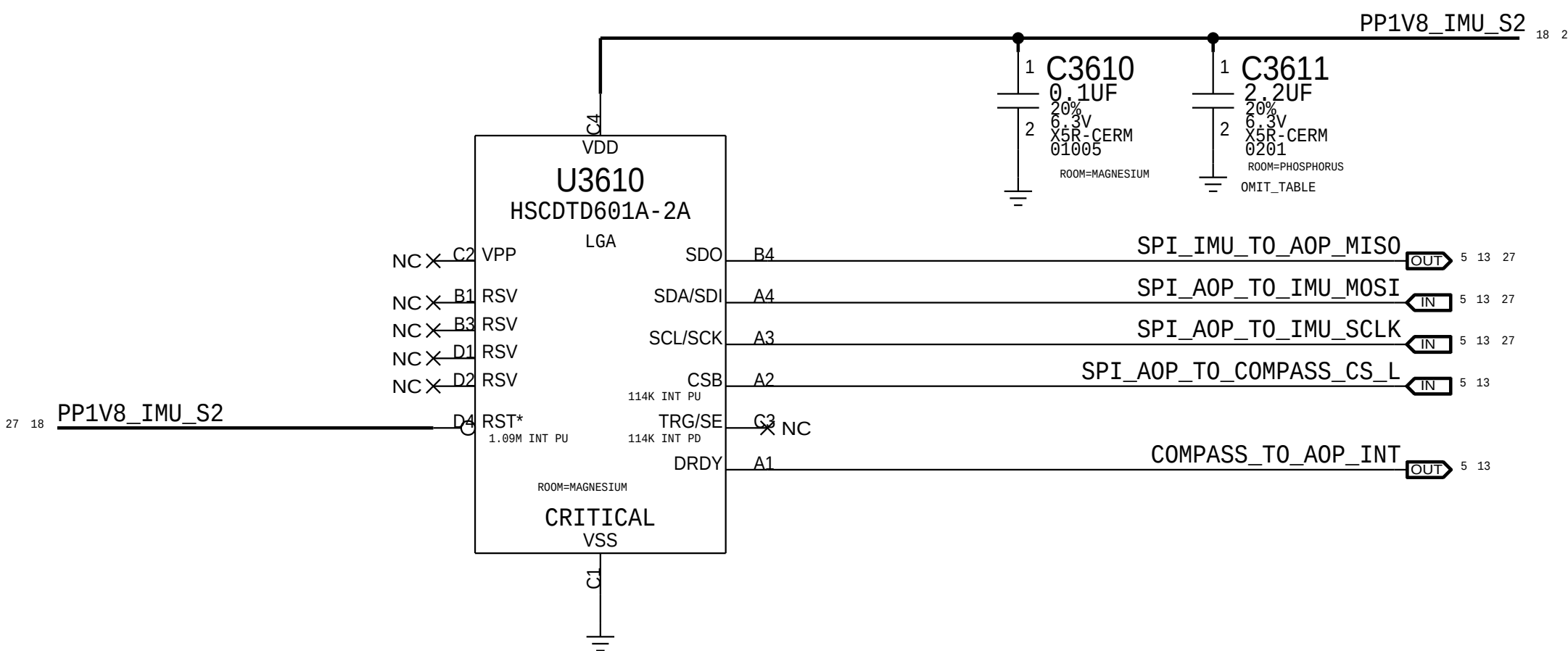
APN: 338S00304 (A1)



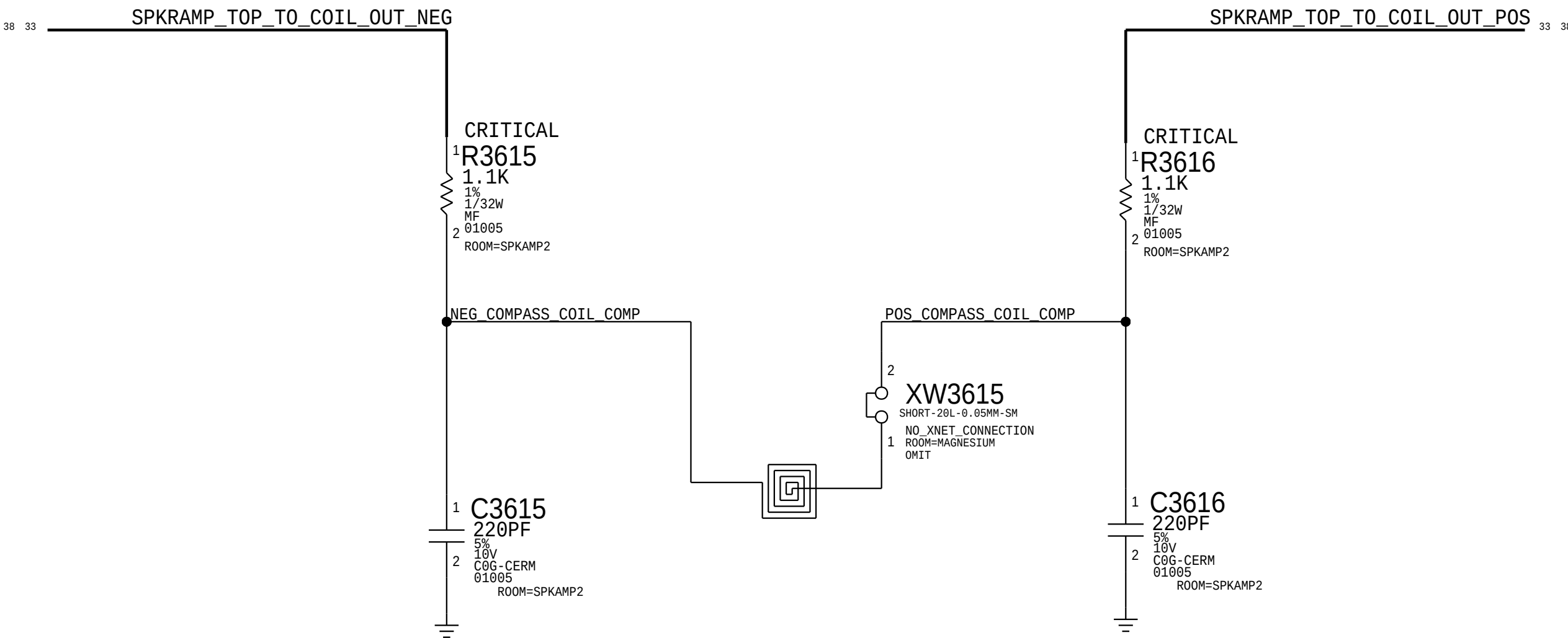
Magnesium - Compass

APN:338S00084

Shared decoupling with Phosphorus

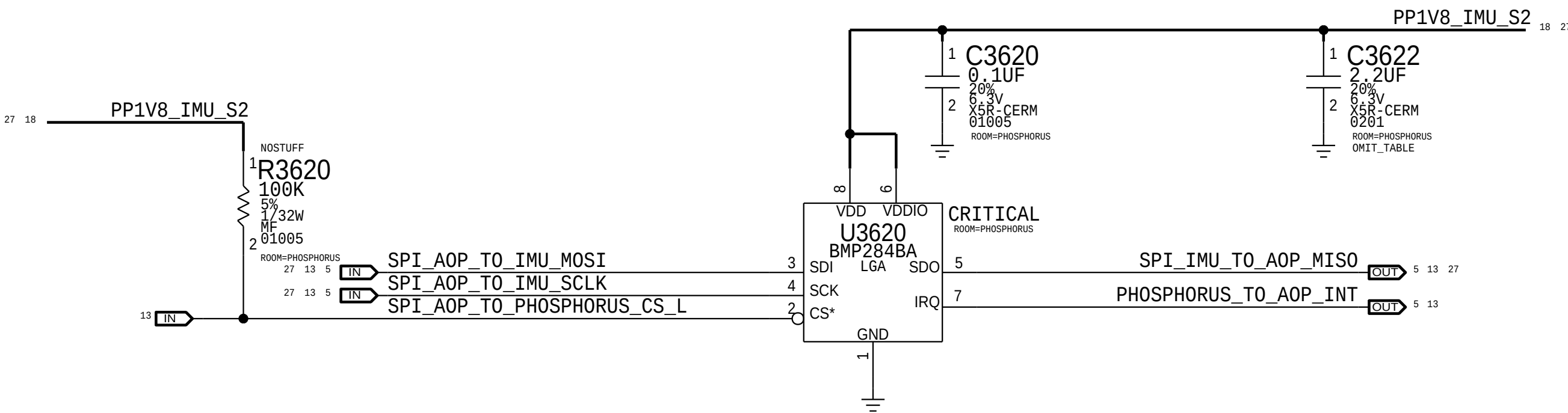


North Speaker Compass Compensation Coil

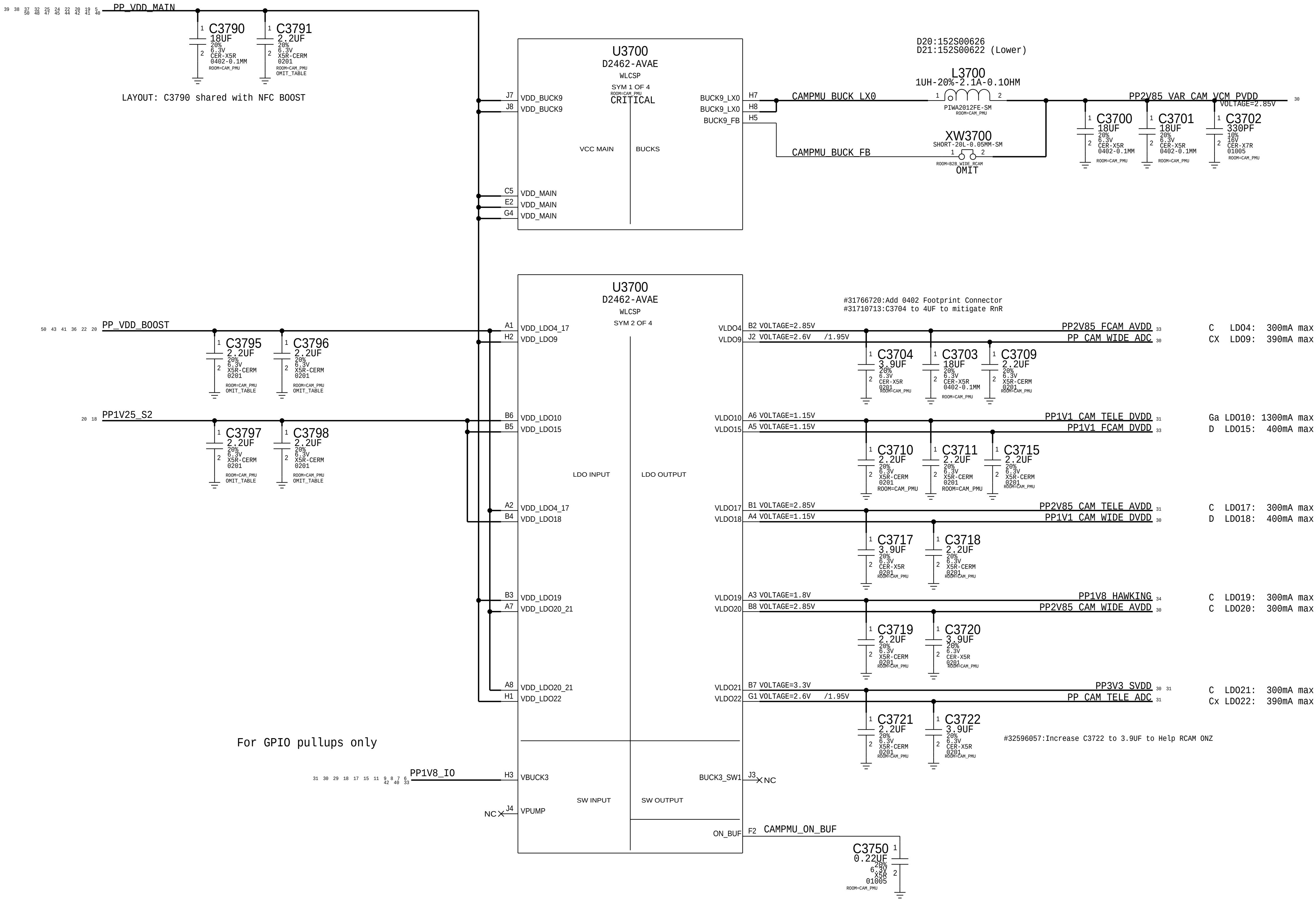


Phosphorus

APN: 338S00334



Camera PMU

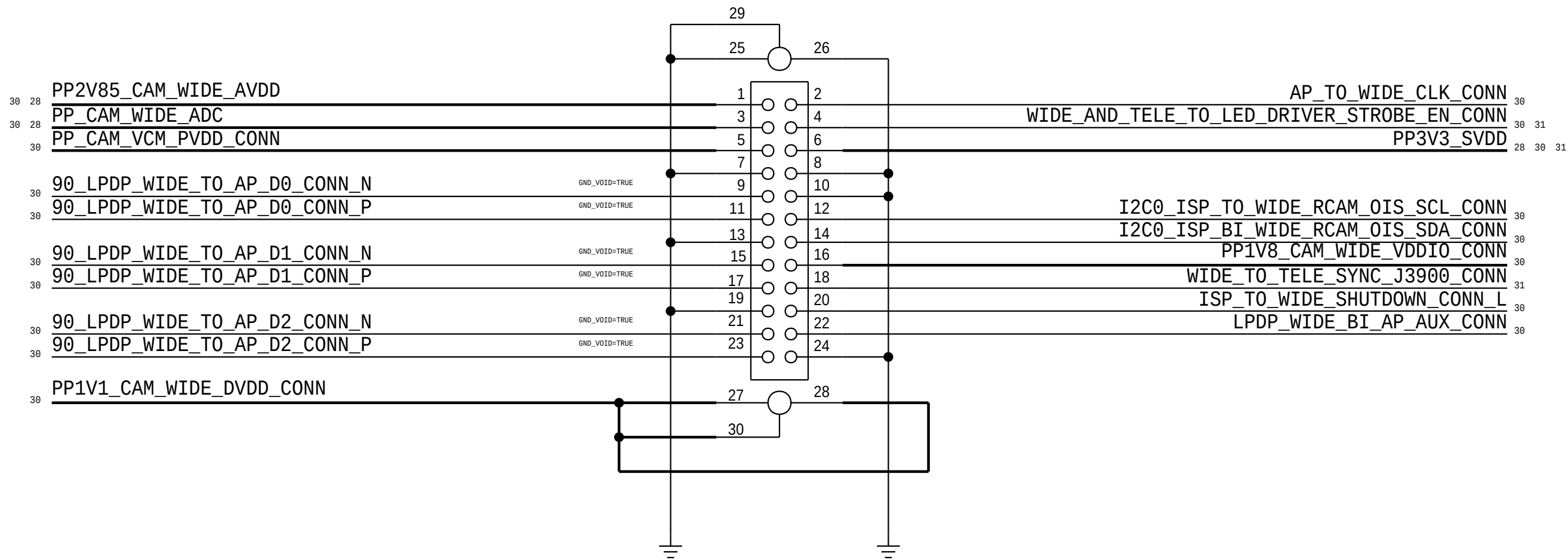


Wide Camera Connector

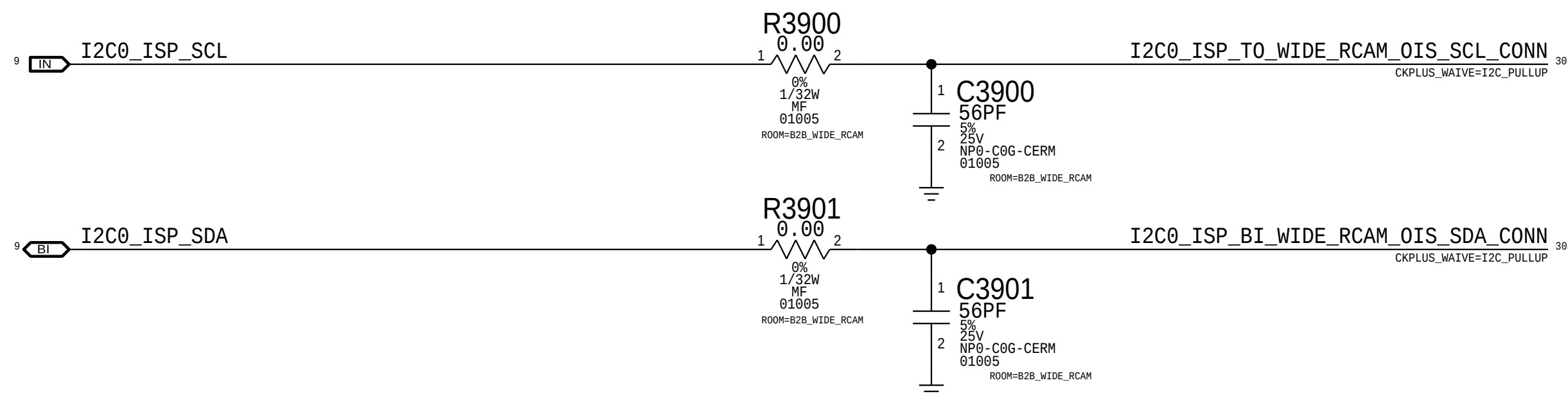
THIS ONE ---->

516S00310/P2 RCPT (USED ON MLB)
516S00309/P2 PLUG (USED ON FLEX)

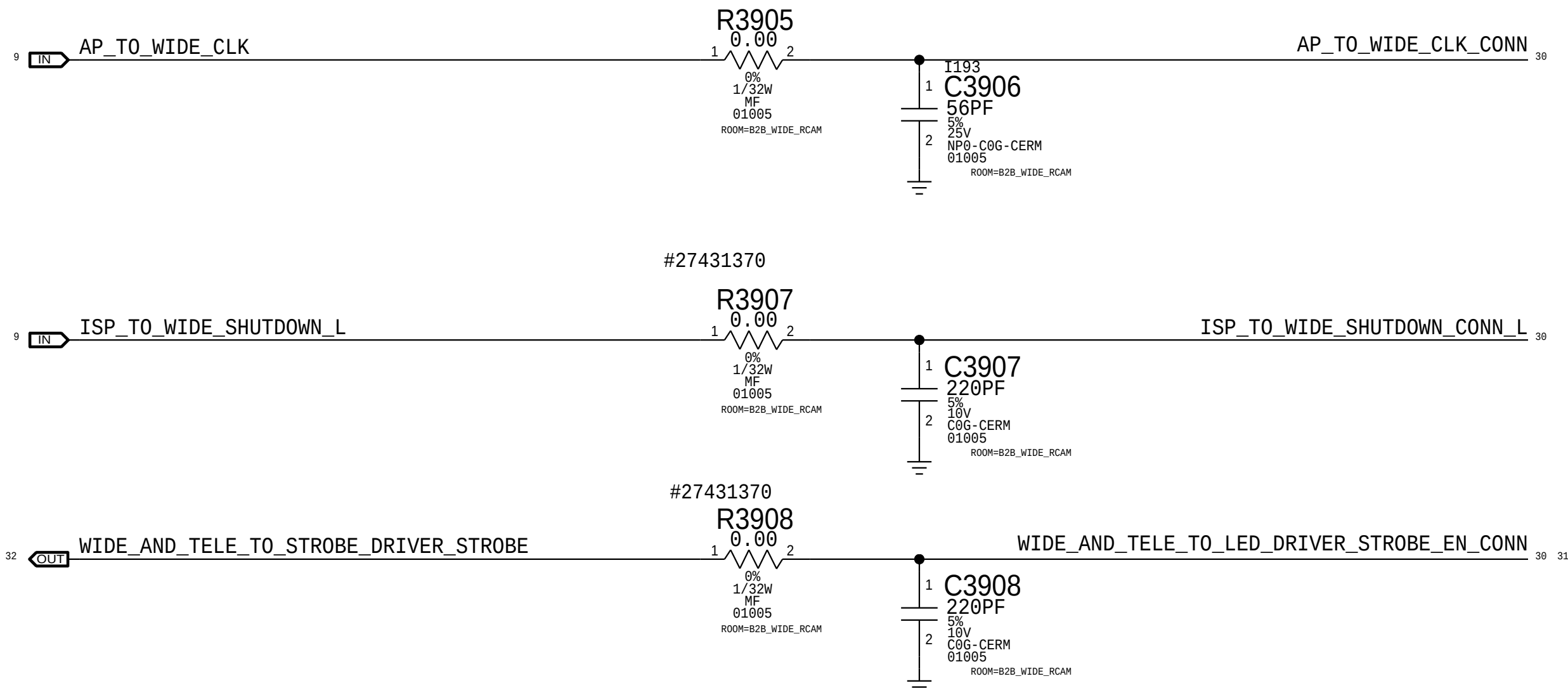
J3900
AA27DK-S024VA1
F-ST-SM



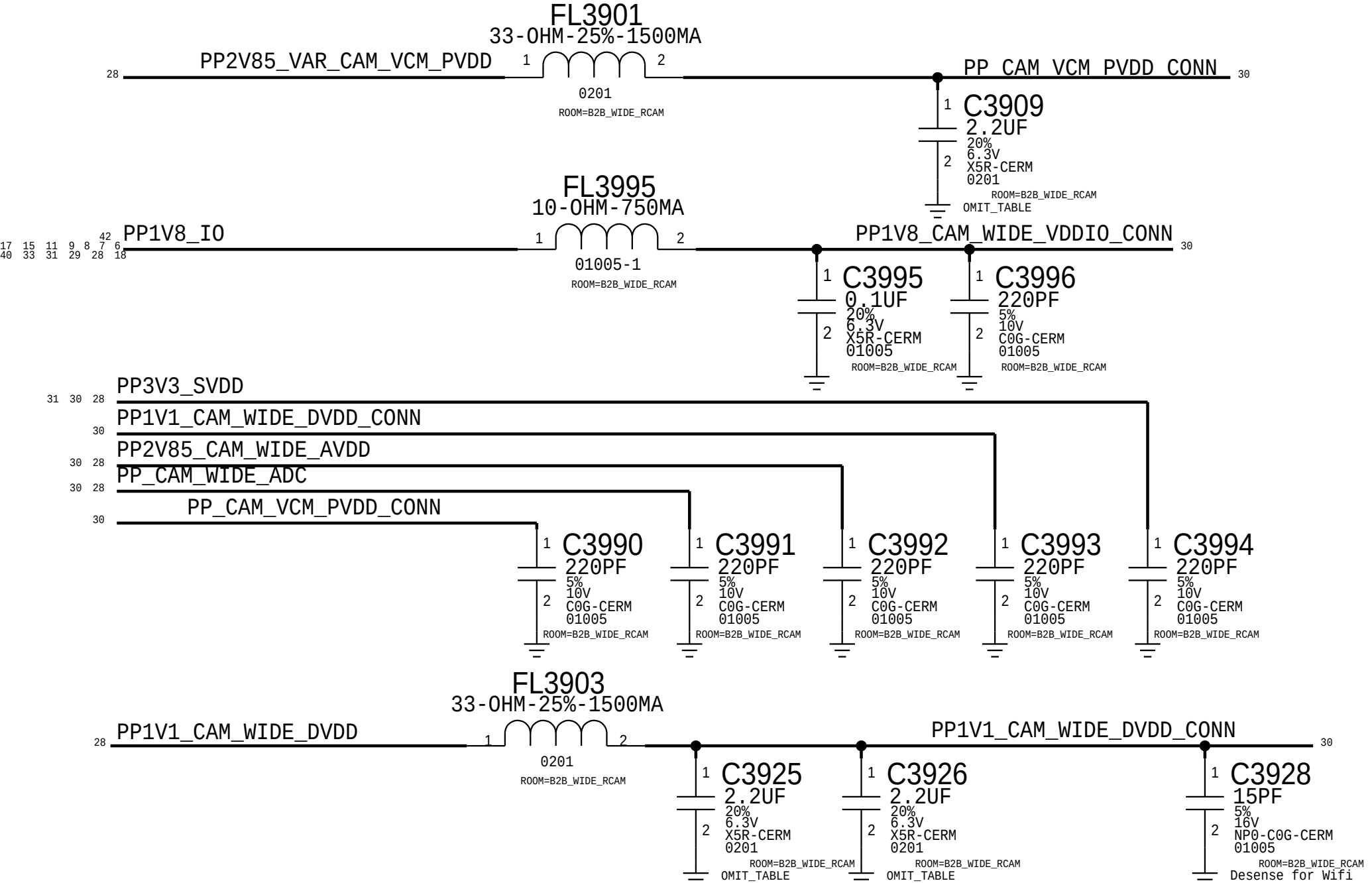
ISP I2C



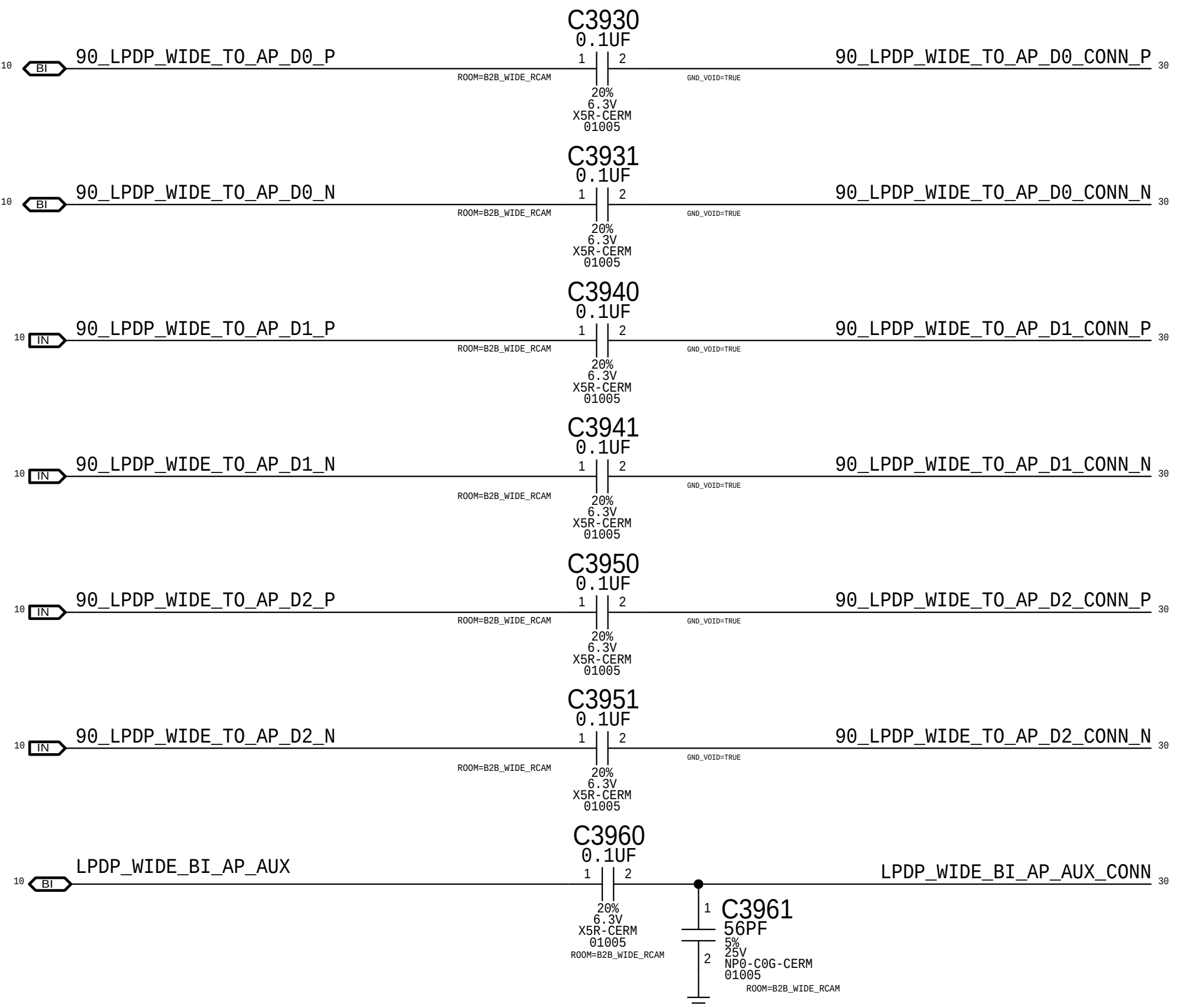
I0 Filters



Power Filtering

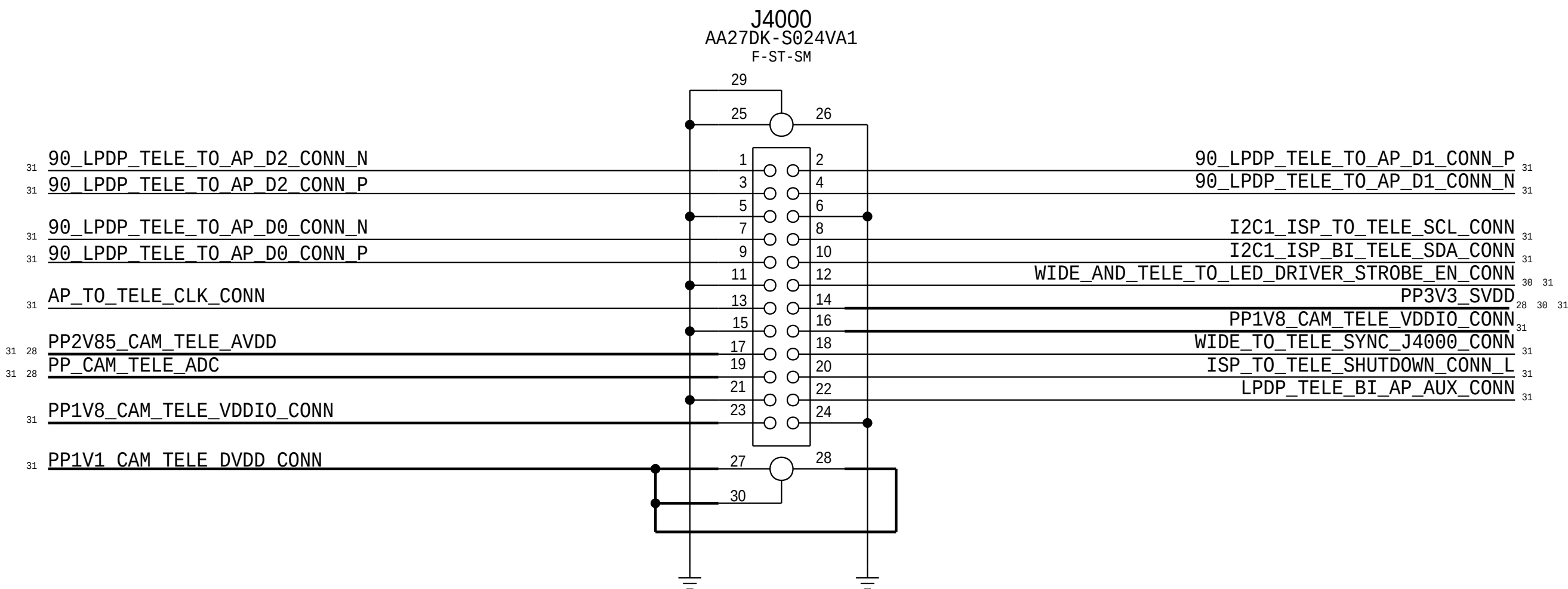


LPDP Filters

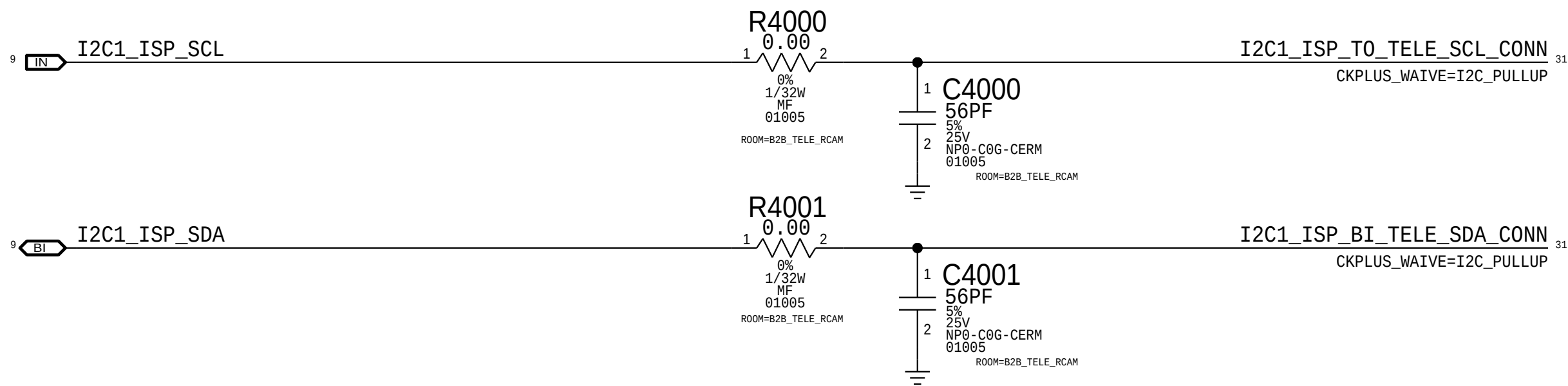


Tele Camera Connector

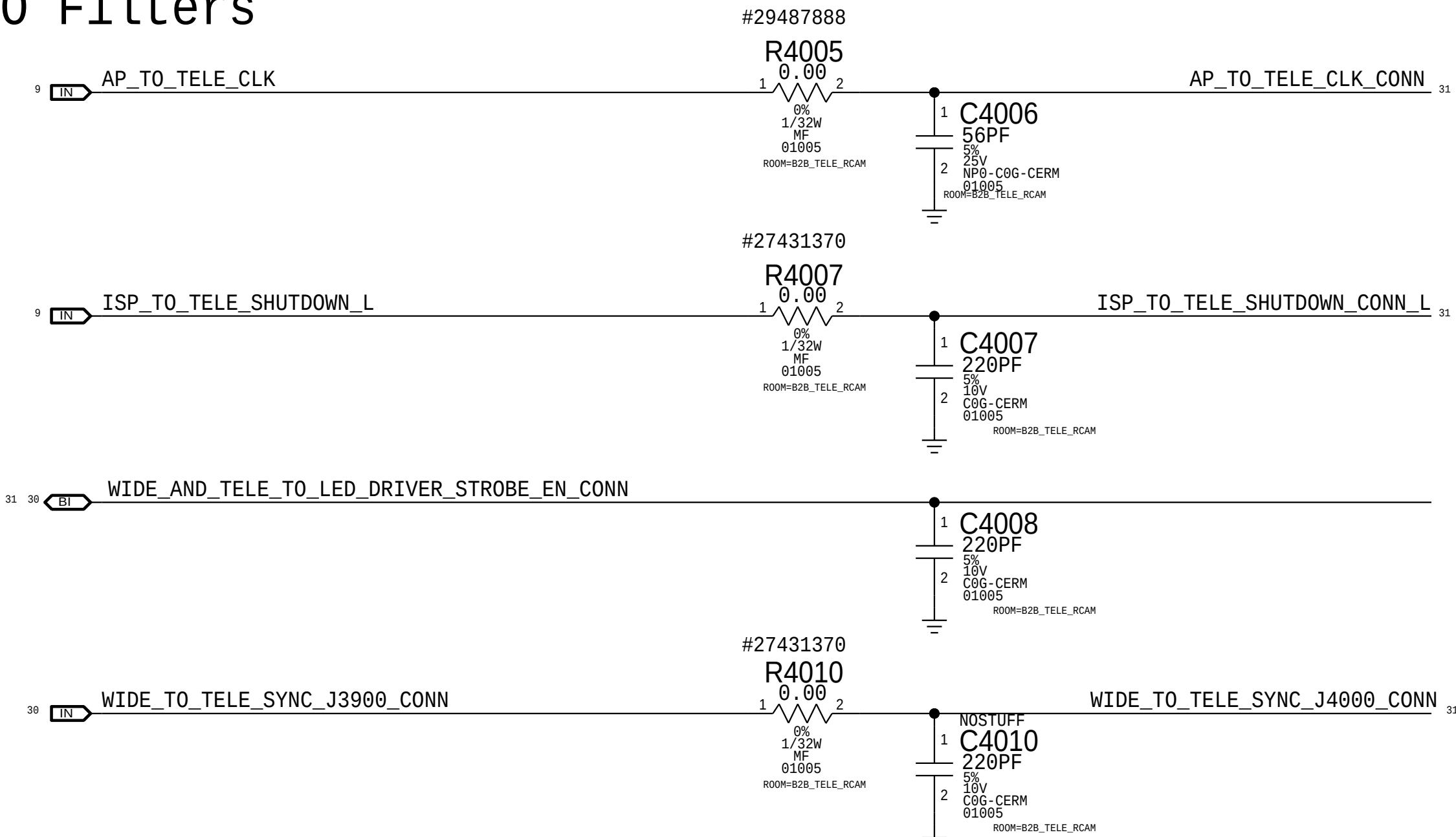
THIS ONE ----> 516S00310/P2 RCPT (USED ON MLB)
516S00309/P2 PLUG (USED ON FLEX)



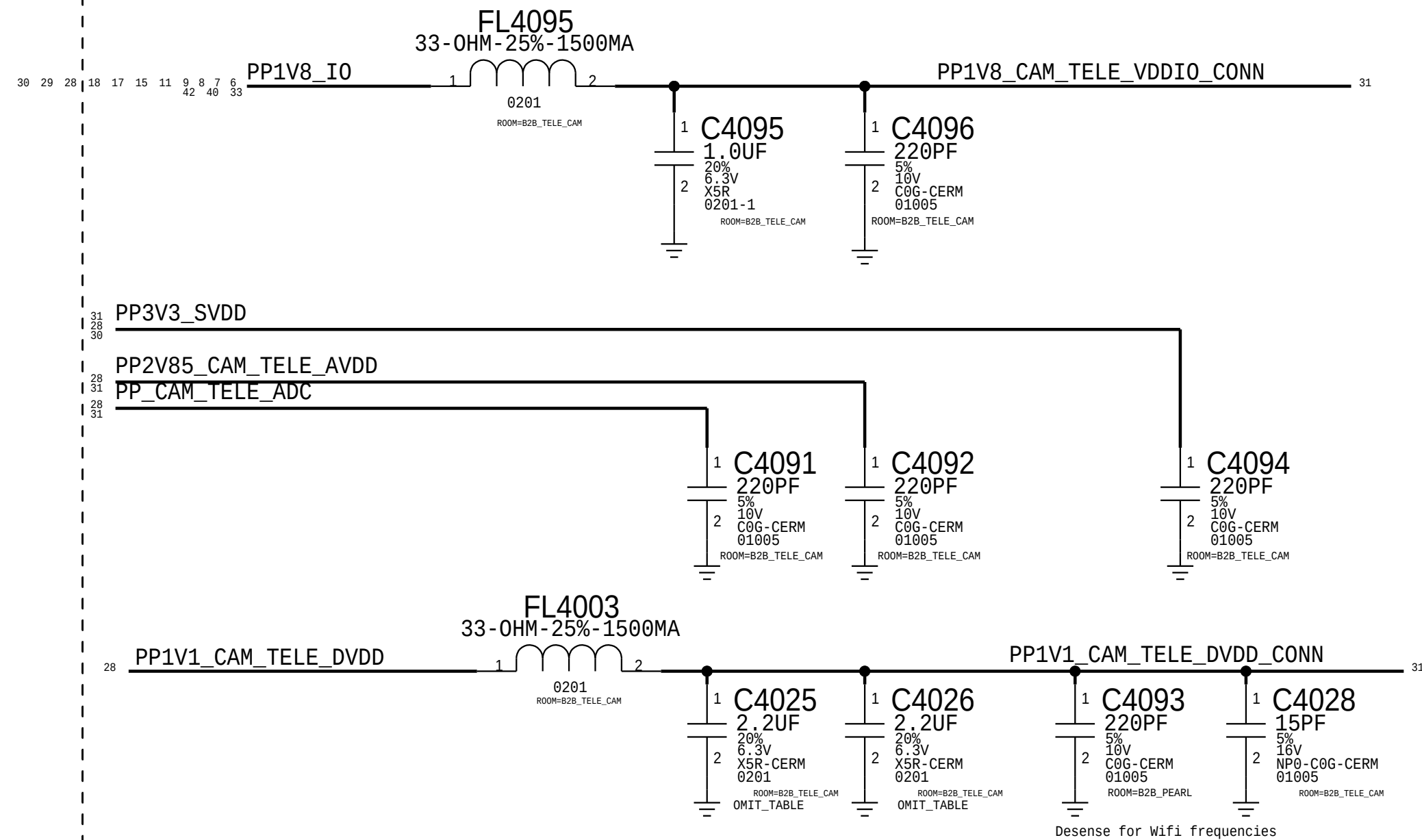
ISP I2C



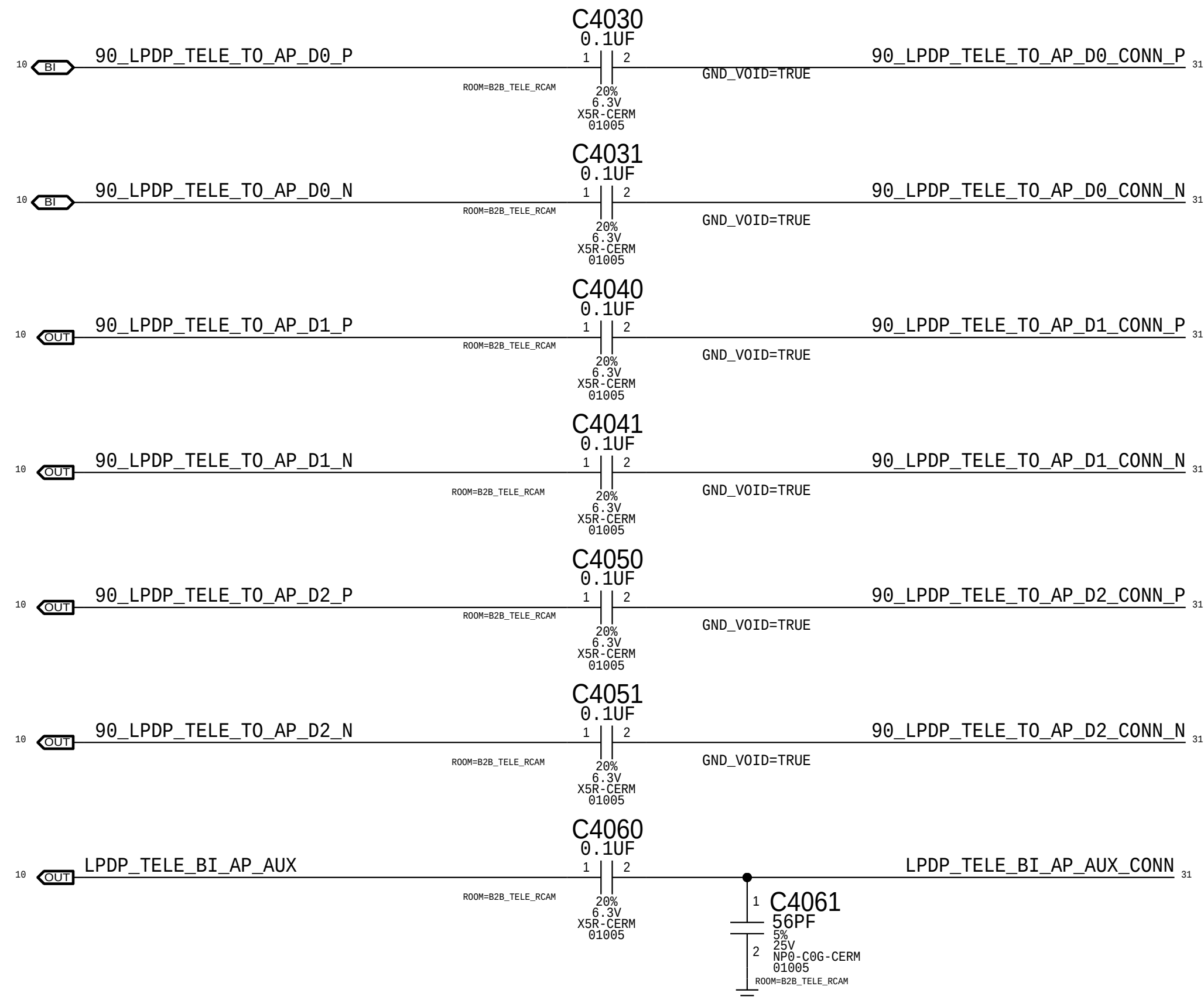
IO Filters



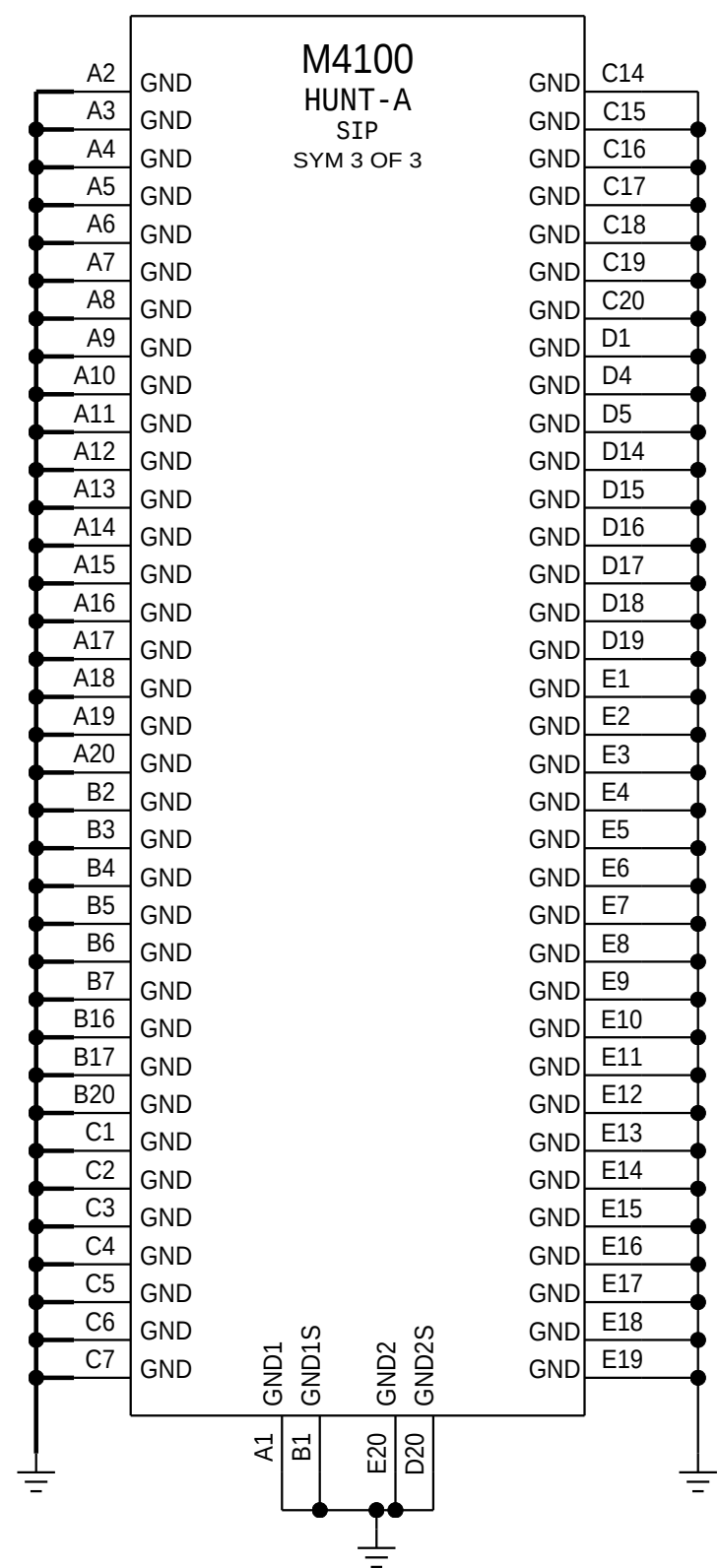
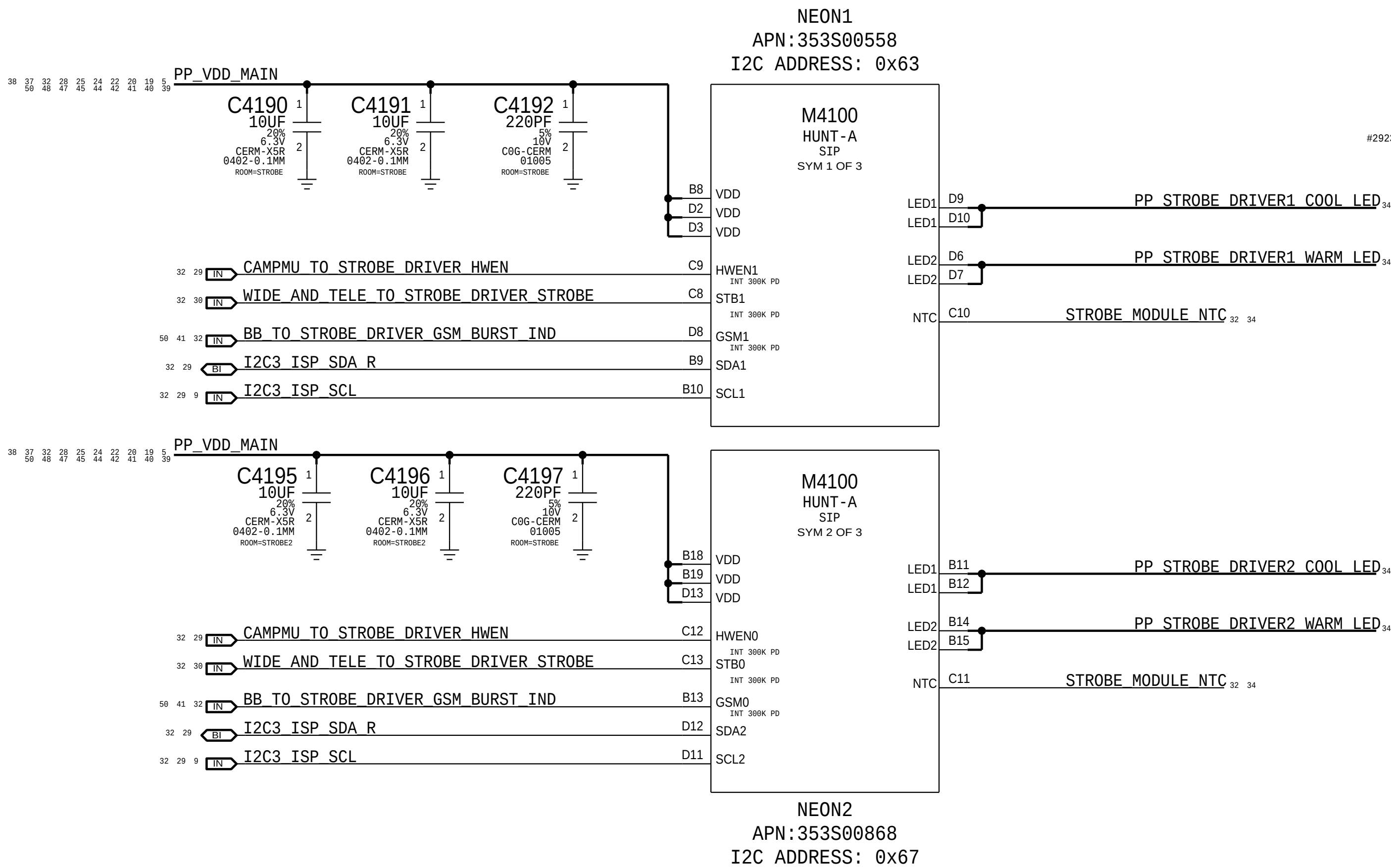
Power Filtering



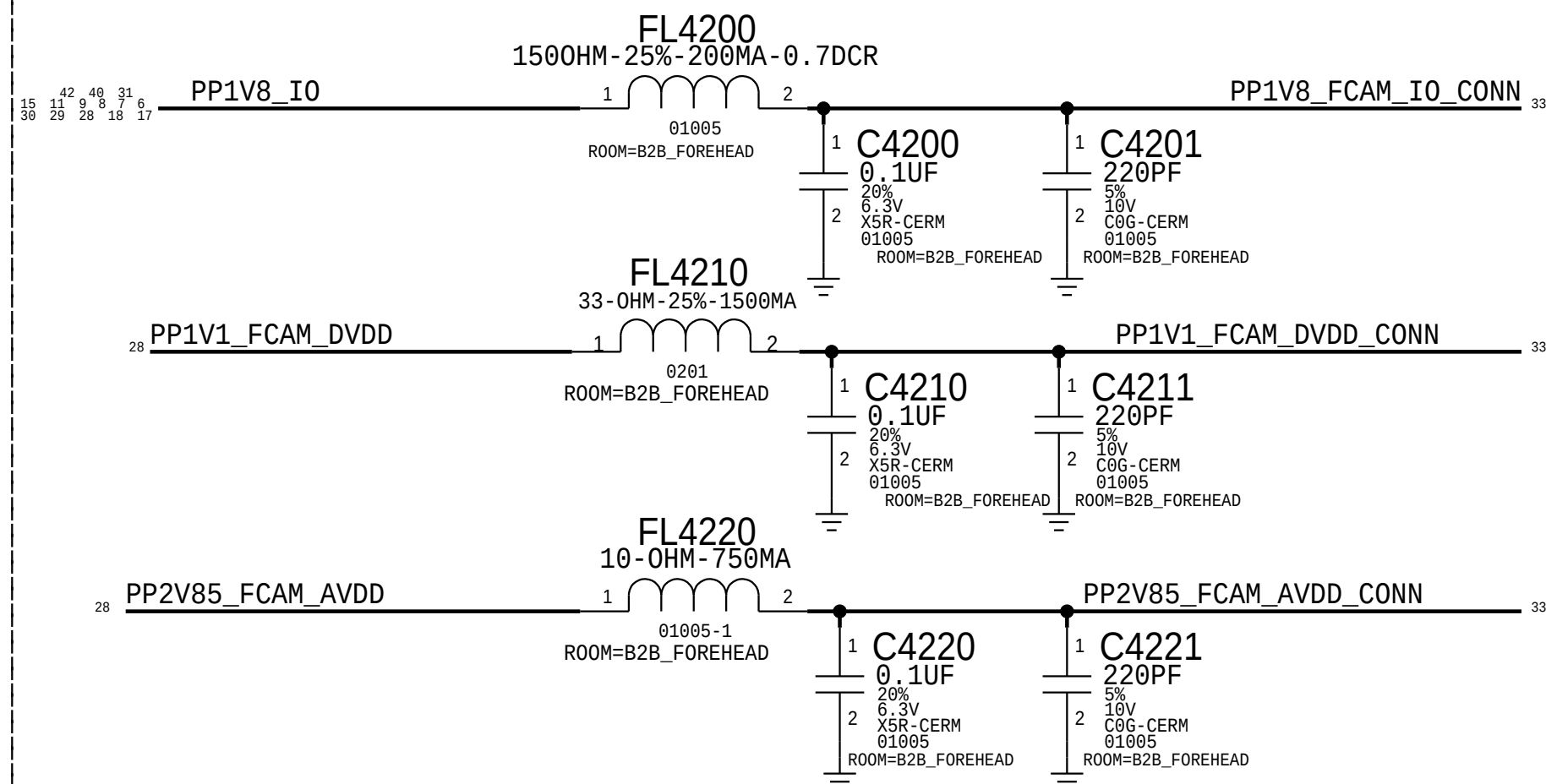
LPDP



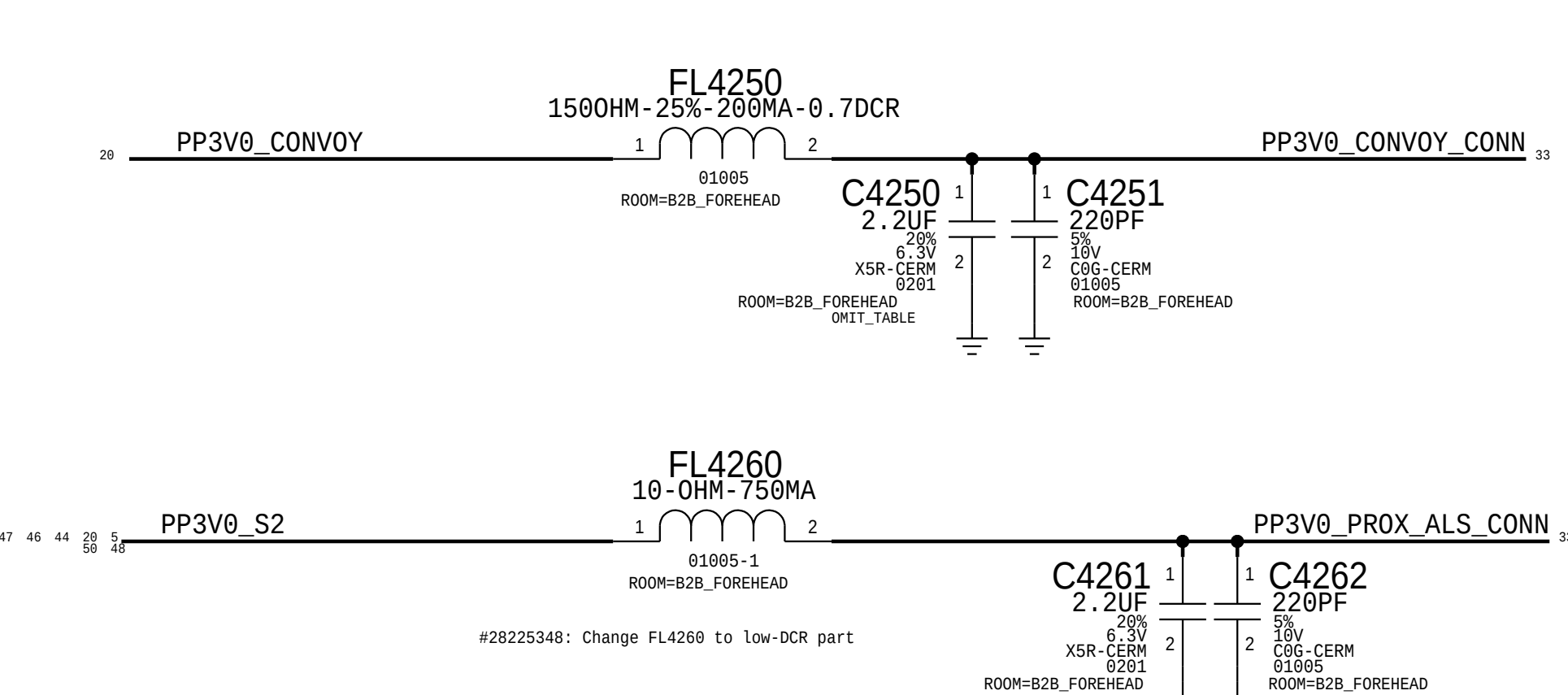
LED STROBE DRIVERS (NEON)



FCAM POWER

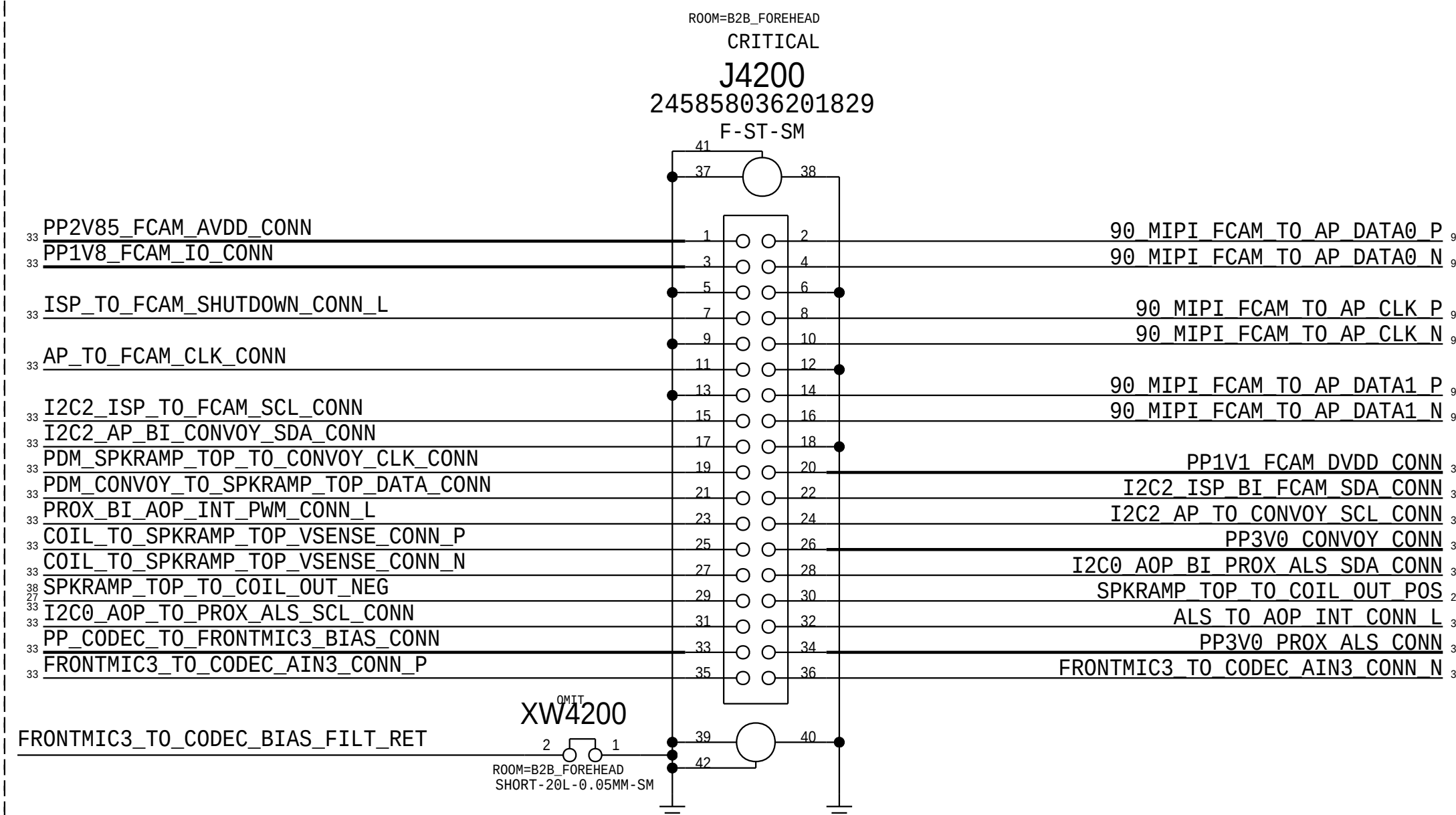


PROX + ALS + CONVOY POWER

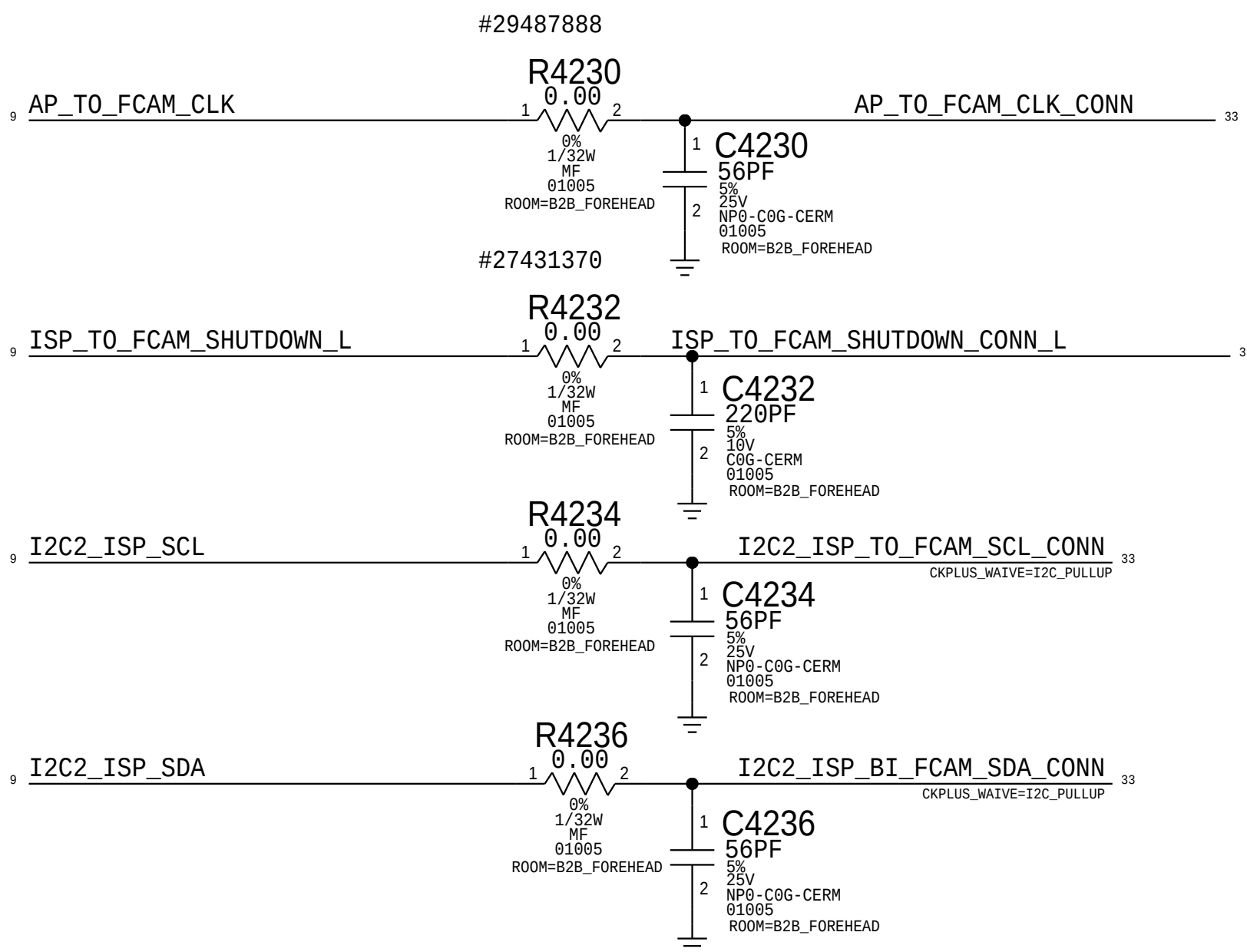


FOREHEAD CONNECTOR

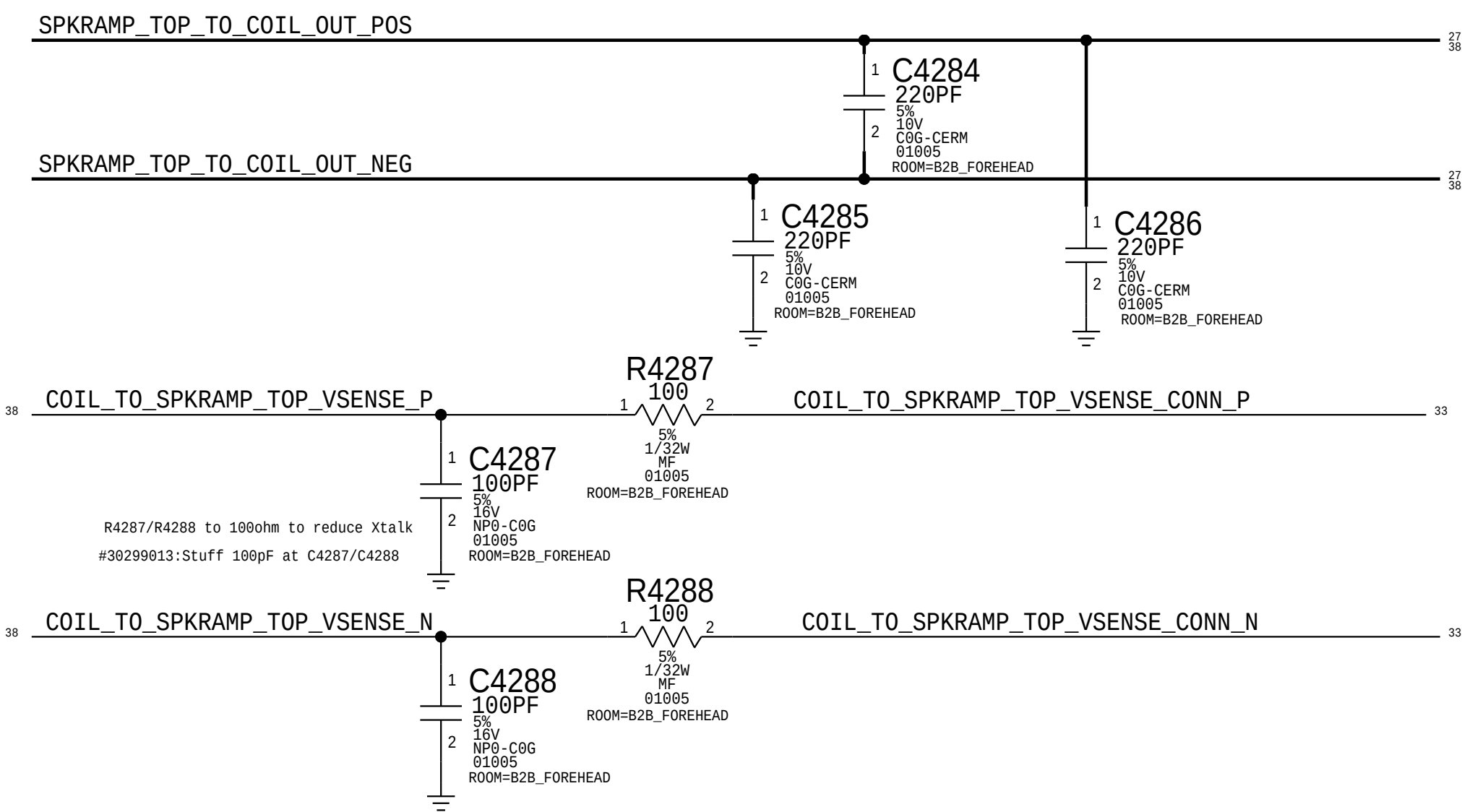
APN: 516S00146



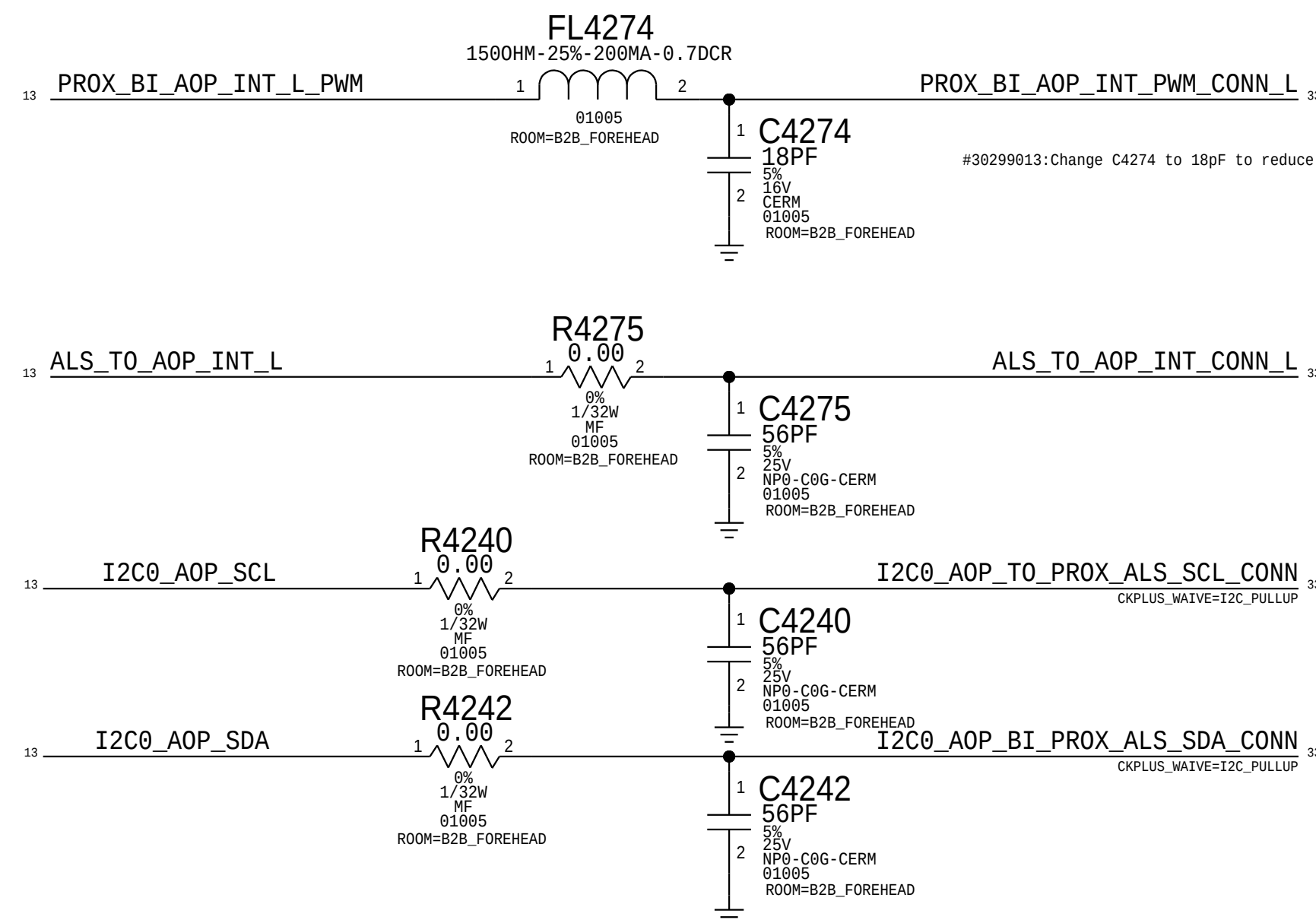
FCAM I/O



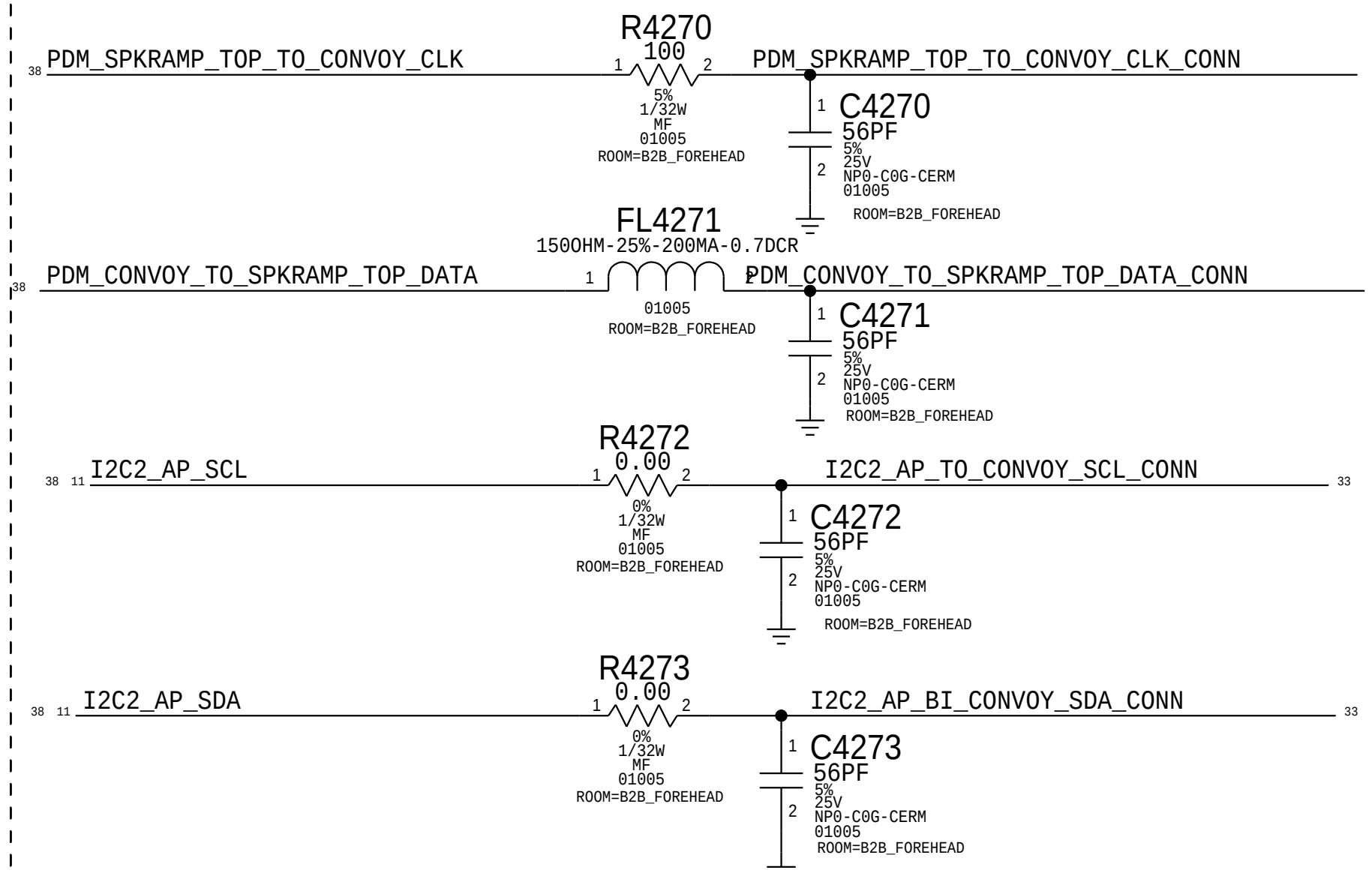
NORTH SPEAKER



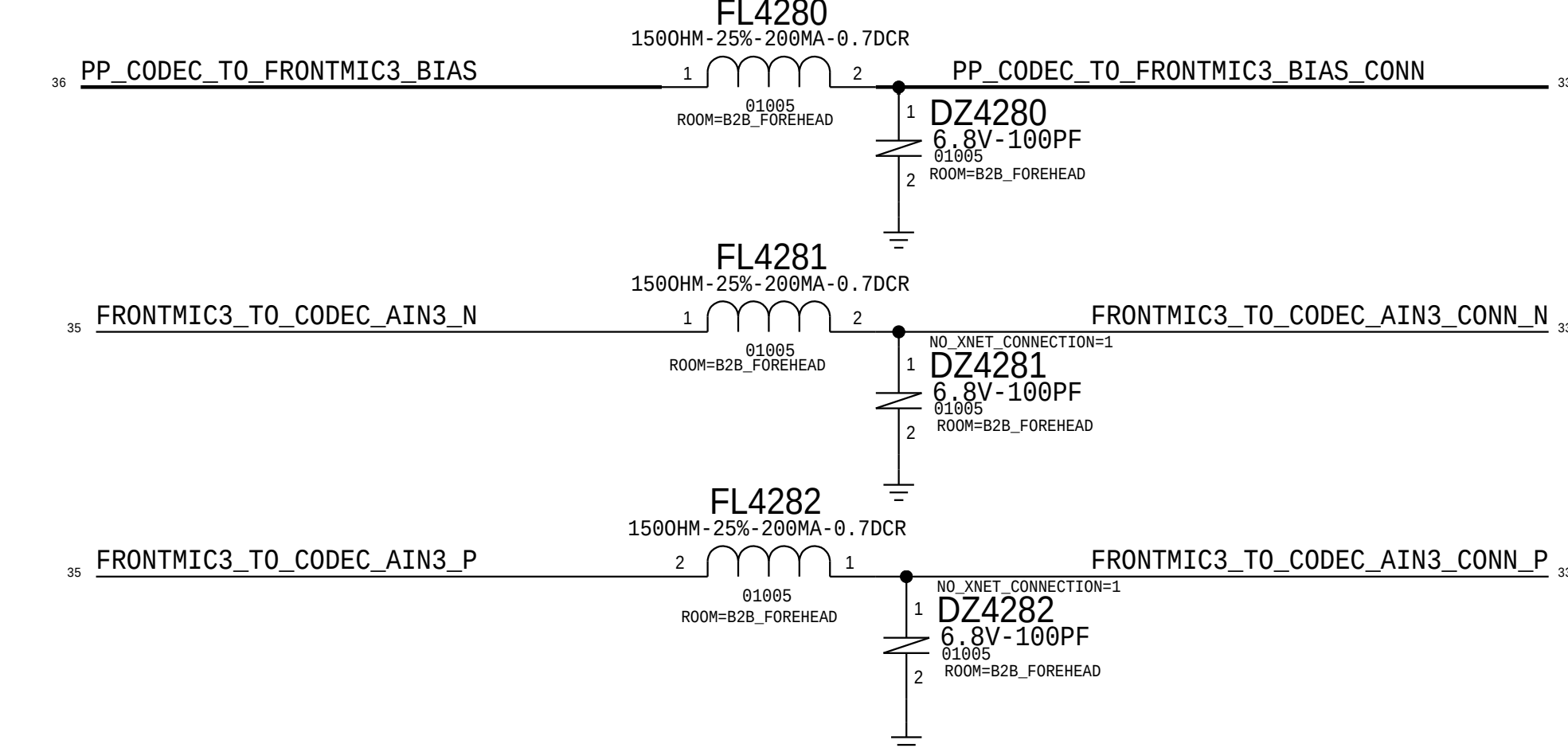
PROX + ALS I/O

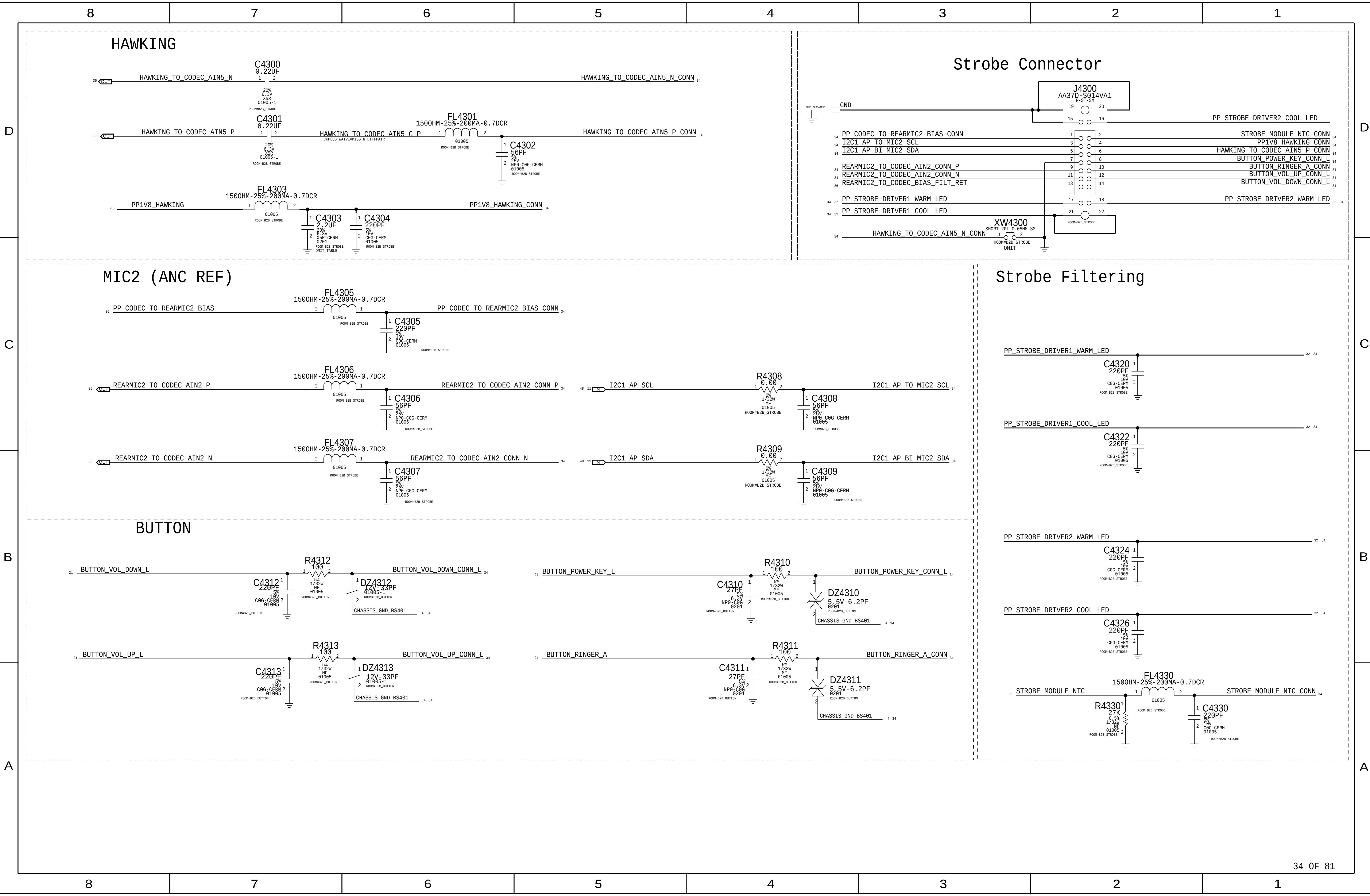


CONVOY I/O

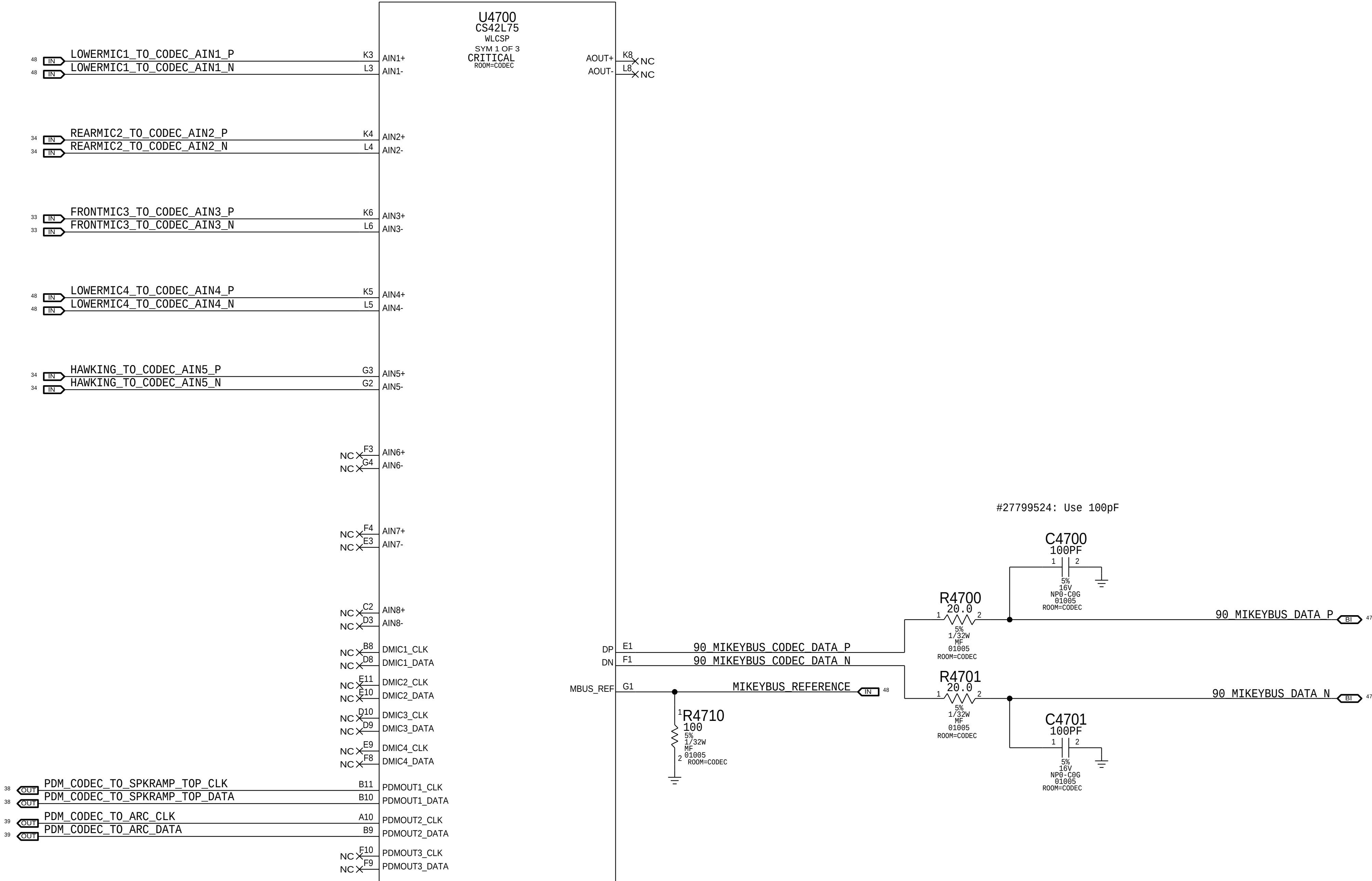


MIC3

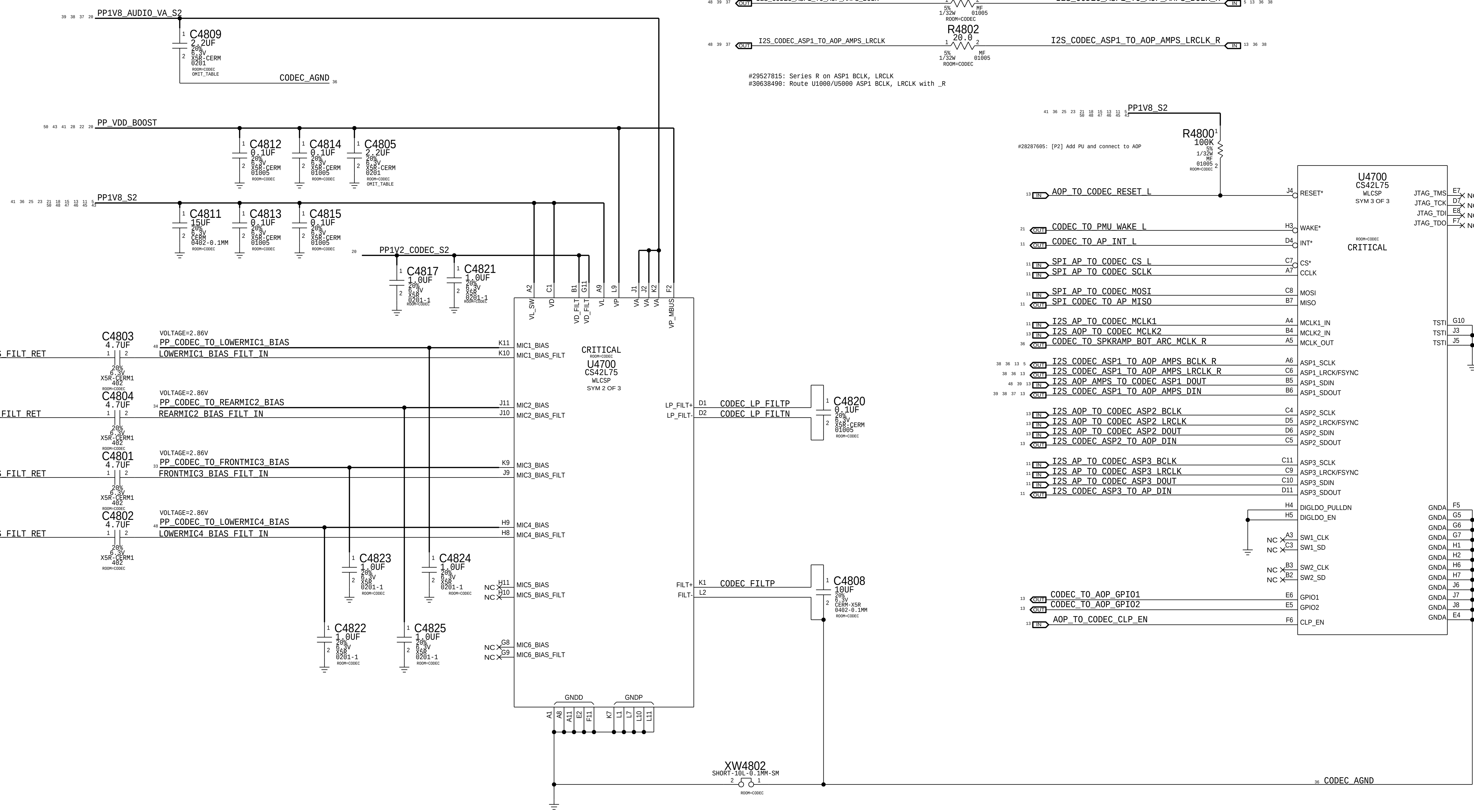




CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



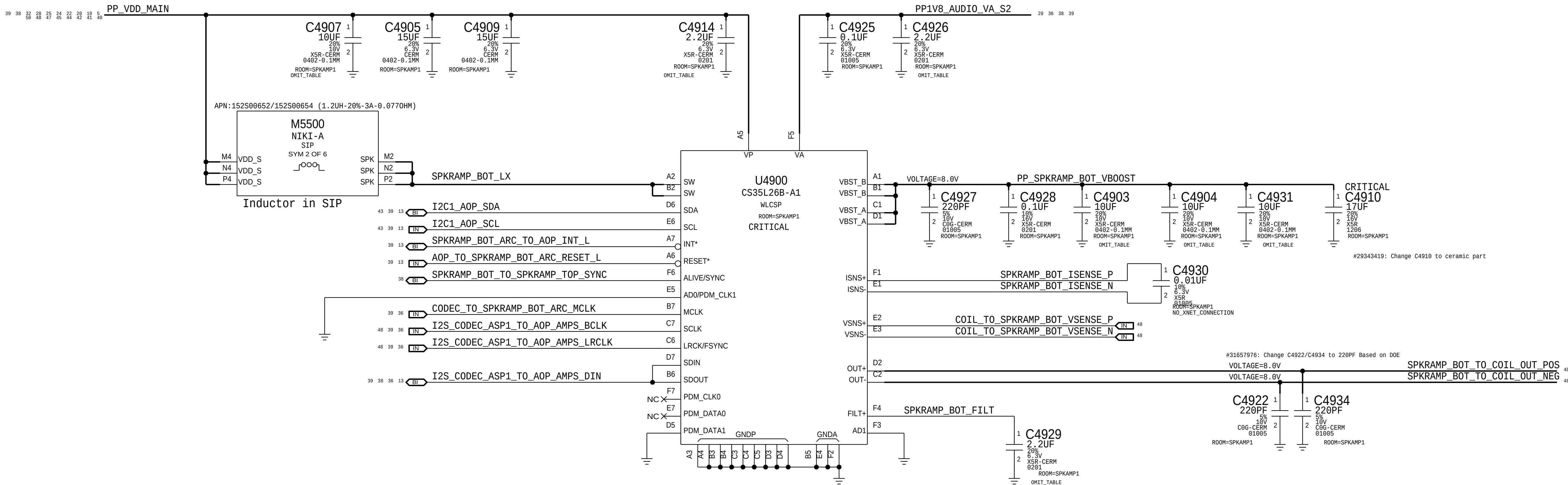
CALLAN AUDIO CODEC (POWER & I/O)



South Speaker Amplifier

APN: 338S00143

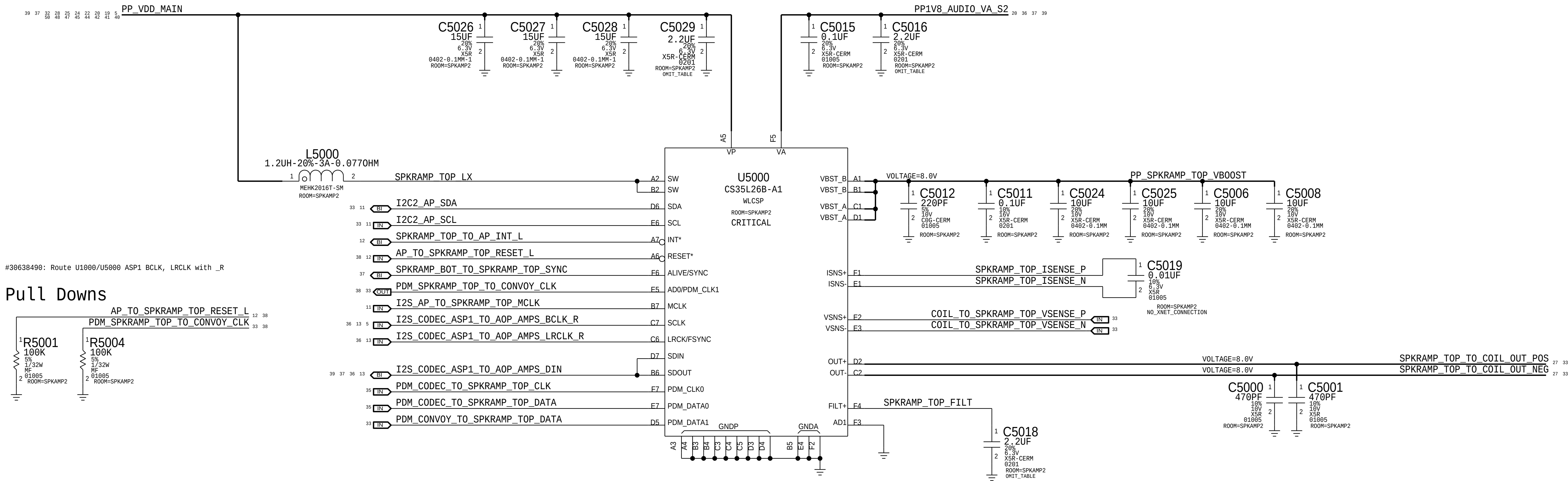
I2C ADDRESS: 1000 000x
0x80



North Speaker Amplifier

APN: 338S00295

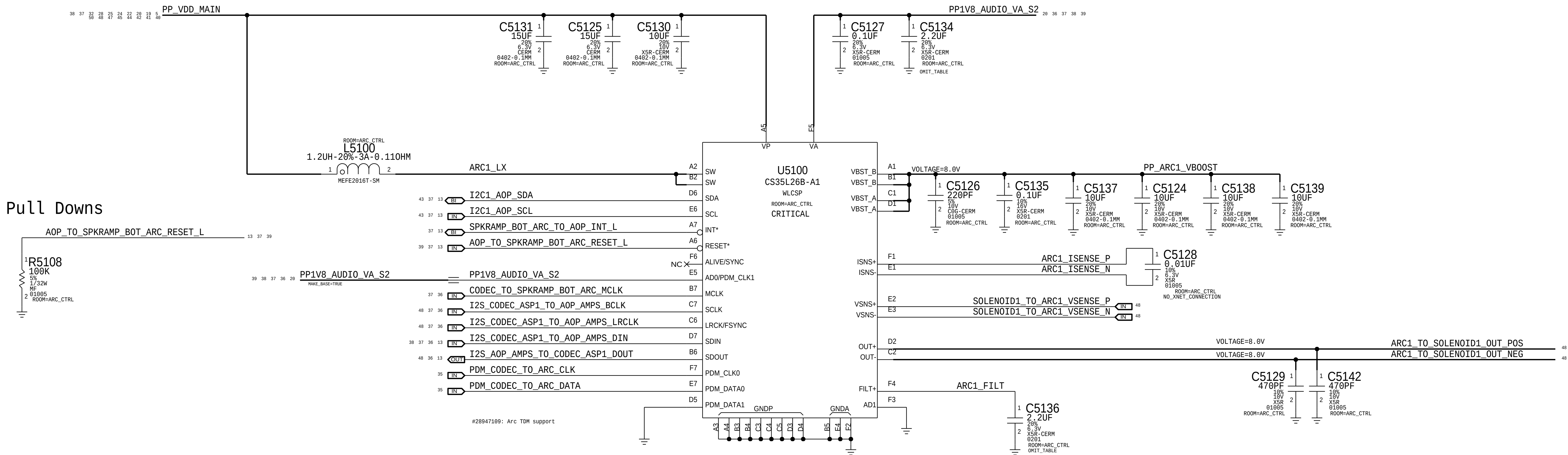
I2C ADDRESS: 1000 000x (0x80)



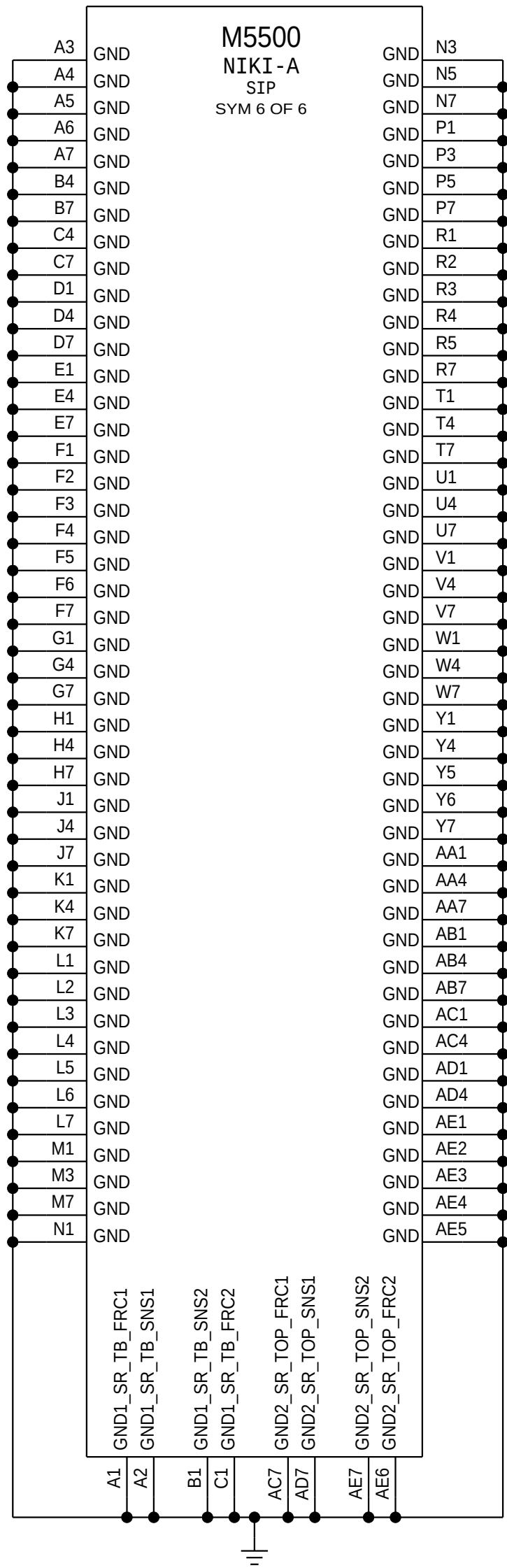
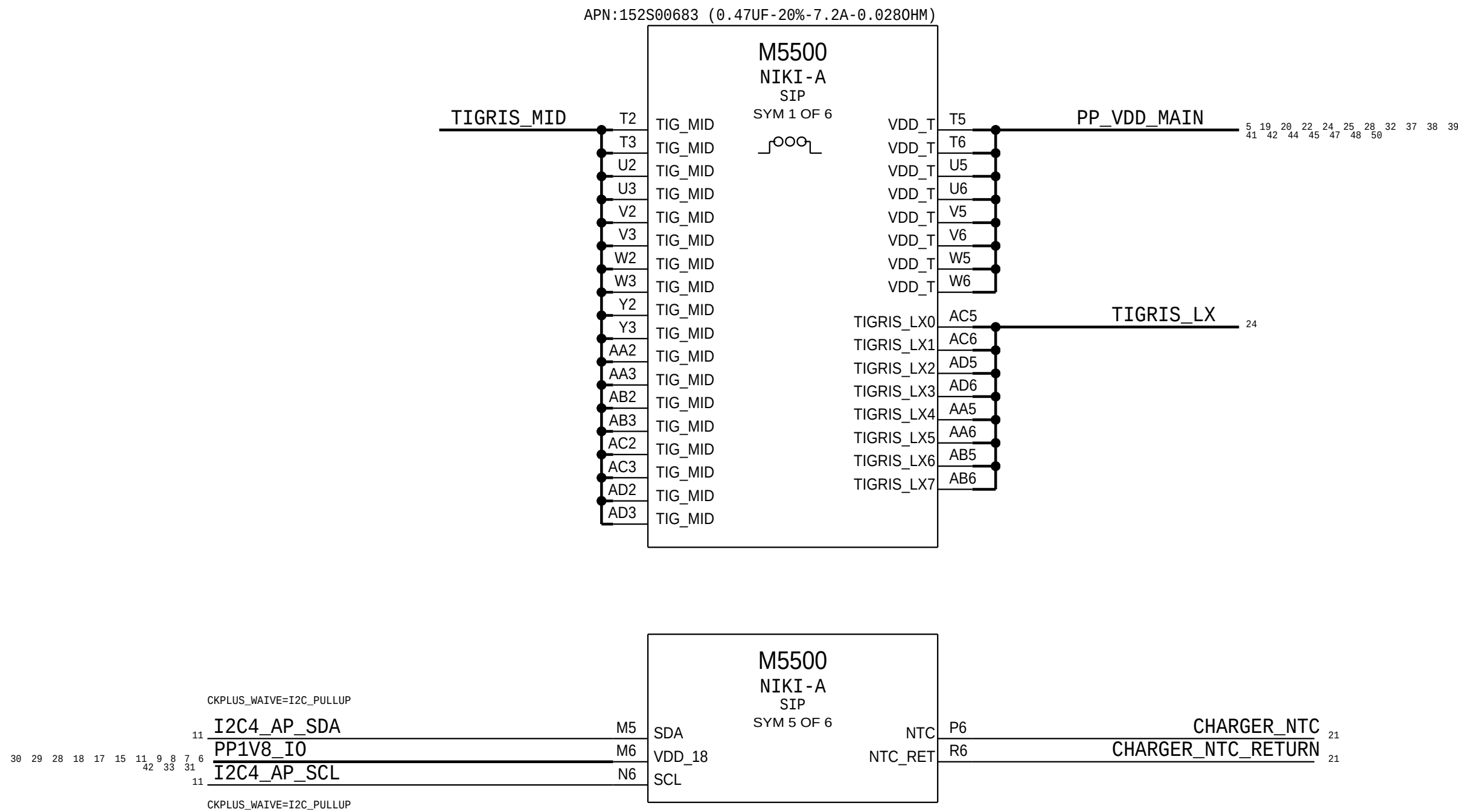
ARC DRIVER

APN: 338S00295

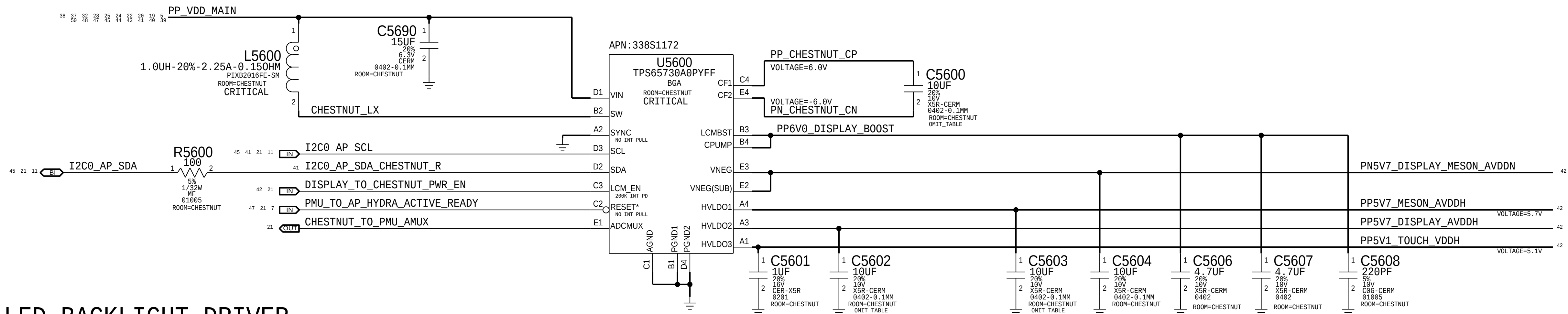
I2C ADDRESS: 1000 001x (0x82)



Niki SIP

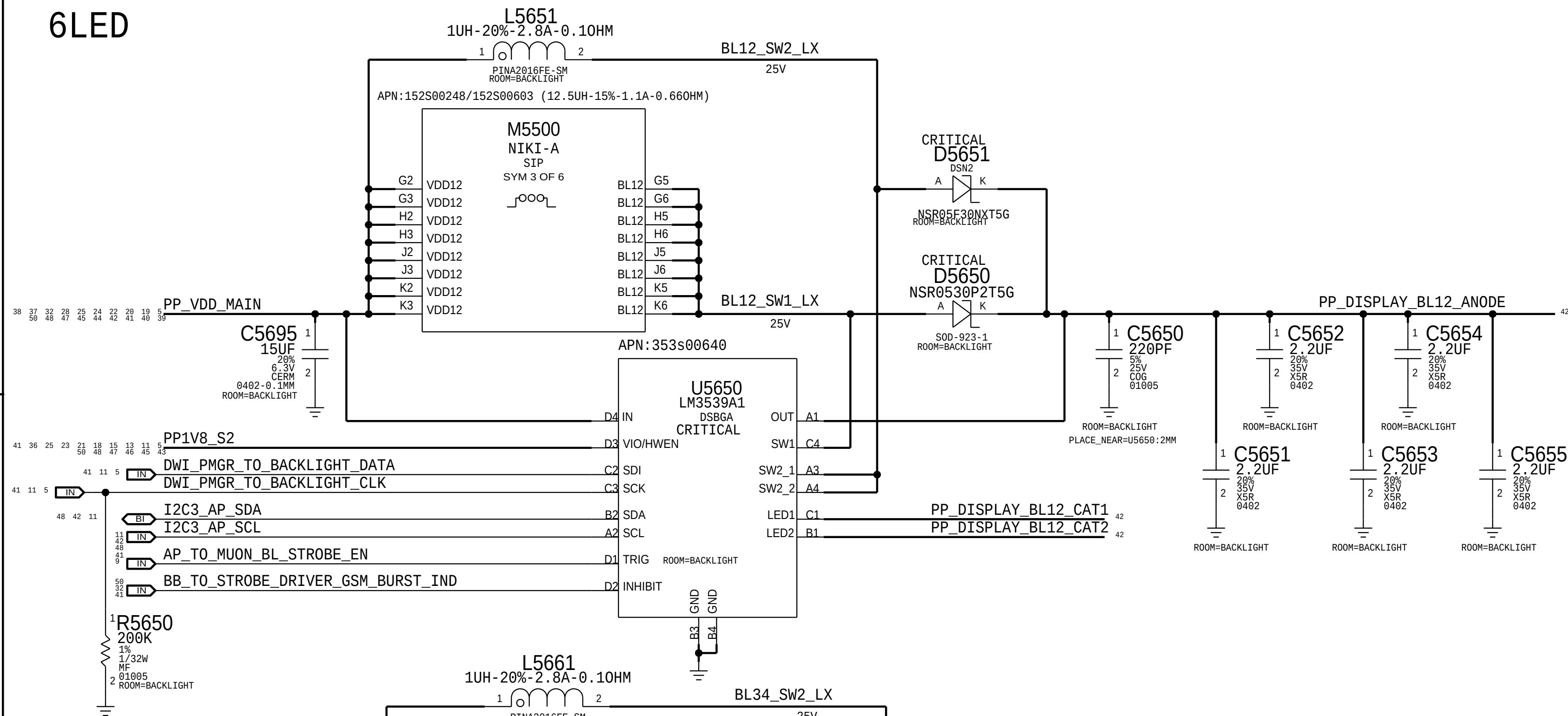


CHESTNUT DISPLAY PMU

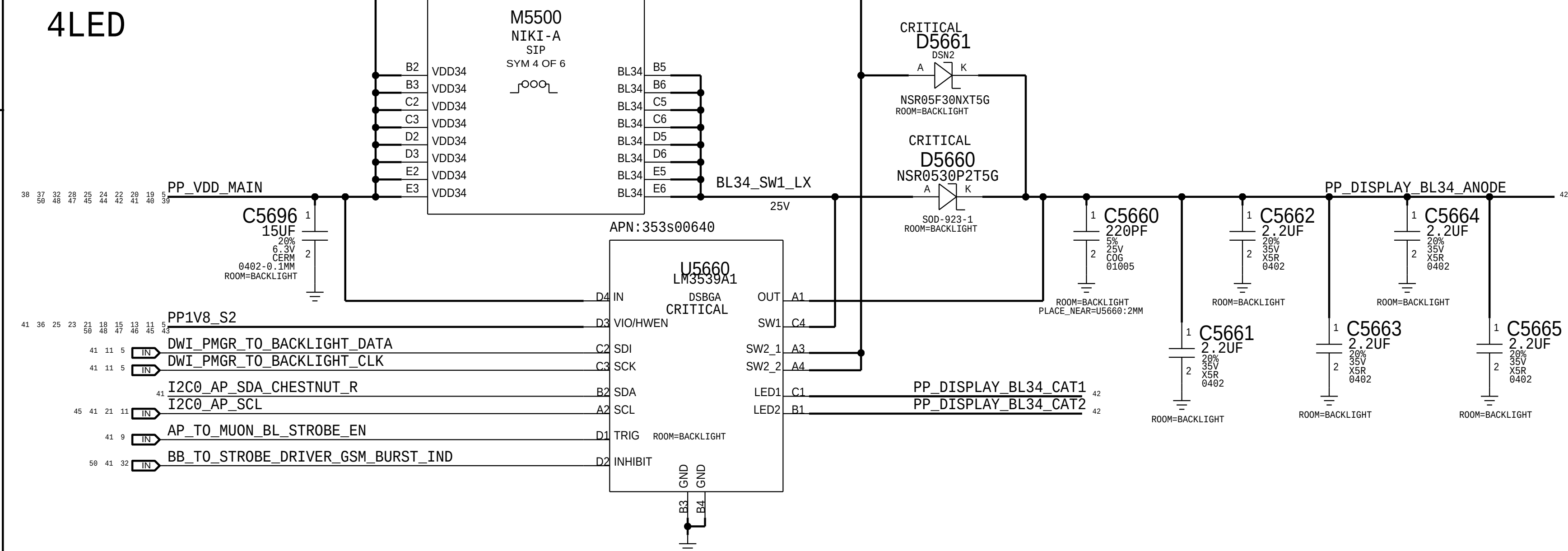


LED BACKLIGHT DRIVER

6LED

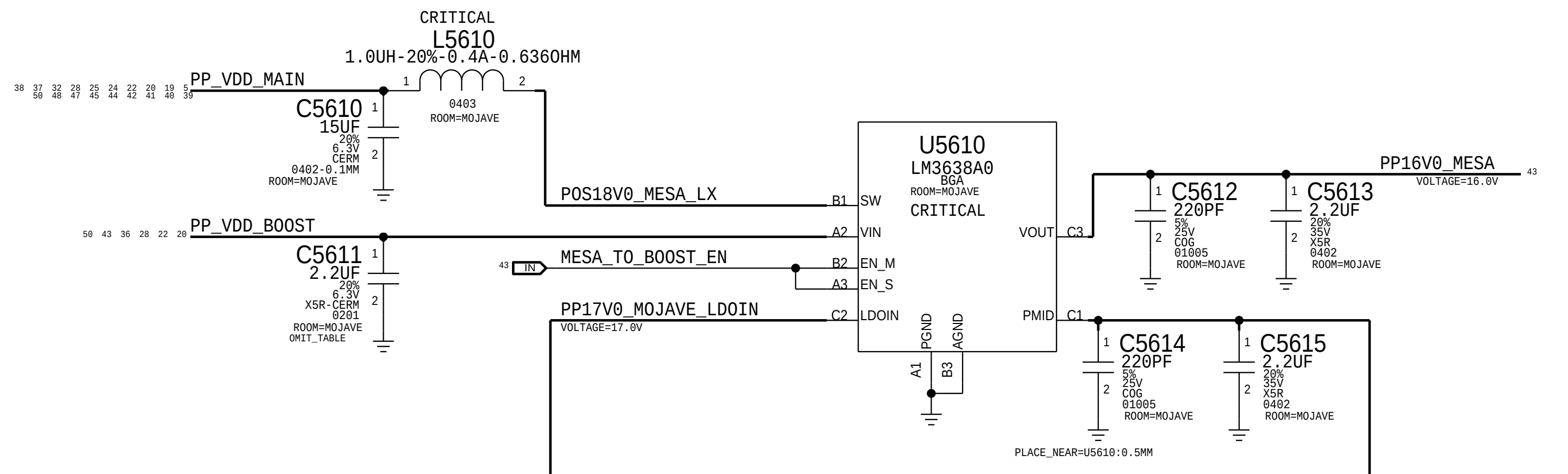


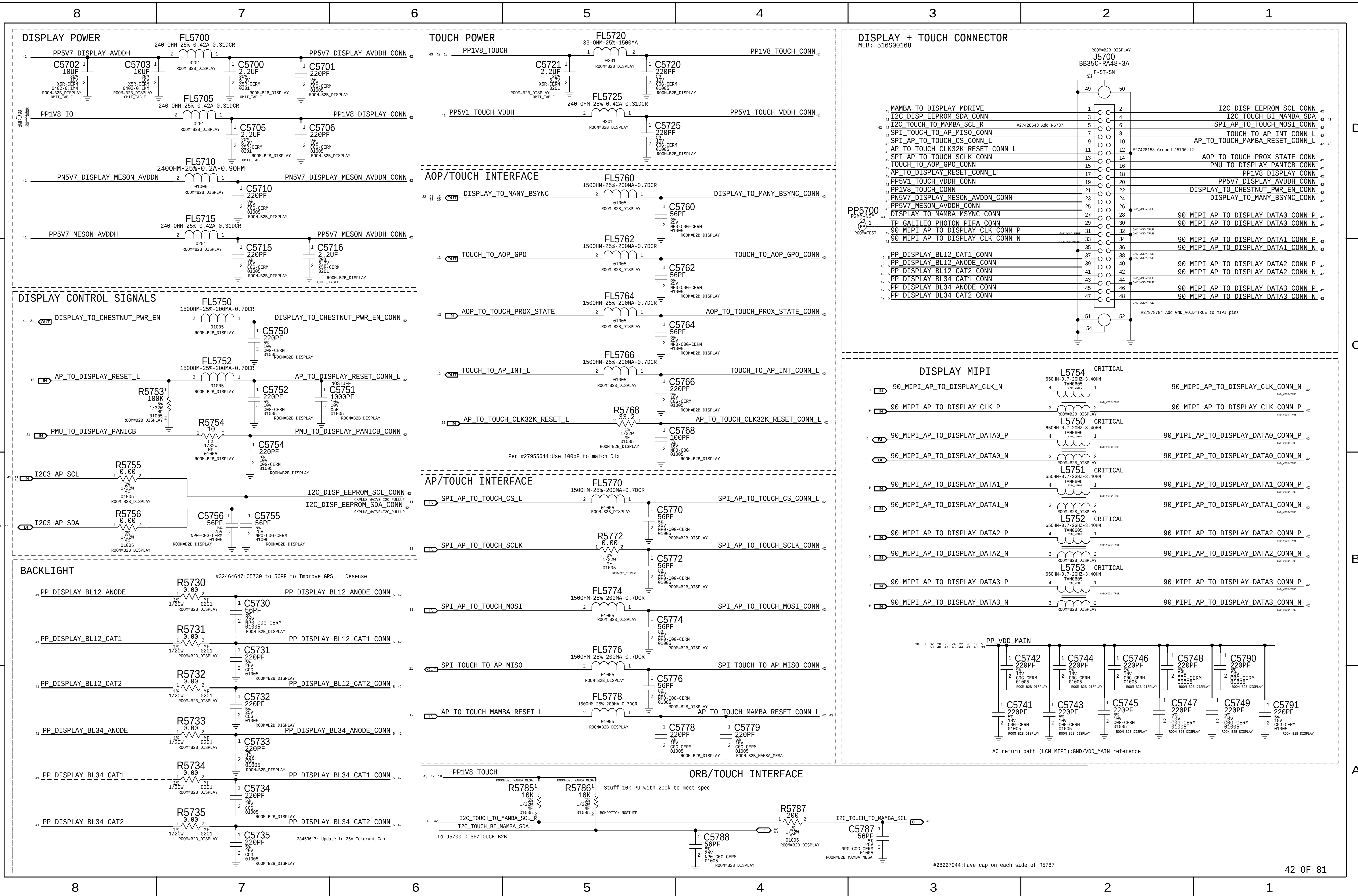
4LED



MOJAVE MESA BOOST

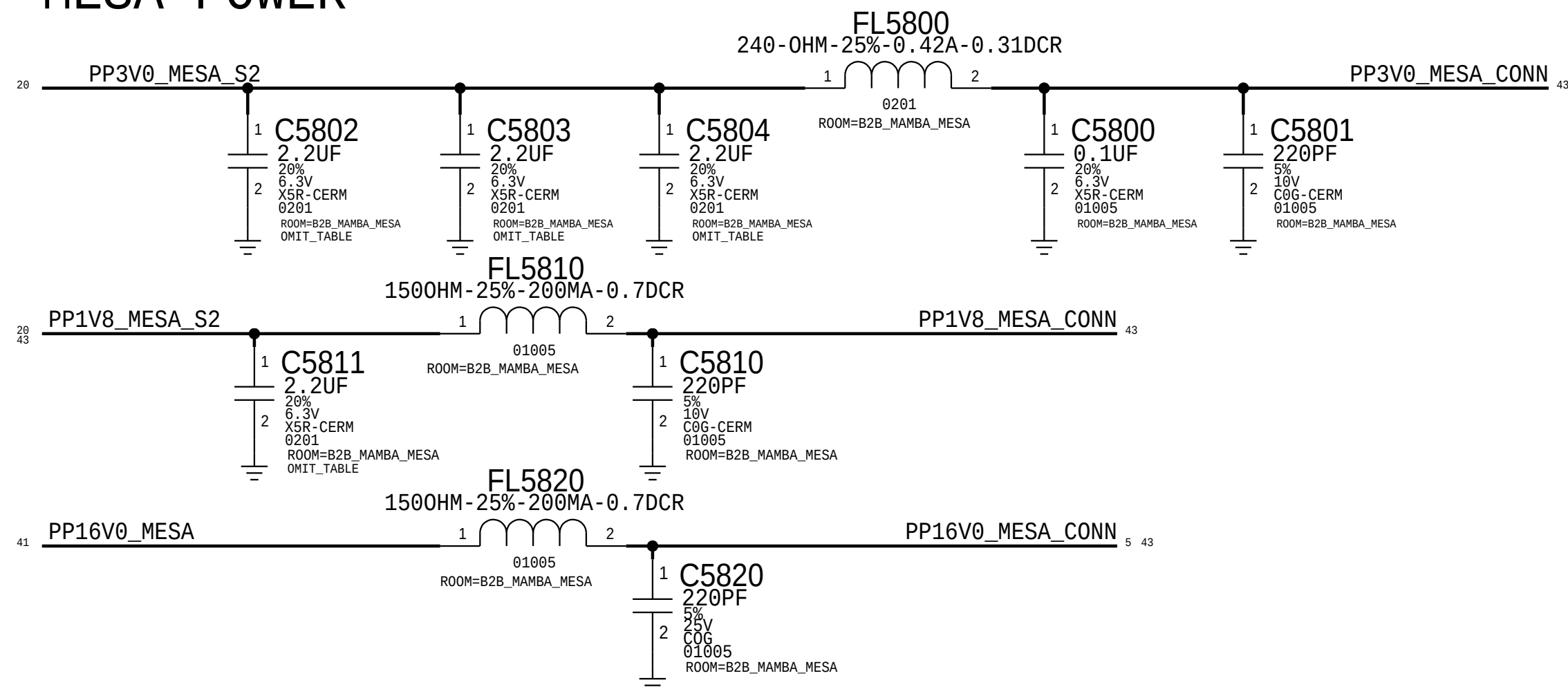
APN: 353S00671





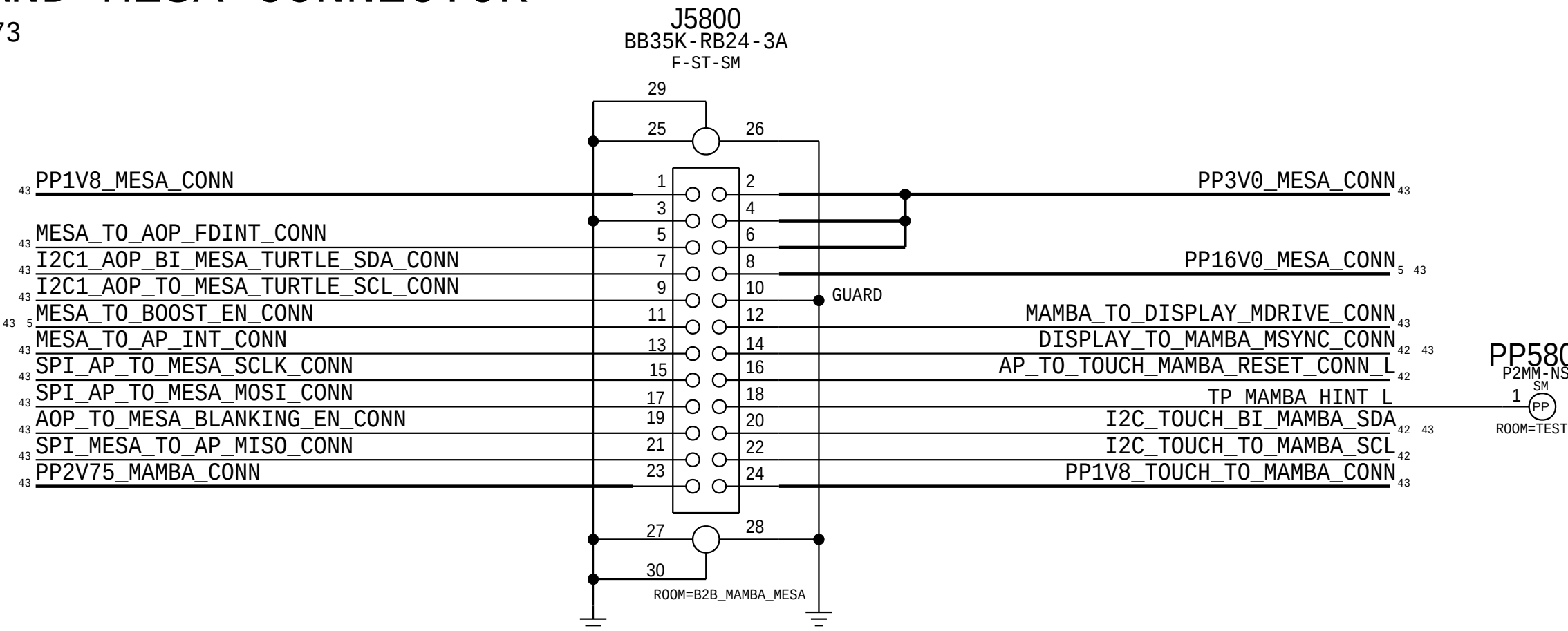
MESA POWER

#30272427: Change FL5800 to 155S00067

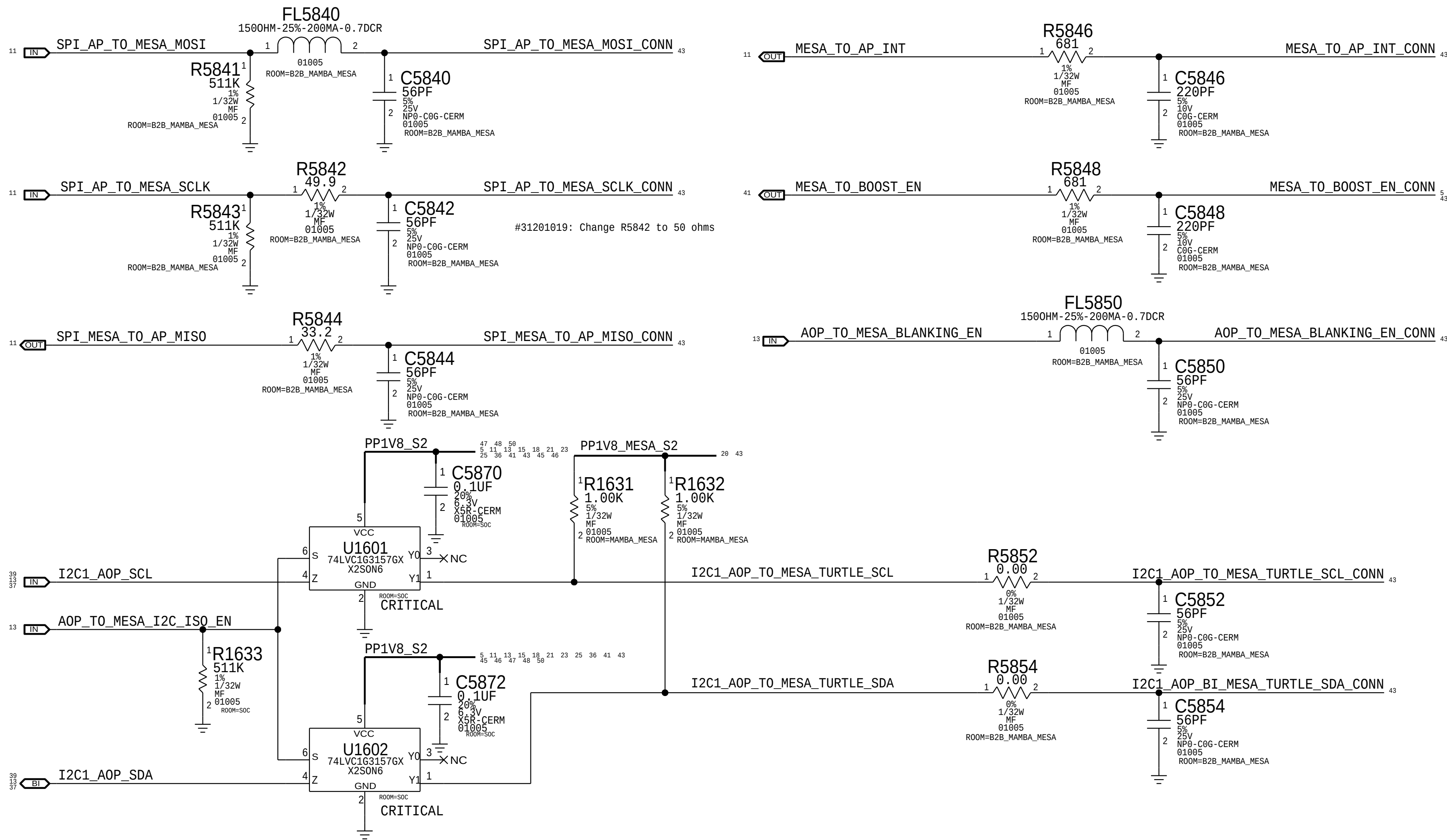


MAMBA AND MESA CONNECTOR

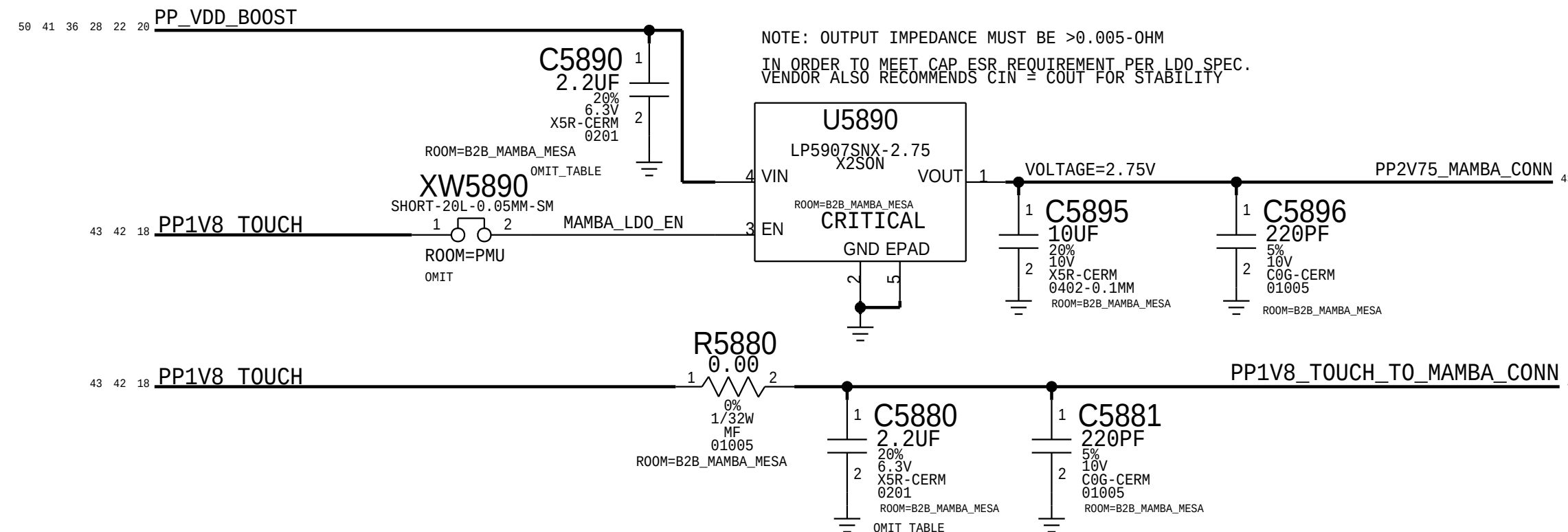
MLB: 516S00273



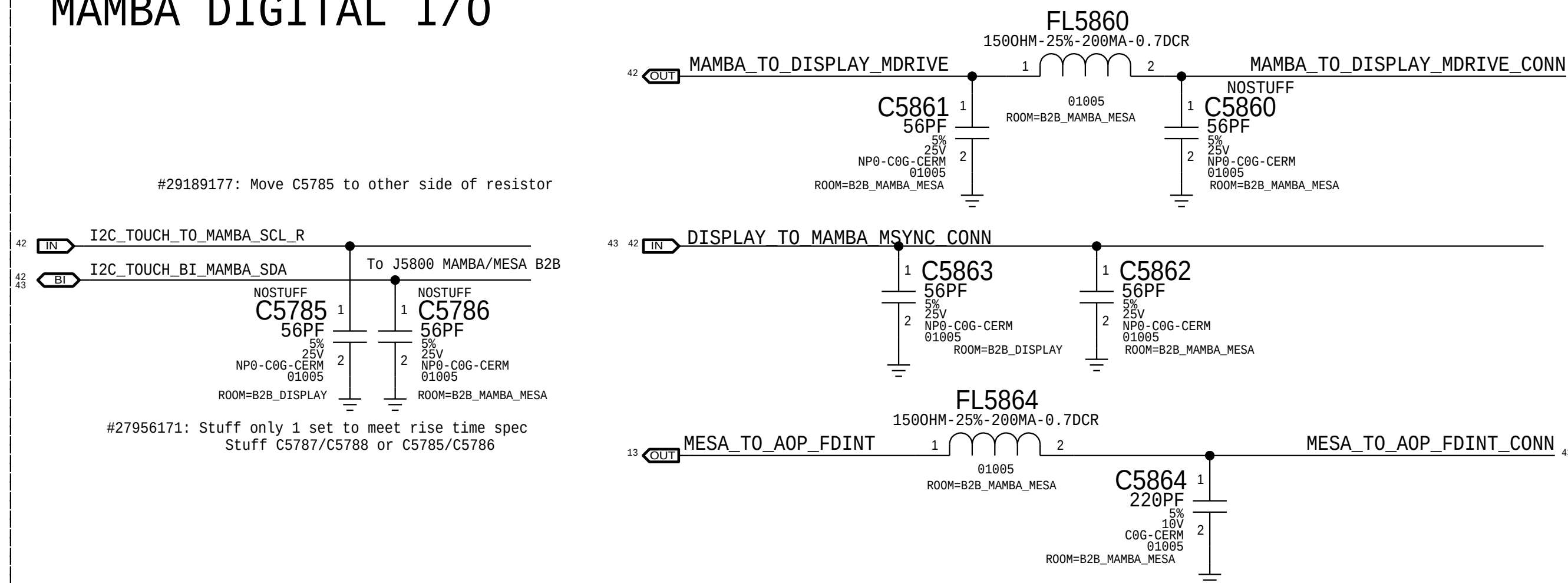
MESA DIGITAL I/O



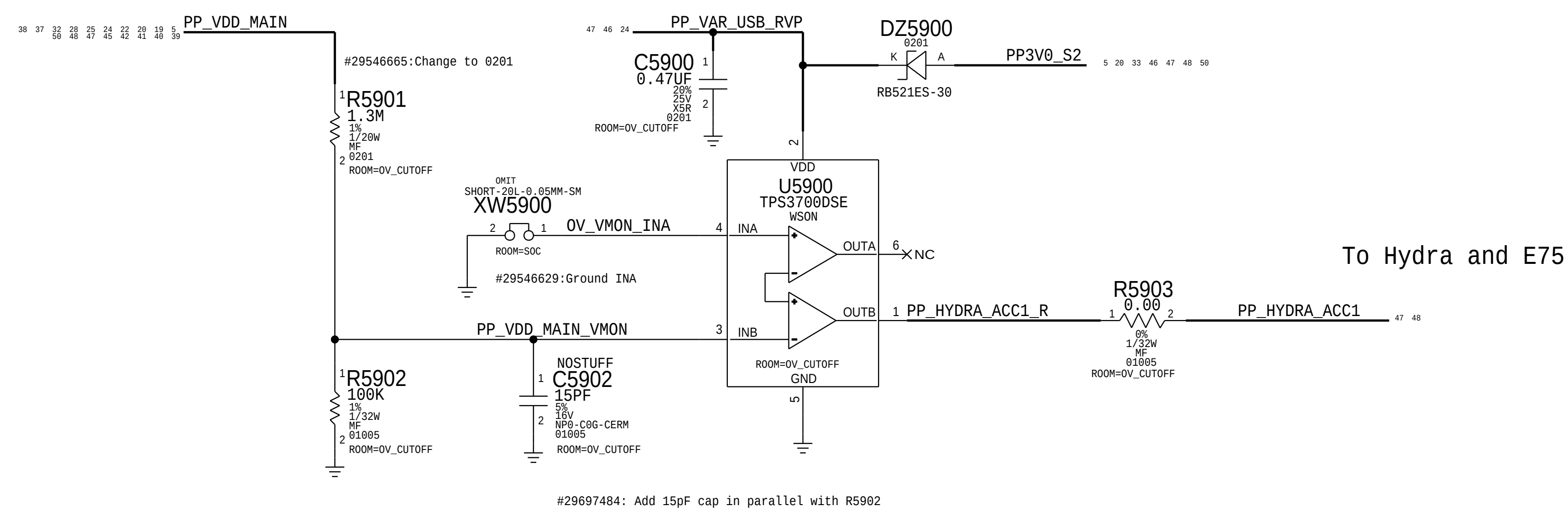
MAMBA POWER

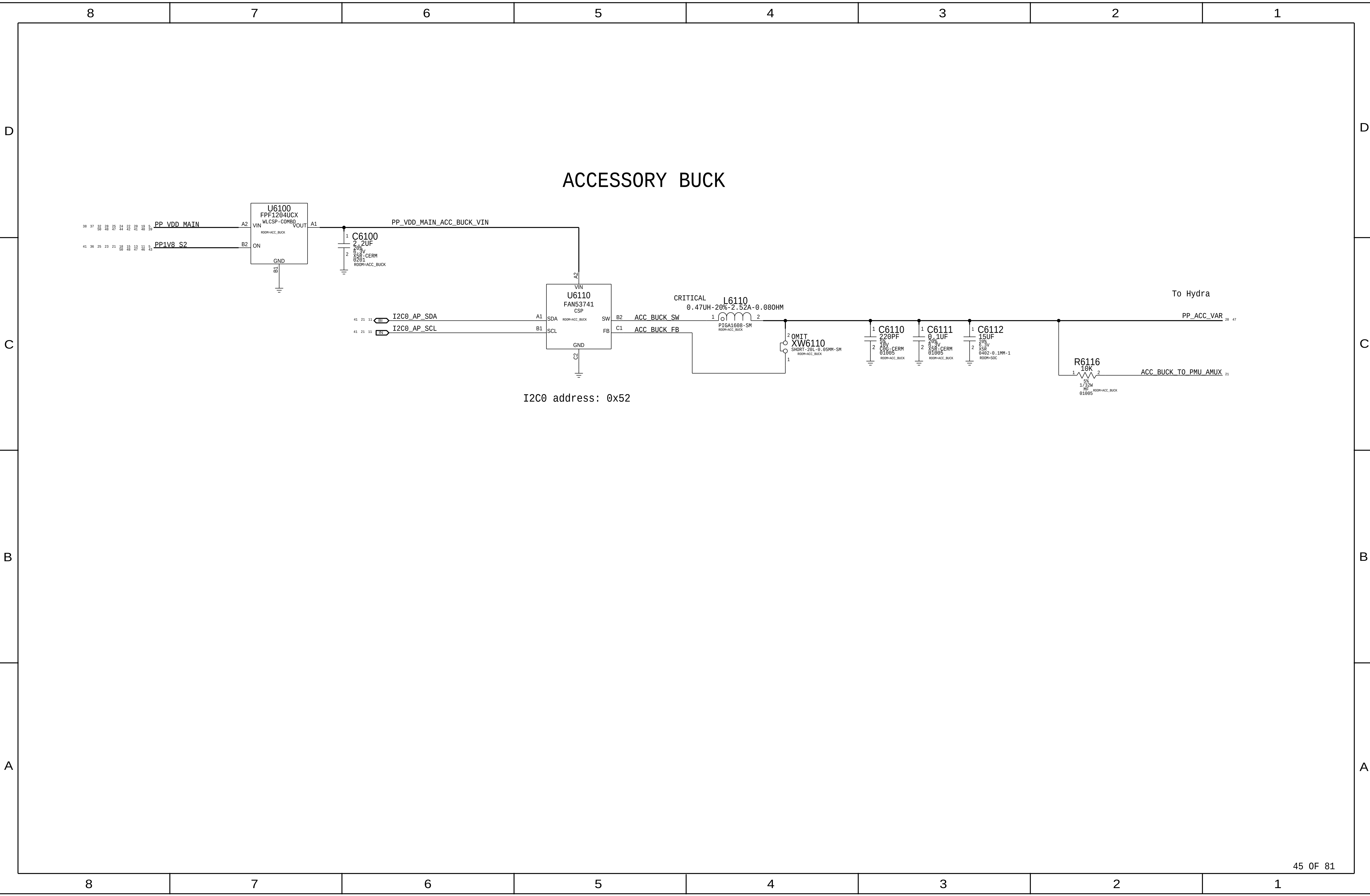


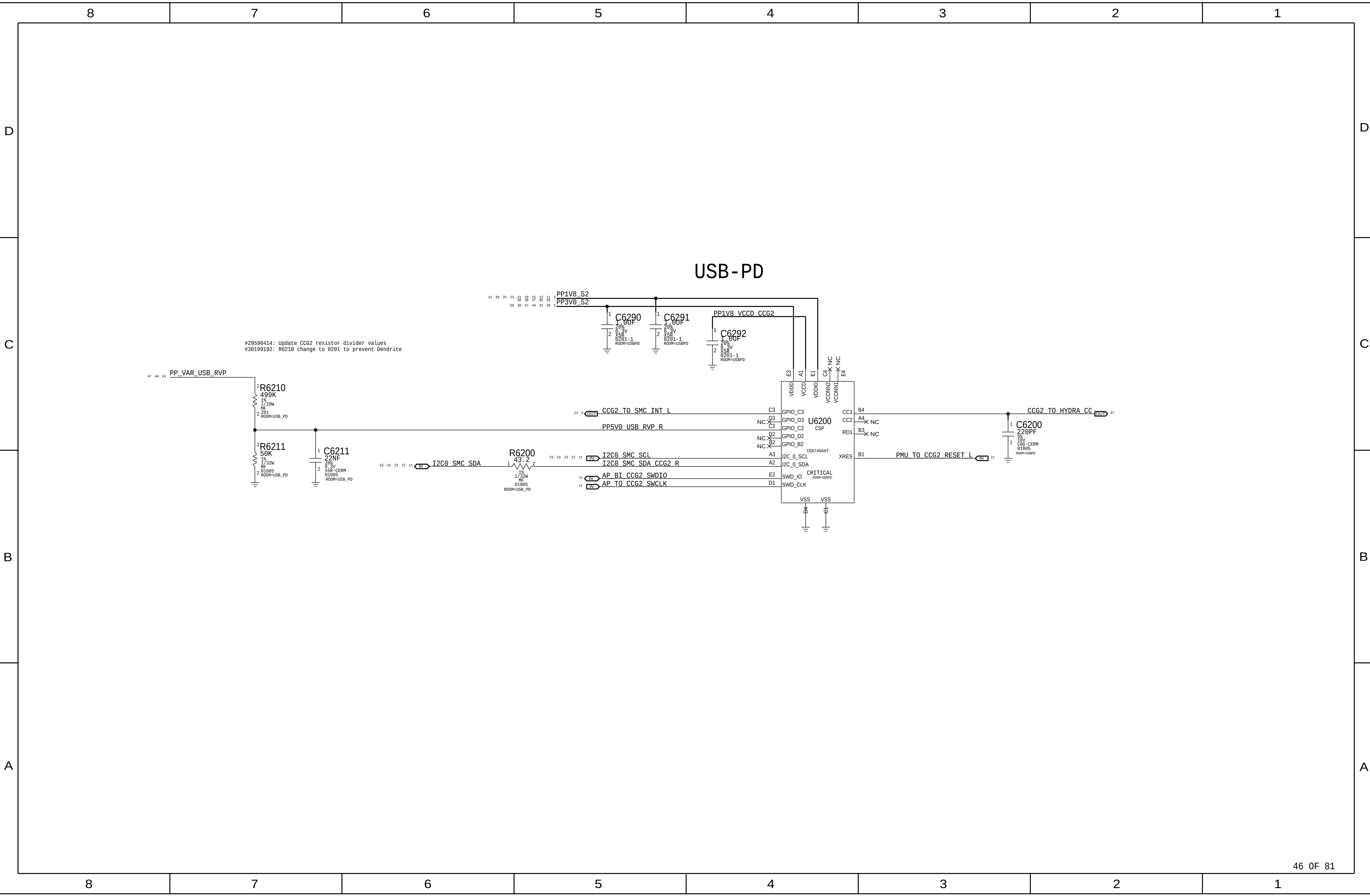
MAMBA DIGITAL I/O



VDD_MAIN 0V CUT-OFF CIRCUIT

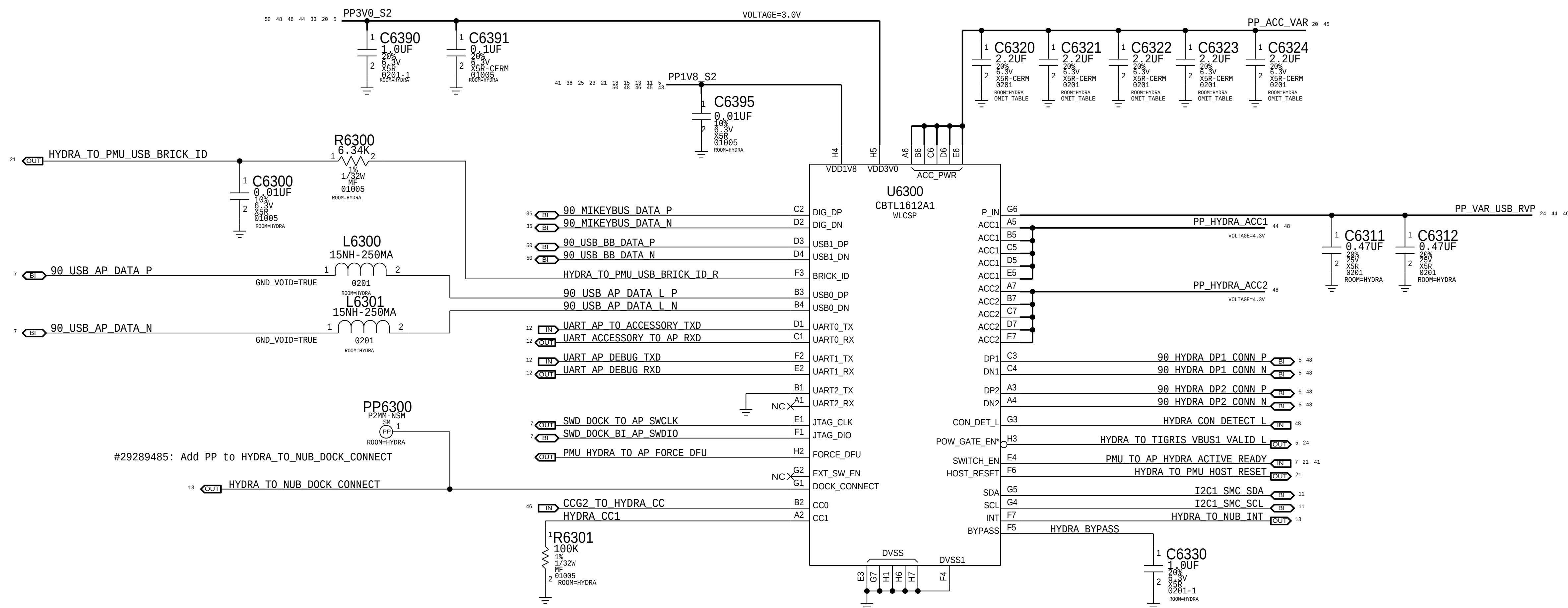




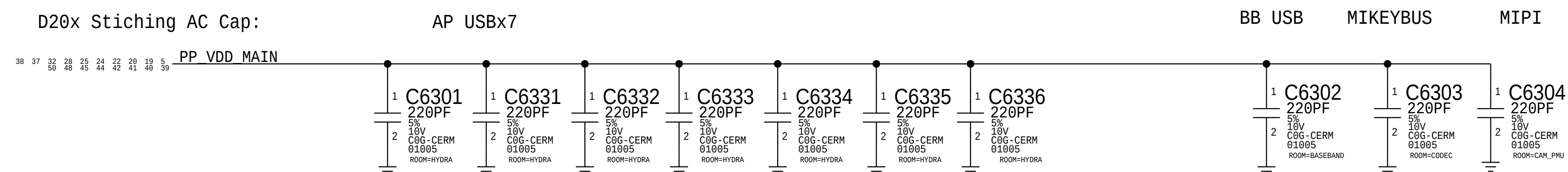


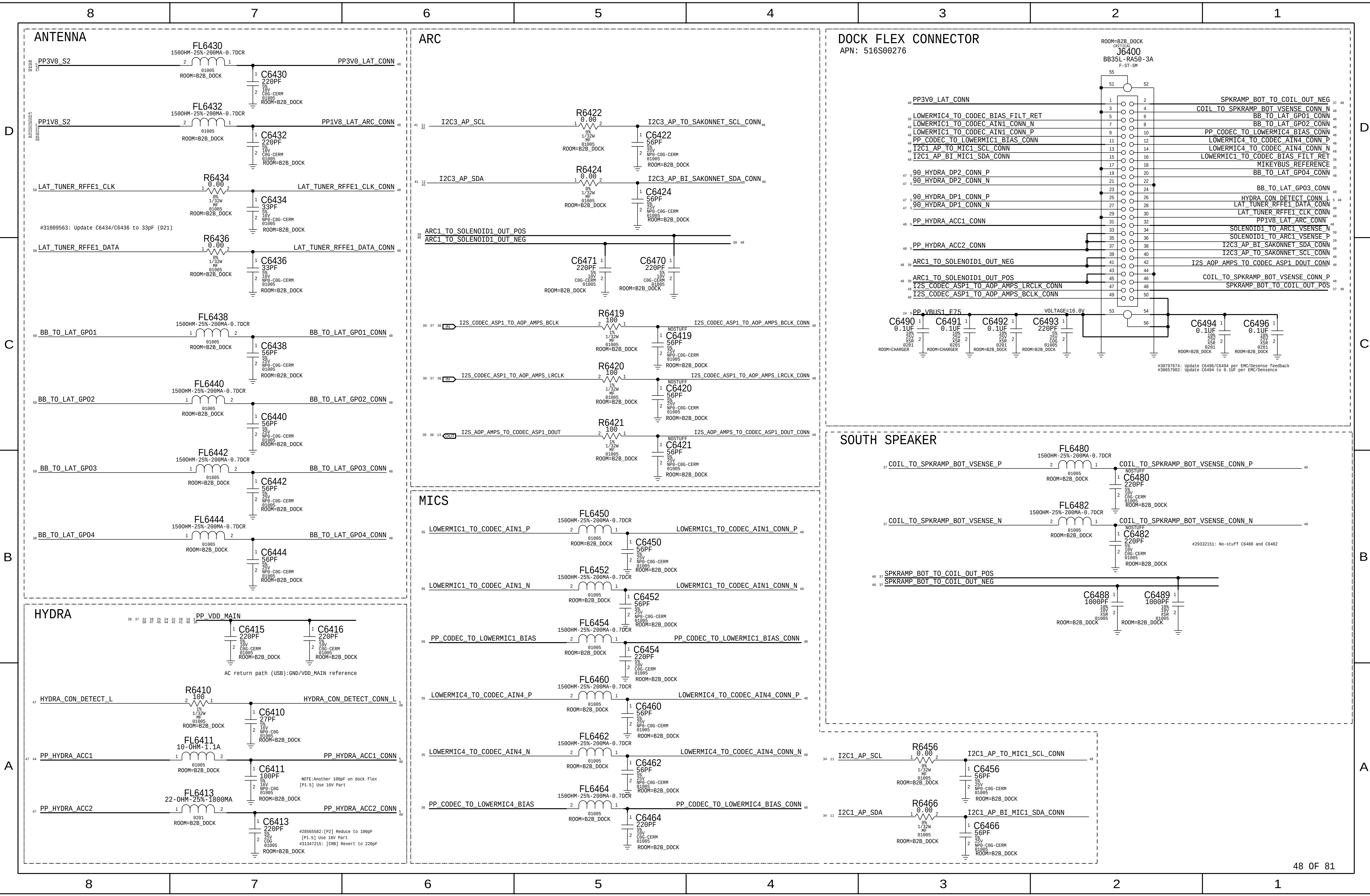
HYDRA

12C ADDRESS: 0011010X



UART TX/RX pin names on Hydra match signal names (i.e. TX pin is input and connects to TX signal)





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I2C MAP

(see #28857723 updates)

Diags ID	Bus	Device	7-Bit Address	8-Bit Address	Device Max Speed	Operating Speed	Net Name
0	AP I2C0	Agnes (System PMU)	0x74	0xE8	400 Kbps	400 Kbps	I2C0_AP
		Chestnut (Display PMU)	0x27	0x4E	400 Kbps		
		AccessoryBuck	0x52	0xA4	400 Kbps		
		Muon (Backlight Controller 2)	0x62	0xC4	1 Mbps		
1	AP I2C1	MIC1 Temp Sensor (South)	0x54	0xA8	1 Mbps	100 Kbps	I2C1_AP
		MIC2 Temp Sensor (North Rear)	0x56	0xAC	1 Mbps		
2	AP I2C2	Adare (Top Speaker Amp)	0x40	0x80	1 Mbps	1 Mbps	I2C2_AP
		Convoy (Top Speaker Cap Sense)	0x21	0x42	1 Mbps		
3	AP I2C3	EEPROM	0x51	0xA2	400 Kbps	400 Kbps	I2C3_AP
		Muon (Backlight Controller 1)	0x62	0xC4	400 Kbps		
		Sakonnet (Arc: Hall Effect Sensor)	0x08	0x10	1 Mbps		
		EEPROM	0x50	0xA0	400 Kbps		
	ISP I2C0	Casper (Rear Wide Camera)	0x10	0x20	1 Mbps	1 Mbps	I2C0_ISP
		Grunberg+ (AF/OIS Wide Camera)	0x3C	0x78	1 Mbps		
	ISP I2C1	Billings (Rear Tele Camera)	0x20	0x40	1 Mbps	1 Mbps	I2C1_ISP
		Edwin (AF Tele Camera)	0x0D	0x1A	1 Mbps		
	ISP I2C2	Concord (Front Camera)	0x10	0x20	1 Mbps	400 Kbps	I2C2_ISP
	ISP I2C3	Neon (Strobe - CW : LM3566)	0x63	0xC6	1 Mbps	400 Kbps	I2C3_ISP
		Ansel (Camera PMU)	0x40	0x80	1 Mbps		
		Neon (Strobe - AM : LM35662)	0x67	0xCE	1 Mbps		
	SEP I2C	EEPROM	0x51	0xA2	400 Kbps	400 Kbps	I2C4_AP
4	AOP I2C0	Doppler (Prox Sensor)	0x58	0xB0	1 Mbps	400 Kbps	I2C0_AOP
		CT709 (ALS : Firefish)	0x29	0x52	400 Kbps		
5	AOP I2C1	Adare (Bottom Speaker Amp)	0x40	0x80	1 Mbps	1 Mbps	I2C1_AOP
		Adare (Arc : Driver)	0x41	0x82	1 Mbps		
		Apache3 (Mesa Power Supply)	0x61	0xC2	1 Mbps		I2C1_AOP_ISO
		Turtle	0x2C	0x58	1 Mbps		
6	SMC I2C0	Tigris2 (Charger)	0x71	0xE2	400 Kbps	400 Kbps	I2C0_SMC
		TPS61280 (VDD Boost)	0x75	0xEA	400 Kbps		
		CCG2 (USB PD)	0x12 (0x08)	0x24 (0x10)	1 Mbps		
		Iktara	0x39	0x72	400 Kbps		
		SN45546 (Gas Gauge)	0x55	0xAA	400 Kbps		
7	SMC I2C1	Hydra	0x1A	0x34	400 Kbps	400 Kbps	I2C1_SMC

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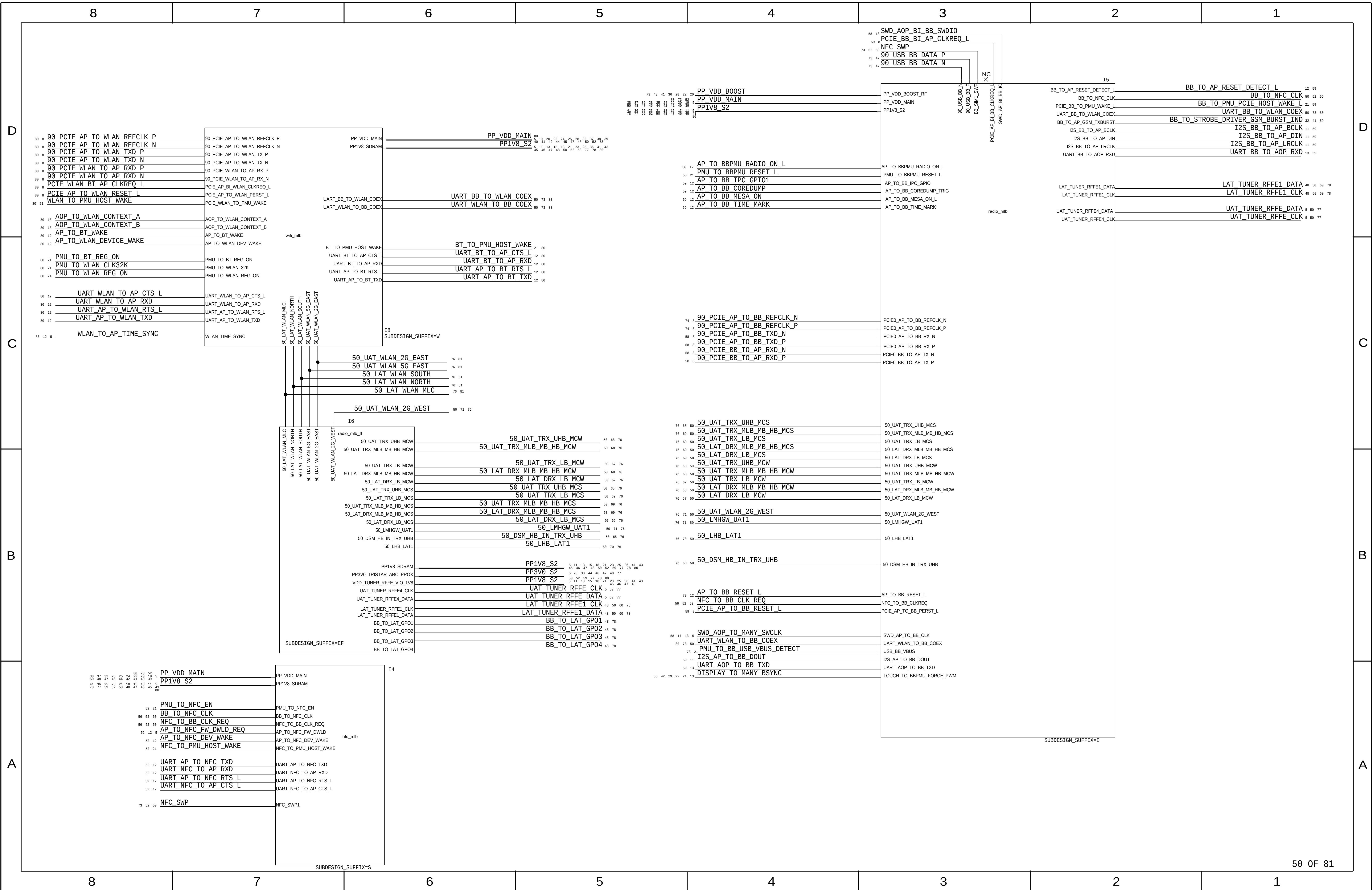
5

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2

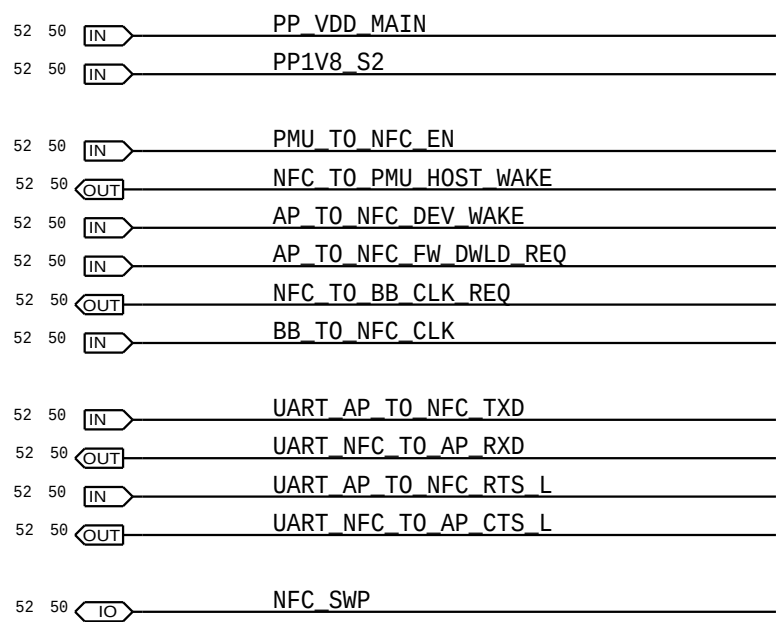
1



NFC_MLB D20/D21

MAY 15, 2017

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	3	CAP, SOFT TERM, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	TYPICAL_CAP



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00882	1	390PF, 0201, 2%, 25V	C7514_S	D20_JPN
131S00026	1	820PF, 0201, 2%, 25V	C7515_S	D20_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D20_JPN
131S00883	1	220PF, 0201, 2%, 50V	C7518_S	D20_JPN
131S00025	1	1000PF, 0201, 2%, 25V	C7510_S	D20_JPN
131S00025	1	1000PF, 0201, 2%, 25V	C7513_S	D20_JPN
157S00022	1	TDK, BALUN, 11T	T7500_S	D20_JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D20_ROW
131S00016	1	470PF, 0201, 2%, 25V	C7514_S	D20_ROW
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D20_ROW
131S00825	1	560PF, 0201, 2%, 25V	C7516_S	D20_ROW
131S00883	1	220PF, 0201, 2%, 50V	C7518_S	D20_ROW
131S00026	1	820PF, 0201, 2%, 25V	C7510_S	D20_ROW
131S00026	1	820PF, 0201, 2%, 25V	C7513_S	D20_ROW
157S00022	1	TDK, BALUN, 11T	T7500_S	D20_ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D201
131S00016	1	470PF, 0201, 2%, 25V	C7514_S	D201
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D201
131S00825	1	560PF, 0201, 2%, 25V	C7516_S	D201
131S00883	1	220PF, 0201, 2%, 50V	C7518_S	D201
131S00026	1	820PF, 0201, 2%, 25V	C7510_S	D201
131S00026	1	820PF, 0201, 2%, 25V	C7513_S	D201
157S00022	1	TDK, BALUN, 11T	T7500_S	D201

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D21_JPN
131S00081	1	270PF, 0201, 2%, 25V	C7514_S	D21_JPN
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D21_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D21_JPN
131S00019	1	150PF, 0201, 2%, 50V	C7518_S	D21_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7510_S	D21_JPN
131S00033	1	680PF, 0201, 2%, 25V	C7513_S	D21_JPN
157S00023	1	MURATA, BALUN, 11T	T7500_S	D21_JPN

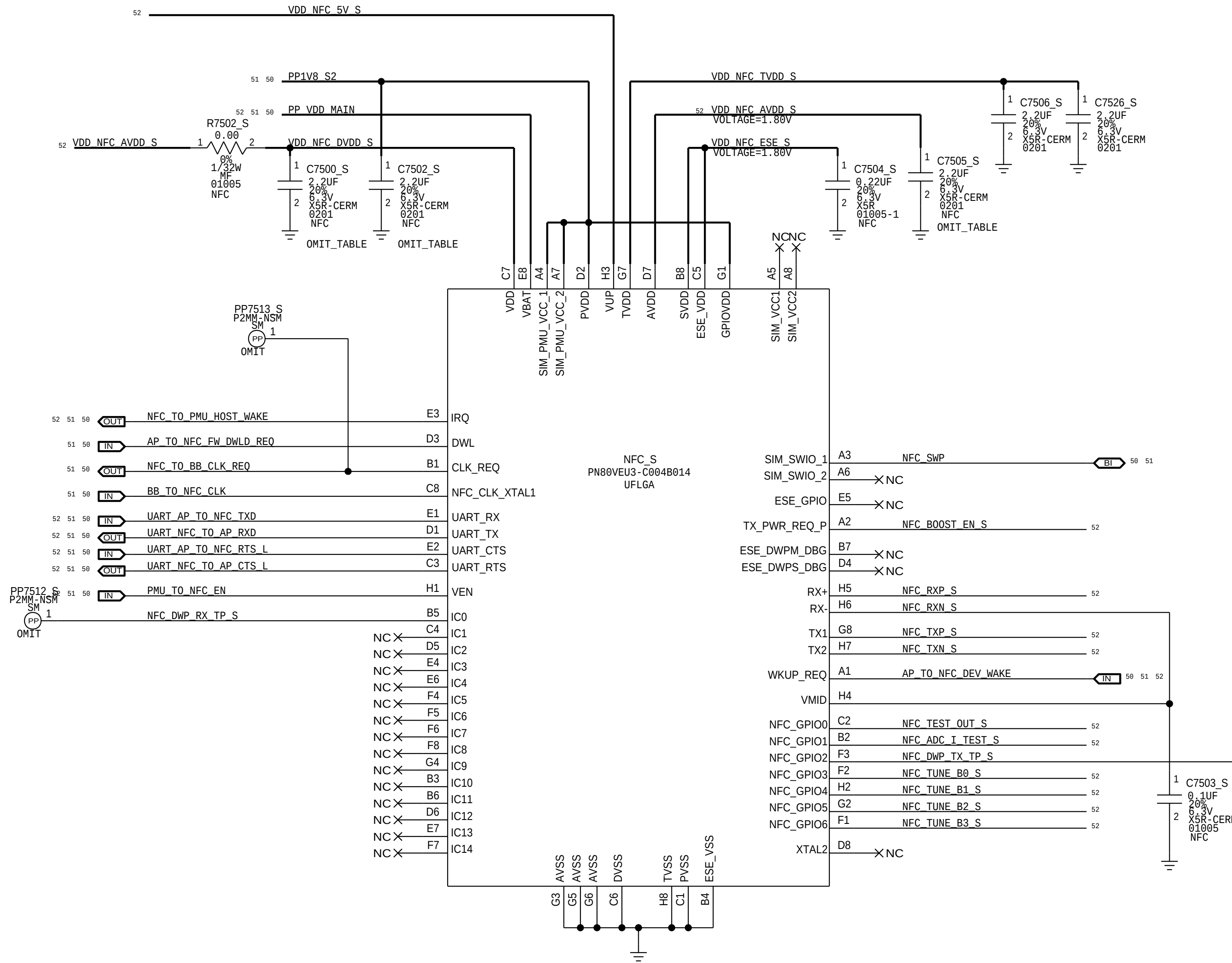
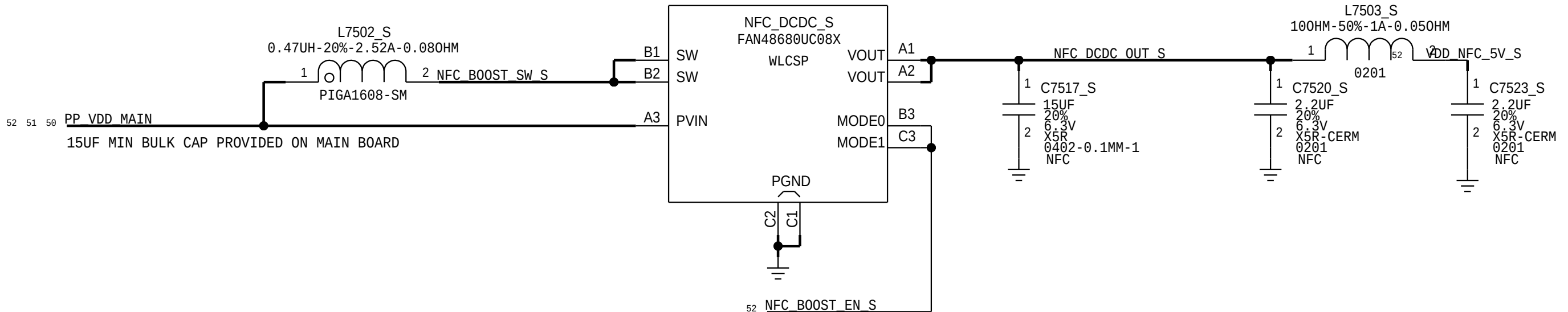
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D21_ROW
131S00081	1	270PF, 0201, 2%, 25V	C7514_S	D21_ROW
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D21_ROW
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D21_ROW
131S00019	1	150PF, 0201, 2%, 50V	C7518_S	D21_ROW
131S00033	1	680PF, 0201, 2%, 25V	C7510_S	D21_ROW
131S00033	1	680PF, 0201, 2%, 25V	C7513_S	D21_ROW
157S00023	1	MURATA, BALUN, 11T	T7500_S	D21_ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00138	1	39PF, 0201, 2%, 50V	C7512_S	D211
131S00081	1	270PF, 0201, 2%, 25V	C7514_S	D211
131S00025	1	1000PF, 0201, 2%, 25V	C7515_S	D211
131S00033	1	680PF, 0201, 2%, 25V	C7516_S	D211
131S00019	1	150PF, 0201, 2%, 50V	C7518_S	D211
131S00033	1	680PF, 0201, 2%, 25V	C7510_S	D211
131S00033	1	680PF, 0201, 2%, 25V	C7513_S	D211
157S00023	1	MURATA, BALUN, 11T	T7500_S	D211

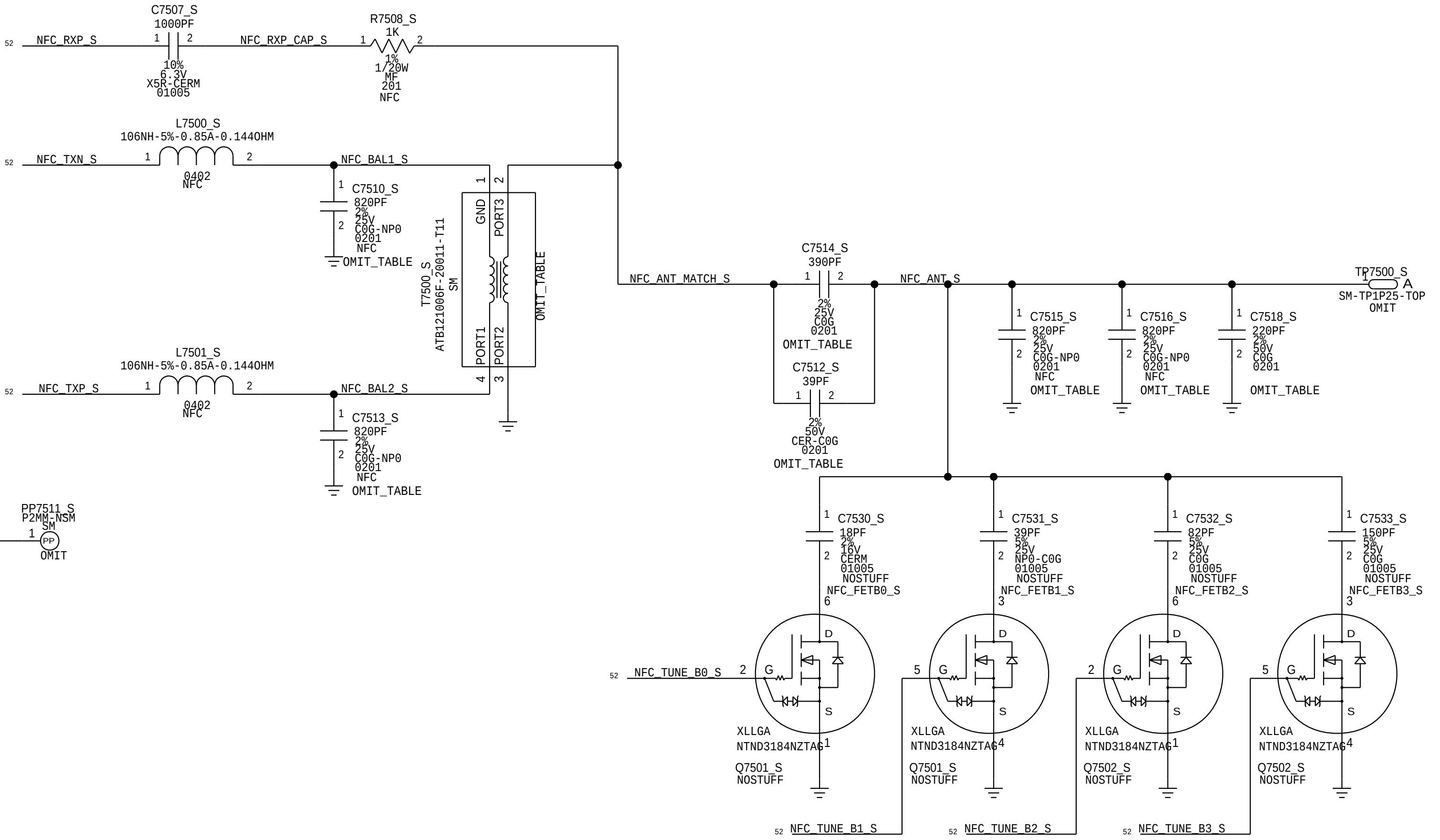
STOCKHOLM

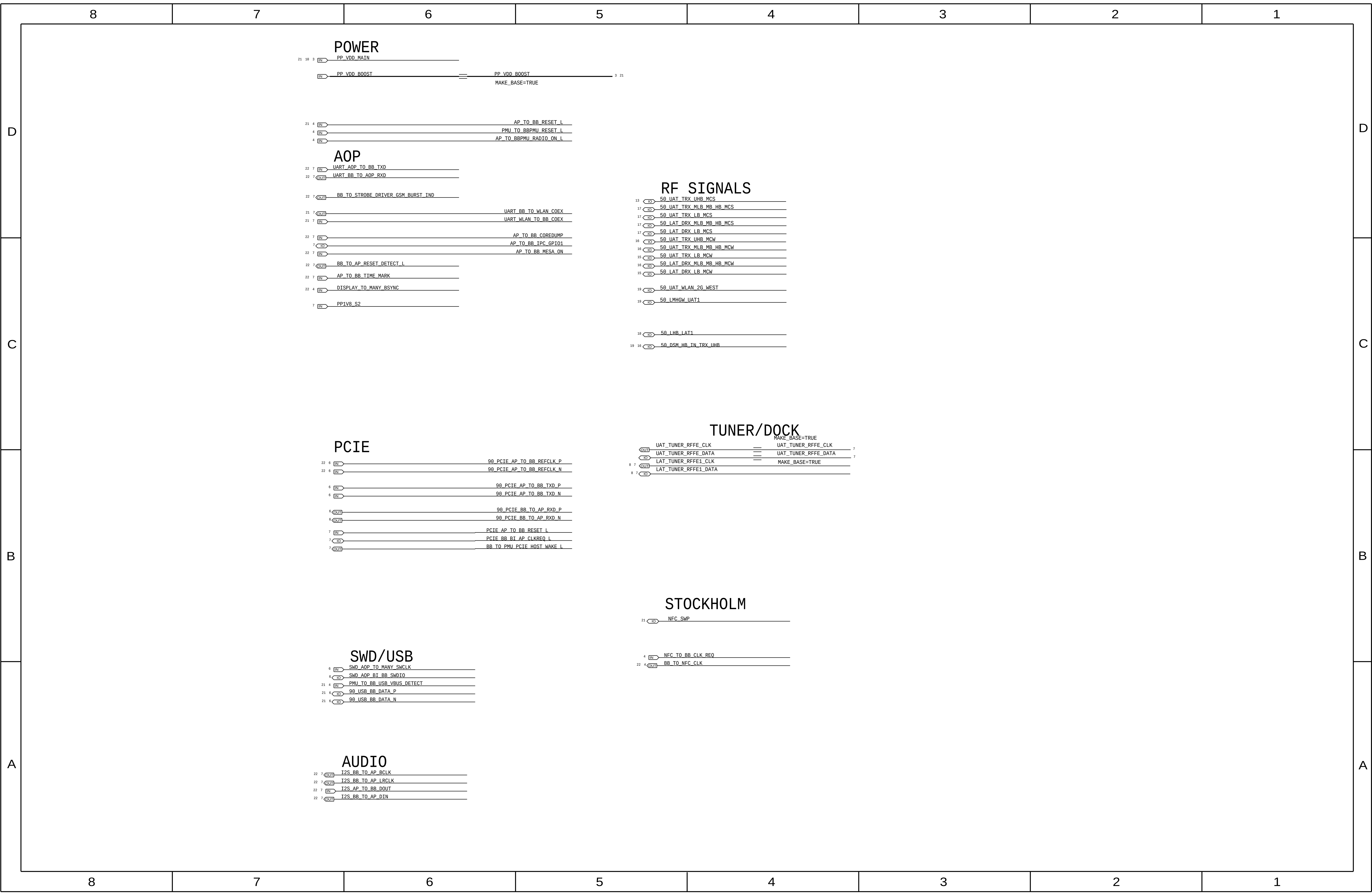
5V BOOSTER

NFC CONTROLLER



NFC FRONT END

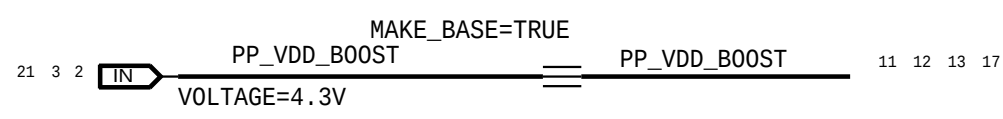




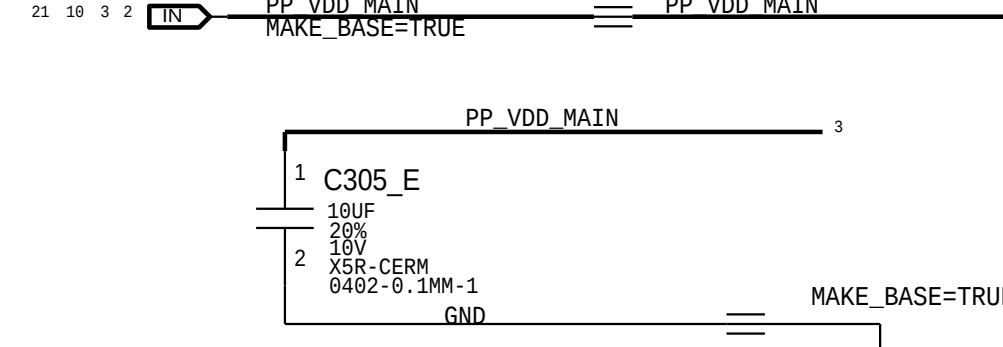
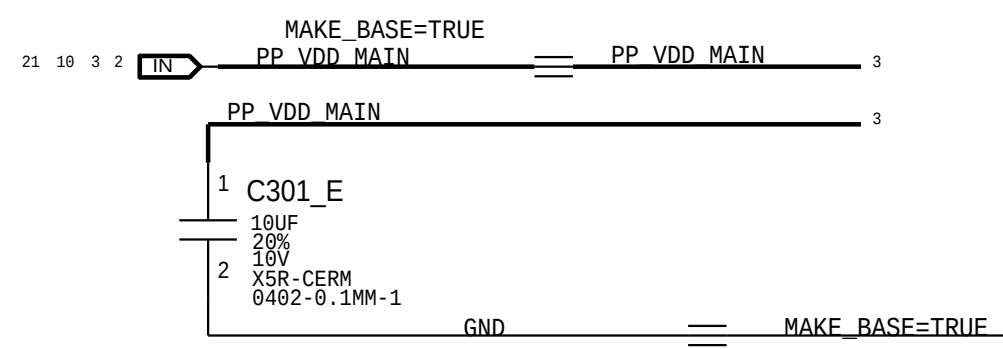
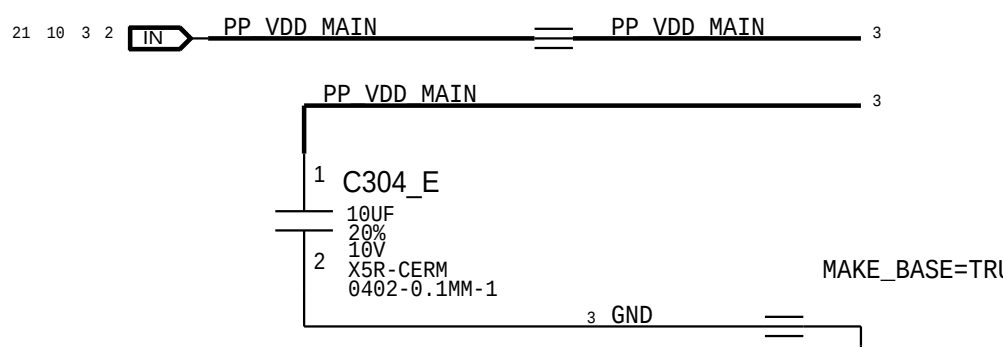
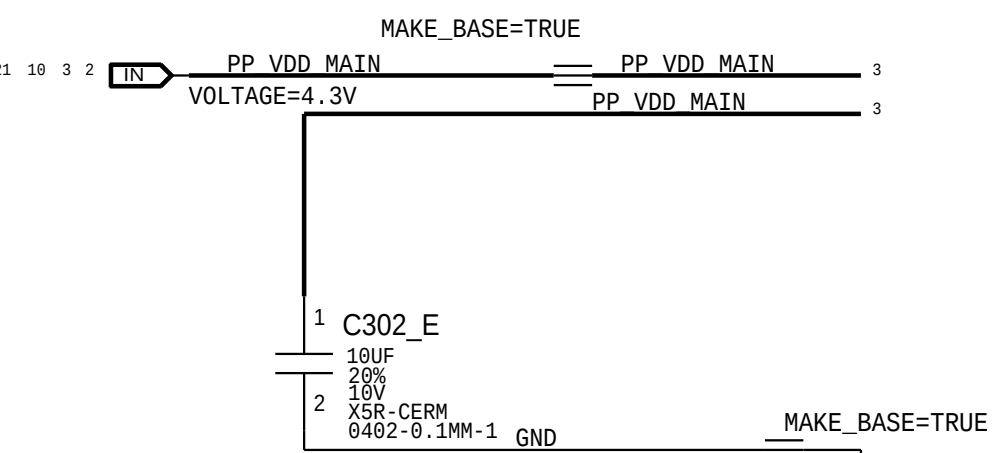
RF PMIC: SWITCHERS & LDOS

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

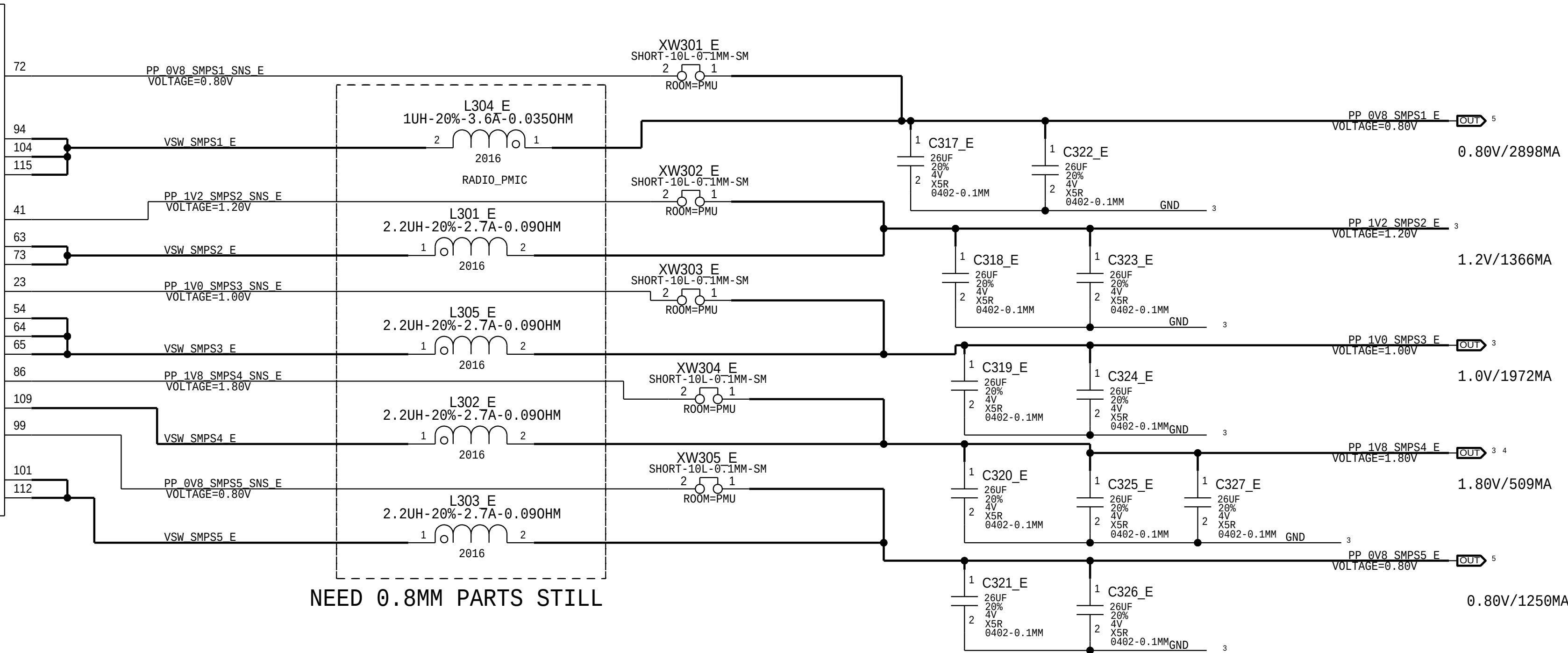
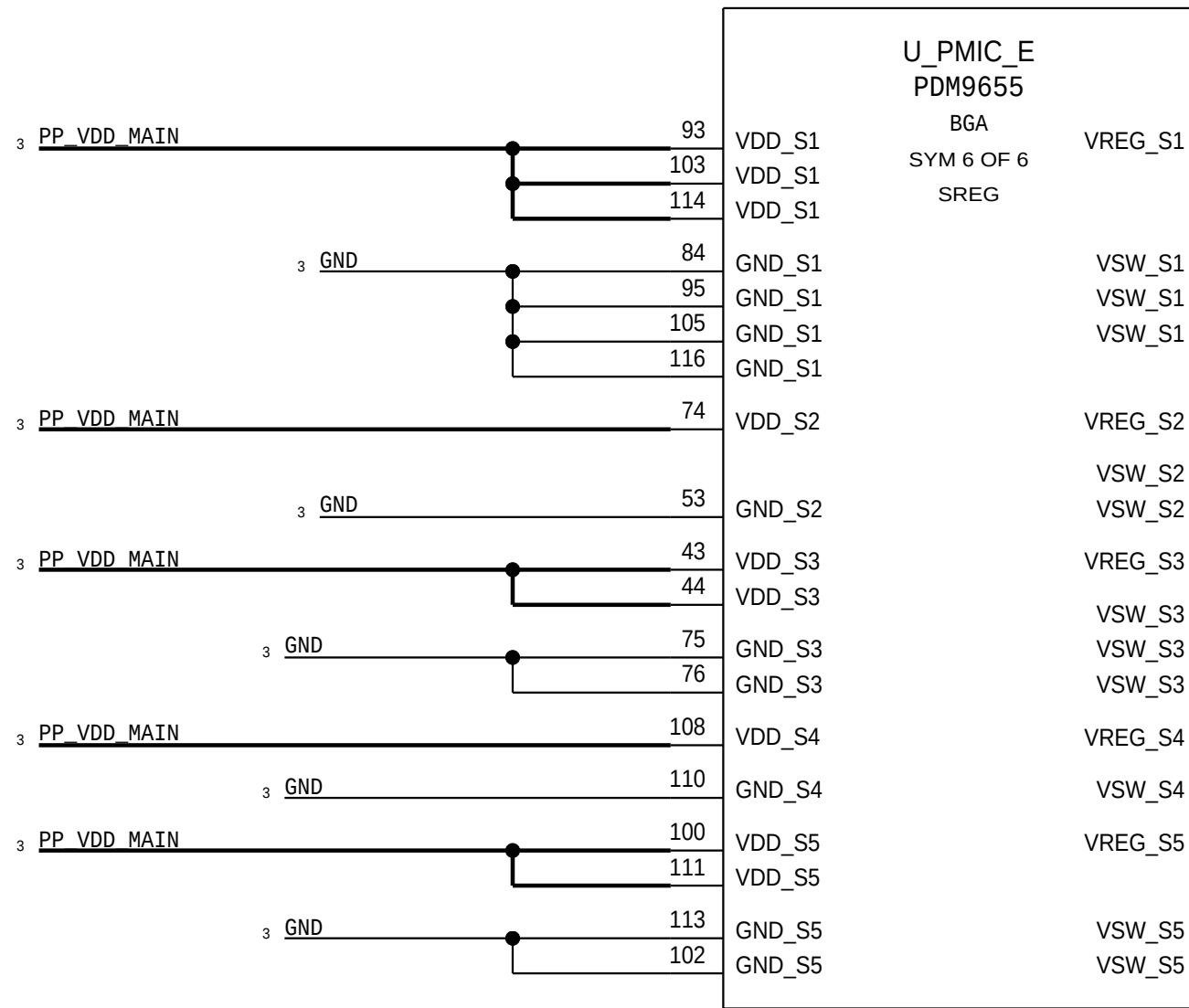
D



SWITCHERS BULK CAPS



A

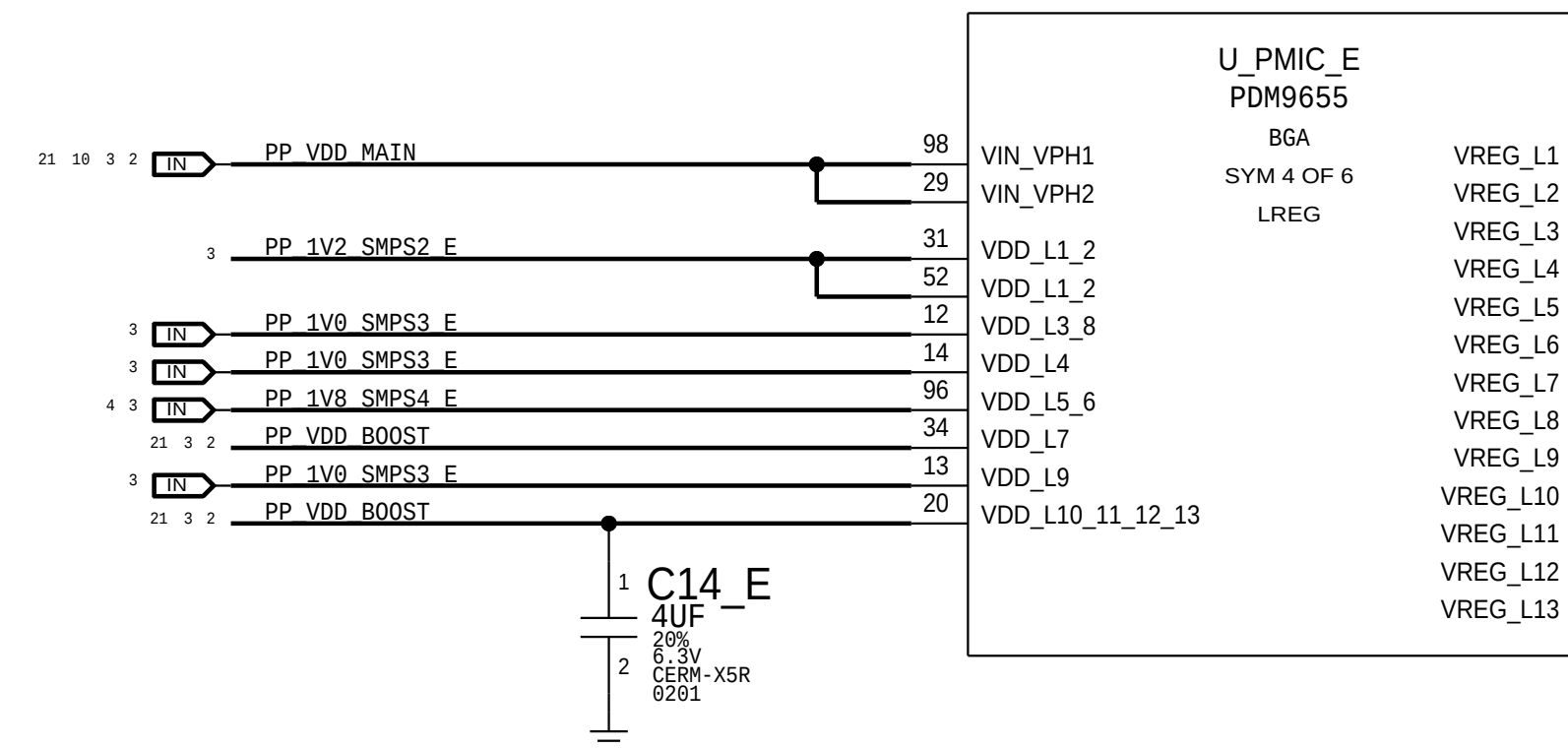


D

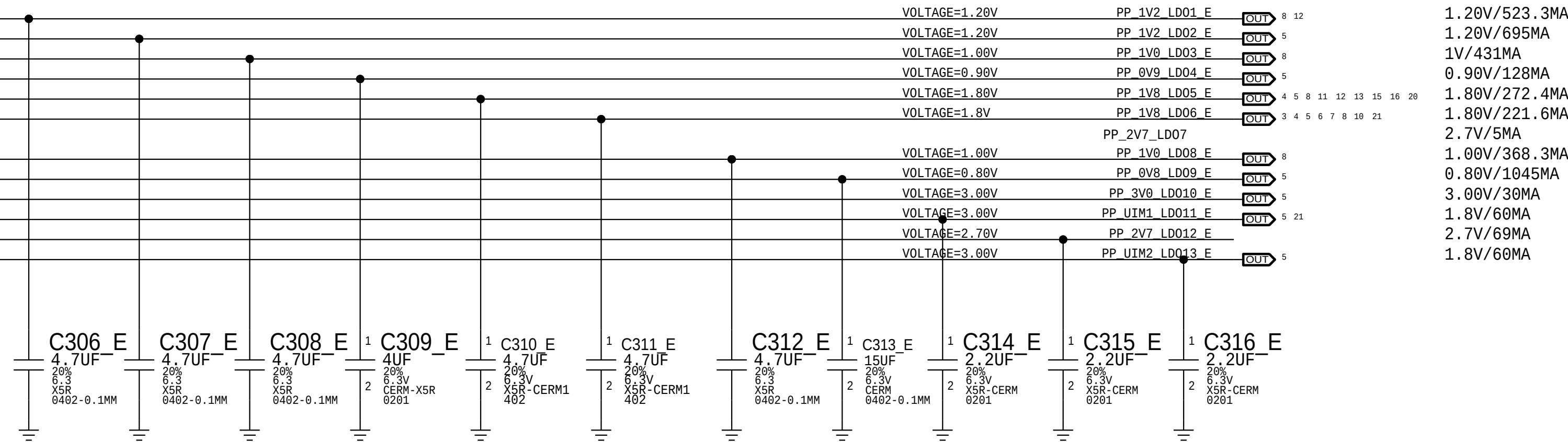
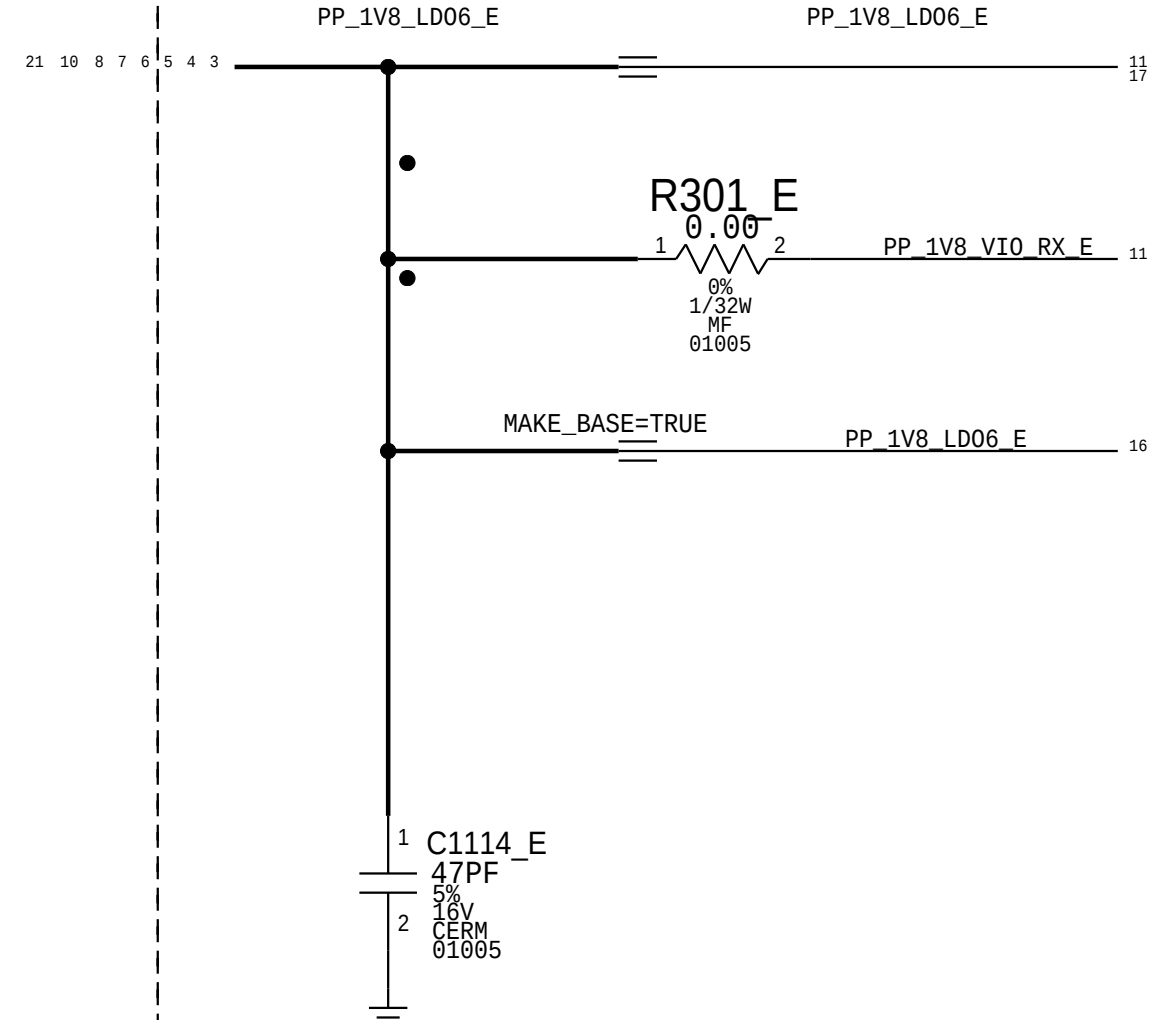
C

B

A



VIO TX & RX



- 1.20V/523.3MA
- 1.20V/695MA
- 1V/431MA
- 0.90V/128MA
- 1.80V/272.4MA
- 1.80V/221.6MA
- 2.7V/5MA
- 1.00V/368.3MA
- 0.80V/1045MA
- 3.00V/30MA
- 1.8V/60MA
- 2.7V/69MA
- 1.8V/60MA

RF PMIC: XTAL, CLOCKS AND MISCELLANEOUS

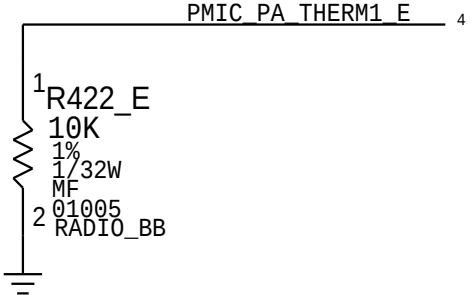
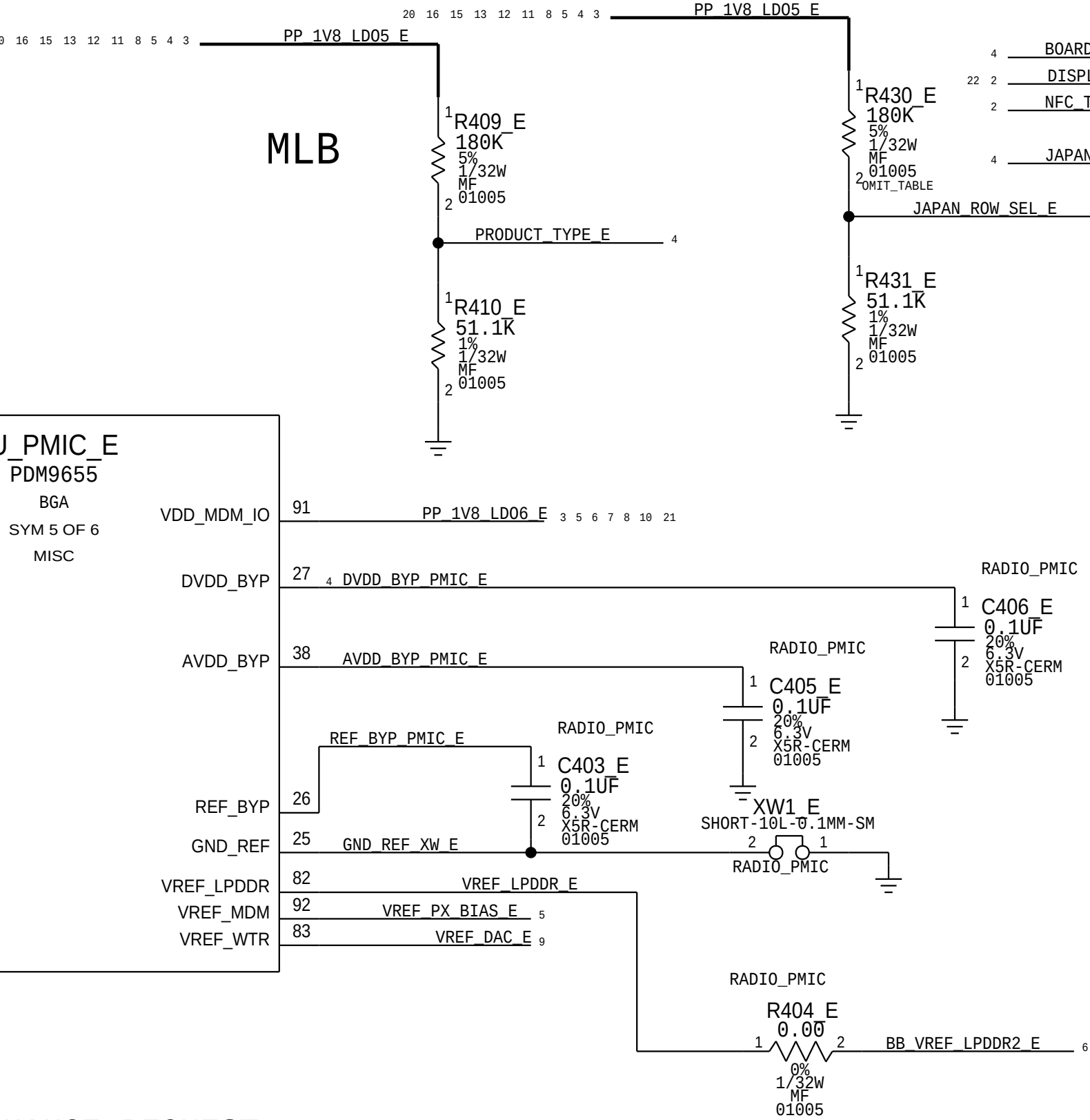
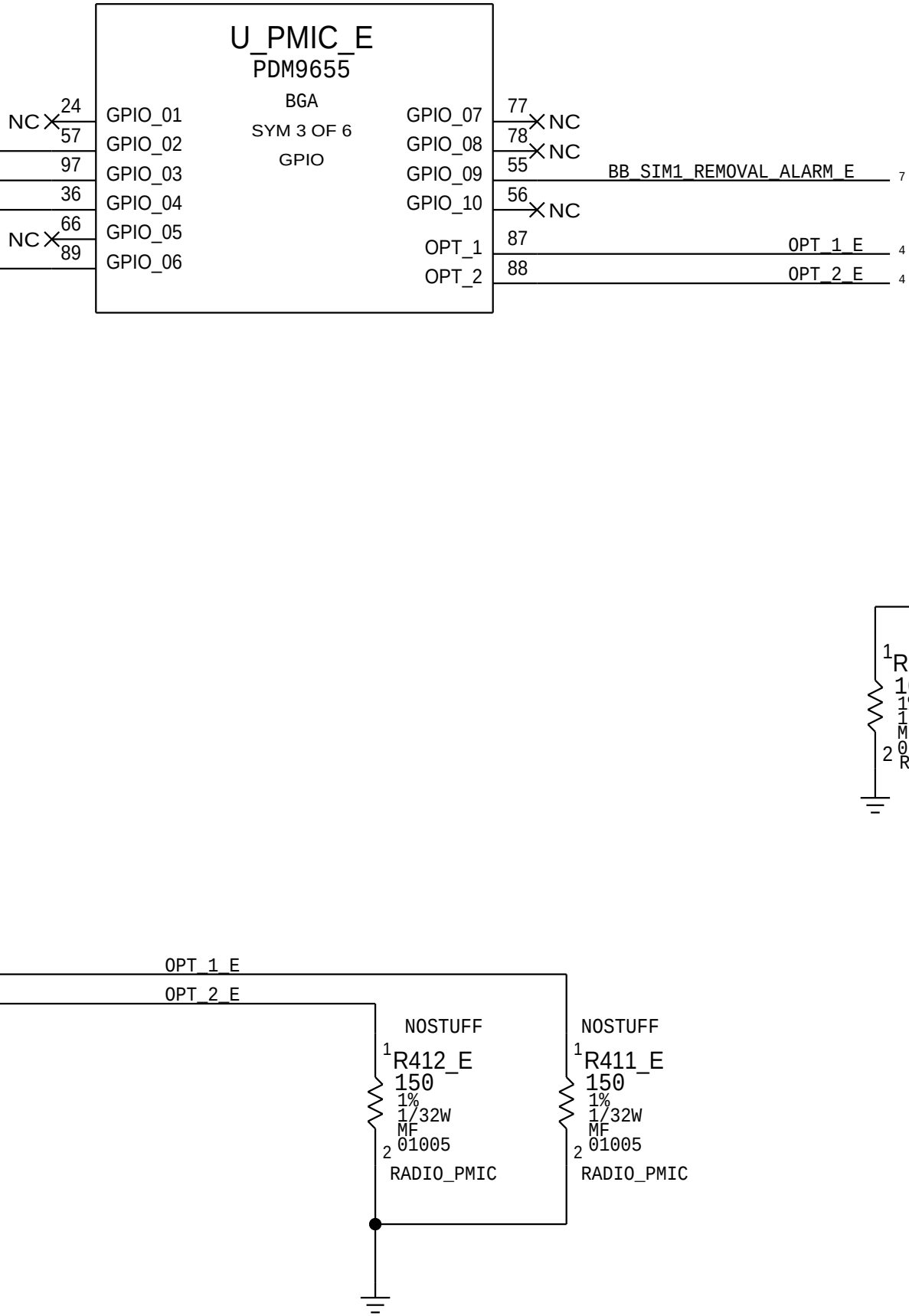
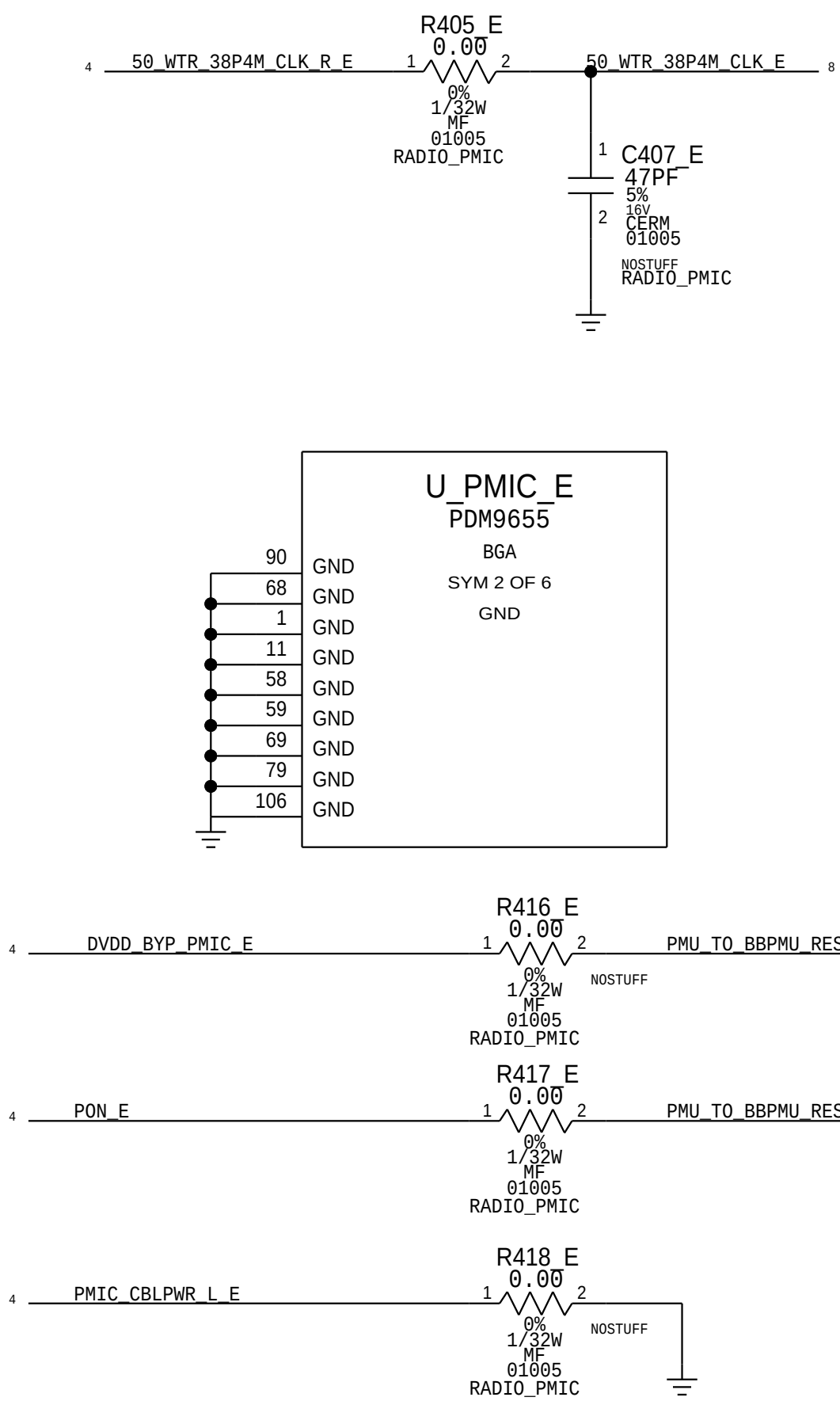
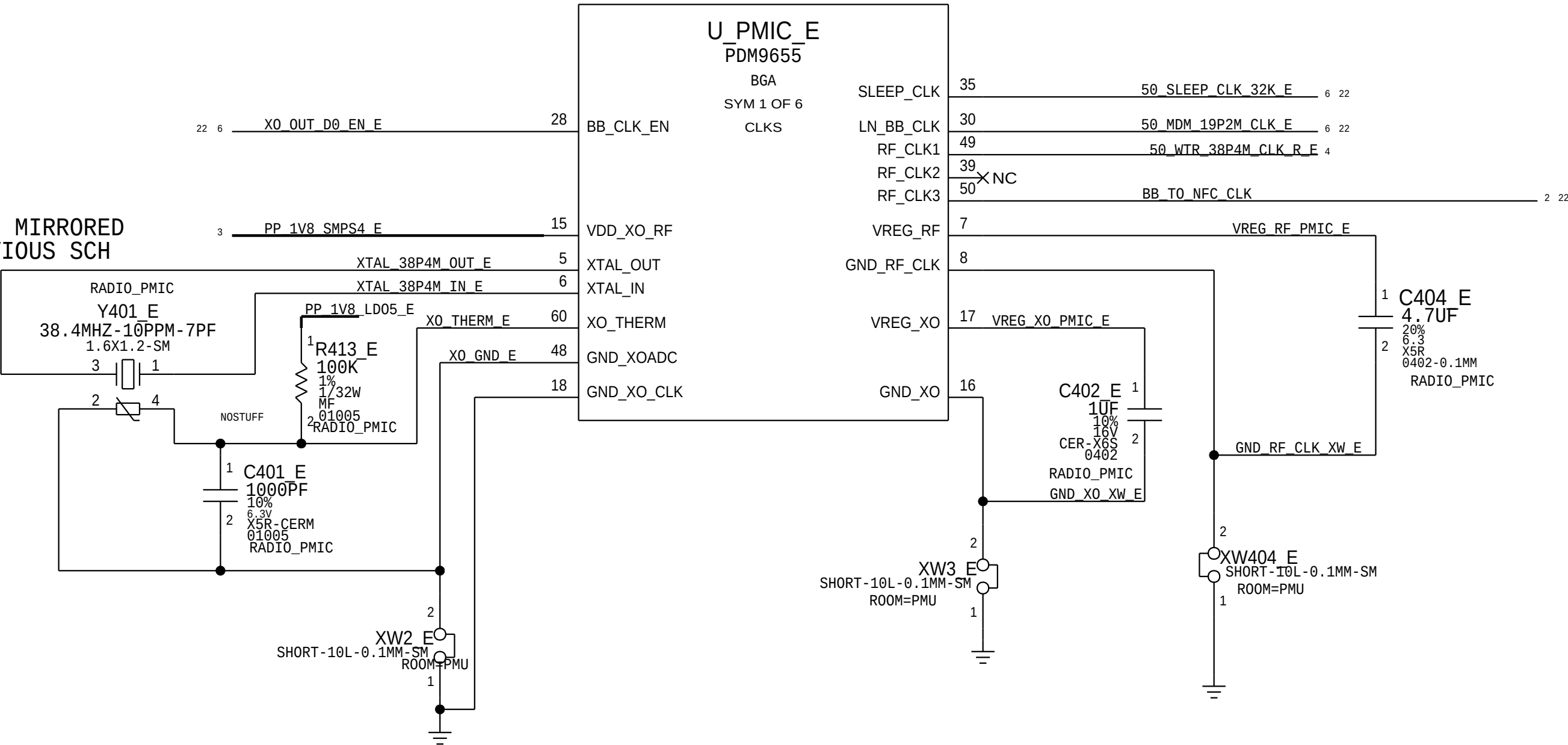
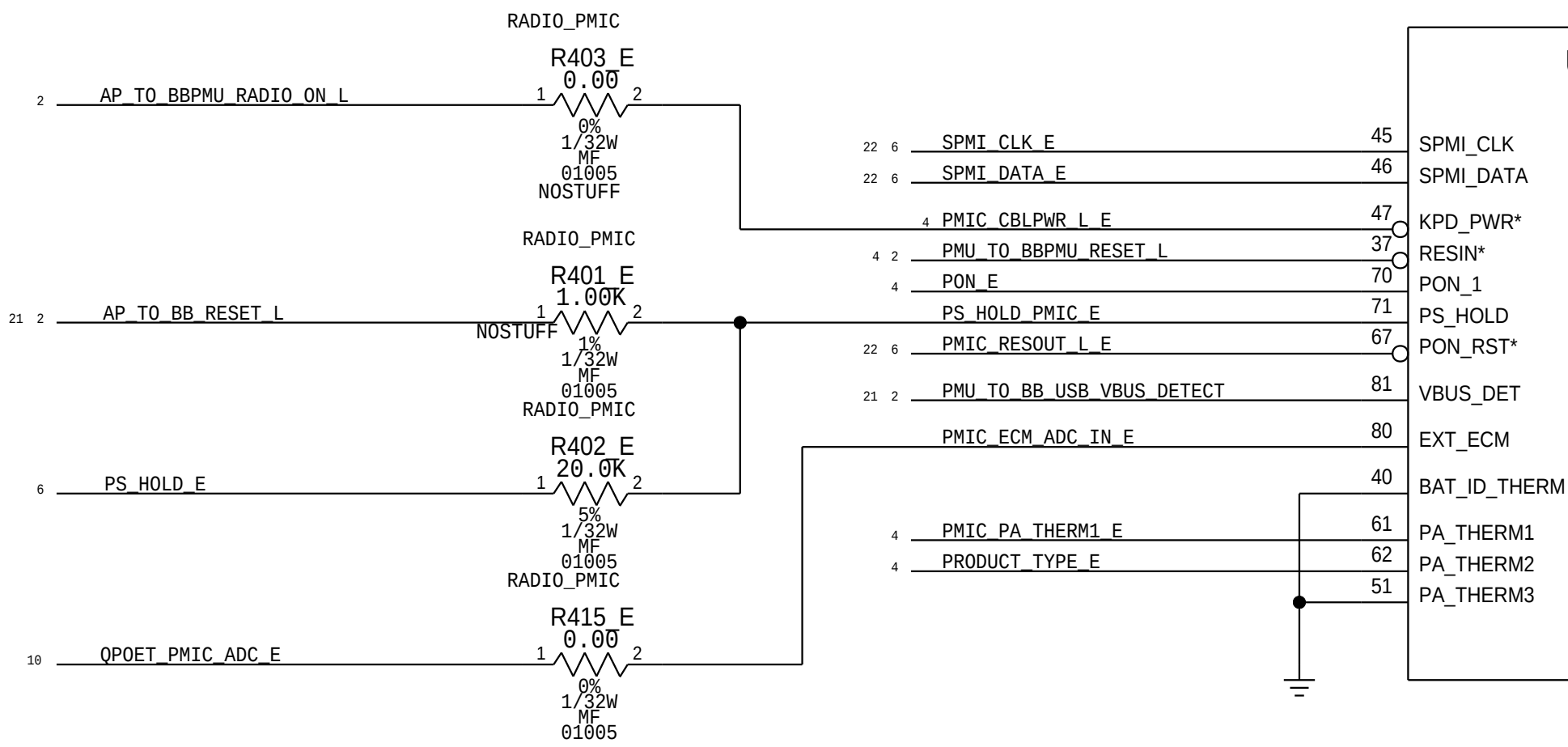
HW_REV_ID	R407	R408	MLB/RADIO_DEV
<0.10V	NOSTUFF	51.1K	T500/DEV1
0.15V-0.25V	422K	51.1K	PROTO1.5
0.35V-0.45V	180K	51.1K	PROTO2.5/DEV2
0.55V-0.65V	105K	51.1K	PROTO1/DEV3
0.75V-0.85V	66.5K	51.1K	PROTO2/DEV4
0.95V-1.05V	44.2K	51.1K	EVT
1.15V-1.25V	28K	51.1K	CARRIER/DEV5
1.35V-1.45V	16.5K	51.1K	DVT/DEV6
1.55V-1.65V	7.87K	51.1K	PVT/DEV7
1.75V-1.85V	10K	NOSTUFF	SPARE

NOTE XTAL SYMBOL MIRRORED COMPARED TO PREVIOUS SCH

PA_THERM2	R409	R410	PRODUCT TYPE	Y
0.00V-0.10V	NOSTUFF	51.1K	RADIO DEV	1
0.15V-0.25V	422K	51.1K	3RD TYPE	2
0.35V-0.45V	180K	51.1K	POR MLB (FF)	3
0.55V-0.65V	105K	51.1K	SPARE	4
0.75V-0.85V	66.5K	51.1K	SPARE	5
0.95V-1.05V	44.2K	51.1K	SPARE	6
1.15V-1.25V	28K	51.1K	SPARE	7
1.35V-1.45V	16.5K	51.1K	SPARE	8
1.55V-1.65V	7.87K	51.1K	SPARE	9
1.75V-1.85V	10K	NOSTUFF	SPARE	A

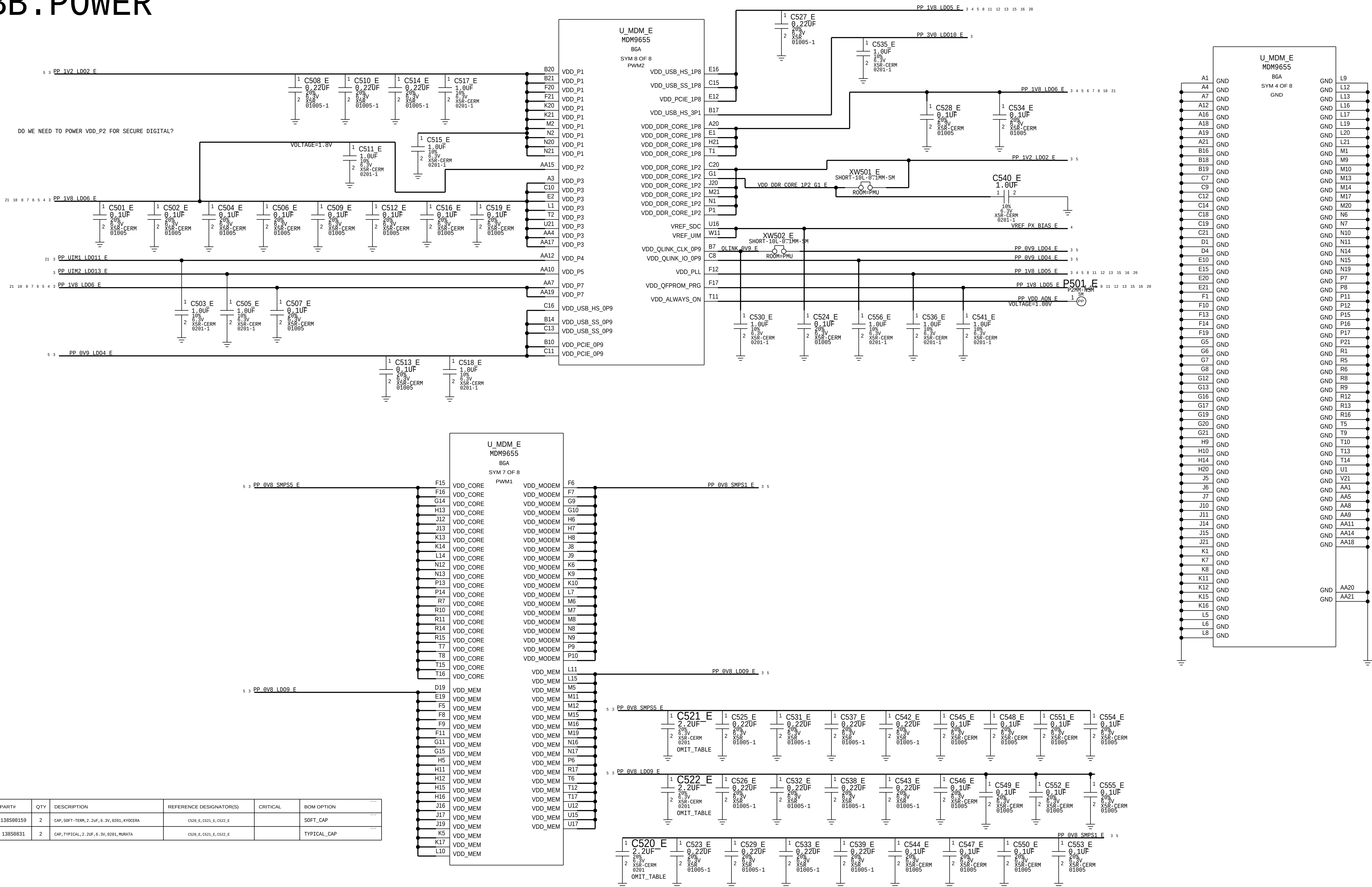
JPN_ROW_SEL	R430	R431	MAV PLATFORM	Y	SKU
<0.10V	NOSTUFF	51.1K	MAV17.0	0	ROW
0.15V-0.25V	422K	51.1K	MAV17.1	1	JPN
0.35V-0.45V	180K	51.1K	MAV17.2	2	ROW
0.55V-0.65V	105K	51.1K	MAV17.3	3	JPN
0.75V-0.85V	66.5K	51.1K	MAV17.4	4	ROW
0.95V-1.05V	44.2K	51.1K	MAV17.5	5	JPN
1.15V-1.25V	28K	51.1K	MAV17.6	6	ROW
1.35V-1.45V	16.5K	51.1K	MAV17.7	7	JPN
1.55V-1.65V	7.87K	51.1K	MAV17.8	8	ROW
1.75V-1.85V	10K	NOSTUFF	MAV17.9	9	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0606	1	RESISTOR, 01005, 105K OHMS	R430_E	CRITICAL	JPN
117S0197	1	RESISTOR, 01005, 100K OHMS	R430_E	CRITICAL	ROW

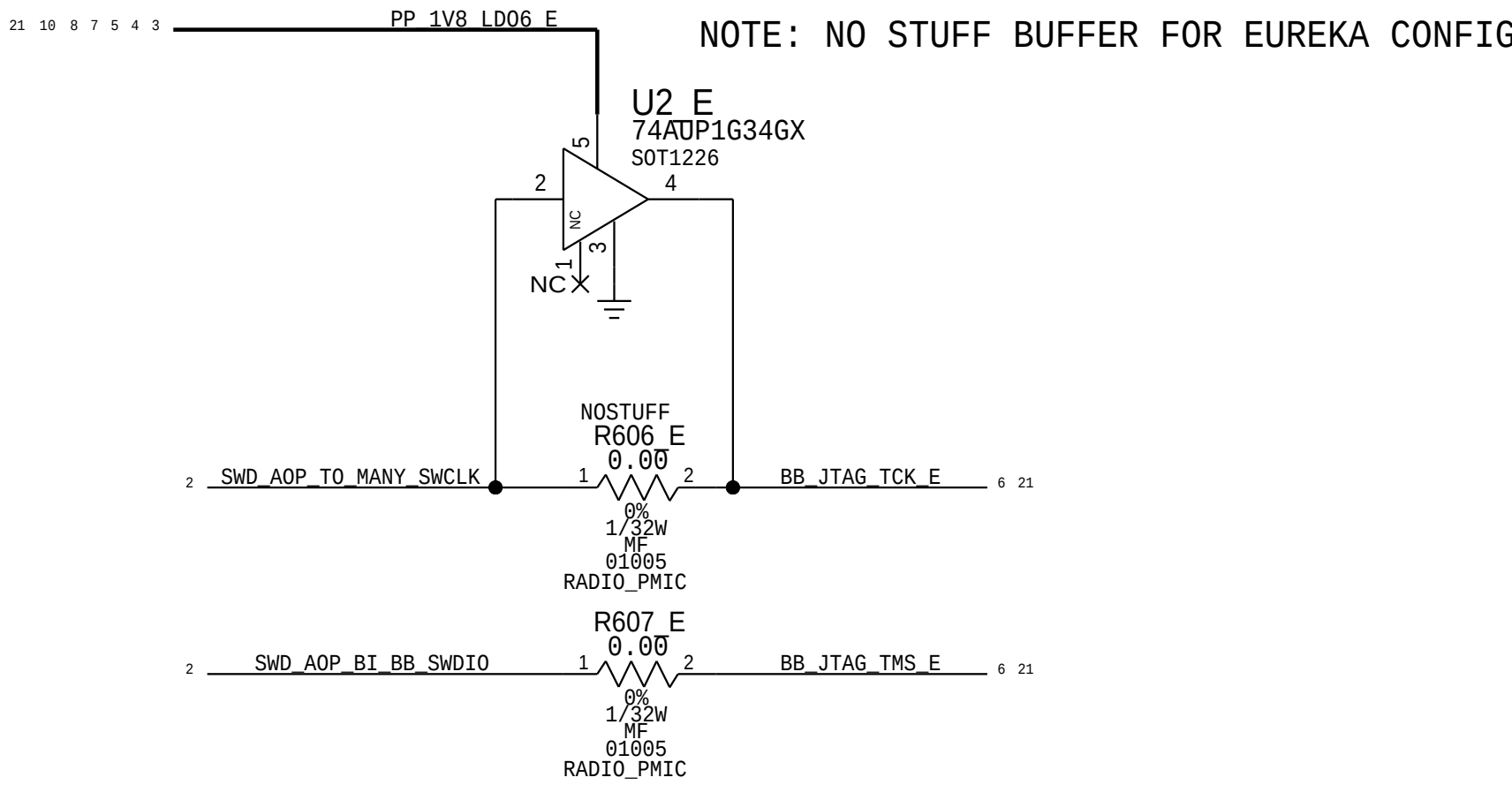
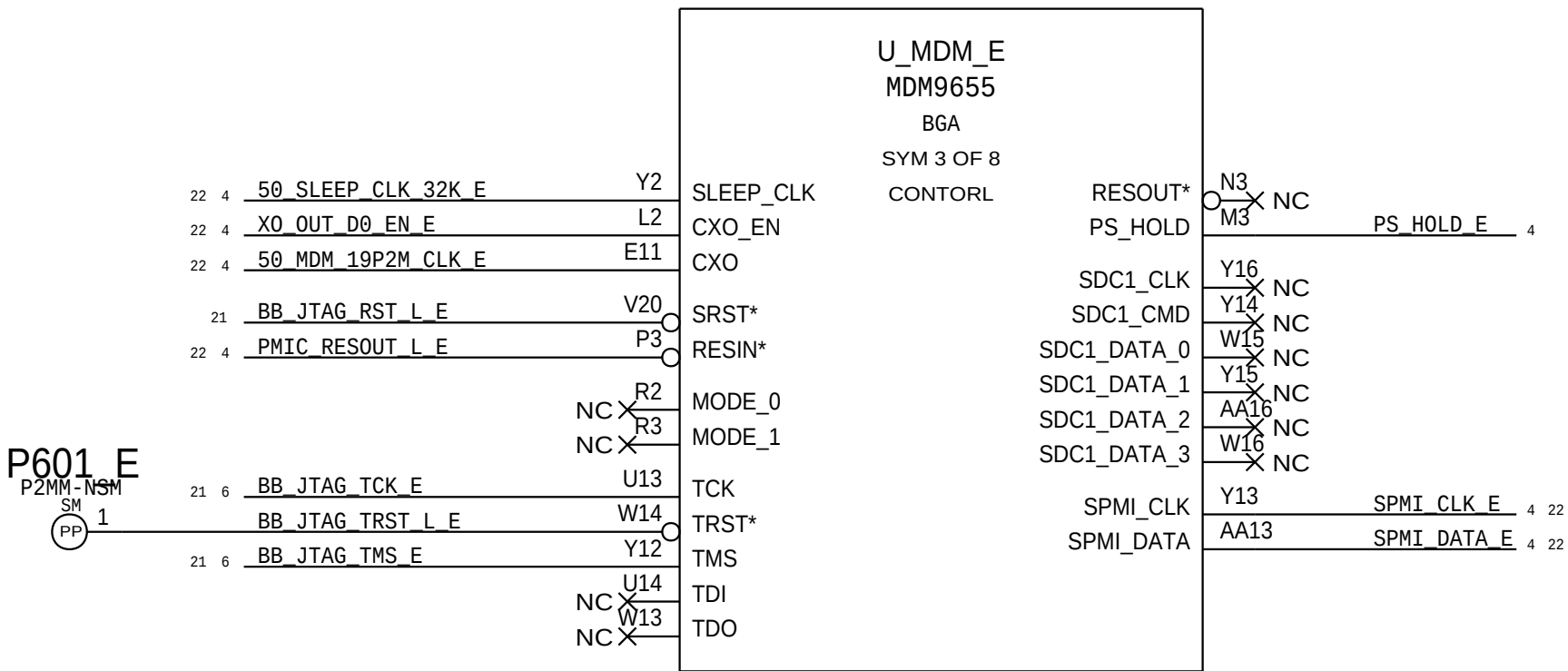
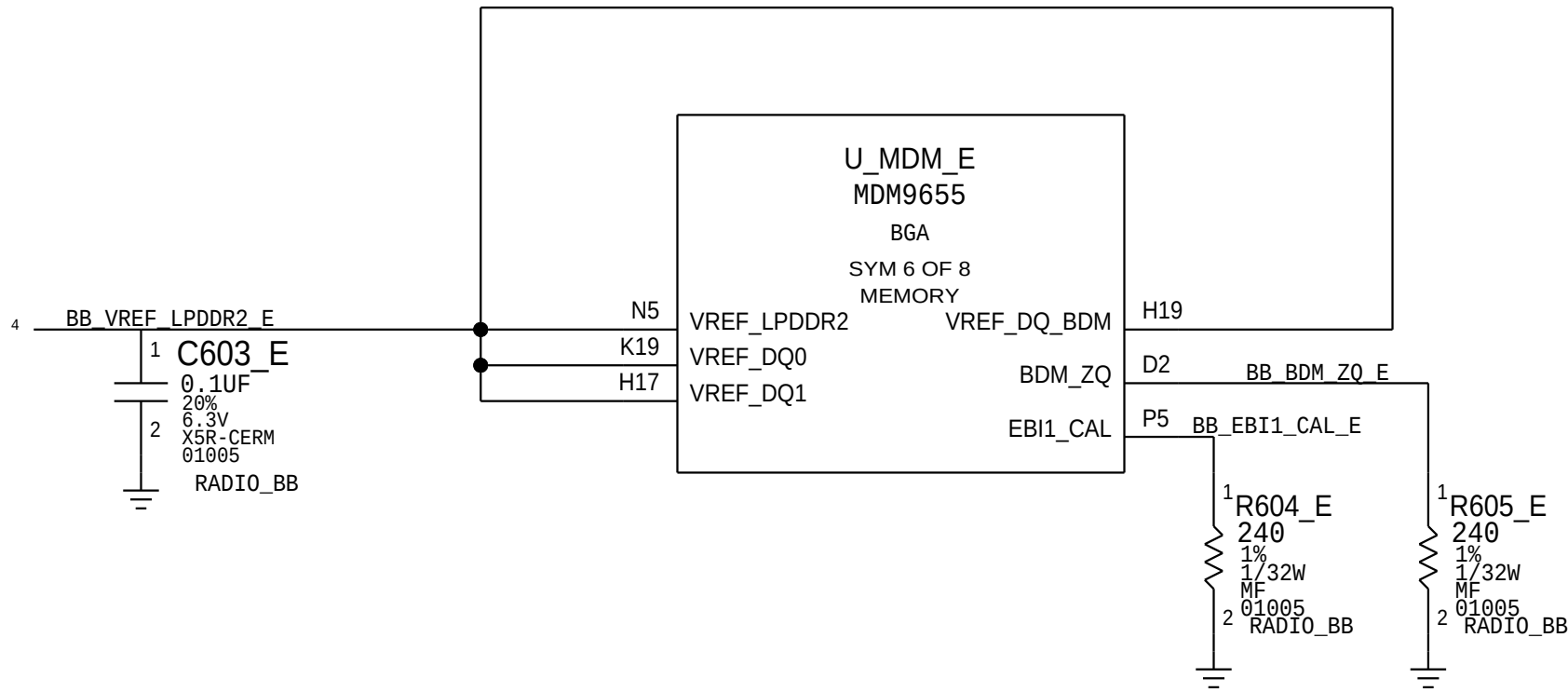
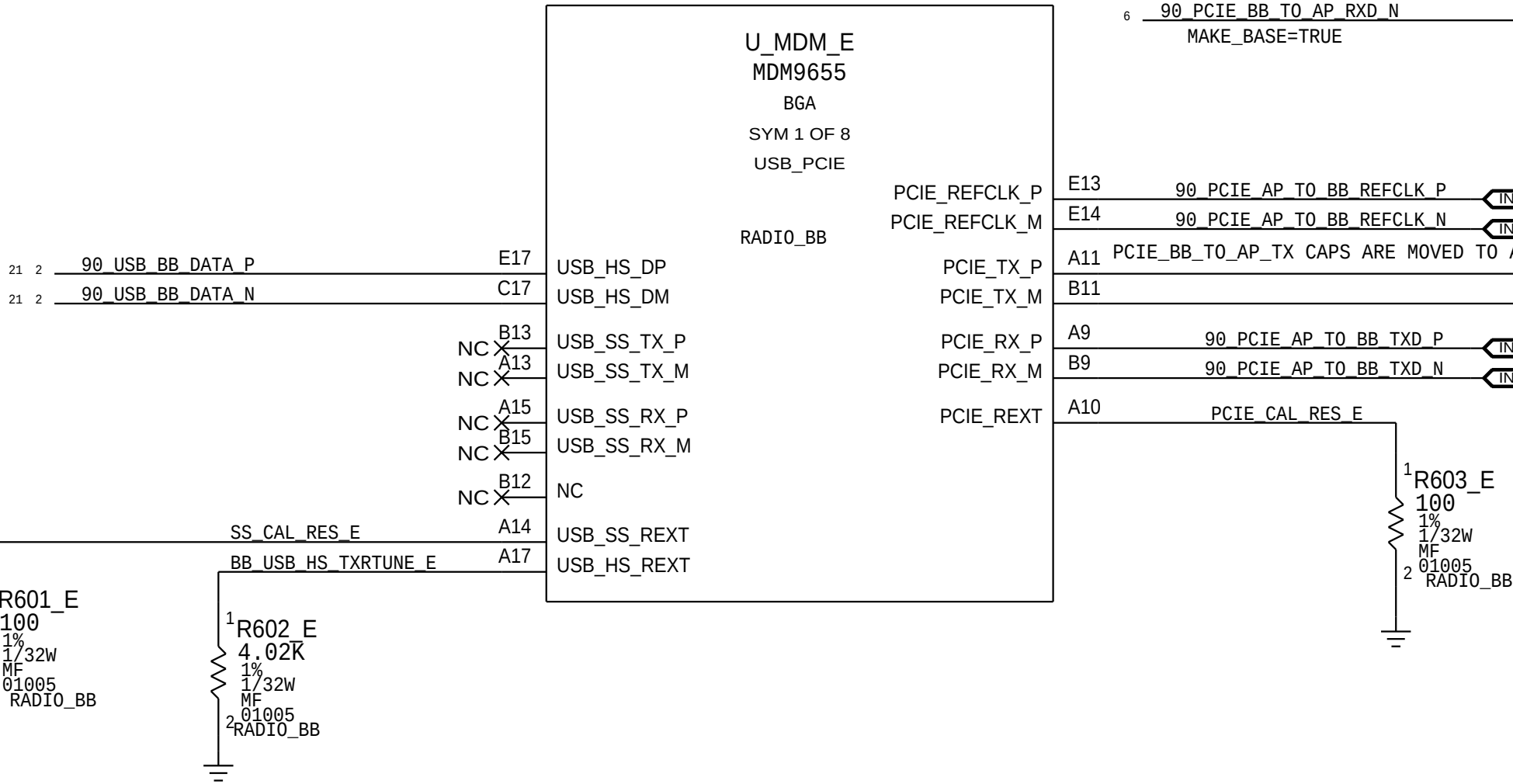
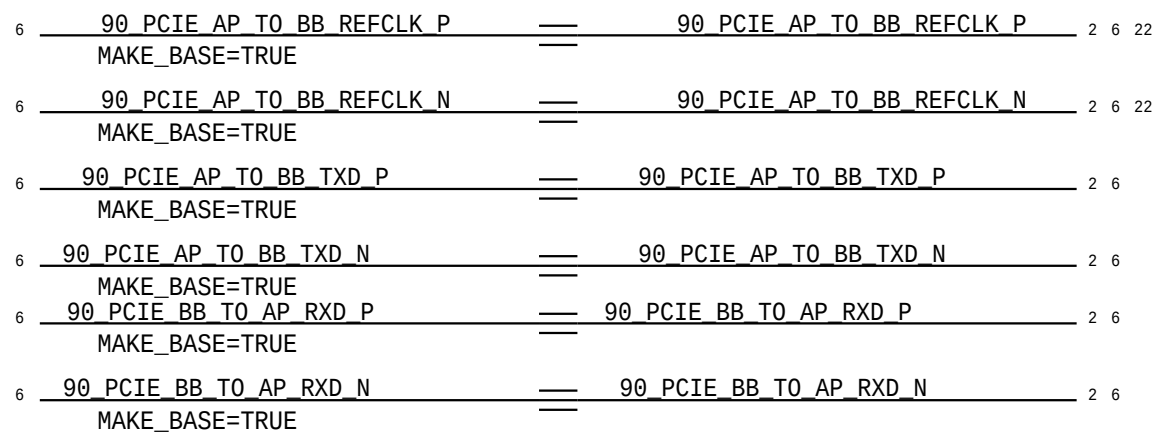
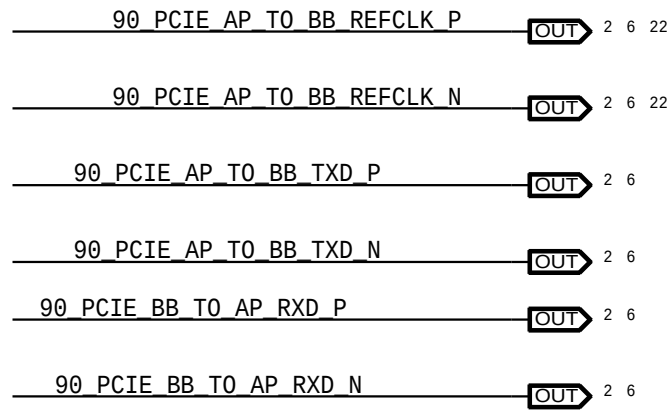


BB : POWER

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

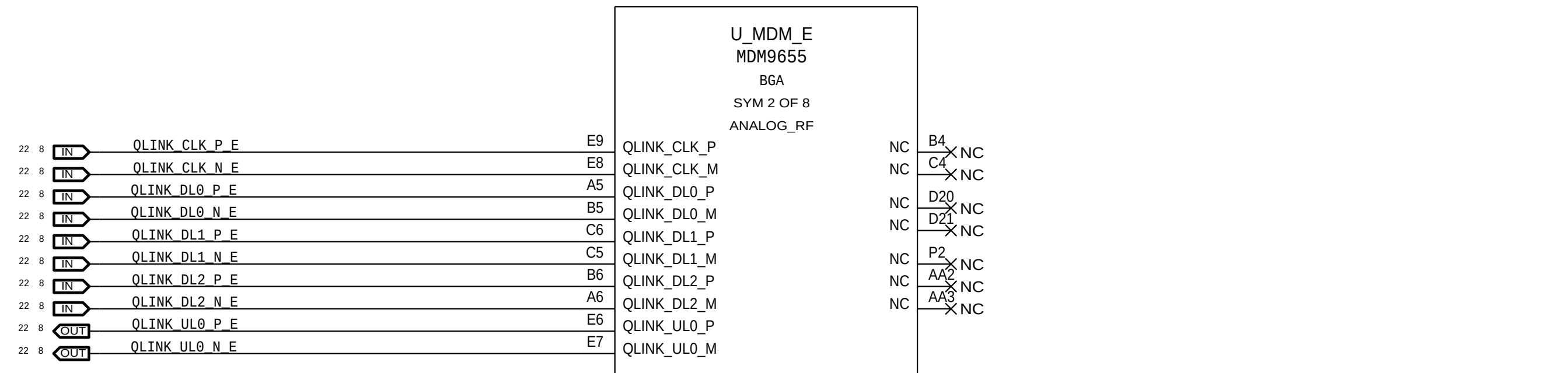
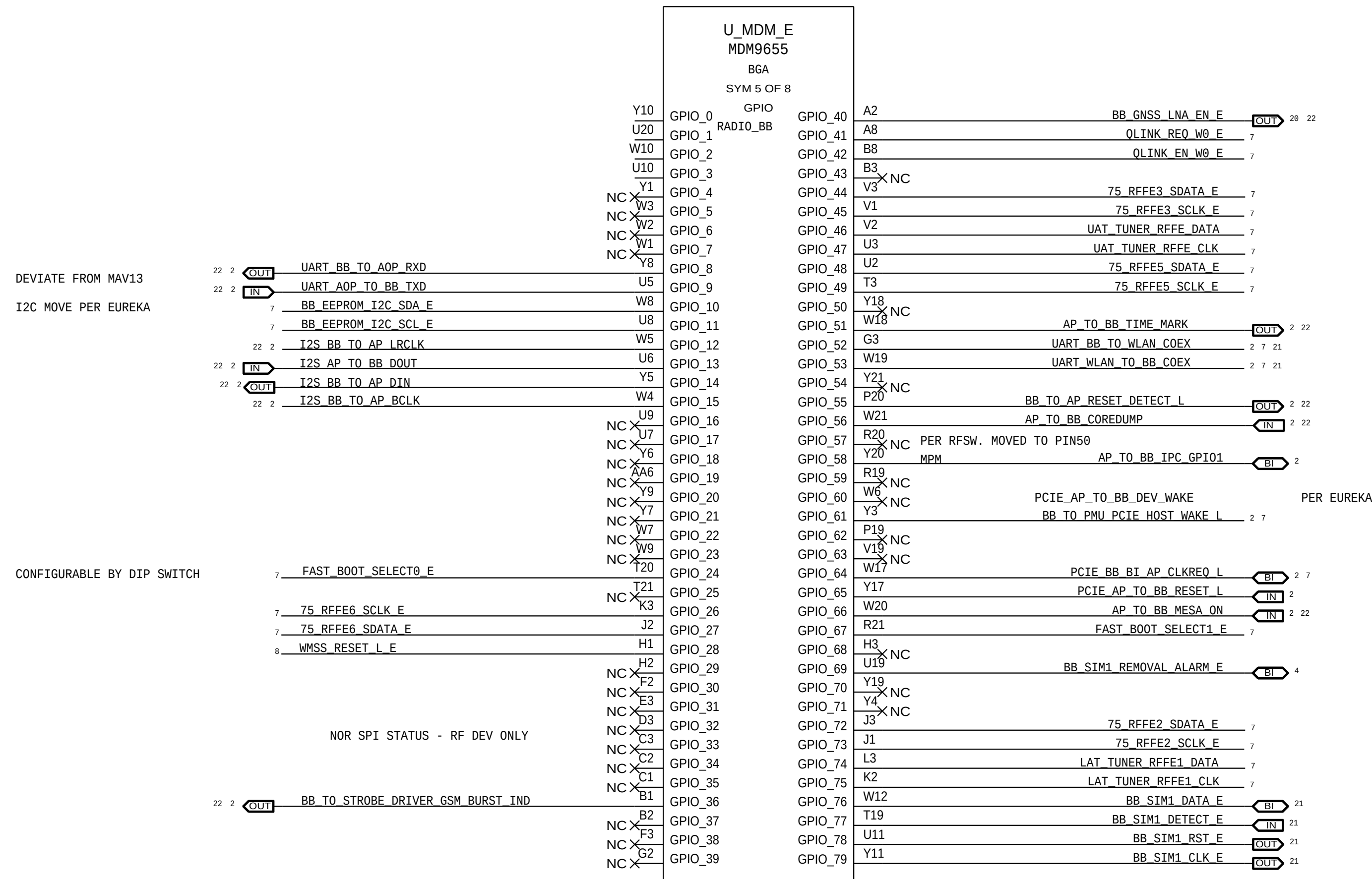


BB: CONTROL & HS PERIPHERALS

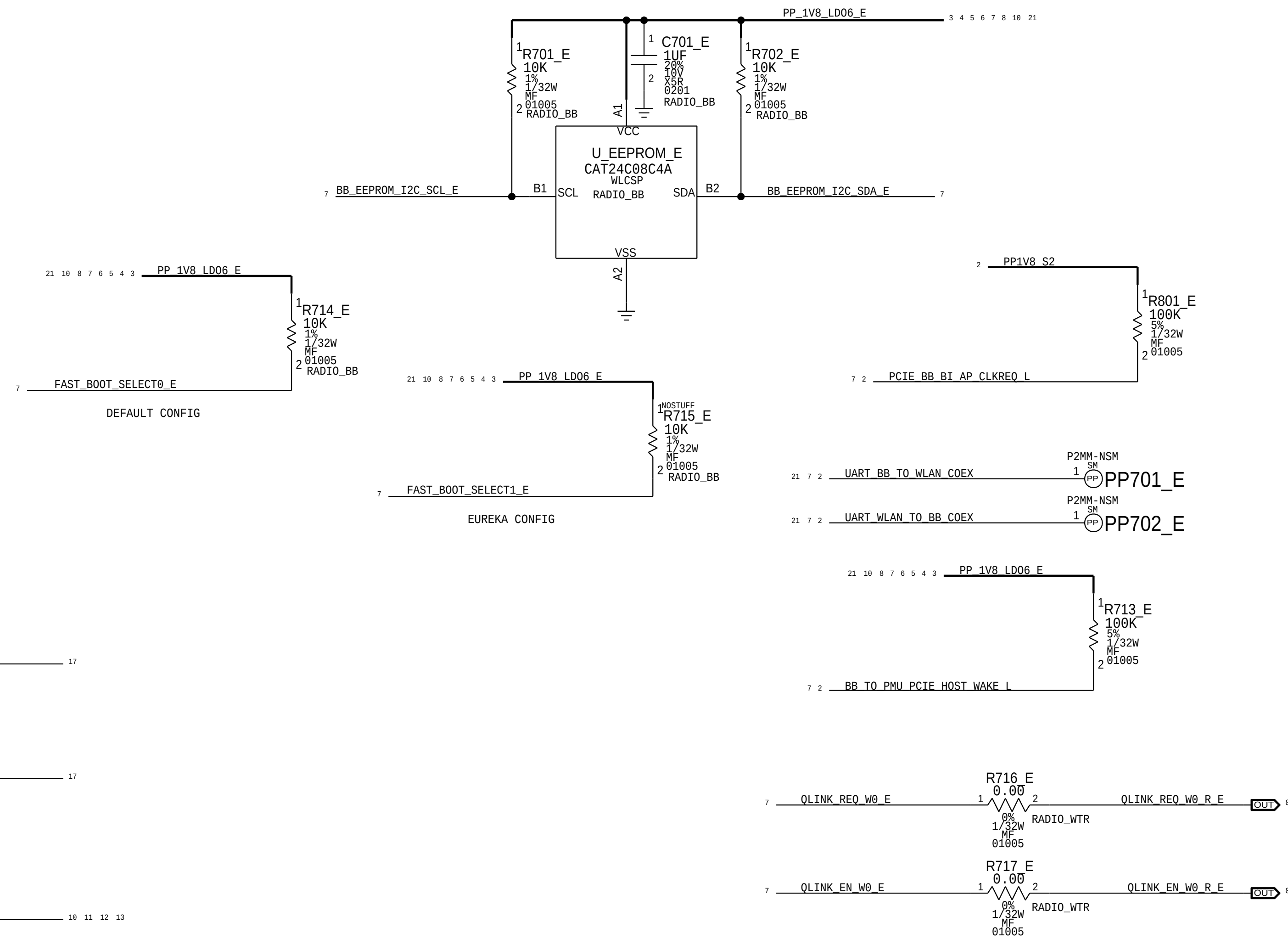


BB:GPIOs & QLINK

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BB EEPROM



D

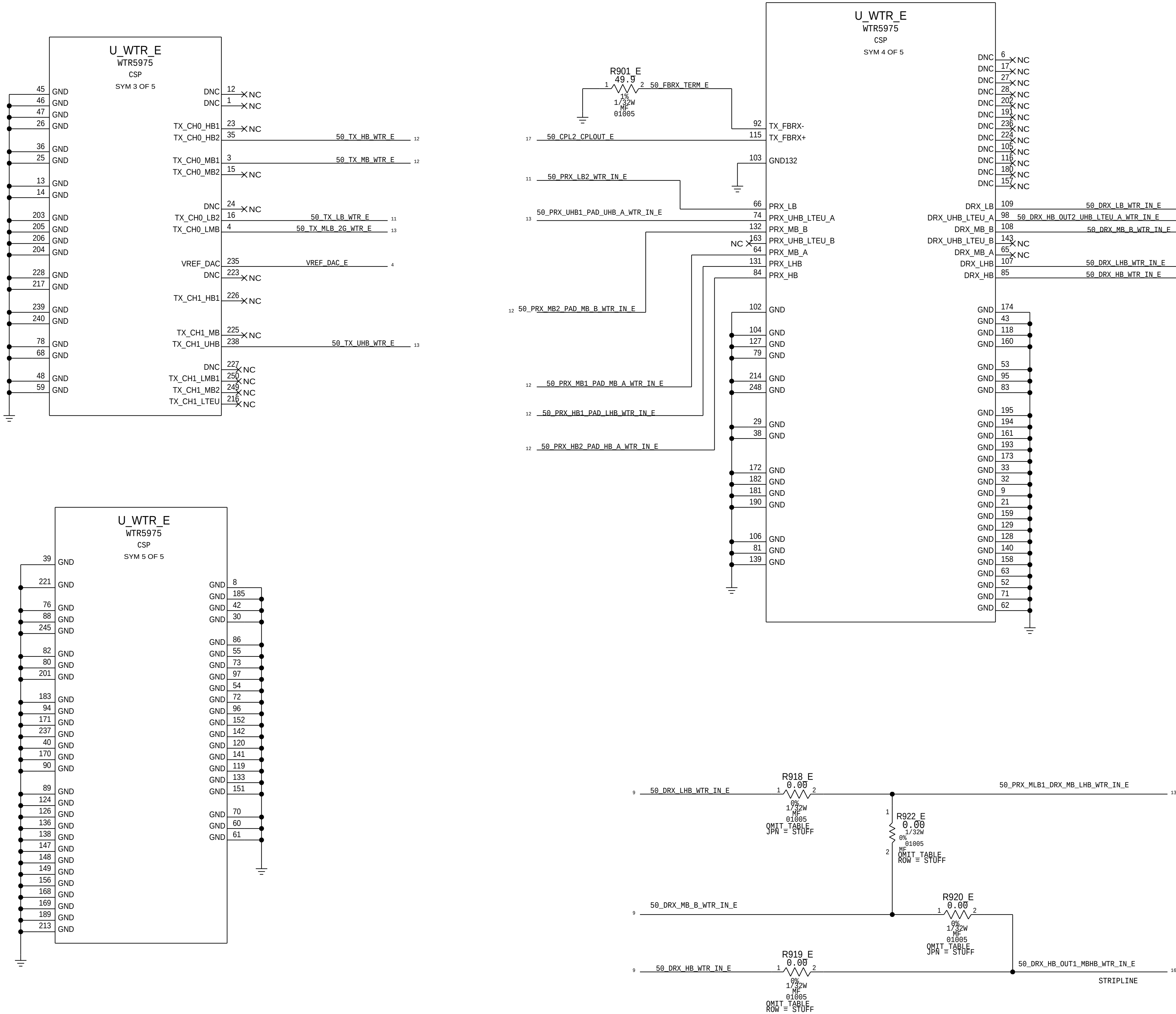


TRANSCEIVER: TX & RX

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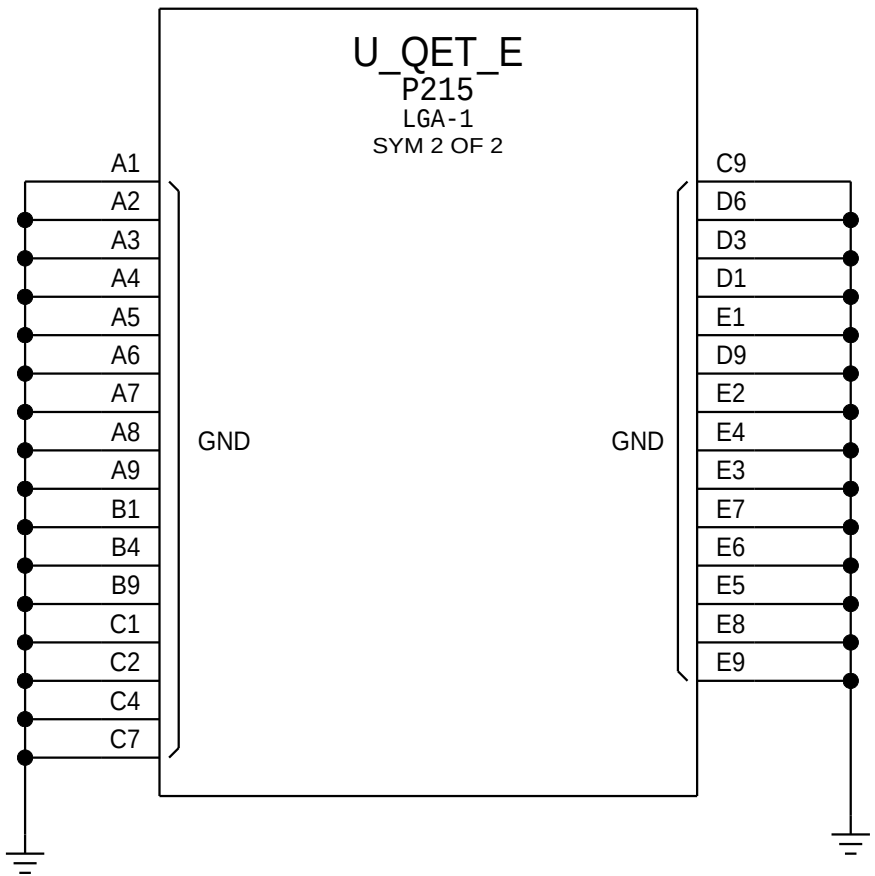
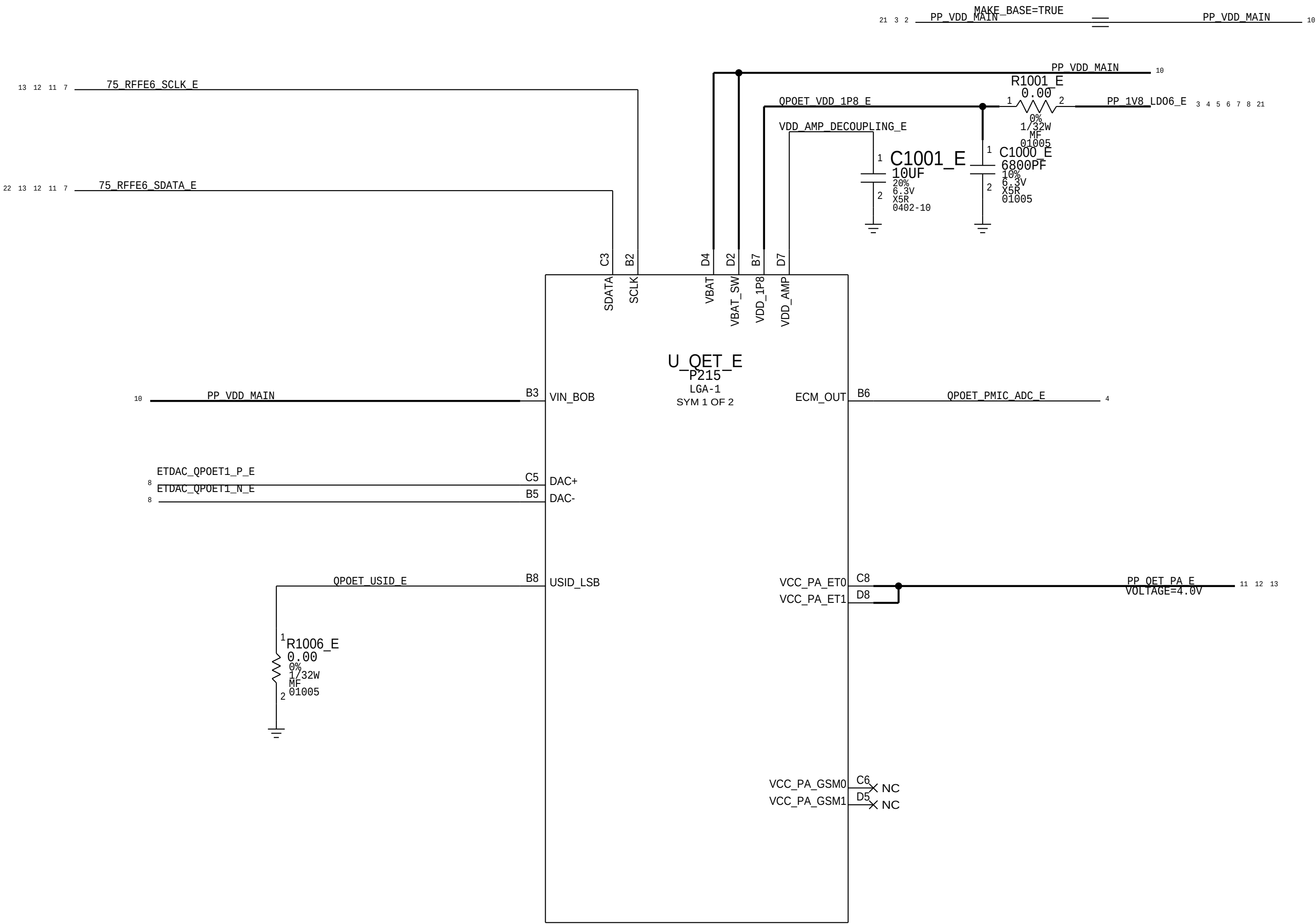
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R919_E	CRITICAL	ROW
117S0161	1	01005 0R	R922_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R918_E	CRITICAL	JPN
117S0161	1	01005 0R	R920_E	CRITICAL	JPN



QET:MODULE

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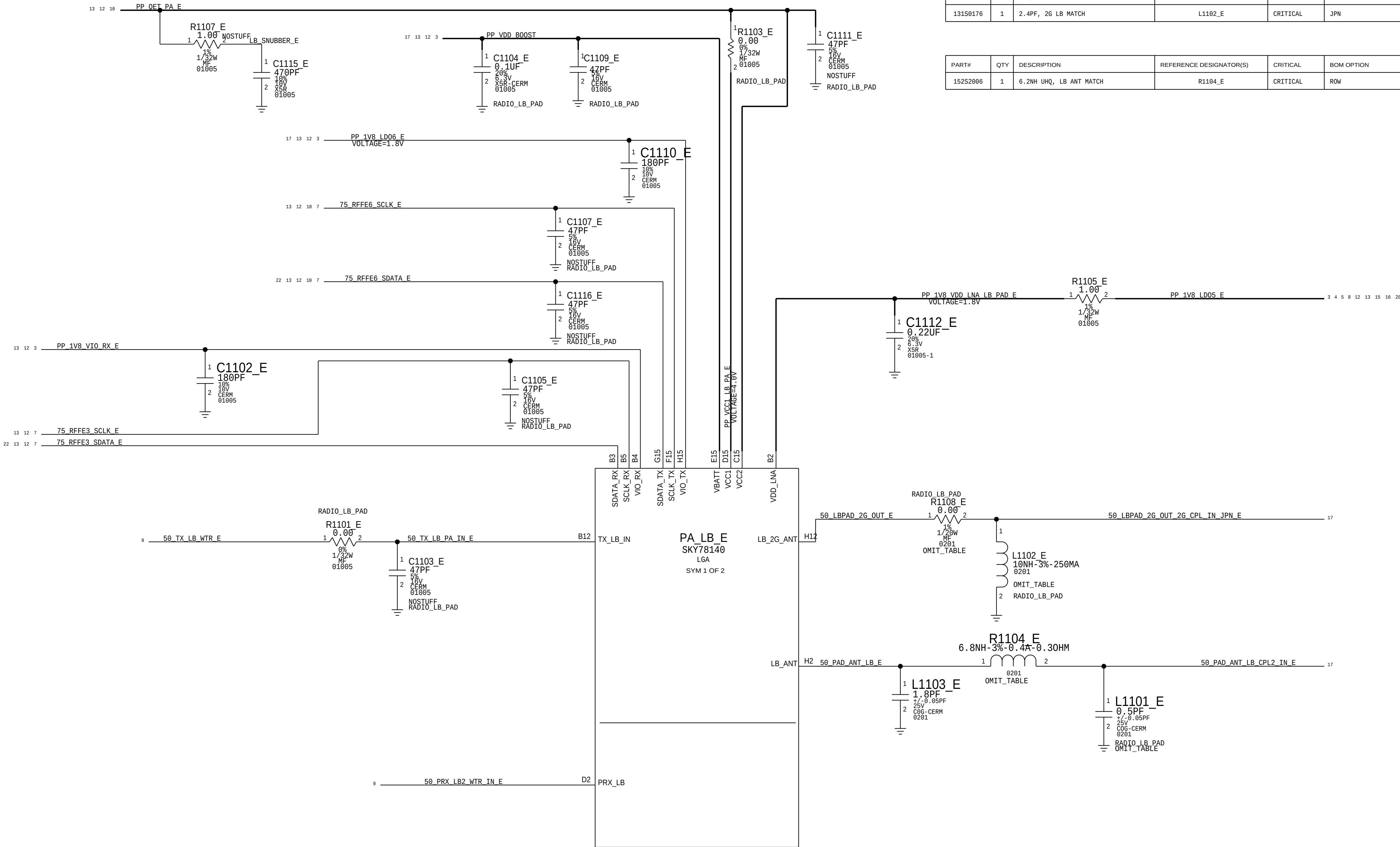


LB PAD

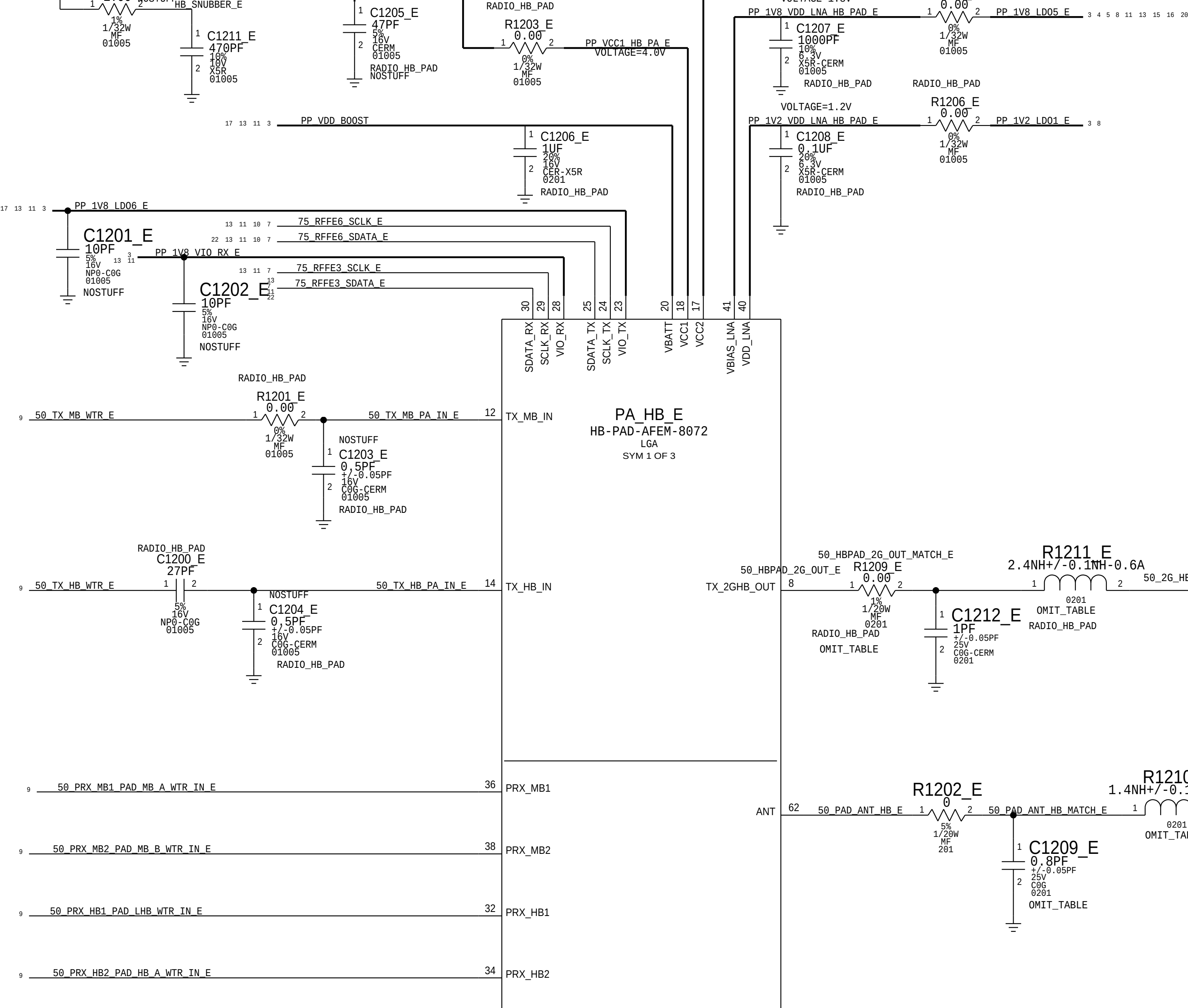
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0724	1	2G LB MATCH, 0 OHMS	R1108_E	CRITICAL	JPN
152S08021	1	6.8NH UHQ, LB ANT MATCH	R1104_E	CRITICAL	JPN
131S0555	1	1PF, LB ANT MATCH	L1101_E	CRITICAL	JPN
131S0176	1	2.4PF, 2G LB MATCH	L1102_E	CRITICAL	JPN

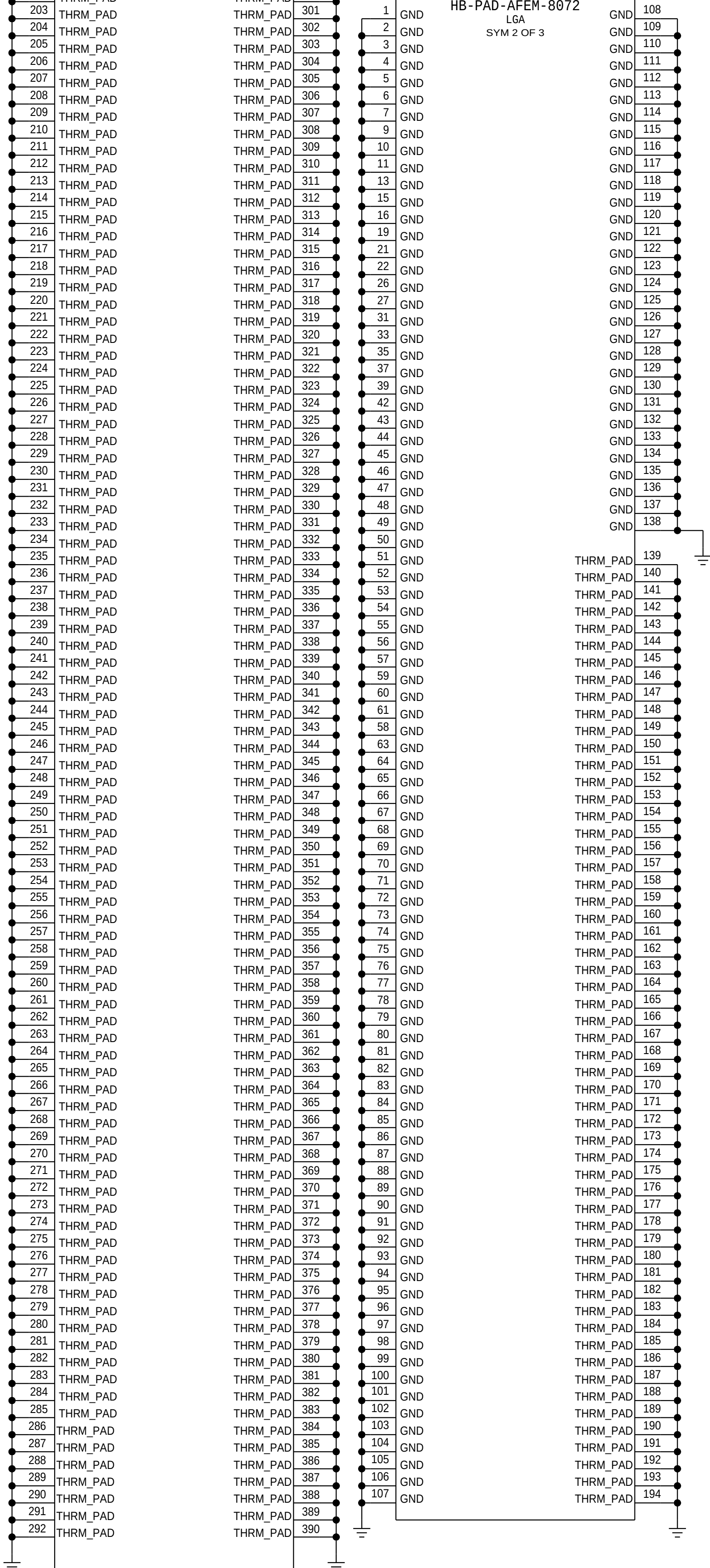
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S2086	1	6.2NH UHQ, LB ANT MATCH	R1104_E	CRITICAL	ROW



1 1% 1/32W 1 C1211 5



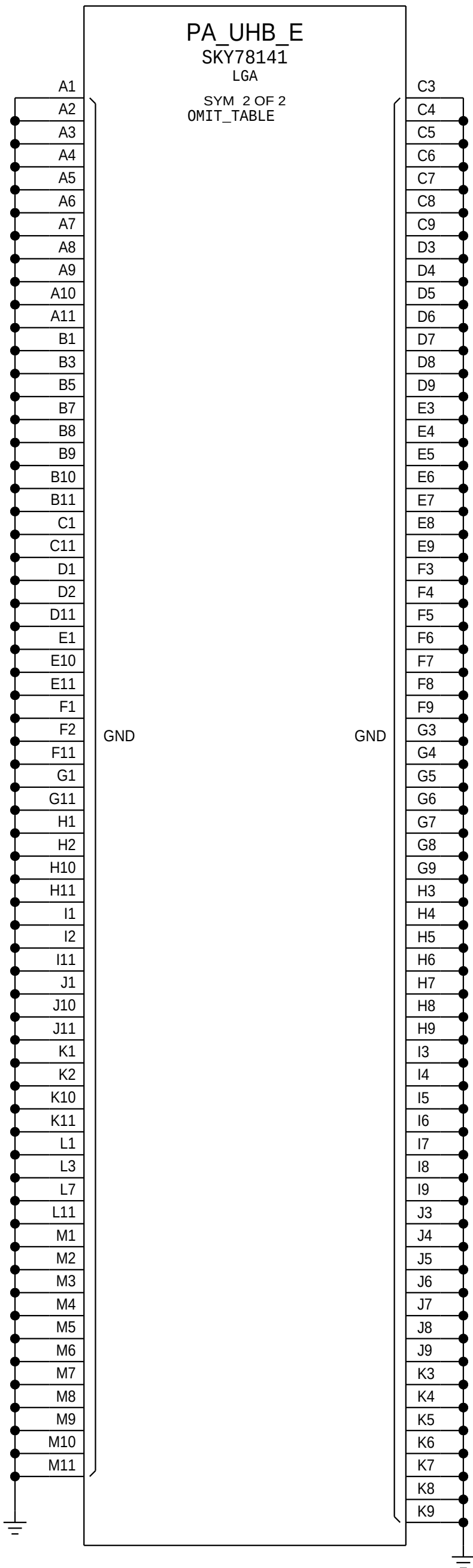
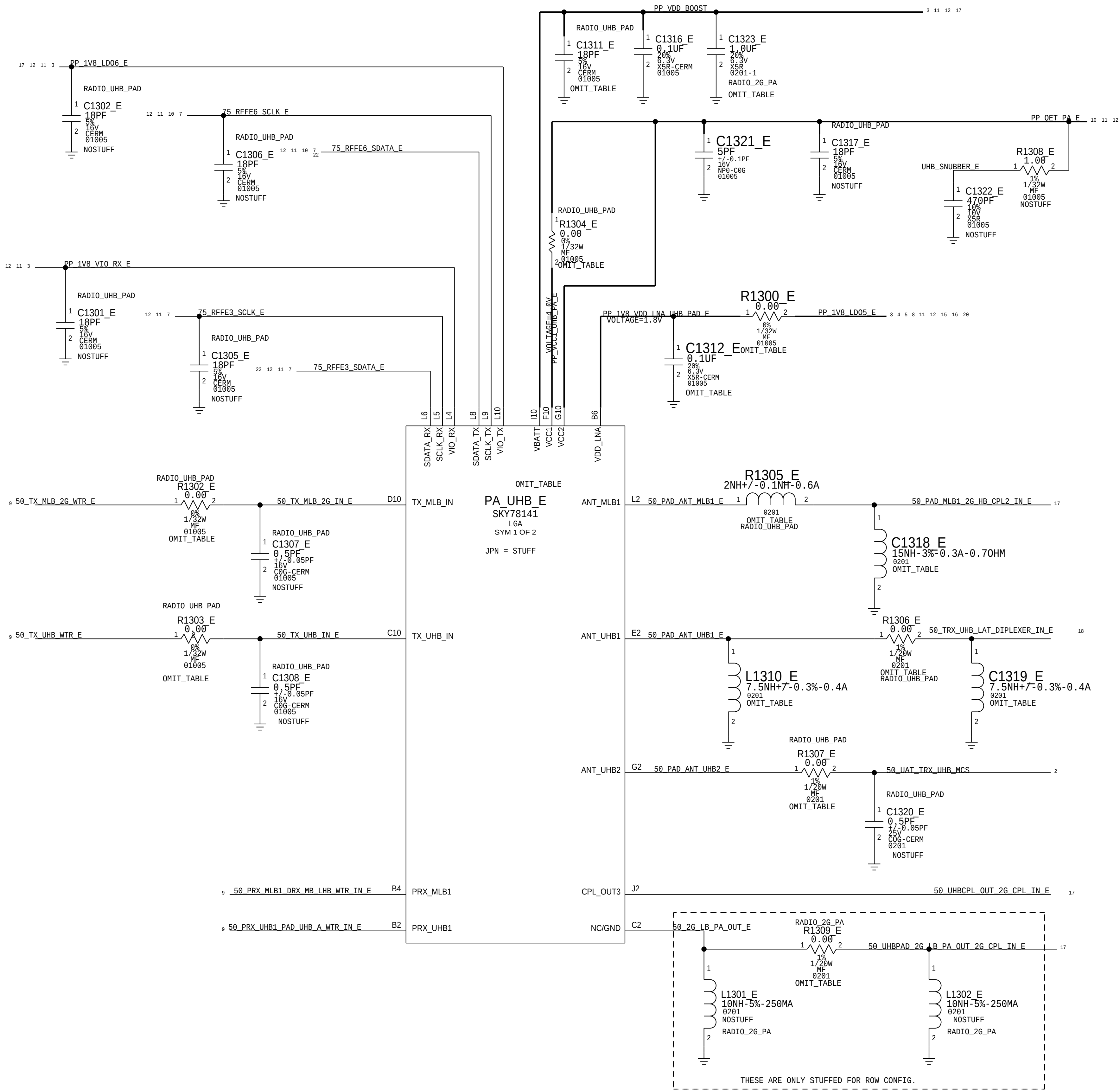
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
13159329	1	0.4PF, HB ANT MATCH	C1269_E	CRITICAL	ROW
15252080	1	2.0NH UHQ, HB ANT MATCH	R1210_E	CRITICAL	ROW



UHB/2G PAD - JPN

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01243	1	UHB PAD	PA_UHB_E	CRITICAL	JPN
117S0161	1	MLB/2GTX IN MCH2, 0 OHMS	R1302_E	CRITICAL	JPN
117S0161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	JPN
117S0092	1	MLB/2G MB OUT MCH1, 0 OHMS	R1305_E	CRITICAL	JPN
131S0552	1	2.7PF UHQ, UHB1 ANT MATCH	R1306_E	CRITICAL	JPN
117S0092	1	0 OHMS UHB2 OUT/DRX BP MCH1	R1307_E	CRITICAL	JPN
132S0316	1	UHB LNA DECOUPLING, 0.1UF	C1312_E	CRITICAL	JPN
117S0161	1	UHB LNA SUPPLY FILTER, 00HMS	R1300_E	CRITICAL	JPN
131S0214	1	VBAT DECOUPLING,18PF	C1311_E	CRITICAL	JPN
138S0692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	JPN
117S0161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	JPN
152S1999	1	22NH, MLB1 ANT MATCH	C1310_E	CRITICAL	JPN
152S1688	1	3.5NH, UHB1 ANT MATCH	L1310_E	CRITICAL	JPN
131S0431	1	0.2PF, UHB TO METROCIRC	C1320_E	CRITICAL	JPN



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2G PA - ROW

D

D

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01247	1	2G PA	PA_UHB_E	CRITICAL	ROW
117S0161	1	MLB/2GTX IN MCH2, 0 OHMS	R1302_E	CRITICAL	ROW
117S0161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	ROW
118S0724	1	MLB/2G MB OUT MCH1, 0 OHMS	R1305_E	CRITICAL	ROW
118S0724	1	UHB2 OUT/DRX BP MCH1, 0 OHMS	R1307_E	CRITICAL	ROW
152S2053	1	4.7NH UHQ, 2G LB OUTPUT MCH2	R1309_E	CRITICAL	ROW
131S0214	1	VBAT DECOUPLING, 18PF	C1311_E	CRITICAL	ROW
138S0692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	ROW
117S0161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	ROW
103S00089	1	50 OHM TERMINATION FOR ROW ONLY	C1319_E	CRITICAL	ROW

C

C

B

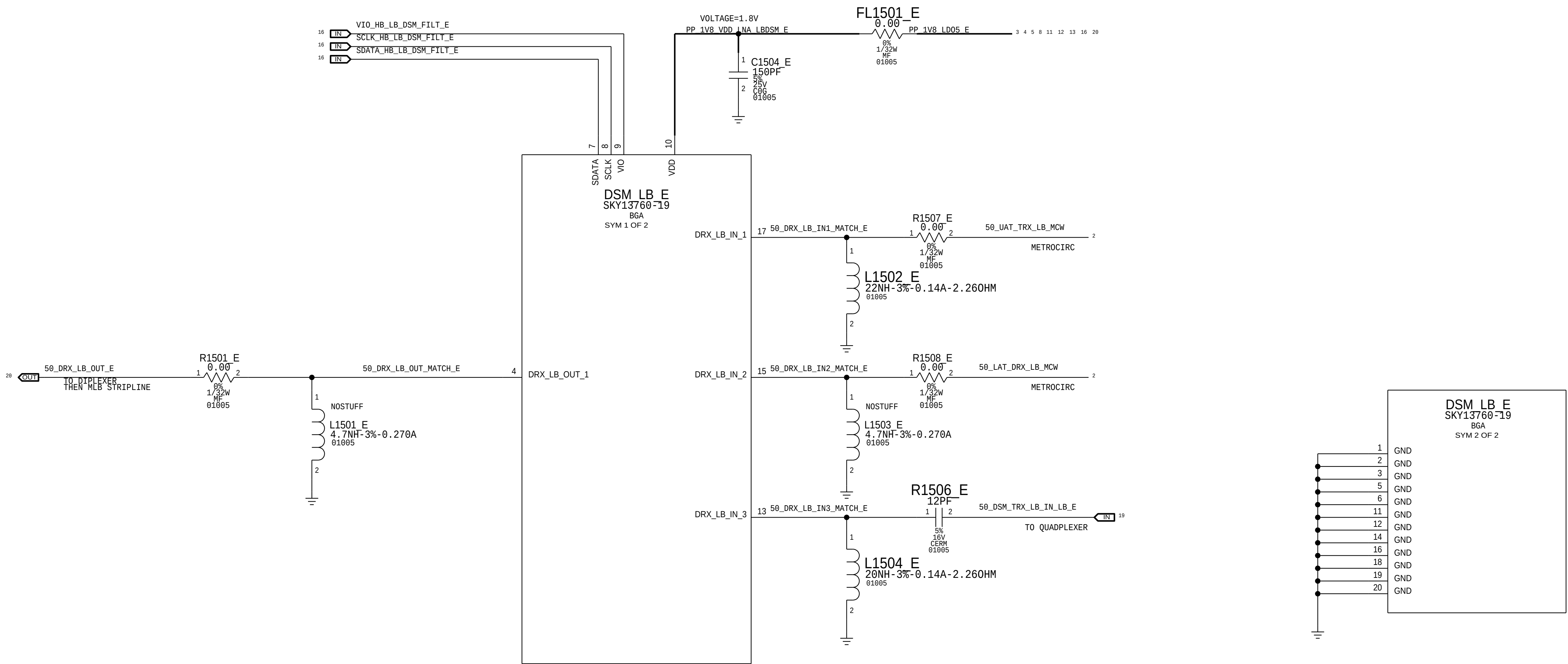
B

A

A

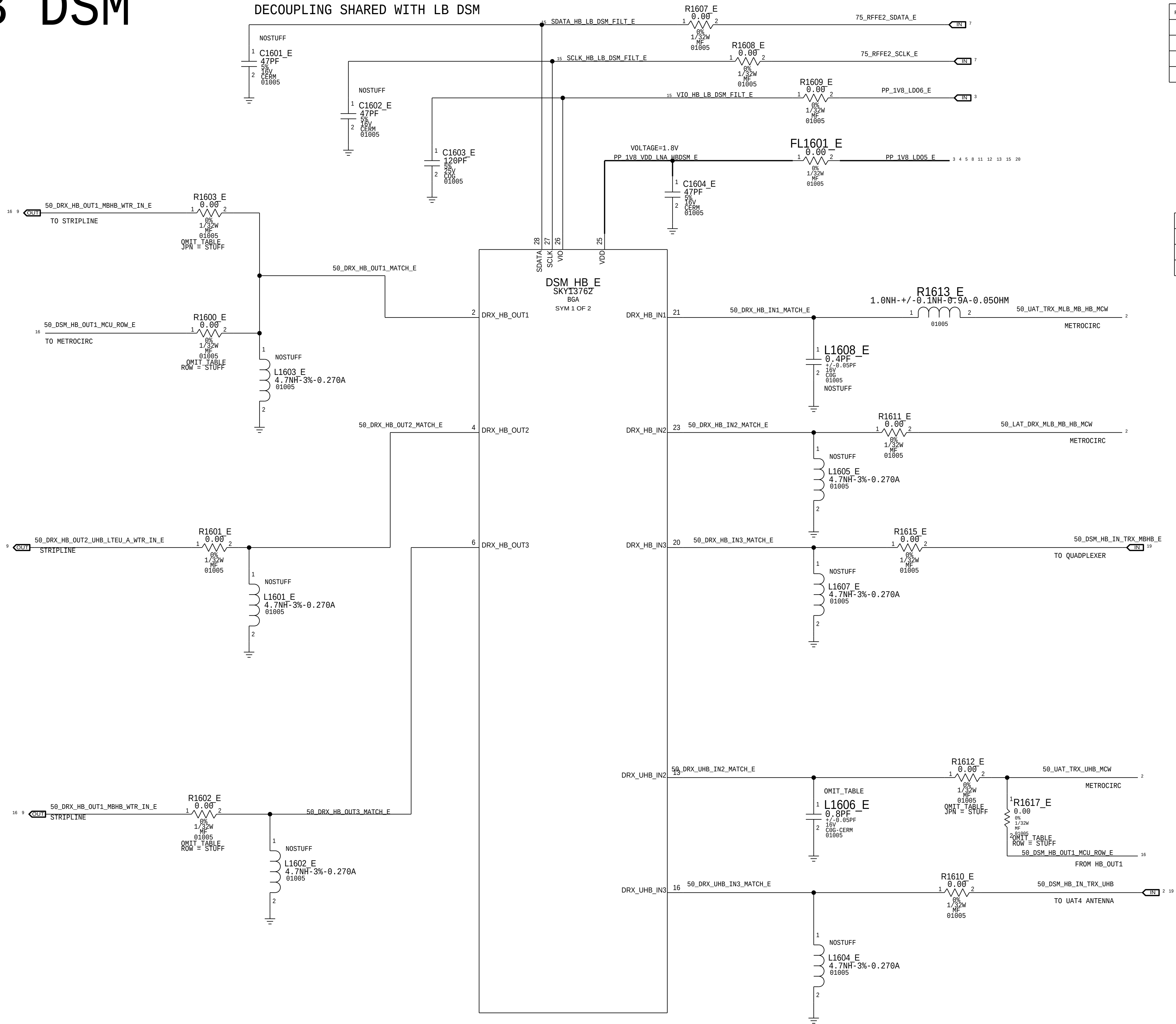
LB DSM

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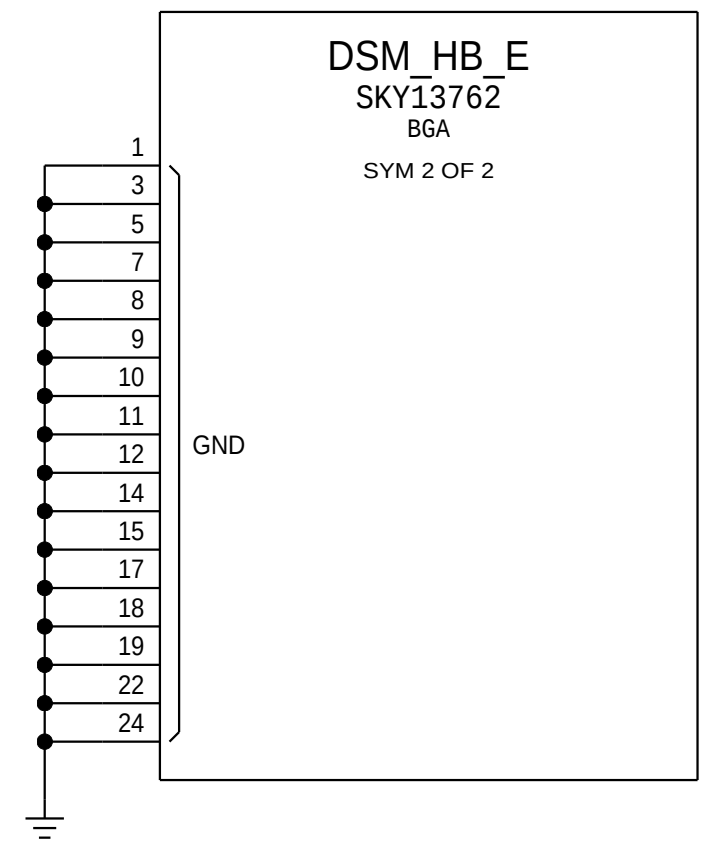
HB DSM

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R1600_E	CRITICAL	ROW
117S0161	1	01005 0R	R1602_E	CRITICAL	ROW
117S0161	1	01005 0R	R1617_E	CRITICAL	ROW
131S0639	1	01005 0.8PF	L1606_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
11750161	1	01005 0R	R1603_E	CRITICAL	JPN
15251985	1	01005 1.7NH SHQ	R1612_E	CRITICAL	JPN
13150616	1	01005 0.6PF	L1606_E	CRITICAL	JPN



COUPLER2

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S9724	1	0201 0R	R1810_E	CRITICAL	ROW
131S90142	1	150PF, DC BLOCK FOR GSM	R1813_E	CRITICAL	ROW
118S9724	1	0201 0R	R1814_E	CRITICAL	ROW
131S90901	1	0.1PF, CPLR HB LAT MATCH	L1801_E	CRITICAL	ROW
152S2044	1	2.2NH, CPLR HB LAT MATCH	R1807_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S9724	1	0201 0R	R1809_E	CRITICAL	JPN
152S2007	1	8.2NH 2G LB MATCH	R1812_E	CRITICAL	JPN
131S9431	1	0.2PF, CPLR HB LAT MATCH	L1801_E	CRITICAL	JPN
152S90955	1	2.6NH, CPLR HB LAT MATCH	R1807_E	CRITICAL	JPN
131S9249	1	2.2PF, 2G LB MATCH	C1810_E	CRITICAL	JPN

D

C

B

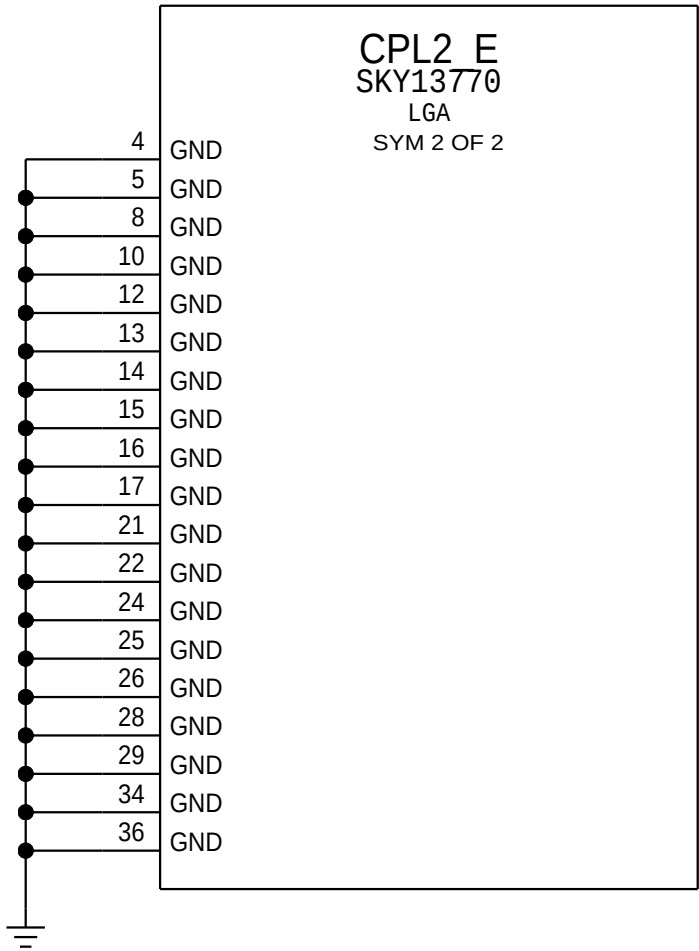
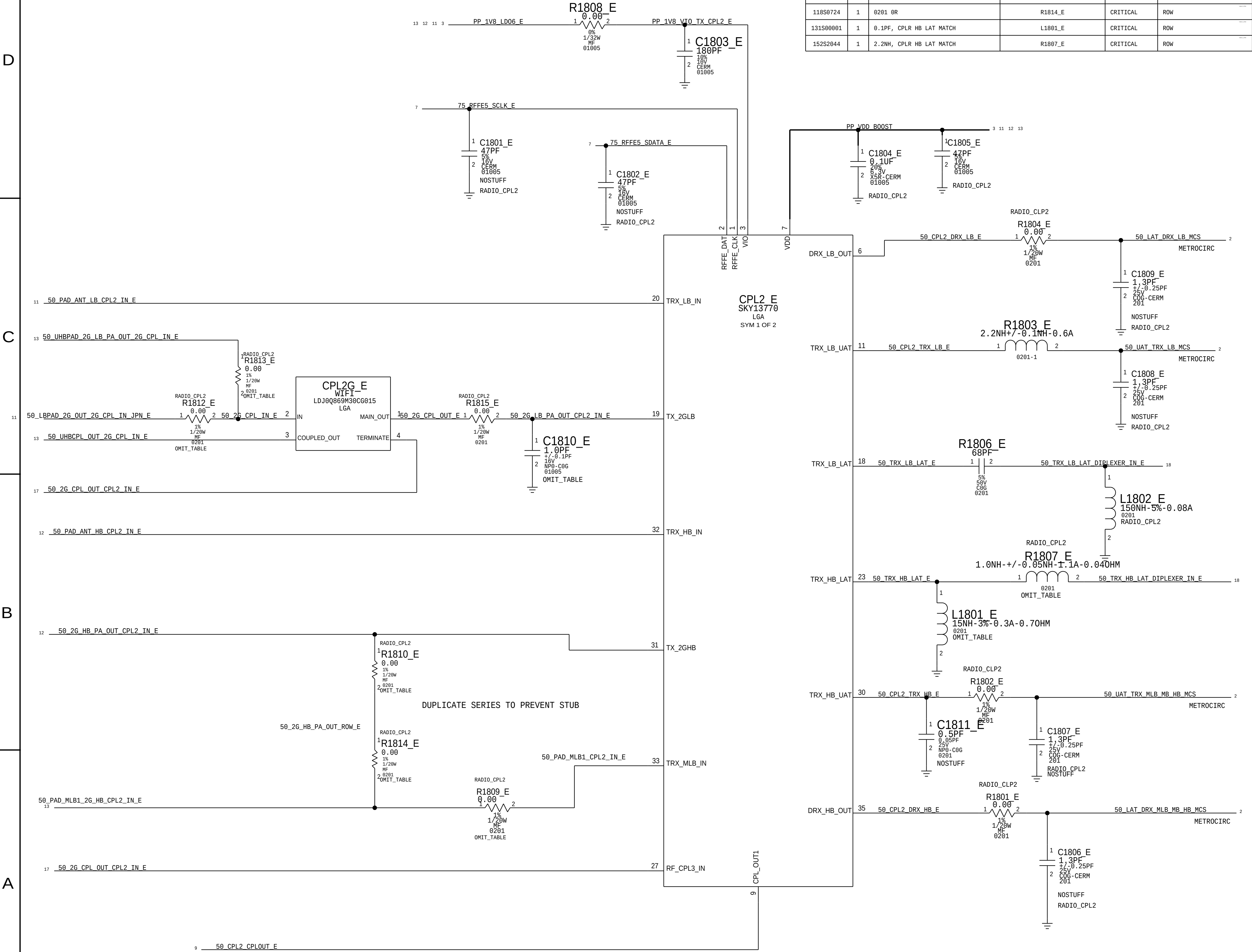
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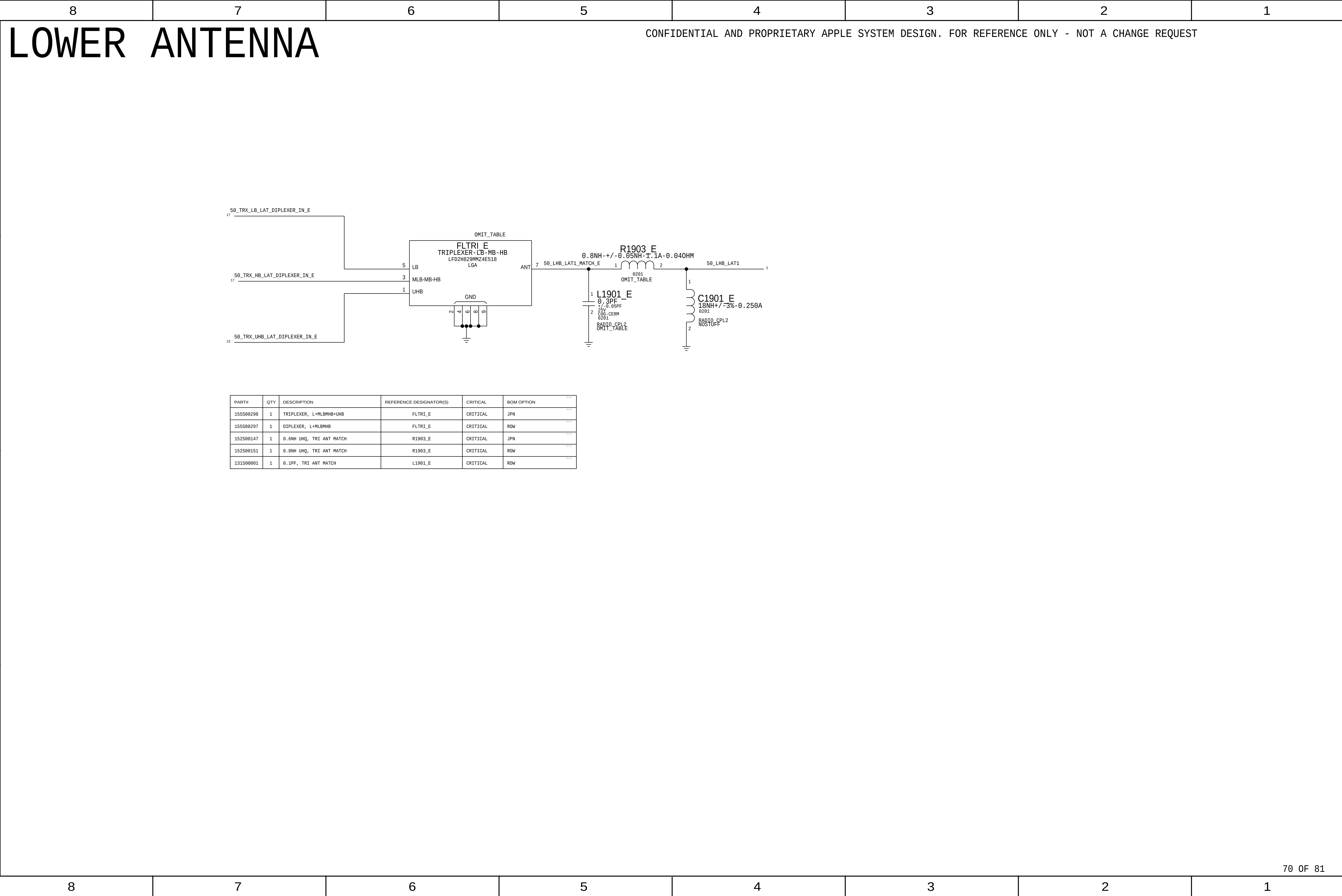
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C

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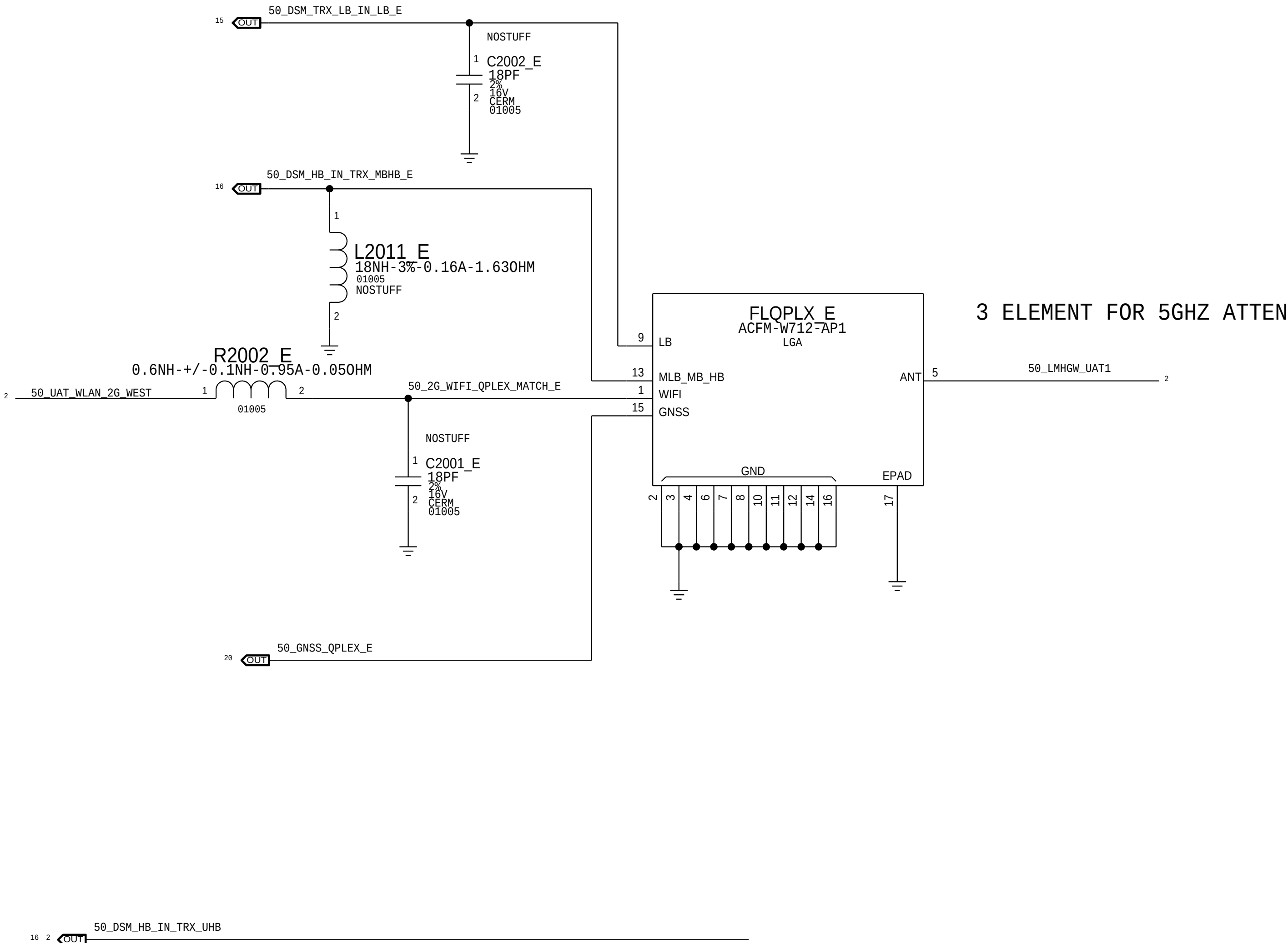
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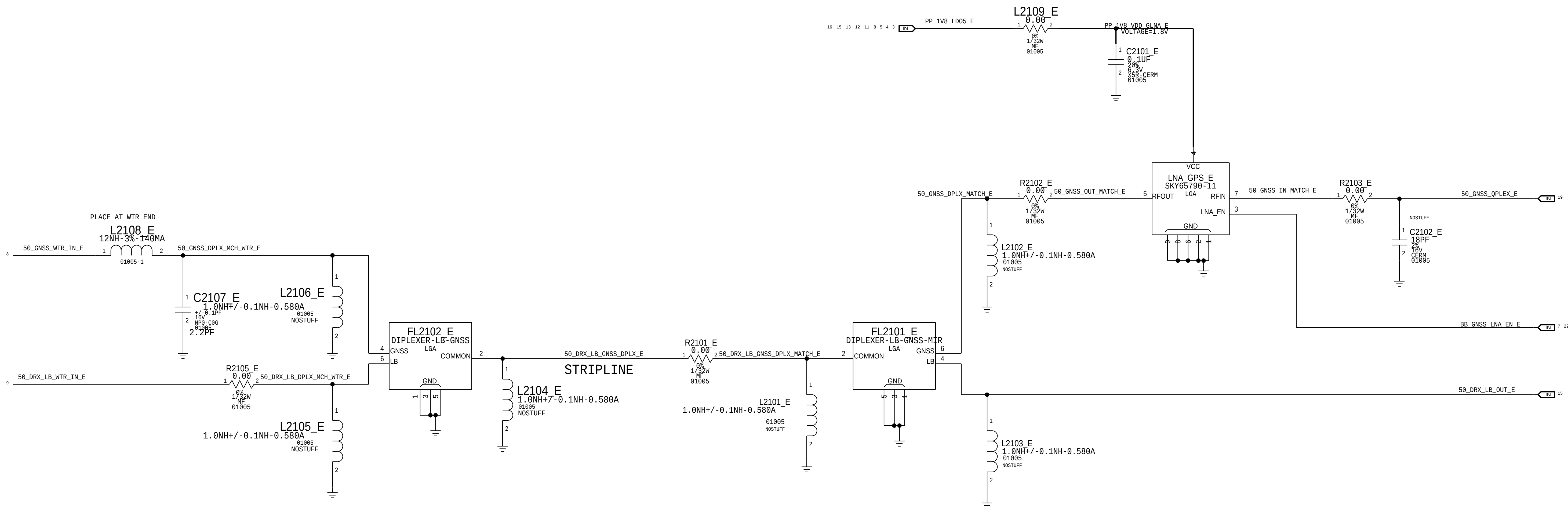


UAT & LB/UHB DIPLEXER

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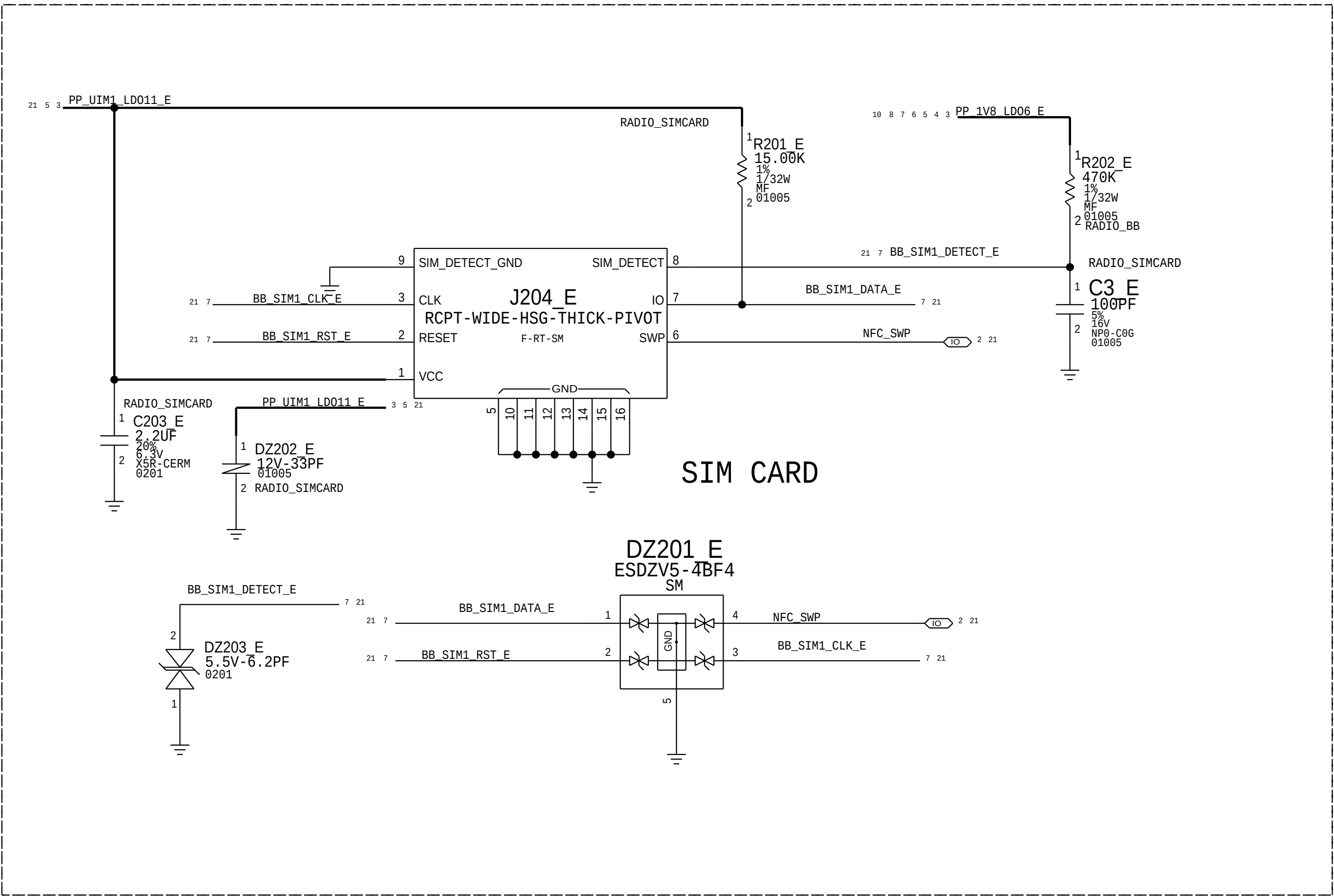


GPS & GPS/LB DIPLEXER



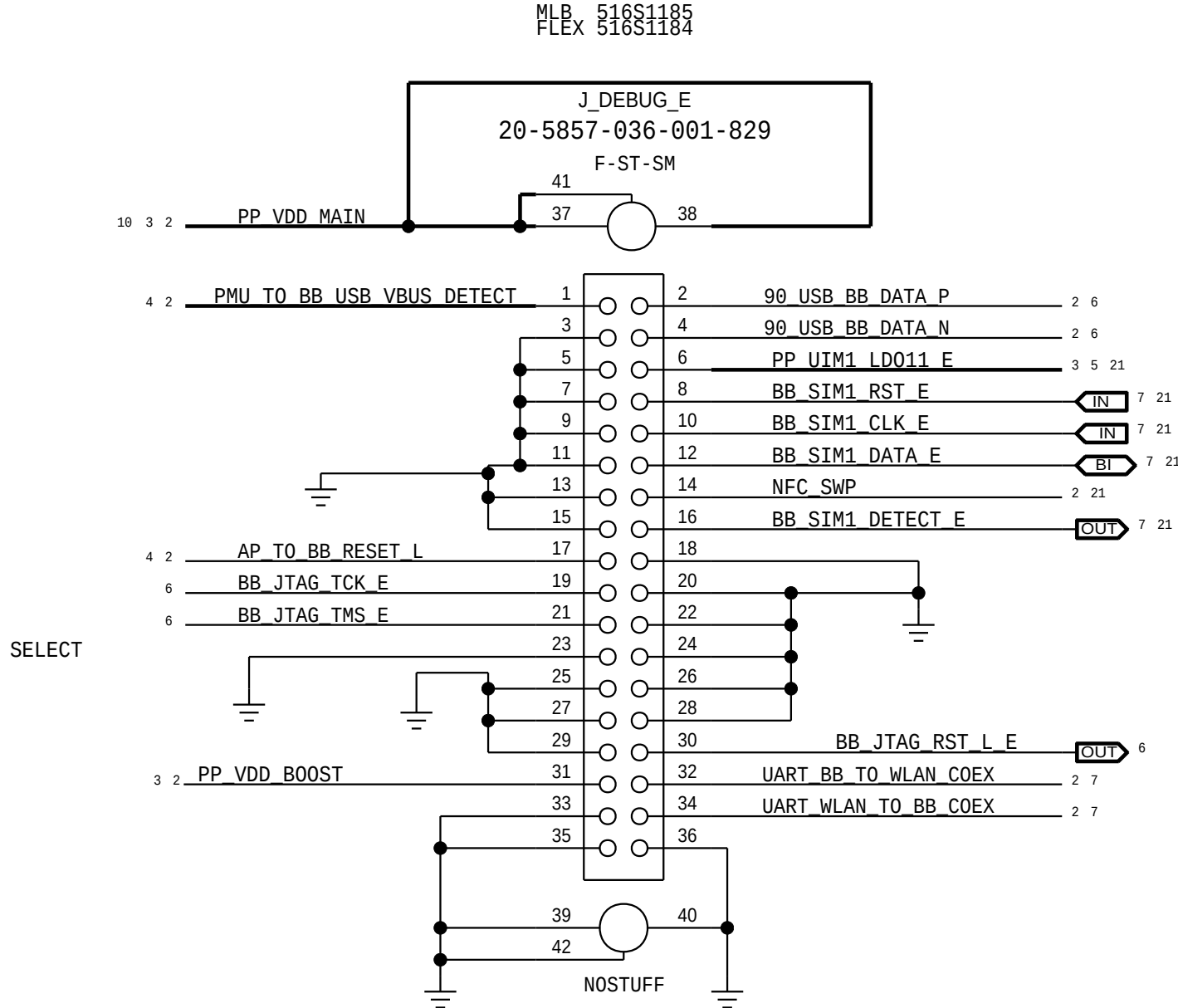
SIM CARD: CONNECTOR

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
S12S00026	S12S00017	BOM_TABLE_ALTS	3204_E	SIMCRO, KYOCERA

DEBUG CONNECTOR



BB PROBE POINTS

QLINK

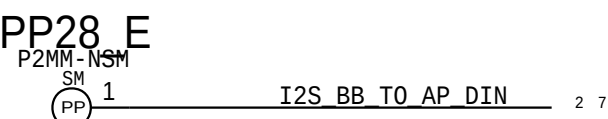
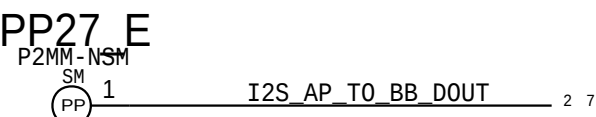
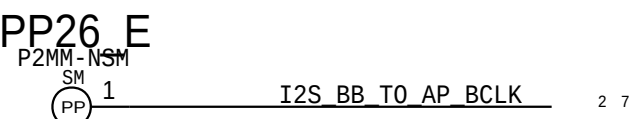
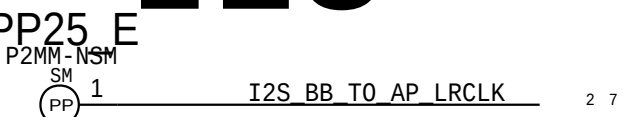
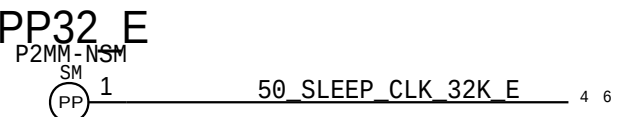
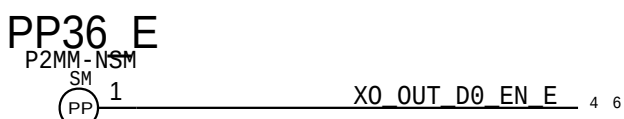
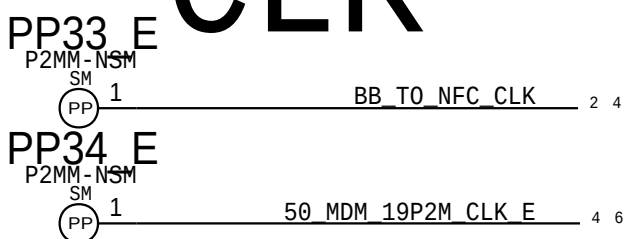
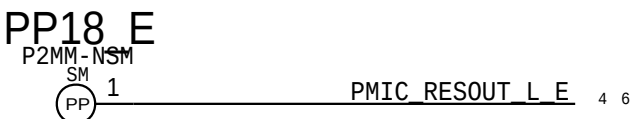
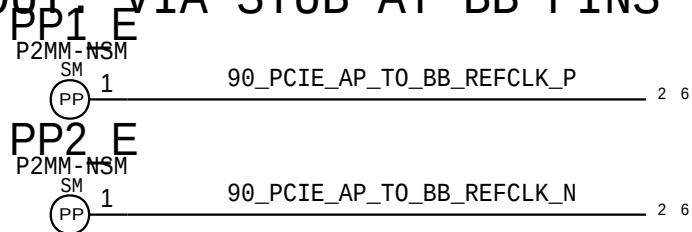
BASEBAND

CLK

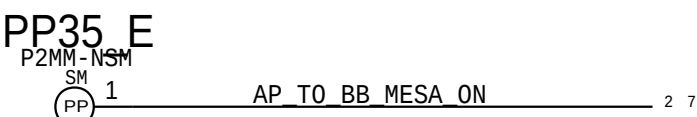
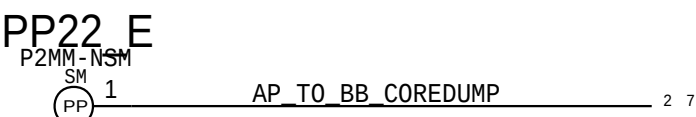
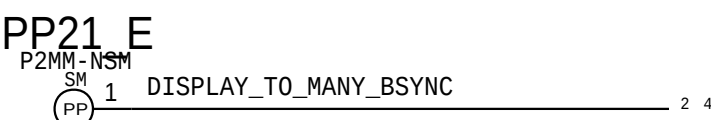
PWR SEQ

PCIE

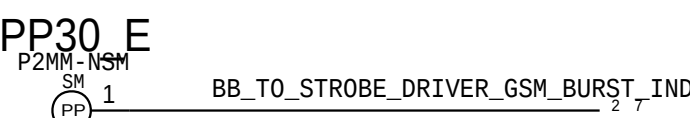
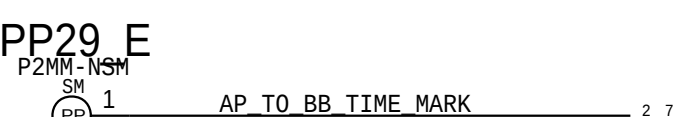
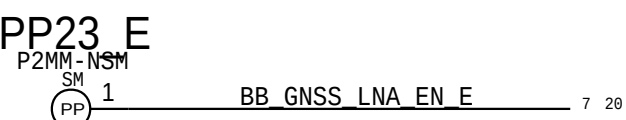
LAYOUT: VIA STUB AT BB PINS



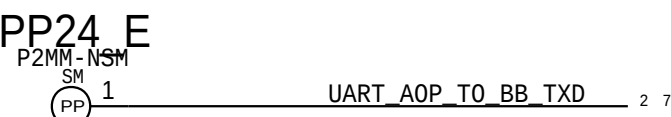
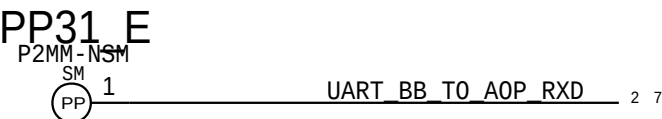
OTHERS



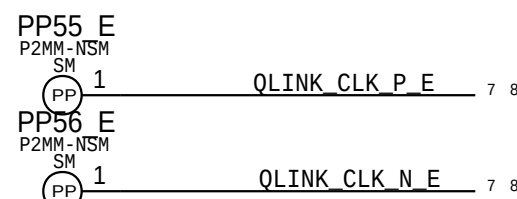
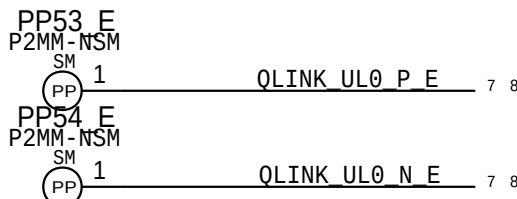
CTRL



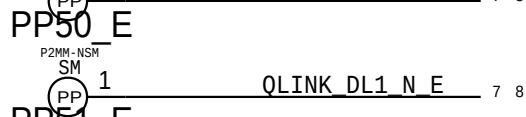
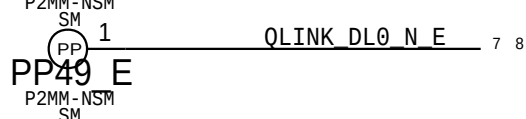
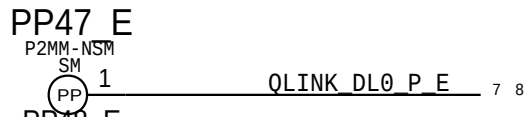
UART



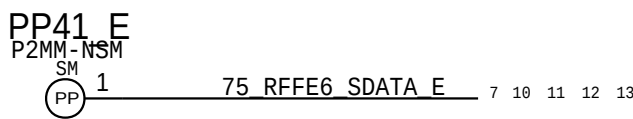
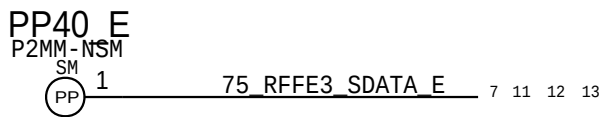
LAYOUT: VIA STUB AT WTR PINS



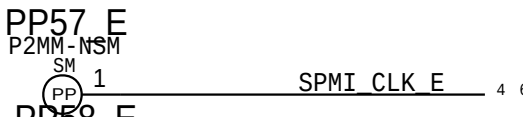
LAYOUT: VIA STUB AT BB PINS



RFFE

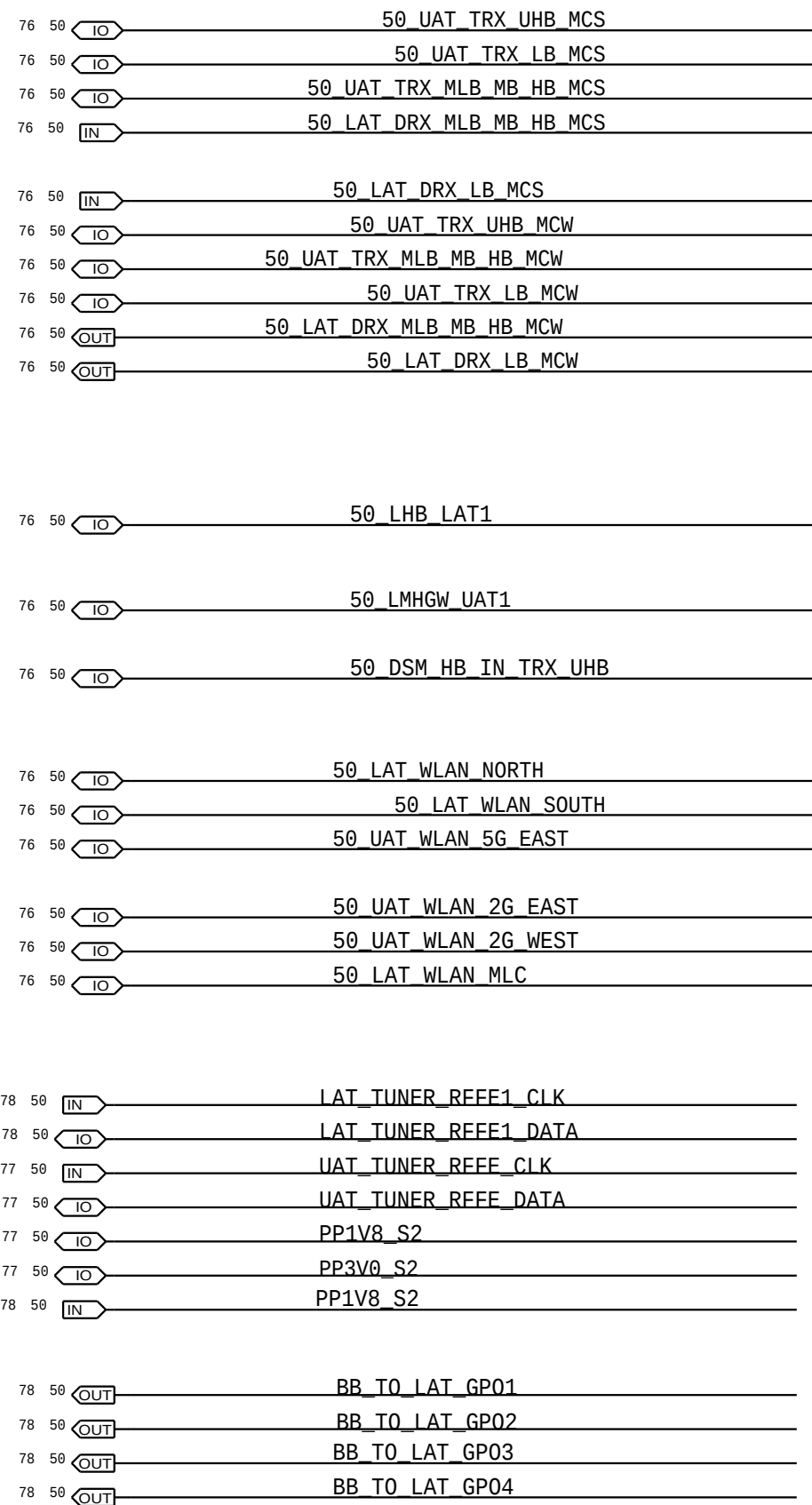


SPMI



D21 RADIO_MLB_FF

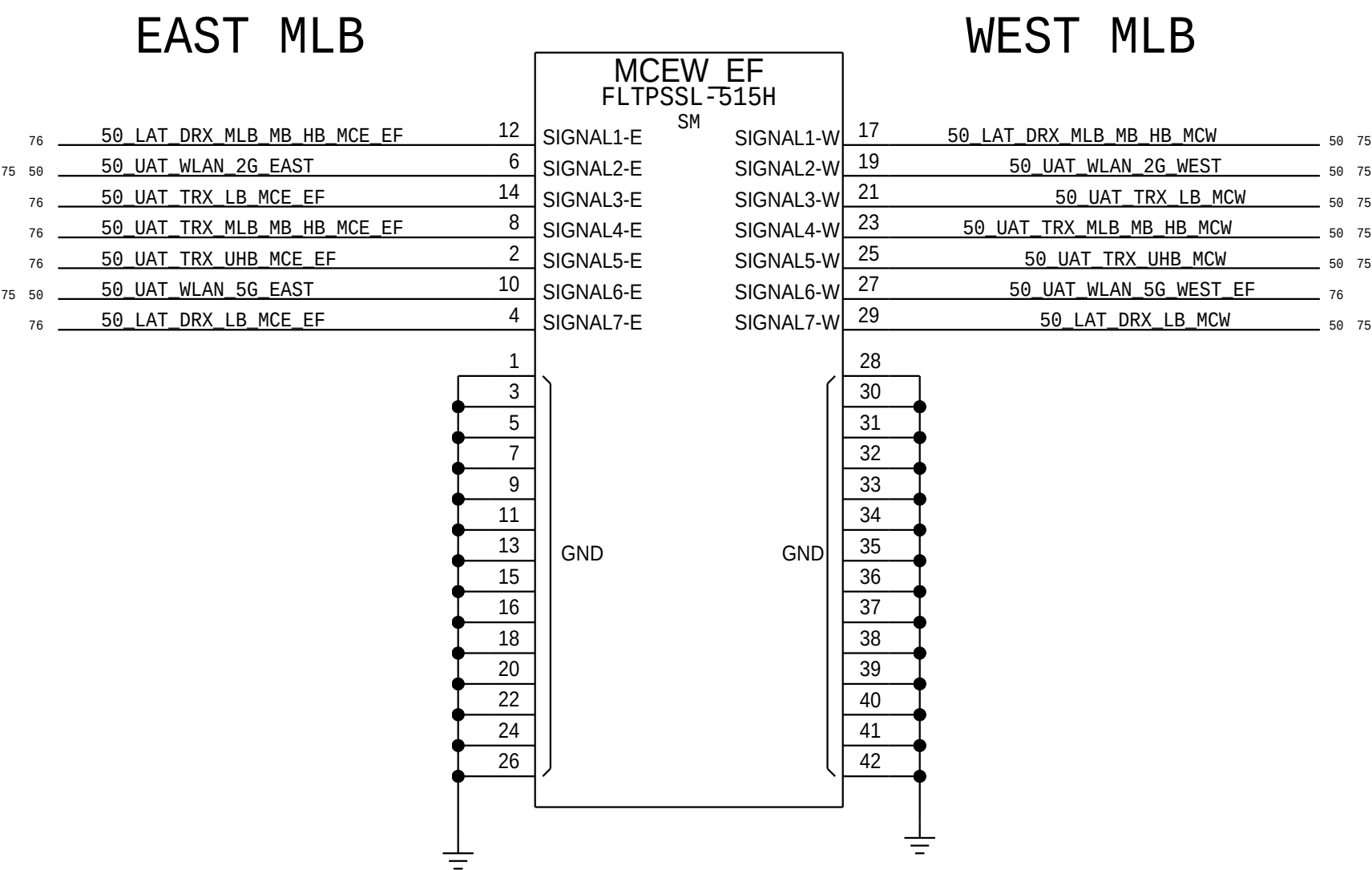
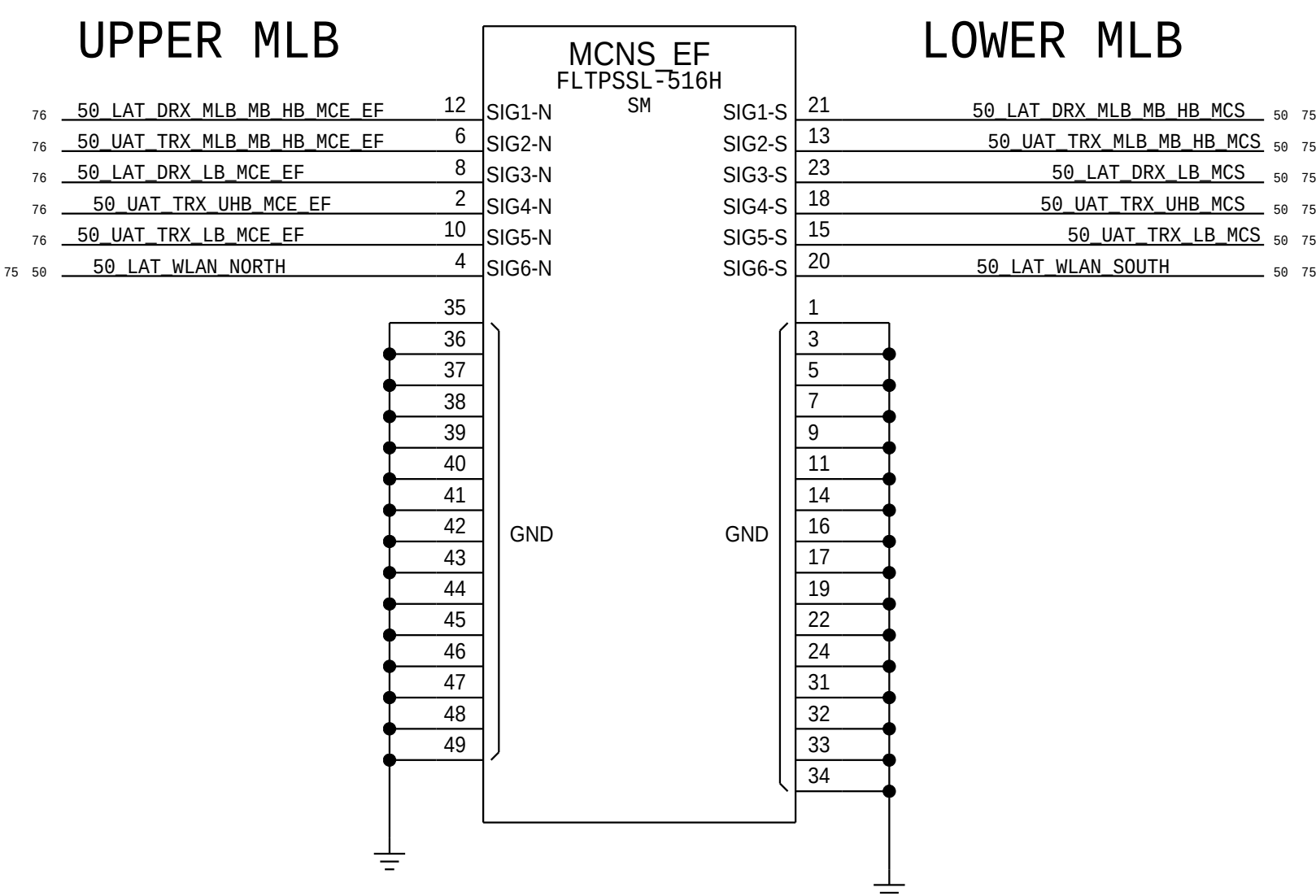
5/22/2017



METROCIRC

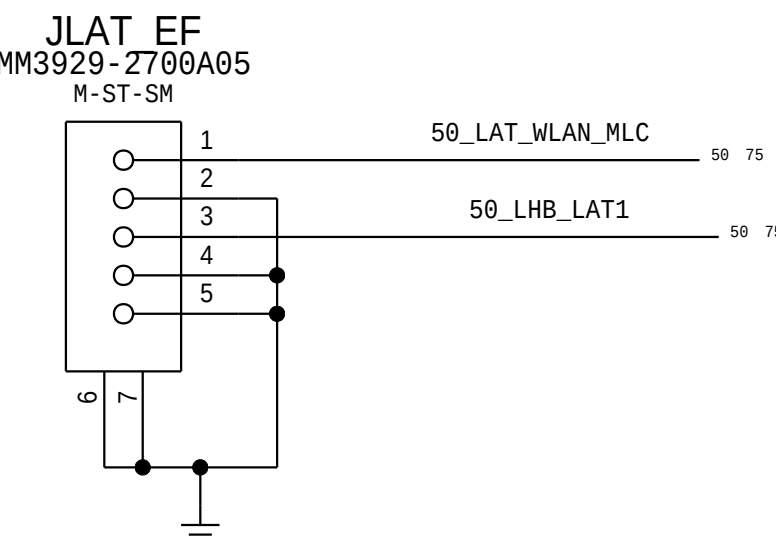
NORTH-SOUTH METROCIRC
339S00297

EAST-WEST METROCIRC
339S00296

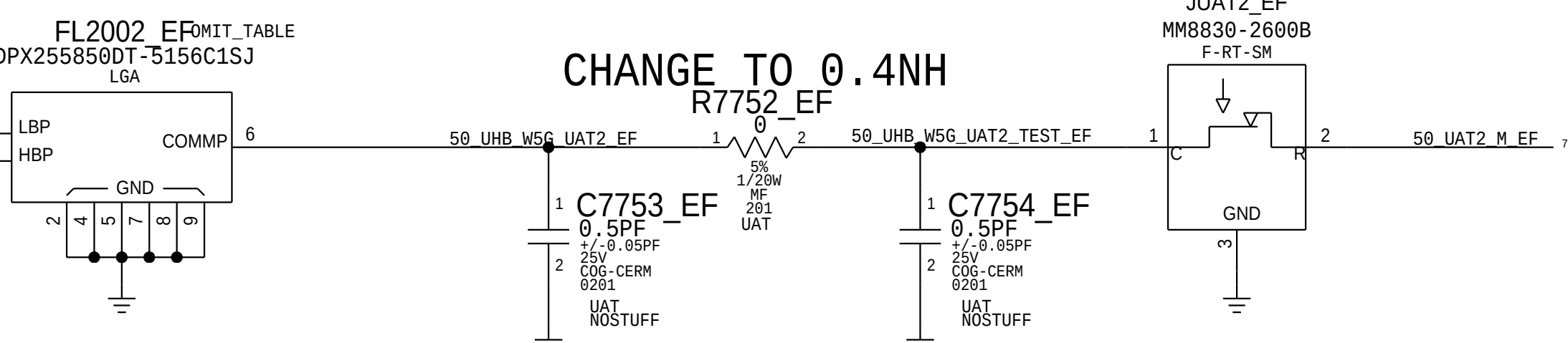
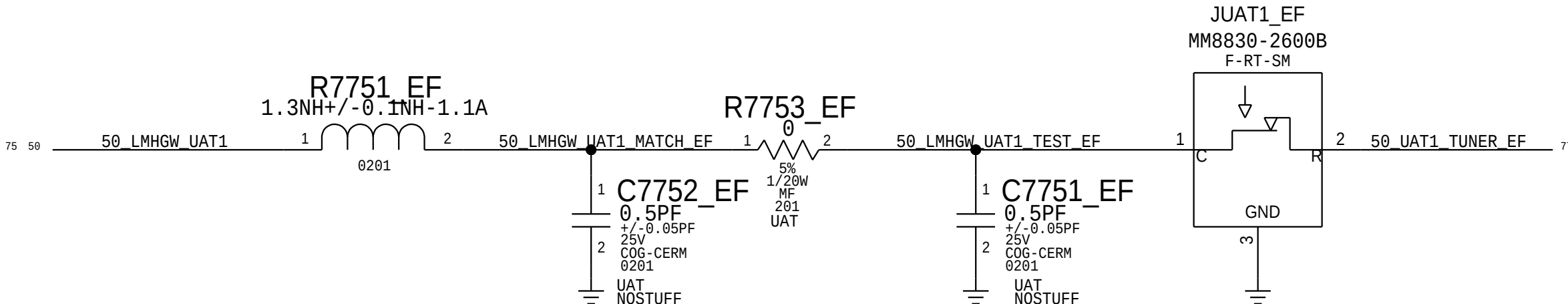


UHB/WIFI 5GHZ TEST

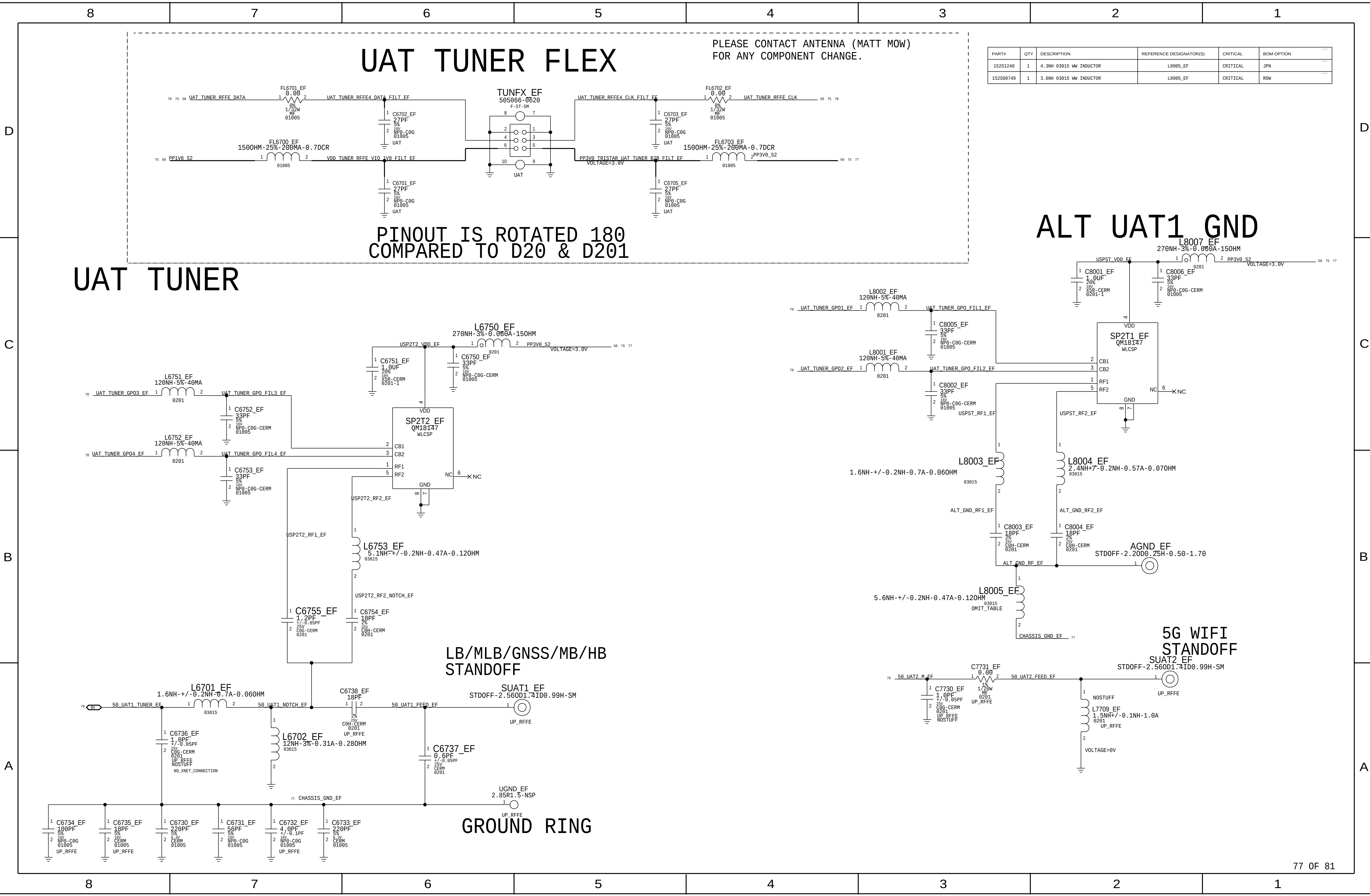
LAT MLC



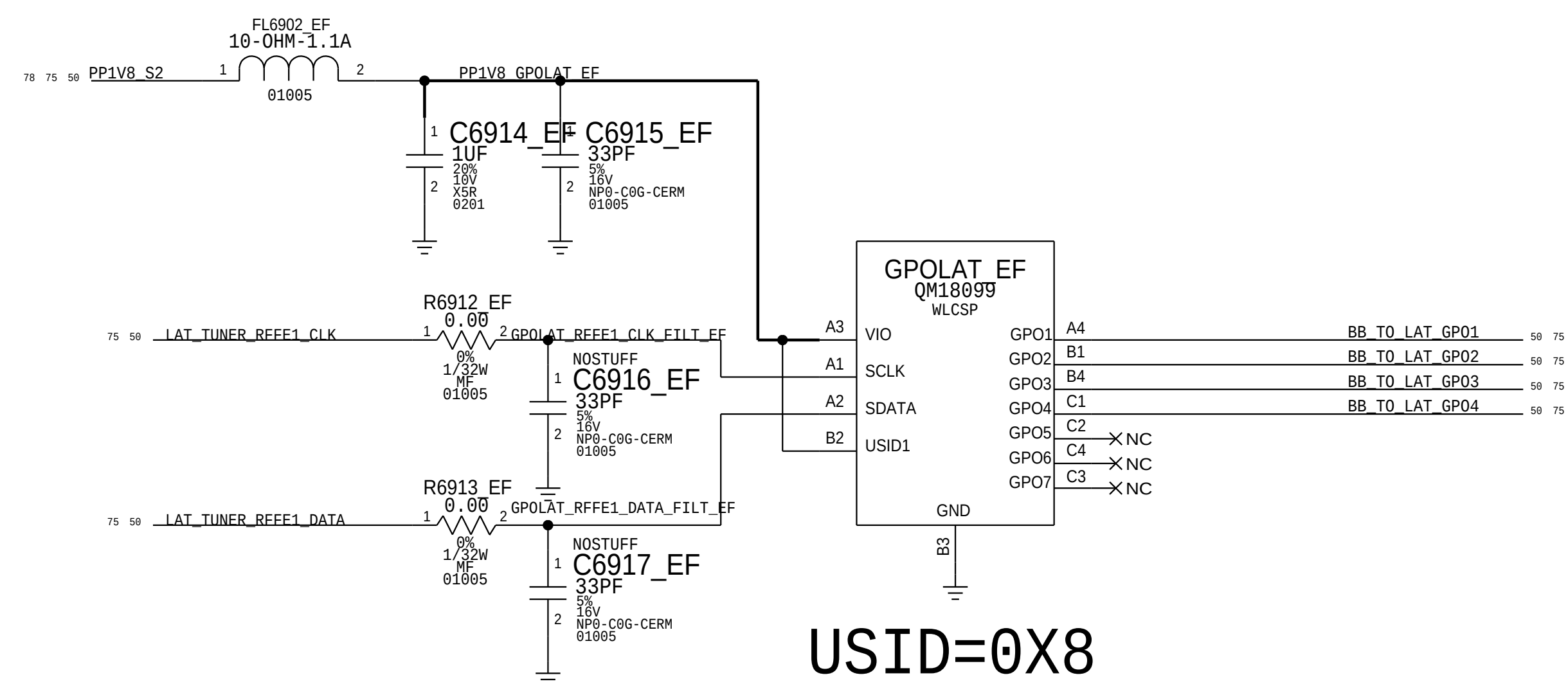
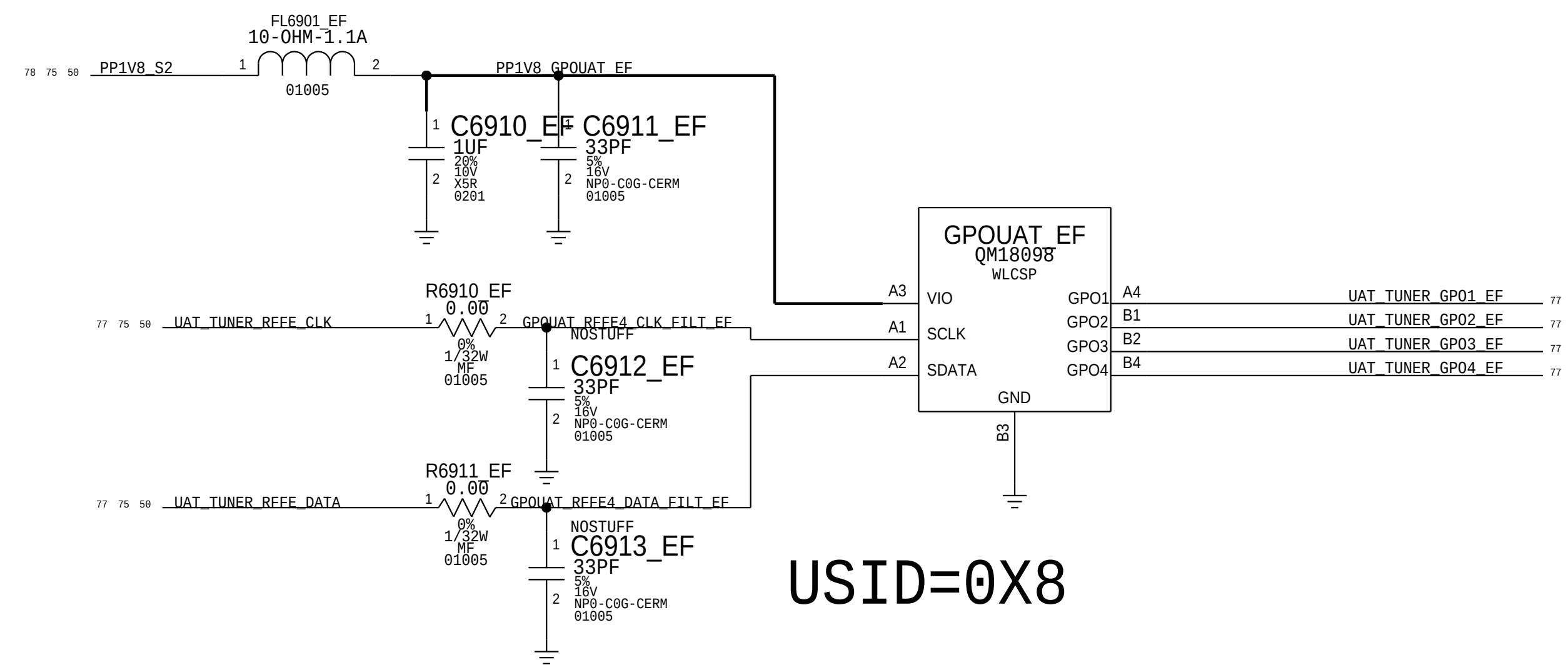
LB/MLB/MB/HB
WIFI 2.4GHZ TEST



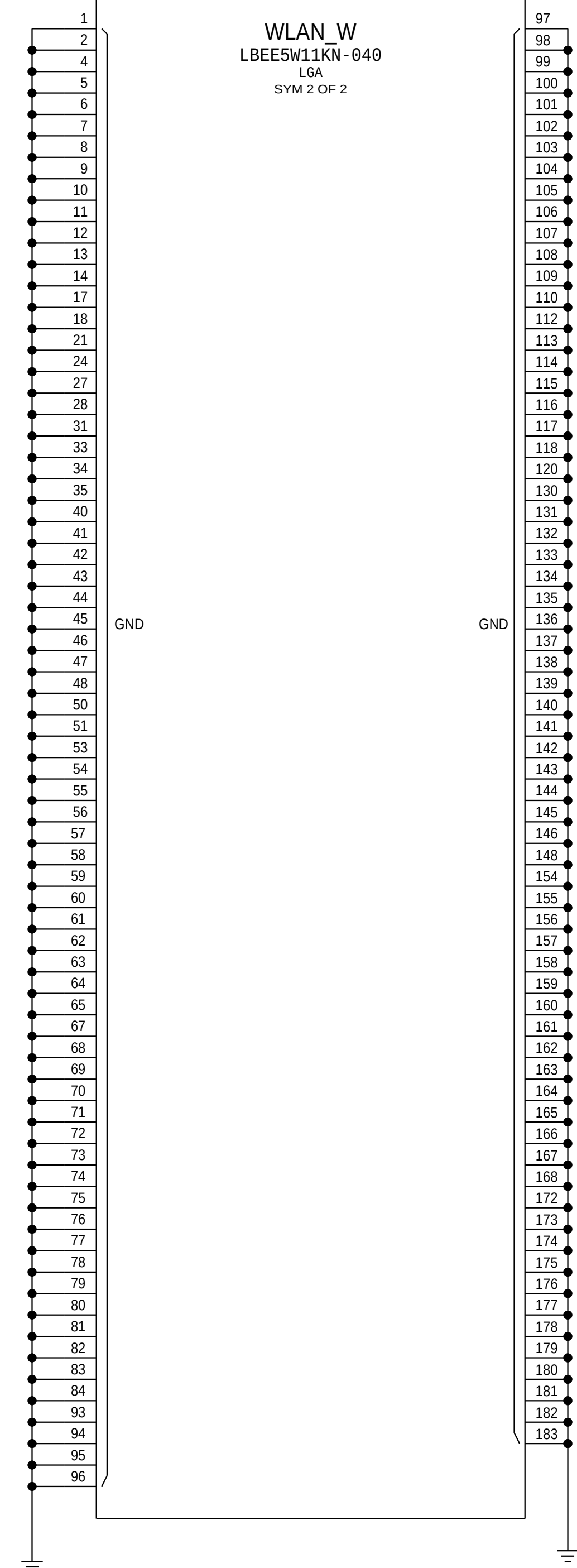
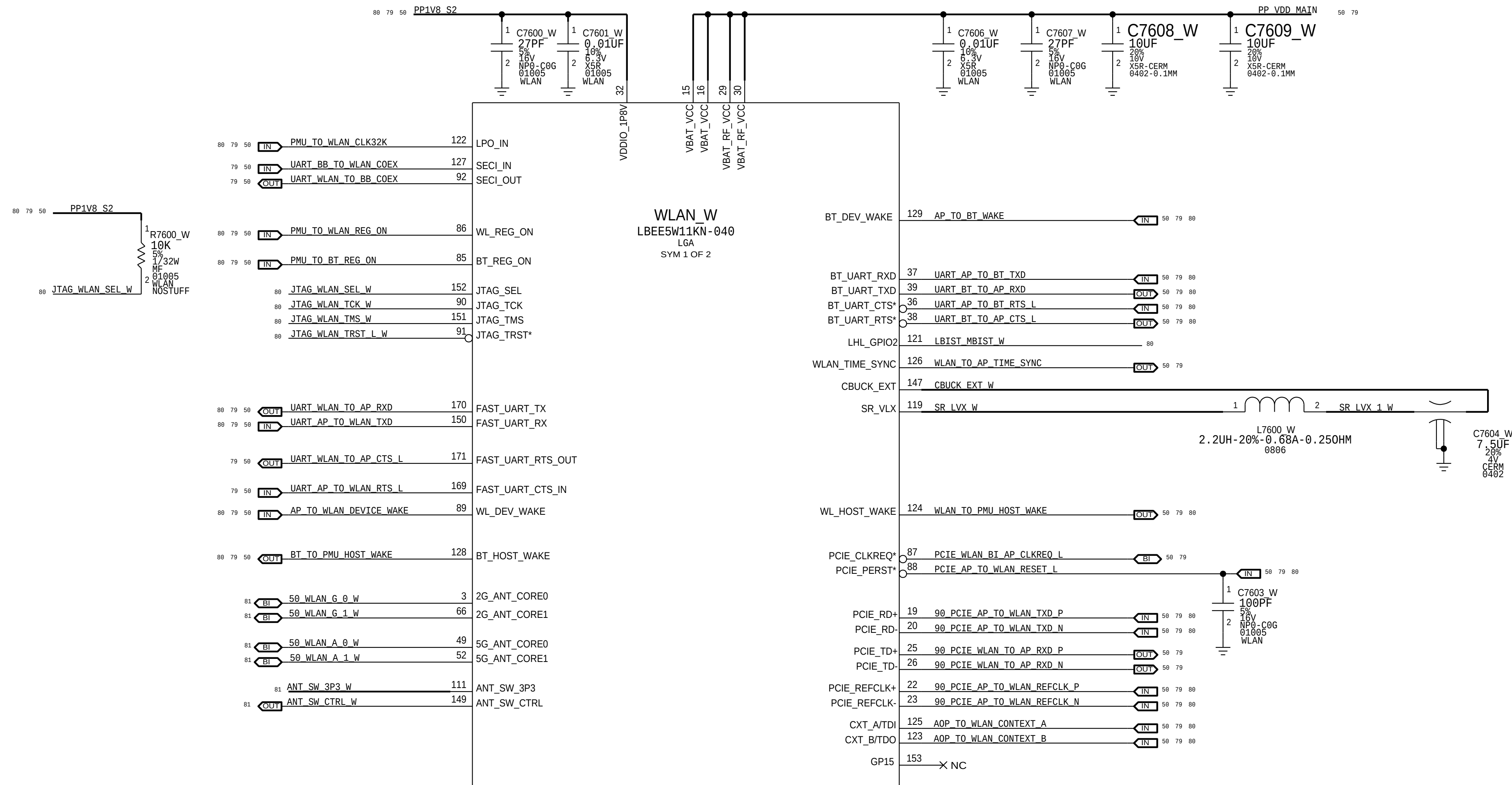
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
155S00274	1	B42/5G DIPLEXER	FL2002_EF	CRITICAL	JPN
155S00275	1	5G PASSTHROUGH	FL2002_EF	CRITICAL	ROW
131S00598	1	B42/5G DIPLEXER 5G_IN MATCH	L7751_EF	CRITICAL	JPN
131S00385	1	5G PASSTHROUGH INPUT MATCH	L7751_EF	CRITICAL	ROW



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1248	1	4.3NH 03015 WW INDUCTOR	L8805_EF	CRITICAL	JPN
152S08749	1	3.6NH 03015 WW INDUCTOR	L8805_EF	CRITICAL	ROW

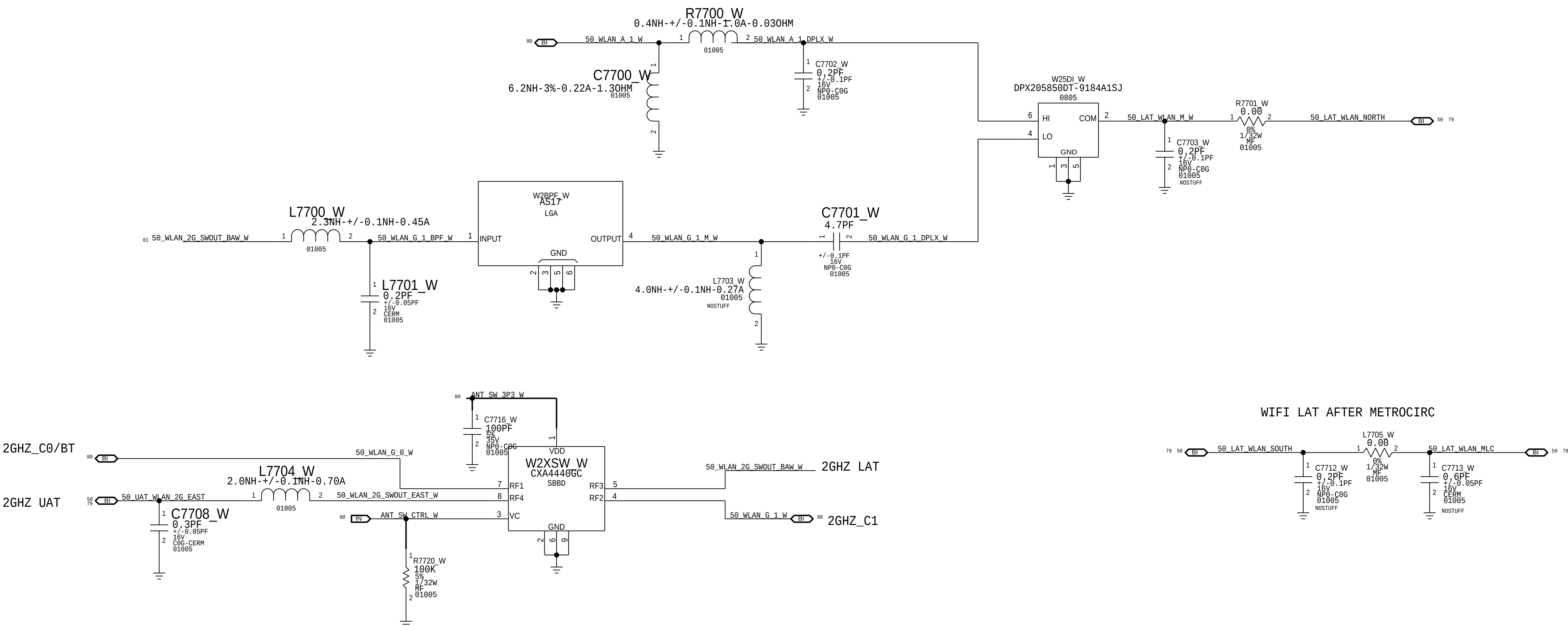


WIFI/BT

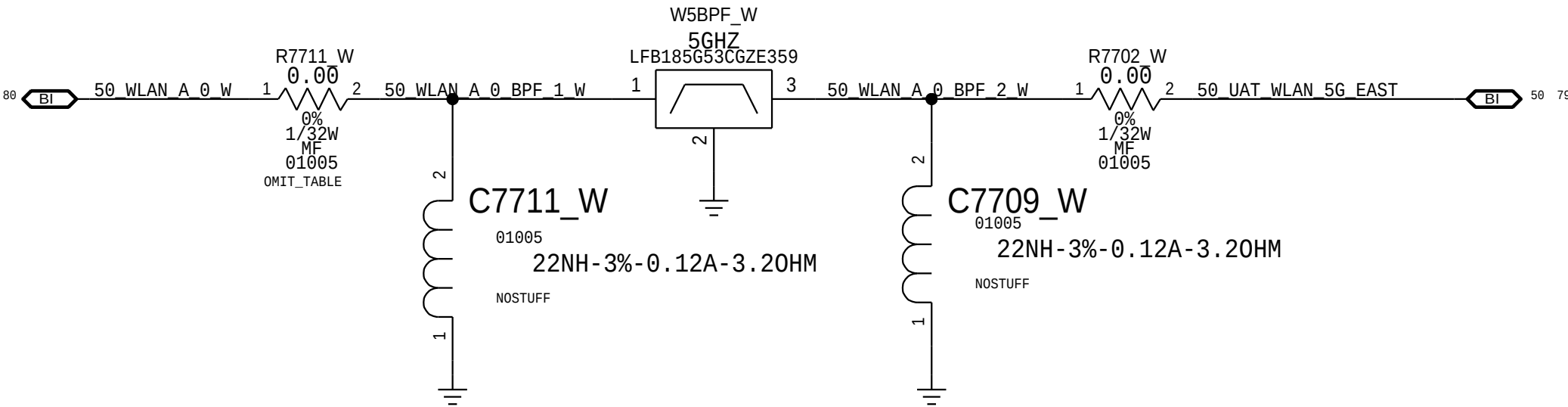


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00397	339S00399	BOM_TABLE_ALTS	WLAN_W	ALT WIFI/BT MODULE

WIFI LOWER ANTENNA FEED



WIFI UPPER ANTENNA FEED



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00498	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	D211
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	JPN
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	ROW